Thue–Morse Frontier Deductions

Boundary Corrections, Dyadic Completion and Mellin Renormalization,
Rational Resonances, Spline and Lattice Corrections, and Nonlinear Prouhet Geometry

A consolidated companion volume to *Thue–Morse Atlas and Frontiers*

Consolidated from six research reports with detailed proofs

September 4, 2026

Abstract

The signed Thue–Morse sequence $\varepsilon_n = (-1)^{s_2(n)}$ factors every block of length 2^m into m binary differences. This volume consolidates six independently written research reports that push that single fact in five directions, each beyond what the corpus atlas already records, and organizes them around one set of preliminaries: the finite product, Prouhet cancellation, the Laplace product $P(t) = \prod_{j\geq 0}(1-e^{-2^j t})$, the dyadic uniform-sum variable $X = \sum_{j\geq 1} 2^{-j}U_j$ with its entire Laplace transform Φ, the centered up-function, and the Bernoulli and Bell expansions that every part uses.

Part I isolates the entire finite-size defect of the Thue–Morse diffraction measure below the block-length cutoff as one signed Stern sequence, cancels it by a positive two-level product that is Fourier-exact through the bandwidth, identifies the universal first correction as a distribution with sharp negative-Sobolev thresholds, factors all remaining modes through a positive-coefficient function governed by two-colored binary partitions, and extends the mechanism to an amplitude family with a sharp positivity threshold; a finite boundary functional then gives the exact dyadic defect of correlations of every order. Part II completes the Laplace product by the Fabius transform, $K = P\Phi$ with $K(2t) = K(t)/t$, so that its Mellin transform obeys $C(s+1) = 2^{s+1}C(s)$; it proves a locally uniform all-orders scaling law for $\Gamma(s-m)D(s-m,a)$ with explicit remainder constants, derives horizontal strings of nonreal zeros of every shifted Thue–Morse Dirichlet function, certifies signed remainders at the half-shift through a positive Laplace representation, classifies every completely monotone solution of the dyadic equation by a scale-invariant measure, exhibits normalized gamma towers that agree beyond all orders, and characterizes the canonical tower by a single boundary slope. Part III studies the complete local profile of the Thue–Morse polynomial in a window of width 2^{-m} about every rational frequency, with closed cumulants, twisted moments, eventual recurrences, zero divisors, a Gaussian–Fabius hierarchy of coalescing-root limits, a cyclotomic aggregation identity $\prod_a \mathcal{H}_{\zeta^a} = \mathcal{U}(qz)/\mathcal{U}(z)$, and a sharp regularity classification of the resulting digit laws. Part IV proves a global all-orders C^k correction expansion for the uniform-convolution splines with exact leading constants, exact polynomial stabilization of spline values at dyadic points with a finite Richardson reconstruction, an exact lattice-to-spline differential conversion, a uniform local-limit expansion with exact dyadic stopping, and the lognormal and theta moment twins of the Fourier-energy measure. Part V transports the signs to nonlinear Boolean-cube coordinates, where an exact covering-family identity makes hafnians and matching polynomials the first surviving moments, and resolves the singular limit of the Thue–Morse-weighted geometric product $\Phi_q(z) = \prod_j (1-zq^j)^{\varepsilon_j}$ through an all-order Lambert-W saddle expansion with a periodic amplitude that is the profile of Part II, and a flat boundary layer recovering the Woods–Robbins products.

Every theorem is proved. Classical ingredients are identified as such, and no historical priority is asserted for the syntheses. The six exact and high-precision verification programs shipped with the sources are retained unchanged.

Relation to the atlas and to the six sources

The atlas [1] develops the binary recurrences, Prouhet cancellation, the dyadic finite products, power moments, Fourier and Mellin representations, rarefied sums, finite autocorrelation recurrences, the finite-block calculus of the Fabius–Rvachev–Thue–Morse system, the zeta–Lambert tail calculus, and geometric extrapolation weights. Every part of this volume starts from that material, cites it as an input, and does not present it again as a discovery. Where a part answers a question the atlas leaves open (the strong spline-correction conjecture of its Part II, the normalization question of its automatic Barnes hierarchy, Allouche's exhaustion question for the zeros of $\sum \varepsilon_n(n+1)^{-s}$), the answer is stated as a consequence of the proofs, not as a claim that the question had remained unresolved elsewhere.

The six sources. This volume replaces six reports filed together on September 4, 2026, each an extension of the atlas: an article on exact boundary corrections for Thue–Morse diffraction (Part I); two articles on the dyadic completion of the Laplace product, one emphasizing the Mellin transform, its nonreal zeros and the resulting Dirichlet zero strings, the other emphasizing the uniform remainder, the classification of completely monotone dyadic solutions and the gamma-tower uniqueness theorem (merged into Part II); an article on rational resonances and coalescing roots of the Thue–Morse polynomials (Part III); an article on new deductions from Thue–Morse cancellation for splines, lattices and Fourier energy (Part IV); and an article on boundary defects, nonlinear Prouhet geometry and flat automatic q-products, whose correlation chapter joins Part I and whose remaining chapters form Part V.

The two dyadic-completion articles built the same object under different letters; Chapter 1 fixes one notation, and the provenance appendix records the dictionary. Their shared theorems are stated once, in the form that carried the more complete proof, and the second article's strengthening (an explicit remainder constant through a tube estimate) is added as a refinement. The four other articles overlap with one another only in the preliminaries, which Chapter 1 states once with proofs; each part otherwise keeps its own notation, listed in Section 1.6, because the objects genuinely differ. Passages kept verbatim keep their internal cross-references; references to shared preliminaries resolve to Chapter 1.

The volume loads the corpus notation file `docs/fabius-notation.tex` for consistency with the other volumes and defines the short macros it uses. No Lean formalization is claimed for any result here; each part ends with the formalization route its source proposed.

Contents

Chapter 1

Common preliminaries and notation

Every part of this volume uses the same finite objects: the signed sequence, its block products, the Prouhet cancellation of moments, the Laplace product P, the dyadic uniform-sum variable X with its Laplace transform Φ, the centered up-function, and the Bernoulli and Bell expansions of the logarithms that occur. They are collected here once, with proofs, in the notation used throughout. Empty products are one, empty sums are zero, and $0^0 = 1$ in coefficient formulae. Throughout, $L = \log 2$.

1.1 The signed sequence and its finite products

Write $s_2(n)$ for the number of ones in the binary expansion of $n \geq 0$ and

$$\varepsilon_n = (-1)^{s_2(n)}, \qquad \varepsilon_0 = 1, \tag{1.1}$$

so that the first sixteen signs are $1, -1, -1, 1, -1, 1, 1, -1, -1, 1, 1, -1, 1, -1, -1, 1$. Appending a binary digit gives

$$\varepsilon_{2n} = \varepsilon_n, \qquad \varepsilon_{2n+1} = -\varepsilon_n. \tag{1.2}$$

Lemma 1.1 (Block and reflection identities)**.** *For $N = 2^m$, $q \geq 0$, and $0 \leq r < N$,*

$$\varepsilon_{qN+r} = \varepsilon_q \varepsilon_r, \qquad \varepsilon_{N-1-r} = (-1)^m \varepsilon_r. \tag{1.3}$$

If $m \geq 1$, then, with the index interpreted modulo N,

$$\varepsilon_{(r+N/2) \bmod N} = -\varepsilon_r. \tag{1.4}$$

Proof. The binary digits of qN and r occupy disjoint positions, so their digit sums add. Subtraction from $N-1$ complements exactly m digits, proving the second formula. Adding $N/2$ modulo N toggles the highest of those digits and leaves the others unchanged, proving the third. □

Lemma 1.2 (Finite products and the generating function)**.** *For $m \geq 0$ put $N = 2^m$ and*

$$P_m(z) = \sum_{n=0}^{2^m-1} \varepsilon_n z^n, \qquad P_0 = 1. \tag{1.5}$$

Then

$$P_m(z) = \prod_{j=0}^{m-1} \left(1 - z^{2^j}\right) \tag{1.6}$$

and

$$P_{m+1}(z) = (1-z)P_m(z^2) = (1-z^N)P_m(z). \tag{1.7}$$

On the open unit disk the infinite product

$$E(z) = \prod_{j=0}^{\infty} \bigl(1 - z^{2^j}\bigr) = \sum_{n=0}^{\infty} \varepsilon_n z^n \tag{1.8}$$

converges locally uniformly, has no zeros, and satisfies

$$E(z) = (1-z)E(z^2), \qquad E(z) = P_m(z)E(z^{2^m}). \tag{1.9}$$

Proof. Choosing the term $-z^{2^j}$ in a subset J of the factors of the product contributes $(-1)^{|J|} z^{\sum_{j\in J} 2^j}$, and every $n < 2^m$ has exactly one binary representation; this proves the product formula. Removing the last factor gives the second form of (1.7), and substituting (1.2) into the sum gives the first: the even coefficients of P_{m+1} are those of $P_m(z^2)$ and the odd ones are their negatives. For $|z| \le r < 1$ one has $\sum_j |z|^{2^j} < \infty$, so the tail product converges to a nonzero value, no individual factor vanishes, and the partial products converge locally uniformly; coefficient stabilization identifies the Taylor coefficients with ε_n. Removing the first factor, or the first m factors, gives (1.9). □

Lemma 1.3 (Prouhet cancellation)**.** *For every $m \ge 0$,*

$$\sum_{n<2^m} \varepsilon_n n^h = 0 \quad (0 \le h < m), \qquad \sum_{n<2^m} \varepsilon_n n^m = (-1)^m m!\, 2^{m(m-1)/2}. \tag{1.10}$$

Proof. Substitute $z = e^t$ in (1.6). Each factor has the expansion $1 - e^{2^j t} = -2^j t + O(t^2)$, so the product starts with $(-1)^m 2^{m(m-1)/2} t^m$. Differentiating $\sum_n \varepsilon_n e^{nt}$ at $t = 0$ proves both assertions. □

The argument controls the order of vanishing, not merely the value; Part III follows that order at every root of unity and through moving roots.

1.2 The Laplace product and its flatness

For $t > 0$, and more generally for $\operatorname{Re} t > 0$, put

$$P(t) = E(e^{-t}) = \prod_{j\ge 0} \bigl(1 - e^{-2^j t}\bigr) = \sum_{n\ge 0} \varepsilon_n e^{-nt}. \tag{1.11}$$

Although the sum has signs, the product is strictly positive for real $t > 0$, and removing its first factor gives

$$P(2t) = \frac{P(t)}{1 - e^{-t}}, \qquad P(t/2) = \bigl(1 - e^{-t/2}\bigr)P(t). \tag{1.12}$$

Lemma 1.4 (Bounds needed for all Mellin operations)**.** *The product P is positive and smooth on $(0,\infty)$; the product and all its differentiated logarithms converge uniformly on compact subintervals. For every integer $r \ge 0$ and every $t > 0$,*

$$0 < P(t) \le 2^{r(r-1)/2} t^r, \tag{1.13}$$

and

$$0 \le 1 - P(t) \le \frac{e^{-t}}{1 - e^{-t}}. \tag{1.14}$$

Thus P is smaller than every fixed power at zero and approaches one exponentially at infinity. For $m \ge 1$ the finite products satisfy

$$0 \le P_m(e^{-t}) \le \min\{t, 1\} \qquad (t > 0), \tag{1.15}$$

and they decrease to $P(t)$ as $m \to \infty$.

Proof. Uniform convergence on compact subintervals follows from the exponential growth of 2^j. For (1.13), bound the first r factors by $1 - e^{-2^j t} \le 2^j t$ and the remaining ones by one. For (1.14), use $1-\prod_j(1-u_j) \le \sum_j u_j$ for $0 \le u_j \le 1$ and $2^j \ge j+1$. The first factor of $P_m(e^{-t})$ is at most $\min\{t, 1\}$ and every further factor lies in $(0, 1)$, which proves (1.15) and the monotone convergence. □

The flatness bound is the analytic counterpart of the vanishing of the first m power moments in (1.10).

1.3 The dyadic uniform sum and its transforms

Let $U_1, U_2, \ldots$ be independent random variables uniformly distributed on $[0, 1]$ and define

$$X = \sum_{j=1}^{\infty} 2^{-j} U_j. \tag{1.16}$$

The series converges surely and $0 \le X \le 1$. Put

$$\psi(z) = \frac{1 - e^{-z}}{z}, \qquad \psi(0) = 1, \tag{1.17}$$

the Laplace transform of one uniform variable, and

$$\Phi(z) = \mathbb{E}e^{-zX} = \prod_{j\ge1} \psi(z/2^j) = \prod_{j\ge1} \frac{1 - e^{-z/2^j}}{z/2^j}. \tag{1.18}$$

The distribution of X is the standard dyadic uniform-sum realization underlying the Fabius function; fixing (1.16) explicitly avoids the factor-of-two ambiguities among Fabius and Rvachev conventions.

Lemma 1.5 (The Fabius transform). *(i) The product* (1.18) *converges normally on compact sets, so* Φ *is entire, and*

$$\Phi(2z) = \psi(z)\Phi(z), \qquad \Phi(z/2) = \frac{\Phi(z)}{\psi(z/2)}, \tag{1.19}$$

$$\Phi(-z) = e^z \Phi(z). \tag{1.20}$$

(ii) The law of X *is symmetric about* $1/2$*, and*

$$\mathbb{E}X = \frac{1}{2}, \qquad \mathrm{Var}(X) = \frac{1}{12} \sum_{j\ge1} 4^{-j} = \frac{1}{36}. \tag{1.21}$$

(iii) The zeros of Φ *are exactly the points*

$$z = 4\pi \mathrm{i} n, \qquad n \in \mathbb{Z} \setminus \{0\}, \quad \textit{with multiplicity } 1 + \nu_2(n), \tag{1.22}$$

where ν_2 *is the* 2*-adic valuation; the nearest zeros* $\pm 4\pi\mathrm{i}$ *are simple.*

(iv) If $F(x) = \mathbb{P}(X \le x)$ *is extended by zero for* $x \le 0$ *and by one for* $x \ge 1$*, then* X *has a continuous density* F' *supported on* $[0, 1]$*,*

$$F'(x) = 2\big(F(2x) - F(2x - 1)\big), \qquad F'(x) = 2F(2x) \ \ (0 < x < 1/2), \tag{1.23}$$

and that density is C^∞*.*

(v) For real $x > 0$ *and every* $0 < \delta < 1$*,*

$$\Phi(x) \ge \mathbb{P}(X \le \delta)\, e^{-\delta x}, \qquad \mathbb{P}(X \le \delta) > 0. \tag{1.24}$$

Proof. (i) Each factor satisfies $\psi(z/2^j) = 1 + O_K(2^{-j})$ on a compact set K, which gives normal convergence and entireness; boundedness of X gives the same conclusion directly. The first dilation law is an index shift, and the second is its rearrangement. The reflection law follows from $X \overset{d}{=} 1 - X$, obtained by replacing every U_j by $1 - U_j$, or directly from $\psi(-z) = e^z\psi(z)$.

(ii) The symmetry was just noted; the mean and the variance follow from independence and $\operatorname{Var}(U_j) = 1/12$.

(iii) The jth factor $\psi(z/2^j)$ vanishes exactly at $z = 2^{j+1}\pi i k$, $k \neq 0$, simply. Away from these points the logarithmic tail converges normally, so the product is nonzero there. At $z = 4\pi i n$ with $n = 2^v u$, u odd, precisely the factors $j = 1, \dots, v+1$ vanish.

(iv) In distribution $X = (U + X')/2$ with U uniform and X' an independent copy of X. Conditioning on X', or convolving with the uniform density, gives the density $2\mathbb{P}(2x - 1 \le X' \le 2x) = 2(F(2x) - F(2x-1))$; it is continuous because X has no atoms, containing an independent continuous summand. For every $N \ge 2$ the sum of the first N terms of (1.16) has a C^{N-2} spline density, and convolution with the law of the remaining terms preserves that regularity; since N is arbitrary the density is smooth. Alternatively, for each fixed N, $|\Phi(it)| \le \prod_{k=1}^N \min\{1, 2^{k+1}/|t|\} = O_N(|t|^{-N})$, so Fourier inversion can be differentiated any prescribed number of times.

(v) Choose J with $\sum_{j>J} 2^{-j} \le \delta/2$ and restrict the finitely many U_j, $j \le J$, to a small interval near zero; this event has positive probability and forces $X \le \delta$. Then $\Phi(x) \ge \mathbb{E}[e^{-xX}\mathbf{1}_{X\le\delta}] \ge \mathbb{P}(X \le \delta)e^{-\delta x}$. □

Lemma 1.6 (Bernoulli expansions). *Let the Bernoulli numbers be fixed by $z/(e^z-1) = \sum_{n\ge0} B_n z^n/n!$, so $B_1 = -1/2$, $B_2 = 1/6$, $B_4 = -1/30$. With the logarithms that vanish at zero,*

$$\log\psi(z) = -\frac{z}{2} + \sum_{j\ge1} \frac{B_{2j}}{2j\,(2j)!} z^{2j} \qquad (|z| < 2\pi), \tag{1.25}$$

$$\log\Phi(z) = -\frac{z}{2} + \sum_{j\ge1} b_j z^{2j}, \qquad b_j = \frac{B_{2j}}{2j\,(2j)!\,(4^j - 1)} \qquad (|z| < 4\pi), \tag{1.26}$$

and in particular

$$\log\Phi(x) = -\frac{x}{2} + \frac{x^2}{72} - \frac{x^4}{43200} + O(x^6). \tag{1.27}$$

Both radii are exact: 2π is the distance to the first zero of ψ and 4π to the first zero of Φ. Consequently, for $\operatorname{Re} z > 0$,

$$\log\Phi(z) = -\frac{z}{2} + \sum_{j\ge1} b_j z^{2j} \quad \textit{and} \quad \frac{\mathrm{d}}{\mathrm{d}z}\log\psi(z) = \frac{1}{e^z - 1} - \frac{1}{z} = -\frac{1}{2} + \sum_{j\ge1} \frac{B_{2j}}{(2j)!} z^{2j-1}. \tag{1.28}$$

Proof. The logarithmic derivative of ψ is $1/(e^z-1) - 1/z$, whose expansion is the Bernoulli generating function with its first term removed; integrating from zero gives (1.25). Substitute $z/2^\ell$ and sum over $\ell \ge 1$: the linear terms sum with $\sum_\ell 2^{-\ell} = 1$, and the even terms with $\sum_\ell 2^{-2j\ell} = 1/(4^j - 1)$. Absolute convergence for $|z| < 4\pi$ permits the interchange, and the nearest zero of Φ fixes the radius. Inserting B_2 and B_4 gives (1.27). □

1.4 The centered variable and the up-function

Let $V_j = 2U_j - 1$, uniform on $[-1, 1]$, and

$$Y = 2X - 1 = \sum_{j=1}^{\infty} 2^{-j} V_j, \qquad Y_m = \sum_{j=1}^{m} 2^{-j} V_j, \qquad h = 2^{-m}. \tag{1.29}$$

Write u for the density of Y and u_m for the density of Y_m. Then u is the Rvachev up-function in the normalization $\operatorname{supp} u = [-1, 1]$, $u(0) = 1$; it is even, smooth, $0 \le u \le 1$, and $u(x) = \frac{1}{2}F'((x+1)/2)$

in terms of the Fabius distribution function of Theorem 1.5. For $m \geq 2$, u_m is a C^{m-2} piecewise polynomial of degree at most $m-1$ supported on $[-1+h, 1-h]$. The moment generating function and the characteristic function of Y are

$$M(z) = \mathbb{E}e^{zY} = e^{-z}\Phi(-2z) = \prod_{j\geq 1} \frac{\sinh(2^{-j}z)}{2^{-j}z}, \tag{1.30}$$

$$\widehat{u}(t) = \mathbb{E}e^{\mathrm{i}tY} = \int_{\mathbb{R}} e^{\mathrm{i}tx} u(x)\,\mathrm{d}x = \prod_{j\geq 1} \operatorname{sinc}(2^{-j}t), \qquad \operatorname{sinc} t = \frac{\sin t}{t},\ \operatorname{sinc} 0 = 1. \tag{1.31}$$

Part IV develops the spline theory of u_m; Part III uses $\mathcal{U}(z) = \Phi(-z) = \mathbb{E}e^{zX}$; Part V uses Φ directly. In the corpus's other volumes the same variable appears as the geometric uniform convolution at base $1/2$.

1.5 Bell polynomials and partition sums

The complete exponential Bell polynomials are the finite sums

$$\mathcal{B}_h(x_1, \ldots, x_h) = h! \sum_{\substack{\nu_1,\ldots,\nu_h \geq 0 \\ \nu_1+2\nu_2+\cdots+h\nu_h = h}} \prod_{j=1}^{h} \frac{1}{\nu_j!} \left(\frac{x_j}{j!}\right)^{\nu_j}, \qquad \mathcal{B}_0 = 1, \tag{1.32}$$

characterized by

$$\exp\left(\sum_{j\geq 1} x_j \frac{z^j}{j!}\right) = \sum_{h\geq 0} \mathcal{B}_h(x_1, \ldots, x_h) \frac{z^h}{h!}. \tag{1.33}$$

For a sequence $v_1, v_2, \ldots$ the partition polynomial

$$\mathcal{E}_j(v) = \sum_{\substack{k_1,\ldots,k_j \geq 0 \\ \sum_{r=1}^{j} r k_r = j}} \prod_{r=1}^{j} \frac{v_r^{k_r}}{k_r!}, \qquad \mathcal{E}_0(v) = 1, \tag{1.34}$$

is characterized by $\exp(\sum_{r\geq 1} v_r z^r) = \sum_{j\geq 0} \mathcal{E}_j(v) z^j$; thus $\mathcal{E}_j(v) = \mathcal{B}_j(1!\,v_1, 2!\,v_2, \ldots)/j!$. Both are obtained by expanding an exponential coefficient by coefficient, so every coefficient formula in this volume is a finite sum and not a hidden recurrence.

1.6 Notation across the parts

Each part keeps the notation of its source where the objects genuinely differ; the shared preliminaries are always denoted as above. The following table records the part-specific letters that a reader may encounter more than once with different meanings.

Part	Notation particular to the part
I	μ_m, μ the finite and limiting diffraction measures on $\mathbb{T}=\mathbb{R}/\mathbb{Z}$; $r_m(k)$, $\eta(k)$ their Fourier coefficients; $g(k)=B_k(-2)$ the signed Stern sequence; $\mathcal{G}$, $\mathcal{G}_m$ the one-sided autocorrelation generating functions; $\mathcal{K}$ the analytic defect; $\mathcal{Q}(z)=zE(z)^2$; ν the universal correction distribution; σ, s_* the Sobolev thresholds. For correlations of every order: $\boldsymbol{h}$ a shift tuple, $D(\boldsymbol{h})$ the boundary functional, S_m, R_m noncyclic and cyclic block sums, $\sigma(Q)$ the alternating digit sum.
II	$K=P\Phi$ the completed product; $C(s)$ its Mellin transform, $C_0=C(0)$; $Q(s)=C(s)/(C_0 2^{s(s+1)/2})$ the periodic factor; $H(v)$ the logarithmic profile of K and h_k its Fourier coefficients; $\Theta(\rho)=C_0Q(-\rho)$; $D(s,a)$ the shifted Dirichlet function and $M(s,a)$ its Mellin numerator; $Z_\rho(a)=M(-\rho,a)$; $B_a(z)=e^{-az}/\Phi(z)$ with coefficients $\beta_j(a)$ and $\alpha_j(a)=2^{j(j+1)/2}\beta_j(a)$; $\Psi=B_{1/2}$, W, $\mathcal{H}$, c_r, A_r the centered objects; ϑ the lattice theta sum.
III	$\mathcal{U}(z)=\Phi(-z)$ the Fabius transform and $g(z)=\psi(-z)$; $\mathcal{H}_{\zeta,s}$ the rational profiles; $\mathcal{Q}_{m,\zeta}$ the finite profiles; $\mathcal{N}_{r,M}$ the zero-removed dyadic profiles; $\mathcal{R}_q=\mathcal{U}(qz)/\mathcal{U}(z)$; $L=\operatorname{ord}_q 2$ (not $\log 2$).
IV	u, u_m the up-function and its splines; $\widehat{u}$ the sinc product; a_j, b_j, $\gamma_j(m)$, $H_j(m)$ the four coefficient families; $K_r=2^{r(r+1)/2}$ the derivative norms; $c_m(k)$, ρ_m the repeated-prefix array; ν the Fourier-energy measure; L a lognormal variable.
V	$Z_x(t)$ the signed exponential sum on a nonlinearly positioned Boolean cube; $\mathcal{H}$, τ, Haf, $\mathcal{M}_G$; $\Phi_q(z)$ the Thue–Morse-weighted geometric product and $F(t,z)=-\log\Phi_{e^{-t}}(z)$; $a=\log 2$; $\mathsf{L}=\log(1/t)$; $\Psi(v)=H(-av)$ the periodic amplitude; w, r_*, θ, $\Pi(t,z)$, $c_n(\theta)$ the saddle data; $\mathcal{H}(\gamma)$, $\mathcal{P}(\gamma)$ the boundary-layer profile.

Part I

Finite diffraction and exact boundary corrections

Chapter 2

Normalizations and the finite diffraction measures

The object of this part is a measure on the frequency circle: the normalized squared modulus of the length-2^m Thue–Morse polynomial. The atlas's finite autocorrelation recurrences make one question natural: can the complete finite-block defect be isolated, rather than bounded, and then removed without losing positivity? The answer given here identifies the entire error below the block-length cutoff with one signed Stern sequence, and then identifies the topology in which the isolated defect has a limit. The Riesz product and the finite and infinite Fourier recurrences are established background [13]; the polynomial family $B_k(t)$ is the Stern polynomial family of Klavžar, Milutinović and Petr [25]; the eigenvalue $1+\sqrt{17}$ and the correlation dimension are due to Zaks, Pikovsky and Kurths [30], the even moments of the generating products to Mauduit, Montgomery and Rivat [27], and the explicit square-correlation sum also appears in Coons, Mazáč, Pincus–Kazmar and Stout [16]; general correlation background is developed by Baake and Coons [12]. What is proposed as new is the joint package: the exact Stern boundary term for finite diffraction and forward correlations, its cancellation by a positive two-level product, the universal non-measure correction with sharp Sobolev convergence and saturation, the analytic defect factorization with its positive binary-partition coefficients, the amplitude family's sharp positivity threshold, and, in the last chapter of this part, the finite boundary functional that gives the exact dyadic defect of correlations of every order. The signs, block products and the generating function E are those of Chapter 1, with $N = 2^m$ throughout.

2.1 Fourier conventions

We use $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ and

$$\widehat{u}(k) = \langle u, \mathrm{e}^{-2\pi \mathrm{i} kx} \rangle, \qquad k \in \mathbb{Z}. \tag{2.1}$$

For a measure this means integration; for a periodic distribution it means its usual pairing with a smooth function. All the measures considered here are real and even, so their Fourier coefficients are real and even. Define

$$f_m(x) = \frac{1}{N} \left| P_m(\mathrm{e}^{2\pi \mathrm{i} x}) \right|^2 = \prod_{j=0}^{m-1} (1 - \cos(2\pi 2^j x)), \qquad \mu_m = f_m(x)\,\mathrm{d}x. \tag{2.2}$$

Orthogonality of the exponentials gives $\int f_m = 1$. Thus μ_m is a probability measure. Its Fourier coefficients are

$$r_m(k) := \widehat{\mu}_m(k) = \begin{cases} \dfrac{1}{N} \displaystyle\sum_{j=0}^{N-k-1} \varepsilon_j \varepsilon_{j+k}, & 0 \le k < N, \\ 0, & k \ge N, \end{cases} \qquad r_m(-k) = r_m(k). \tag{2.3}$$

In particular, the coefficient at the endpoint $k = N$ is zero. Keeping this endpoint in later statements will be useful.

Lemma 2.1 (Finite Fourier recursion)**.** *For $m \geq 0$ and $r \geq 0$,*

$$r_{m+1}(2r) = r_m(r), \qquad r_{m+1}(2r+1) = -\frac{1}{2}\big(r_m(r) + r_m(r+1)\big). \tag{2.4}$$

The initial sequence is $r_0(k) = \mathbf{1}_{k=0}$.

Proof. Equation (1.7) gives $f_{m+1}(x) = (1 - \cos(2\pi x)) f_m(2x)$. The Fourier frequencies of $f_m(2x)$ are even. Multiplication by $1 - (\mathrm{e}^{2\pi \mathrm{i} x} + \mathrm{e}^{-2\pi \mathrm{i} x})/2$ leaves even coefficients unchanged and creates each odd coefficient from its two neighbors. This is exactly (2.4). □

Proposition 2.2 (The limiting measure)**.** *The measures μ_m converge weakly to a unique probability measure μ. Writing $\eta(k) = \widehat{\mu}(k)$ for $k \geq 0$, its coefficients satisfy*

$$\eta(0) = 1, \qquad \eta(1) = -\frac{1}{3}, \tag{2.5}$$

$$\eta(2r) = \eta(r), \qquad \eta(2r+1) = -\frac{1}{2}\big(\eta(r) + \eta(r+1)\big). \tag{2.6}$$

They are rational and obey $|\eta(k)| \leq 1/3$ for $k \geq 1$.

Proof. The recursion at $k = 1$ is $r_{m+1}(1) = -(1 + r_m(1))/2$, whence

$$r_m(1) = -\frac{1}{3} + \frac{1}{3}\left(-\frac{1}{2}\right)^m. \tag{2.7}$$

For every $k \geq 2$, the right sides in (2.4) involve smaller nonnegative indices. Induction on k therefore proves convergence of every fixed coefficient, with limits (2.5)–(2.6). Probabilities on the compact circle have weakly convergent subsequences. Every subsequential limit has these same Fourier coefficients. Trigonometric polynomials are uniformly dense in continuous functions, so those coefficients determine the limit uniquely and the entire sequence converges. Rationality and the bound follow by induction: for $k \geq 2$ each recursion either copies one previously bounded coefficient or takes minus the average of two such coefficients. □

The measure μ is the classical singular-continuous Thue–Morse diffraction measure [13]. The results below require the elementary characterization just proved; singular continuity is only used when contrasting weak convergence with total-variation convergence.

Chapter 3

The signed Stern sequence hidden in the boundary

Definition 3.1. Set $g(0) = 0$, and for $k \geq 1$ set

$$g(k) = \sum_{j=0}^{k-1} \varepsilon_j \varepsilon_{k-1-j}. \tag{3.1}$$

The shift by one makes the binary recursion particularly simple. Ordinary convolution of the series (1.8) gives

$$\mathcal{Q}(z) := \sum_{k \geq 1} g(k) z^k = zE(z)^2. \tag{3.2}$$

Lemma 3.2 (Stern recursion and bounds)**.** *The sequence in* (3.1) *satisfies*

$$g(1) = 1, \qquad g(2r) = -2g(r), \qquad g(2r+1) = g(r) + g(r+1). \tag{3.3}$$

The last recursion is an identity at $r = 0$ and a descending computation for $r \geq 1$. Moreover, for $k \geq 1$,

$$|g(k)| \leq k, \qquad g(2^j) = (-2)^j, \qquad \nu_2(g(k)) = \nu_2(k). \tag{3.4}$$

Here ν_2 of a nonzero negative integer means the valuation of its absolute value.

Proof. Let $c_n = \sum_{j=0}^{n} \varepsilon_j \varepsilon_{n-j}$, so $g(k) = c_{k-1}$. For an odd index $2r-1$, each term has one even and one odd subscript. Splitting by parity yields $c_{2r-1} = -2c_{r-1}$. For an even index $2r$, the even-even terms sum to c_r, and the odd-odd terms sum to c_{r-1} (with $c_{-1} = 0$). Thus $c_{2r} = c_r + c_{r-1}$, proving (3.3).

The bound follows because (3.1) is a sum of k signs. Iteration of the even recursion proves the power-of-two value. To prove the valuation statement, first show $g(k) \equiv k \pmod 2$ by induction: even indices give an even value, while at an odd index the two smaller adjacent indices have opposite parity. Hence every odd-index value is odd. Writing $k = 2^j u$ with u odd now gives $g(k) = (-2)^j g(u)$ and the exact valuation. $\square$

The Stern polynomials [25] are defined by $B_0(t) = 0$, $B_1(t) = 1$, $B_{2r}(t) = tB_r(t)$, and $B_{2r+1}(t) = B_r(t) + B_{r+1}(t)$. Therefore

$$g(k) = B_k(-2). \tag{3.5}$$

A useful binary evaluation scheme keeps two adjacent values. For $v(k) = (g(k), g(k+1))^{\mathsf{T}}$,

$$v(2k) = \begin{pmatrix} -2 & 0 \\ 1 & 1 \end{pmatrix} v(k), \qquad v(2k+1) = \begin{pmatrix} 1 & 1 \\ 0 & -2 \end{pmatrix} v(k), \qquad v(0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{3.6}$$

Thus a single value can be computed using a number of small matrix updates linear in the bit length of k.

k	1	2	3	4	5	6	7	8
$g(k)$	1	-2	-1	4	-3	2	3	-8
$3\eta(k)$	-1	-1	1	-1	0	1	0	-1
k	9	10	11	12	13	14	15	16
$g(k)$	1	6	-1	-4	5	-6	-5	16
$3\eta(k)$	$1/2$	0	$-1/2$	1	$-1/2$	0	$1/2$	-1

Chapter 4

An exact finite-block error, including the endpoint

Theorem 4.1 (Exact Stern boundary correction)**.** *For $m \geq 0$, $N = 2^m$, and $0 \leq k \leq N$,*

$$\boxed{r_m(k) = \eta(k) + \frac{(-1)^m}{3N} g(k).} \tag{4.1}$$

In particular,

$$|r_m(k) - \eta(k)| \leq \frac{k}{3N}, \tag{4.2}$$

with equality whenever k is a power of two in this interval.

Proof. At $m = 0$, the only indices are 0 and 1. Their identities are respectively $1 = 1$ and $0 = -1/3 + 1/3$. Suppose the theorem holds at level m. For an even index $2r \leq 2N$, the difference at the next level equals the old difference at r. Since

$$\frac{(-1)^{m+1}}{6N} g(2r) = \frac{(-1)^m}{3N} g(r),$$

the claimed formula follows. For an odd index $2r + 1 \leq 2N$, both r and $r + 1$ lie in $[0, N]$. Subtracting (2.6) from (2.4) gives

$$-\frac{1}{2} \frac{(-1)^m}{3N} \big(g(r) + g(r+1)\big) = \frac{(-1)^{m+1}}{6N} g(2r+1).$$

This completes the induction, including the zero coefficient at the moving endpoint. Equation (3.4) proves the bound and its sharpness. □

Corollary 4.2 (No nonzero stabilized harmonic is accidentally exact)**.** *For every $k \geq 1$ and every m with $2^m \geq k$, $r_m(k) \neq \eta(k)$. For fixed k, the error changes sign with m and has exactly half the magnitude at the next level. Its rational two-adic valuation is $\nu_2(k) - m$.*

Proof. Use $g(k) \neq 0$ and its valuation in (4.1). The denominator factor 3 is odd. □

The conclusion is about each fixed coefficient, not uniform convergence on the entire expanding band: at $k = N$, the error has magnitude $1/3$ for every m. The motion of the cutoff is exactly why function-space estimates later have a smoothness threshold.

4.1 Aperiodic versus forward correlations

The normalized coefficients r_m use a truncated sum, because both indices have to remain inside the polynomial block. The atlas also considers a forward sum whose second index may cross the right boundary. Define

$$C_m(k) = \sum_{j=0}^{N-k-1} \varepsilon_j \varepsilon_{j+k}, \qquad A_m(k) = \sum_{j=0}^{N-1} \varepsilon_j \varepsilon_{j+k}, \qquad 0 \le k \le N. \tag{4.3}$$

The sum defining $C_m(N)$ is empty.

Theorem 4.3 (Both finite-window conventions)**.** *For* $0 \le k \le N$*,*

$$C_m(k) = N\eta(k) + \frac{(-1)^m}{3} g(k), \tag{4.4}$$

$$A_m(k) = N\eta(k) - \frac{2(-1)^m}{3} g(k). \tag{4.5}$$

Proof. The first formula is Theorem 4.1 multiplied by N. For the k terms crossing the right boundary, write $j = N - k + r$, $0 \le r < k$. Binary complement inside an m-bit block gives $\varepsilon_{N-1-u} = (-1)^m \varepsilon_u$, and concatenation gives $\varepsilon_{N+r} = -\varepsilon_r$. Consequently

$$A_m(k) - C_m(k) = -(-1)^m \sum_{r=0}^{k-1} \varepsilon_{k-1-r} \varepsilon_r = -(-1)^m g(k).$$

Combining the two identities proves (4.5). □

This theorem upgrades the finite autocorrelation recurrences to a closed finite-size expression valid simultaneously for every shift through the block length. In particular, the different constants $1/3$ and $-2/3$ are not conflicting normalizations: they correspond to distinct window conventions.

Example 4.4 (A length-four block)**.** For $m = 2$, the block is $(1, -1, -1, 1)$ and

$$(r_2(0), \dots, r_2(4)) = (1, -1/4, -1/2, 1/4, 0).$$

Subtracting $(\eta(0), \dots, \eta(4))$ gives $(0, 1, -2, -1, 4)/12$, exactly the first five Stern values divided by $3N$.

Chapter 5

Positive extrapolation with an exact Fourier band

Theorem 5.1 (The two-level positive correction)**.** *For $m \geq 0$, define*

$$\widetilde{\mu}_m = \frac{1}{3}\mu_m + \frac{2}{3}\mu_{m+1}. \tag{5.1}$$

It is a probability measure with nonnegative trigonometric-polynomial density

$$\boxed{\widetilde{f}_m(x) = f_m(x)\left(1 - \frac{2}{3}\cos(2\pi N x)\right), \quad N = 2^m.} \tag{5.2}$$

Its Fourier coefficients satisfy

$$\widehat{\widetilde{\mu}_m}(k) = \eta(k) \qquad (|k| \leq N). \tag{5.3}$$

Among affine combinations $w\mu_m + (1-w)\mu_{m+1}$, it is the unique one agreeing with the limit at $k = 1$. Its density has degree at most $2N - 1$.

Proof. The coefficient error at level $m + 1$ is $-1/2$ times the error at level m on the common band. The weights $1/3$ and $2/3$ cancel it exactly. The error at $k = 1$ is nonzero; hence any cancelling affine weight must solve $w - (1 - w)/2 = 0$, which gives $w = 1/3$. The second factorization in (1.7) yields $f_{m+1} = f_m(1 - \cos(2\pi N x))$, proving (5.2). Its new factor lies between $1/3$ and $5/3$. Positivity and total mass one also follow directly from the convex combination. The degree bound is the sum of $N - 1$ and N. □

For every trigonometric polynomial ϕ of degree at most N, this gives an exact integration identity

$$\int_{\mathbb{T}} \phi \, \mathrm{d}\mu = \int_0^1 \phi(x)\widetilde{f}_m(x) \, \mathrm{d}x. \tag{5.4}$$

Thus a singular measure has a positive, explicitly factored density representing its entire low-frequency moment vector. This is not an assertion that the two measures agree on arbitrary sets or arbitrary continuous functions.

5.1 Smooth and analytic observables

For $r \geq 0$, write

$$\mathcal{A}_r(\phi) = \sum_{k \in \mathbb{Z}\setminus\{0\}} |k|^r |\widehat{\phi}(k)|. \tag{5.5}$$

Every smooth periodic function has finite $\mathcal{A}_r$ for every fixed r: repeated integration by parts bounds its Fourier coefficients by any desired power.

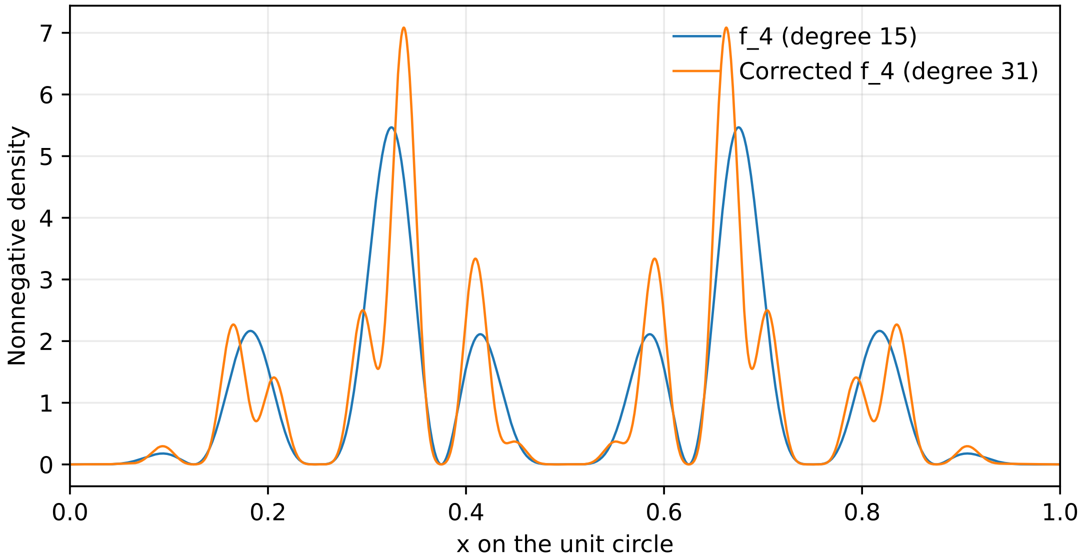

Figure 5.1: The raw and corrected densities at $m = 4$. Both are nonnegative. The corrected curve reproduces the limiting measure's Fourier coefficients through $|k| = 16$ exactly. The figure makes no assertion of pointwise convergence to a density of the limit.

Corollary 5.2 (Tail-only integration error)**.** *For an absolutely convergent Fourier series,*

$$|\langle \widetilde{\mu}_m - \mu, \phi \rangle| \leq 2 \sum_{|k|>N} |\widehat{\phi}(k)|. \tag{5.6}$$

Consequently, whenever $\mathcal{A}_r(\phi) < \infty$,

$$|\langle \widetilde{\mu}_m - \mu, \phi \rangle| \leq 2N^{-r}\mathcal{A}_r(\phi). \tag{5.7}$$

If $|\widehat{\phi}(k)| \leq M\rho^{|k|}$ for $0 < \rho < 1$, then

$$|\langle \widetilde{\mu}_m - \mu, \phi \rangle| \leq \frac{4M\rho^{N+1}}{1-\rho}. \tag{5.8}$$

Proof. Termwise pairing is justified by absolute convergence and finite total variation. Every coefficient on the low band vanishes by (5.3); every other coefficient of the difference has magnitude at most two because both measures are probabilities. Summing the Fourier tail gives (5.6); comparison with $|k|^r/N^r$ and a geometric series gives the other two statements. □

For a fixed C^∞ observable, the corrected error decays faster than every power of N^{-1}. For a fixed analytic strip it decays exponentially in N. The constants depend on the observable or the strip, so these statements must not be applied uniformly to probes whose frequencies grow with N.

Remark 5.3 (Why positivity is a genuine constraint). Ordinary Richardson extrapolation often uses a negative weight. Here the raw error alternates in m, so the extrapolation weights are both positive. It is therefore exact on an expanding spectral band without sacrificing the measure interpretation. This does not imply convergence in total variation: since μ is singular and each $\widetilde{\mu}_m$ is absolutely continuous, their total-variation norm distance is two, or one under the convention $\sup_A |\widetilde{\mu}_m(A) - \mu(A)|$.

Chapter 6

The universal first correction is a distribution, not a measure

Definition 6.1. Let ν be the real even periodic distribution specified by

$$\widehat{\nu}(0) = 0, \qquad \widehat{\nu}(k) = \frac{1}{3} g(|k|) \quad (k \neq 0). \tag{6.1}$$

The bound $|g(k)| \leq k$ makes this definition legitimate: pairing its Fourier coefficients with those of a smooth function is absolutely convergent and continuous in an appropriate smooth seminorm.

Theorem 6.2 (One algebraic correction, with a rapidly small weak remainder)**.** *There are distributions R_m such that*

$$\mu_m = \mu + \frac{(-1)^m}{N} \nu + R_m, \qquad \widehat{R}_m(k) = 0 \quad (|k| \leq N). \tag{6.2}$$

For $r \geq 1$ and $\mathcal{A}_r(\phi) < \infty$,

$$|\langle R_m, \phi \rangle| \leq \frac{4}{3} N^{-r} \mathcal{A}_r(\phi). \tag{6.3}$$

In particular $(-1)^m N(\mu_m - \mu) \to \nu$ in the space of periodic distributions. The distribution ν is not a finite signed measure.

Proof. Define R_m by (6.2). The low coefficients vanish by Theorem 4.1. For $|k| > N$ the finite product has no coefficient, so

$$\widehat{R}_m(k) = -\eta(|k|) - \frac{(-1)^m}{3N} g(|k|).$$

Its magnitude is at most $1 + |k|/(3N) \leq 4|k|/(3N)$. Hence

$$|\langle R_m, \phi \rangle| \leq \frac{4}{3N} \sum_{|k| > N} |k| |\widehat{\phi}(k)| \leq \frac{4}{3} N^{-r} \mathcal{A}_r(\phi),$$

where $|k| \leq N^{1-r} |k|^r$ on this tail. Taking any fixed $r > 1$ proves the renormalized distributional convergence. Finally $|\widehat{\nu}(2^j)| = 2^j/3 \to \infty$. Fourier coefficients of a finite signed measure are bounded by its total variation, so ν cannot be one. $\square$

For each fixed smooth observable the expansion has just one nonzero algebraic finite-size term. All subsequent powers can be assigned zero coefficients, with a remainder smaller than every prescribed power. This is a weak statement, not a pointwise expansion of f_m. The negative Sobolev spaces in Section 10 quantify exactly how much smoothness is required to promote it to norm convergence.

Chapter 7

An exact analytic factorization of all remaining modes

7.1 The affine Mahler equation and its homogeneous defect

Define the one-sided Fourier generating functions

$$\mathcal{G}(z) = \sum_{k=0}^{\infty} \eta(k) z^k, \qquad \mathcal{G}_m(z) = \sum_{k=0}^{N-1} r_m(k) z^k, \qquad |z| < 1. \tag{7.1}$$

These are not the same functions as E and P_m: E and P_m encode the sign sequence, while $\mathcal{G}$ and $\mathcal{G}_m$ encode autocorrelations.

Lemma 7.1 (Affine functional equations)**.** *Writing* $A(z) = 1 - (z + z^{-1})/2 = -(1-z)^2/(2z)$, *one has*

$$\mathcal{G}(z) = A(z)\mathcal{G}(z^2) + \frac{1}{2z}, \qquad \mathcal{G}_{m+1}(z) = A(z)\mathcal{G}_m(z^2) + \frac{1}{2z}. \tag{7.2}$$

The singularities displayed at zero are removable.

Proof. The even part of either series is the previous series at z^2. The odd part is

$$-\frac{z}{2}\left(\mathcal{G}(z^2) + \frac{\mathcal{G}(z^2) - 1}{z^2}\right)$$

for the limiting series, and the identical expression with $\mathcal{G}_m$ for a finite one. Adding the even part proves the equations. The constant coefficient is one on both sides, which also exhibits the cancellation at zero. □

Theorem 7.2 (Exact analytic defect)**.** *The function*

$$\mathcal{K}(w) = \frac{3(1 - \mathcal{G}(w))}{wE(w)^2}, \qquad \mathcal{K}(0) = 1, \tag{7.3}$$

is analytic on $\mathbb{D}$*. For every* $m \geq 0$ *and* $z \in \mathbb{D}$*,*

$$\boxed{\mathcal{G}_m(z) - \mathcal{G}(z) = \frac{(-1)^m}{3N}\, zE(z)^2\mathcal{K}(z^N).} \tag{7.4}$$

If $\widetilde{\mathcal{G}}_m = (\mathcal{G}_m + 2\mathcal{G}_{m+1})/3$*, then*

$$\widetilde{\mathcal{G}}_m(z) - \mathcal{G}(z) = \frac{(-1)^m}{9N}\, zE(z)^2\big(\mathcal{K}(z^N) - \mathcal{K}(z^{2N})\big). \tag{7.5}$$

Proof. Subtract the two equations in (7.2). Iterating the resulting homogeneous equation from $\mathcal{G}_0 = 1$ gives

$$\begin{aligned}\mathcal{G}_m(z) - \mathcal{G}(z) &= \left(\prod_{j=0}^{m-1} A(z^{2^j})\right)(1 - \mathcal{G}(z^N)) \\ &= \frac{(-1)^m}{N} z^{1-N} P_m(z)^2 (1 - \mathcal{G}(z^N)).\end{aligned}$$

The infinite product splits as $E(z) = P_m(z)E(z^N)$. Substitute $1 - \mathcal{G}(w) = wE(w)^2\mathcal{K}(w)/3$ to obtain (7.4). The denominator in (7.3) has no zeros except its explicit first-order zero at zero, while $1 - \mathcal{G}(w) = w/3 + O(w^2)$, so $\mathcal{K}$ is analytic and $\mathcal{K}(0) = 1$. Combining two consecutive levels gives (7.5). □

The proof also supplies a second derivation of the low-band theorem: in $\mathcal{Q}(z)\mathcal{K}(z^N)$, the nonconstant terms of $\mathcal{K}$ can contribute only from degree $N + 1$ onward. The exact boundary correction is precisely the constant term $\mathcal{K}(0) = 1$.

7.2 Two-colored binary partitions and positive defect coefficients

Let $p_2(n)$ count partitions of n into powers of two, with each part available in two distinguishable colors. Equivalently,

$$\mathcal{P}(z) = \frac{1}{E(z)^2} = \prod_{j\geq 0}(1 - z^{2^j})^{-2} = \sum_{n\geq 0} p_2(n)z^n. \tag{7.6}$$

This product converges locally uniformly on the disk. The combinatorial interpretation follows by assigning independently a nonnegative multiplicity to each color of each power of two.

Theorem 7.3 (Finite coefficient formula and positivity)**.** *The analytic function $\mathcal{K}$ satisfies*

$$2\mathcal{K}(z) + \mathcal{K}(z^2) = 3\mathcal{P}(z). \tag{7.7}$$

Writing $\mathcal{K}(z) = \sum_{n\geq 0} \kappa_n z^n$, one has $\kappa_0 = 1$ and, for $n \geq 1$,

$$\boxed{\kappa_n = \frac{3}{2}\sum_{j=0}^{\nu_2(n)} \left(-\frac{1}{2}\right)^j p_2(n/2^j).} \tag{7.8}$$

All κ_n are positive, and

$$\frac{3}{4}p_2(n) \leq \kappa_n \leq \frac{3}{2}p_2(n) \qquad (n \geq 1). \tag{7.9}$$

Proof. From (7.2),

$$1 - \mathcal{G}(z) = A(z)(1 - \mathcal{G}(z^2)) + \frac{z}{2}.$$

Using $E(z) = (1-z)E(z^2)$ and the definition of $\mathcal{K}$ transforms this into $\mathcal{K}(z) = 3/(2E(z)^2) - \mathcal{K}(z^2)/2$, proving (7.7). Comparison of coefficients gives

$$2\kappa_n + \mathbf{1}_{2|n}\kappa_{n/2} = 3p_2(n) \quad (n \geq 1).$$

The descending recursion ends at an odd integer; expanding it proves the finite formula. Adding a part of size one of a fixed color is an injection from the partitions of n into those of $n + 1$, so $p_2(n)$ is nondecreasing. In the alternating finite sum the magnitude of each term is at most half that of the previous term. It lies between its first term minus its second and its first term. Those bounds, multiplied by $3/2$, yield (7.9); if the second term is absent the upper bound is attained. □

For reference, the first coefficients are

$$\begin{aligned}(p_2(n))_{n=0}^{8} &= (1, 2, 5, 8, 16, 24, 40, 56, 88),\\ (\kappa_n)_{n=0}^{8} &= (1, 3, 6, 12, 21, 36, 54, 84, 243/2).\end{aligned}$$

The κ_n need not be integers. Their positivity is particularly useful: an alternating expression involving binary partitions becomes an analytic function with strictly positive Taylor coefficients.

Corollary 7.4 (A convergent finite-size expansion). *For* $|z| < 1$*,*

$$\mathcal{G}_m(z) - \mathcal{G}(z) = \frac{(-1)^m}{3N} \mathcal{Q}(z) \sum_{j=0}^{\infty} \kappa_j z^{jN}. \tag{7.10}$$

For fixed $0 < r < 1$ *and* $J \geq 0$*, truncation after* $j = J$ *has error* $O_{J,r}(N^{-1} r^{(J+1)N})$*, uniformly for* $|z| \leq r$*. The first mode missed by the positive correction is exactly*

$$\widehat{\widetilde{\mu}_m}(N + 1) - \eta(N + 1) = \frac{(-1)^m}{3N}. \tag{7.11}$$

Proof. Expand the analytic function $\mathcal{K}$ in (7.4). On $|w| \leq r$, its Taylor remainder after J is bounded by $C_{J,r}|w|^{J+1}$; for example, factor out that power and use continuity of the resulting analytic remainder on the compact disk. Put $w = z^N$ and bound $\mathcal{Q}$ on $|z| \leq r$. Finally $\mathcal{Q}(z) = z + O(z^2)$, and $\mathcal{K}(z^N) - \mathcal{K}(z^{2N}) = 3z^N + O(z^{2N})$. Extracting the coefficient of z^{N+1} in (7.5) proves (7.11). □

This is a convergent analytic expansion in the exponentially small variable z^N, multiplied by the alternating algebraic factor N^{-1}. It may be viewed as a finite-size transseries on compact subsets of the disk. No assertion of convergence on $|z| = 1$, nor an interchange with a boundary limit, is included. The distinction is essential because the circle supports the singular limiting measure and the non-measure correction.

Chapter 8

Alternating bounds for Poisson observables

For $0 < \rho < 1$, consider the positive Poisson kernel centered at zero,

$$\Pi_\rho(x) = \frac{1-\rho^2}{1-2\rho\cos(2\pi x)+\rho^2} = 1 + 2\sum_{k\geq 1}\rho^k\cos(2\pi kx). \tag{8.1}$$

Its expectation under the limiting measure is $2\mathcal{G}(\rho) - 1$. The positivity of $\mathcal{K}$ turns the analytic factorization into a signed, not merely absolute, error certificate.

Theorem 8.1 (Poisson bracketing and strict improvement)**.** *For every $m \geq 0$, the two errors $\langle \mu_m - \mu, \Pi_\rho\rangle$ and $\langle \widetilde{\mu}_m - \mu, \Pi_\rho\rangle$ have sign $(-1)^m$ and are nonzero. Writing $t = \rho^N$, their ratio is*

$$\frac{\langle \widetilde{\mu}_m - \mu, \Pi_\rho\rangle}{\langle \mu_m - \mu, \Pi_\rho\rangle} = \frac{\mathcal{K}(t) - \mathcal{K}(t^2)}{3\mathcal{K}(t)} \in (0, 1/3). \tag{8.2}$$

As $m \to \infty$ this ratio is asymptotic to ρ^N. The even raw approximations decrease to the limit and the odd raw approximations increase to it. Both raw and corrected approximations give upper bounds at even levels and lower bounds at odd levels.

Proof. For real $\rho \in (0,1)$, $\mathcal{Q}(\rho) = \rho E(\rho)^2 > 0$. Theorems 7.2 and 7.3 give the exact errors

$$\langle \mu_m - \mu, \Pi_\rho\rangle = \frac{2(-1)^m}{3N}\mathcal{Q}(\rho)\mathcal{K}(t), \tag{8.3}$$

$$\langle \widetilde{\mu}_m - \mu, \Pi_\rho\rangle = \frac{2(-1)^m}{9N}\mathcal{Q}(\rho)\big(\mathcal{K}(t) - \mathcal{K}(t^2)\big). \tag{8.4}$$

Strict positivity of the coefficients implies $\mathcal{K}(t) > \mathcal{K}(t^2) > 0$. This proves all sign and ratio statements. Since $\mathcal{K}(t) = 1 + 3t + O(t^2)$, the ratio equals $t + O(t^2)$. Finally the magnitude of the raw error at level $m+2$, divided by that at level m, is $\mathcal{K}(t^4)/(4\mathcal{K}(t)) < 1/4$. Its sign is unchanged. This proves the nesting of each raw parity subsequence. ☐

The theorem concerns Poisson kernels centered at zero. A spatially shifted kernel introduces oscillatory phases in its Fourier coefficients, so the same positivity argument does not automatically provide alternating upper and lower bounds. Likewise, nesting of the *corrected* parity subsequences is not needed and is not asserted here.

m	N	Raw absolute error	Corrected absolute error
0	1	7.849×10^{-1}	2.5157×10^{-1}
1	2	1.5101×10^{-2}	4.4344×10^{-3}
2	4	8.9896×10^{-4}	2.0776×10^{-4}
3	8	1.3784×10^{-4}	1.7373×10^{-5}
4	16	4.286×10^{-5}	1.1421×10^{-6}
5	32	1.9717×10^{-5}	1.5597×10^{-8}
6	64	9.835×10^{-6}	6.1735×10^{-12}
7	128	4.9175×10^{-6}	1.9376×10^{-18}
8	256	2.4587×10^{-6}	3.8172×10^{-31}
9	512	1.2294×10^{-6}	2.9632×10^{-56}

Table 8.1: Absolute errors for $\Pi_{4/5}$. Values were evaluated with 160-digit arithmetic from the exact factorizations and independently checked against finite Laurent coefficients. Rounded decimal values are illustrations, not interval-arithmetic certificates. The signs are rigorously $(-1)^m$.

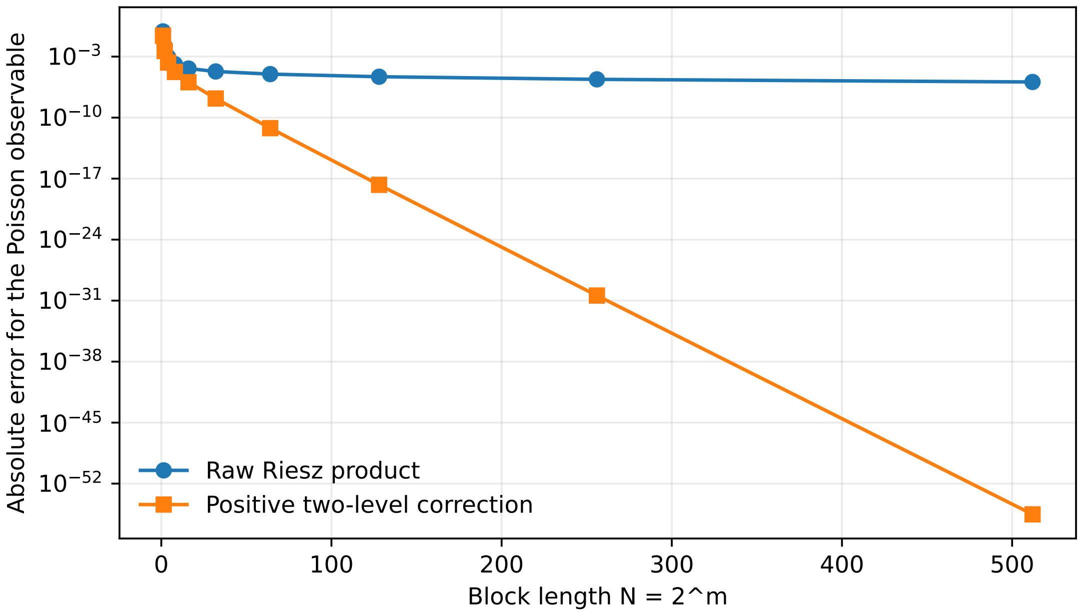

Figure 8.1: The same Poisson errors. The raw error eventually follows N^{-1}; the corrected error has the additional factor $(4/5)^N$. This dramatic gain is for a fixed analytic observable, not a claim of total-variation convergence.

Chapter 9

Exact dyadic energies and the classical moment exponent

9.1 A common two-dimensional transfer matrix

The growth of $g(k)$ along powers of two alone does not determine its Sobolev regularity. We need the total squared energy in a dyadic shell. Endpoint-weighted sums eliminate otherwise distracting boundary terms. Define

$$U_m = \sum_{k=1}^{N-1} g(k)^2 + \frac{1}{2}g(N)^2, \qquad V_m = \sum_{k=0}^{N-1} g(k)g(k+1), \tag{9.1}$$

$$X_m = \sum_{k=1}^{N-1} \eta(k)^2 + \frac{5}{9}, \qquad Y_m = \sum_{k=0}^{N-1} \eta(k)\eta(k+1). \tag{9.2}$$

The constant $5/9$ is the trapezoidal endpoint contribution $(\eta(0)^2 + \eta(N)^2)/2 = (1+1/9)/2$.

Proposition 9.1 (Exact energy recurrences)**.** *Let*

$$M = \begin{pmatrix} 6 & 2 \\ -4 & -4 \end{pmatrix}. \tag{9.3}$$

Then

$$\begin{pmatrix} U_{m+1} \\ V_{m+1} \end{pmatrix} = M \begin{pmatrix} U_m \\ V_m \end{pmatrix}, \qquad \begin{pmatrix} X_{m+1} \\ Y_{m+1} \end{pmatrix} = \frac{1}{4} M \begin{pmatrix} X_m \\ Y_m \end{pmatrix}, \tag{9.4}$$

with $(U_0, V_0) = (1/2, 0)$ *and* $(X_0, Y_0) = (5/9, -1/3)$.

Proof. For g, the even-index contribution to U_{m+1} is $4U_m$, including the half-weighted endpoint. The odd-index contribution is

$$\sum_{r=0}^{N-1} (g(r) + g(r+1))^2 = 2U_m + 2V_m,$$

because $g(0) = 0$. Thus $U_{m+1} = 6U_m + 2V_m$. Adjacent pairs at the two parities contribute

$$\sum_{r=0}^{N-1} \big[-2g(r)(g(r)+g(r+1)) - 2(g(r)+g(r+1))g(r+1)\big] = -4U_m - 4V_m.$$

This proves the first matrix equation.

For η, the even contribution to the trapezoidal square sum is X_m. The odd contribution is $(2X_m + 2Y_m)/4$, giving $X_{m+1} = (3X_m + Y_m)/2$. The adjacent-pair sum is $-(2X_m + 2Y_m)/2 = -X_m - Y_m$. This is precisely $M/4$. The initial values follow from $N = 1$, $g(1) = 1$, and $\eta(1) = -1/3$. ☐

Corollary 9.2 (Closed forms). *Put $d = \sqrt{17}$, $\Lambda = 1 + d$, and $q = \Lambda/4$. Then*

$$U_m = \frac{5+d}{4d}\Lambda^m + \frac{d-5}{4d}(1-d)^m, \tag{9.5}$$

$$X_m = \frac{19+5d}{18d}q^m + \frac{5d-19}{18d}\left(\frac{1-d}{4}\right)^m. \tag{9.6}$$

In particular, $U_m \asymp \Lambda^m$ and $X_m \asymp q^m$.

Proof. The characteristic polynomial of M is $t^2 - 2t - 16$. Hence $U_{m+2} = 2U_{m+1} + 16U_m$ with $U_0 = 1/2$, $U_1 = 3$. For $M/4$ it is $t^2 - t/2 - 1$, and $X_0 = 5/9$, $X_1 = 2/3$. Solving these second-order recurrences gives the displayed formulas. The coefficients of the dominant positive roots are positive and the other roots have smaller modulus, proving the comparisons. □

For clarity about prior work, $\sum_{k=0}^{N-1} \eta(k)^2 = X_m + 4/9$ is the square-correlation formula recorded in [16]. It is not a new exponent calculation. What matters here is that the boundary sequence has the *same* energy matrix, without the factor $1/4$.

9.2 The fourth moment as a Stern energy

Proposition 9.3. *For $N = 2^m$,*

$$\int_0^1 |P_m(\mathrm{e}^{2\pi \mathrm{i}x})|^4 \,\mathrm{d}x = 2U_m, \qquad \|f_m\|_{L^2}^2 = \frac{2U_m}{N^2}. \tag{9.7}$$

Proof. The coefficients of $P_m(z)^2$ through degree $N-1$ equal $g(1), \dots, g(N)$, since truncating E at degree $N-1$ cannot change these convolution coefficients. The reflection identity $z^{N-1}P_m(z^{-1}) = (-1)^m P_m(z)$ makes P_m^2 palindromic. Its coefficient squares therefore sum to $2\sum_{k=1}^{N-1} g(k)^2 + g(N)^2 = 2U_m$. Parseval applied to P_m^2 proves the first equation, and division by N^2 proves the second. □

This connects our transfer calculation to the classical even-moment setting of [27]. It also avoids numerical integration of highly oscillatory products.

9.3 Dyadic shell estimates

Define

$$\sigma = \frac{1}{2}\log_2 \frac{1+\sqrt{17}}{4} = 0.178509318430288\dots, \qquad s_* = 1 + \sigma = 1.178509318430288\dots. \tag{9.8}$$

The familiar correlation dimension is $D_2 = 1 - 2\sigma = 3 - \log_2(1+\sqrt{17})$ [30]. We will use only the energy estimates, not a new dimension theorem.

Lemma 9.4 (Two-sided shell energies). *With $N = 2^m$, constants independent of m give*

$$\sum_{N<k\le 2N} g(k)^2 \asymp N^{2s_*}, \qquad \sum_{N<k\le 2N} \eta(k)^2 \asymp N^{2\sigma}. \tag{9.9}$$

Proof. Let $G_m = \sum_{k=1}^{N} g(k)^2 = U_m + 4^m/2$. Then the first shell sum is $G_{m+1} - G_m = U_{m+1} - U_m + 3 \cdot 4^m/2$. Since $\Lambda > 4$ and the dominant coefficient in (9.5) is positive, this is a positive constant times Λ^m asymptotically. The remaining finitely many shells have positive energy, so constants can be adjusted to cover them all. For η, the corresponding prefix sum through N is $X_m - 4/9$. Its shell difference is $X_{m+1} - X_m$, asymptotic to a positive constant times q^m, since $q > 1$. Each shell contains a power of two, where $\eta = -1/3$, so again every finite initial shell is nonzero. Finally $\Lambda^m = N^{2s_*}$ and $q^m = N^{2\sigma}$. □

Chapter 10

Sharp Sobolev regularity and finite-size convergence

10.1 The function spaces and a shell-summation lemma

For a periodic distribution u, use the norm

$$\|u\|_{H^{-s}}^2 = \sum_{k\in\mathbb{Z}}(1+k^2)^{-s}|\widehat{u}(k)|^2. \tag{10.1}$$

The use of k rather than $2\pi k$ fixes constants but not thresholds. A pairing with $\phi \in H^s$ is bounded by $\|u\|_{H^{-s}}\,\|\phi\|_{H^s}$.

Lemma 10.1 (Weighted dyadic sums)**.** *Suppose $a_k \ge 0$ and $\sum_{2^j<k\le 2^{j+1}} a_k \asymp 2^{2tj}$, with a nonzero fixed low coefficient. For $N = 2^m$ and $s > t$,*

$$\sum_{k>N}(1+k^2)^{-s}a_k \asymp N^{2(t-s)}. \tag{10.2}$$

The full weighted series converges if and only if $s > t$. Moreover, for positive s the prefix sum through N is, as $N \to \infty$,

$$\sum_{1\le k\le N}(1+k^2)^{-s}a_k \asymp \begin{cases} N^{2(t-s)}, & s<t,\\ \log N, & s=t,\\ 1, & s>t. \end{cases} \tag{10.3}$$

Proof. On the jth shell, the weight is comparable to 2^{-2sj}, with comparison constants depending only on s. The shell's weighted contribution is therefore comparable to $2^{2(t-s)j}$. Summing this geometric progression over $j \ge m$ gives (10.2); convergence fails at equality because each shell then contributes a positive amount. Summing over $j < m$ gives respectively a last-term-dominated progression, m comparable summands, or a bounded convergent series. The nonzero low coefficient gives the last case's positive lower bound. □

Theorem 10.2 (Exact regularity thresholds)**.** *The limiting measure and its boundary correction satisfy*

$$\mu \in H^{-s} \iff s > \sigma, \qquad \nu \in H^{-s} \iff s > s_*. \tag{10.4}$$

Neither endpoint is included.

Proof. Apply Lemma 10.1 to the two shell estimates in Lemma 9.4. Evenness duplicates the positive frequencies, and the constant Fourier coefficient does not affect convergence. The factor $1/3$ in $\widehat{\nu}$ changes only the norm constant. □

The difference of exactly one in the thresholds expresses a genuine derivative's worth of additional roughness in the universal correction. It is compatible with, but substantially more precise than, the pointwise bound $|g(k)| \le k$.

10.2 Norm convergence of the renormalized error

Theorem 10.3 (Sharp topology for the first correction). *Set $Z_m = (-1)^m N(\mu_m - \mu)$. Then*

$$Z_m \longrightarrow \nu \quad \text{in } H^{-s} \qquad \Longleftrightarrow \qquad s > s_*. \tag{10.5}$$

For $s > s_*$,

$$\|Z_m - \nu\|_{H^{-s}} \ll_s N^{s_* - s}. \tag{10.6}$$

Proof. On $|k| \le N$ the difference vanishes. On the complement it equals $-(-1)^m N\eta(|k|) - g(|k|)/3$. The squared norm is bounded by a fixed constant times the sum of N^2 times the η tail energy and the g tail energy. By (10.2), both have size at most $N^{2(s_*-s)}$. This proves the norm bound and convergence. Conversely, a norm limit in H^{-s} must have every Fourier coefficient equal to the eventually exact coefficient $g(|k|)/3$, hence must be ν. That distribution belongs to H^{-s} only for $s > s_*$. For $s \le \sigma$, already $\mu_m - \mu$ is not an element of H^{-s}; the finite polynomial cannot remove the divergent tail. □

10.3 The raw approximation: a phase transition and saturation

Theorem 10.4 (Sharp raw error rates). *For every fixed $s > \sigma$, as $N = 2^m \to \infty$,*

$$\boxed{\|\mu_m - \mu\|_{H^{-s}} \asymp_s \begin{cases} N^{\sigma - s}, & \sigma < s < s_*, \\ N^{-1}\sqrt{\log N}, & s = s_*, \\ N^{-1}, & s > s_*. \end{cases}} \tag{10.7}$$

For $s > s_*$ there is also the exact leading norm limit

$$N\,\|\mu_m - \mu\|_{H^{-s}} \longrightarrow \|\nu\|_{H^{-s}} > 0. \tag{10.8}$$

Proof. Because the finite coefficients vanish beyond their support, the squared norm has the exact decomposition

$$\|\mu_m - \mu\|_{H^{-s}}^2 = \frac{2}{9N^2}\sum_{k=1}^{N}(1+k^2)^{-s} g(k)^2 + 2\sum_{k>N}(1+k^2)^{-s}\eta(k)^2. \tag{10.9}$$

The endpoint $k = N$ is included in the first sum by Theorem 4.1. Use (10.3) with $t = s_*$ for the first term and (10.2) with $t = \sigma$ for the second. If $s < s_*$, both are of order $N^{2(\sigma - s)}$. At $s = s_*$ the first is of order $N^{-2}\log N$, while the second is of order N^{-2}. Above s_* the first is of order N^{-2} and the second is smaller. Taking square roots proves the three regimes. Equation (10.8) follows from Theorem 10.3 and continuity of the norm; $\nu \neq 0$ because its first coefficient is $1/3$. □

Thus additional smoothness ceases to improve the raw rate after s_*. The obstruction is not an estimate artifact: it is a specific nonzero distribution, with an exactly known Fourier series.

10.4 The corrected approximation removes saturation

Theorem 10.5 (Sharp corrected rate and bandwidth optimality). *For every fixed $s > \sigma$,*

$$\boxed{\|\widetilde{\mu}_m - \mu\|_{H^{-s}} \asymp_s N^{\sigma - s}.} \tag{10.10}$$

More generally, fix $C > 0$. Every approximant v_N whose Fourier support is contained in $|k| \le CN$ satisfies, for all sufficiently large dyadic N,

$$\|v_N - \mu\|_{H^{-s}} \ge c_{s,C} N^{\sigma - s}. \tag{10.11}$$

No positivity assumption on v_N is required for this lower bound.

Proof. The corrected error vanishes through N. For the middle band $N < |k| < 2N$, use $(1+k^2)^{-s} \le N^{-2s}$ and $|a-b|^2 \le 2|a|^2 + 2|b|^2$. By (5.2) and Parseval,

$$\sum_k |\widehat{\widetilde{\mu}_m}(k)|^2 = \left\|\widetilde{f}_m\right\|_{L^2}^2 \le \frac{25}{9}\left\|f_m\right\|_{L^2}^2 = \frac{50U_m}{9N^2} \ll N^{2\sigma}.$$

The η energy on the middle band has the same upper order by Lemma 9.4. Thus that band's weighted contribution is $O_s(N^{2(\sigma-s)})$. Beyond $2N$ the approximant has no coefficients, and Lemma 10.1 gives the same order for the tail. This proves the upper bound. The unchanged η tail beyond $2N$ alone proves the matching lower bound.

For the general assertion, choose a fixed integer j with $2^j > C$ and take the shell $2^jN < |k| \le 2^{j+1}N$. All coefficients of v_N there vanish. Its weighted η energy is bounded below by a constant times $N^{2(\sigma-s)}$. Taking a square root proves (10.11). □

For rough admissible tests, $\sigma < s < s_*$, the correction improves the constant and exactness band but cannot improve the order dictated by the unresolved spectral tail. At the threshold it removes $\sqrt{\log N}$, and above the threshold it removes the N^{-1} saturation completely. These are different regimes of a single exact identity, rather than competing convergence claims.

Chapter 11

A parameter family and the sharp positivity threshold

11.1 Amplitude-deformed binary Riesz products

For $-1 \le a \le 1$, define

$$f_m^{(a)}(x) = \prod_{j=0}^{m-1} (1 - a\cos(2\pi 2^j x)), \qquad \mu_m^{(a)} = f_m^{(a)}(x)\,\mathrm{d}x. \tag{11.1}$$

These are amplitude deformations; no identification with a differently parameterized phase-deformation family is intended. Every factor is nonnegative. The constant coefficient remains one: in $f_{m+1}^{(a)} = (1 - a\cos(2\pi x))f_m^{(a)}(2x)$, the last factor has only even frequencies. Hence all $\mu_m^{(a)}$ are probabilities. Their Fourier recursion is

$$r_{m+1}^{(a)}(2r) = r_m^{(a)}(r), \qquad r_{m+1}^{(a)}(2r+1) = -\frac{a}{2}\big(r_m^{(a)}(r) + r_m^{(a)}(r+1)\big). \tag{11.2}$$

The same fixed-index induction as in Proposition 2.2 proves a weak limit $\mu^{(a)}$, whose coefficients η_a obey this recursion with

$$\eta_a(0) = 1, \qquad \eta_a(1) = -\frac{a}{2+a}. \tag{11.3}$$

The coefficient bound $|\eta_a(k)| \le 1$ follows from the probability measure.

When $a = 0$, every measure is Lebesgue measure. When $a = 1$, we recover the Thue–Morse case. At $a = -1$,

$$f_m^{(-1)}(x) = \frac{1}{N}\left|\sum_{n=0}^{N-1} \mathrm{e}^{2\pi i n x}\right|^2, \tag{11.4}$$

the dyadic Fejér kernel; its coefficients tend to one, so its limit is the point mass at zero. This endpoint is a useful exact check on the general formulas.

Theorem 11.1 (Parameterized Stern boundary). *Fix $a \in [-1,1] \setminus \{0\}$, and write*

$$\alpha = -\frac{a}{2}, \qquad c = \frac{a}{2+a}, \qquad g_a(k) = B_k(-2/a). \tag{11.5}$$

Then, exactly for $0 \le k \le N = 2^m$,

$$r_m^{(a)}(k) - \eta_a(k) = c\alpha^m g_a(k). \tag{11.6}$$

Proof. The initial coefficient at $k = 1$ has error c, agreeing with $g_a(1) = 1$. The even recursion for g_a multiplies by $-2/a = \alpha^{-1}$, compensating for the extra α at the next level. Its odd recursion is the sum of two adjacent values, exactly matching the factor α in the Fourier recursion. The induction of Theorem 4.1, with these constants substituted, therefore proves the statement, including the endpoint. □

Theorem 11.2 (Exactness and a sharp positivity interval)**.** *For $a \in [-1, 1]$, put*

$$\widetilde{\mu}_m^{(a)} = \frac{a\mu_m^{(a)} + 2\mu_{m+1}^{(a)}}{2 + a}. \tag{11.7}$$

It has total mass one and agrees with $\mu^{(a)}$ at every Fourier mode $|k| \leq N$. Its density is

$$\widetilde{f}_m^{(a)}(x) = f_m^{(a)}(x)\left(1 - \frac{2a}{2 + a}\cos(2\pi Nx)\right). \tag{11.8}$$

For every m, this density is nonnegative on the whole circle if and only if

$$\boxed{-\frac{2}{3} \leq a \leq 1.} \tag{11.9}$$

For $a \neq 0$, the affine combination is uniquely determined by exactness at the first mode. At $a = 0$ all levels coincide and uniqueness is not asserted.

Proof. For $a \neq 0$, the common-band errors have ratio $\alpha = -a/2$. Their cancellation condition is $w + (1 - w)\alpha = 0$, giving $w = a/(2 + a)$. The density identity follows by factoring $f_{m+1}^{(a)} = f_m^{(a)}(1 - a\cos(2\pi Nx))$. The second factor is nonnegative exactly when $|2a/(2 + a)| \leq 1$, which on $[-1, 1]$ is $a \geq -2/3$. This condition is sufficient since $f_m^{(a)} \geq 0$. If $a < -2/3$, the second factor is negative where $\cos(2\pi Nx) = -1$, for example at $x = 1/(2N)$: its value there is $(2 + 3a)/(2 + a) < 0$. Every factor of $f_m^{(a)}$ at this point is strictly positive, since for $j < m$, $0 < 2\pi 2^j x \leq \pi/2$ and $1 - a\cos(2\pi 2^j x) \geq 1$. Thus the corrected density is strictly negative on a neighborhood of this point. This proves necessity, including the endpoint $a = -1$. The case $a = 0$ is immediate. □

Remark 11.3 (Negative weights need not force a negative density). For $-2/3 \leq a < 0$, the coefficient of $\mu_m^{(a)}$ in the affine combination is negative, yet the resulting density is nonnegative. The factorization, rather than the signs of the mixing weights alone, decides positivity. Below $-2/3$ this exact two-level construction genuinely becomes signed. The theorem is a sharp threshold for this construction, not a no-go theorem for every possible positive approximation with a larger number of levels.

11.2 The full fixed-parameter Sobolev rate law

The same mechanism has a continuous family of exponents. The proof is included because the amplitude deformation separates two roles that coincide at $a = 1$: the boundary's geometric decay factor and the limiting measure's energy growth. Set

$$T_a = 1 - a + \frac{a^2}{2}, \qquad \rho_a = \frac{T_a + \sqrt{T_a^2 + 4a}}{2}, \tag{11.10}$$

$$\beta_a = \log_2 \frac{2}{|a|}, \qquad \sigma_a = \frac{1}{2}\log_2 \rho_a, \qquad s_a = \sigma_a + \beta_a. \tag{11.11}$$

These quantities are defined for fixed $a \neq 0$ in $[-1, 1]$.

Theorem 11.4 (Parameterized regularity and sharp rates). *Let $\widehat{\nu}_a(0) = 0$ and $\widehat{\nu}_a(k) = c\, g_a(|k|)$ for $k \neq 0$, with c as in (11.5). Then*

$$\mu^{(a)} \in H^{-s} \iff s > \sigma_a, \qquad \nu_a \in H^{-s} \iff s > s_a. \tag{11.12}$$

For every fixed $s > \sigma_a$,

$$\left\|\mu_m^{(a)} - \mu^{(a)}\right\|_{H^{-s}} \asymp_{a,s} \begin{cases} N^{\sigma_a - s}, & \sigma_a < s < s_a, \\ N^{-\beta_a}\sqrt{\log N}, & s = s_a, \\ N^{-\beta_a}, & s > s_a, \end{cases} \tag{11.13}$$

$$\left\|\widetilde{\mu}_m^{(a)} - \mu^{(a)}\right\|_{H^{-s}} \asymp_{a,s} N^{\sigma_a - s}. \tag{11.14}$$

The corrected estimate remains valid below the positivity threshold, where the approximants are signed. For $s > s_a$, the renormalized errors converge to ν_a in H^{-s} after multiplication by α^{-m}, with error $O_{a,s}(N^{s_a - s})$.

Proof. Put $b = -2/a$ and define trapezoidal energies

$$U_m^{(a)} = \sum_{k=1}^{N-1} g_a(k)^2 + \tfrac{1}{2} g_a(N)^2, \qquad V_m^{(a)} = \sum_{k=0}^{N-1} g_a(k) g_a(k+1),$$

$$X_m^{(a)} = \sum_{k=1}^{N-1} \eta_a(k)^2 + \tfrac{1}{2}(1 + c^2), \qquad Y_m^{(a)} = \sum_{k=0}^{N-1} \eta_a(k)\eta_a(k+1).$$

The same even-odd splitting used in Proposition 9.1 gives

$$\begin{pmatrix} X_{m+1}^{(a)} \\ Y_{m+1}^{(a)} \end{pmatrix} = M_a \begin{pmatrix} X_m^{(a)} \\ Y_m^{(a)} \end{pmatrix}, \qquad M_a = \begin{pmatrix} 1 + a^2/2 & a^2/2 \\ -a & -a \end{pmatrix}, \tag{11.15}$$

and

$$\begin{pmatrix} U_{m+1}^{(a)} \\ V_{m+1}^{(a)} \end{pmatrix} = \begin{pmatrix} b^2 + 2 & 2 \\ 2b & 2b \end{pmatrix} \begin{pmatrix} U_m^{(a)} \\ V_m^{(a)} \end{pmatrix} = \frac{4}{a^2} M_a \begin{pmatrix} U_m^{(a)} \\ V_m^{(a)} \end{pmatrix}. \tag{11.16}$$

For example the odd square contribution for g_a is $2U_m^{(a)} + 2V_m^{(a)}$; the adjacent-pair sum is $2b(U_m^{(a)} + V_m^{(a)})$. For η_a, the odd square contribution is $a^2(X_m^{(a)} + Y_m^{(a)})/2$, and the adjacent-pair sum is $-a(X_m^{(a)} + Y_m^{(a)})$. These establish both matrices without omitted boundary terms.

The characteristic polynomial of M_a is $x^2 - T_a x - a$. Its discriminant is positive. For $a \geq 0$ this is immediate. For $a = -u < 0$ it is $(1-u)^2 + u^2 + u^3 + u^4/4 > 0$. Evaluating the polynomial at one gives $-a^2/2 < 0$, so the larger root ρ_a exceeds one. The other root has modulus $|a|/\rho_a < 1$. Thus each first-coordinate energy is a combination of two distinct real geometric sequences. The coefficient of ρ_a^m in $X_m^{(a)}$ is strictly positive: otherwise nonnegativity rules out a negative coefficient, while a zero coefficient would make the energy bounded, contrary to the m nonzero values $\eta_a(2^j) = -c$ in its prefix. For $U_m^{(a)}$, the larger eigenvalue is $(4/a^2)\rho_a > b^2$. The coefficient of that root is again positive: a zero dominant coefficient would contradict $U_m^{(a)} \geq b^{2m}/2$, because the smaller root has modulus strictly less than b^2.

Consequently, taking differences of prefix energies as in Lemma 9.4,

$$\sum_{N < k \leq 2N} \eta_a(k)^2 \asymp_a N^{2\sigma_a}, \qquad \sum_{N < k \leq 2N} g_a(k)^2 \asymp_a N^{2 s_a}.$$

The endpoints $g_a(N)^2 = b^{2m}$ are lower order than the dominant energy; the endpoints $\eta_a(N)^2 = c^2$ are constant. All finite shells have positive energy because they contain powers of two. The sequence g_a

also has polynomial growth: $|g_a(k)| \le k^{\beta_a}$ follows by induction from $|b| = 2^{\beta_a}$ and $r^{\beta_a} + (r+1)^{\beta_a} \le (2r+1)^{\beta_a}$, since $\beta_a \ge 1$. Thus ν_a is a distribution.

The shell-summation lemma now proves (11.12). On the low band the raw error is exactly $c\alpha^m g_a(k)$, with $|\alpha|^m = N^{-\beta_a}$; outside the support it is $-\eta_a(k)$. Applying the weighted prefix and tail estimates gives (11.13), including the logarithm at the critical exponent. The same tail argument proves the renormalized convergence and its rate.

For the corrected upper bound, its support is contained in $|k| < 2N$ and its error vanishes through N. The low-band identity also gives

$$\sum_k |r_m^{(a)}(k)|^2 \ll_a \sum_{k=0}^{N} \eta_a(k)^2 + |\alpha|^{2m} \sum_{k=0}^{N} g_a(k)^2 \ll_a N^{2\sigma_a}.$$

The multiplicative factor in (11.8) is bounded in absolute value by $1 + 2|a|/(2+a)$. Hence its corrected L^2 norm has the same upper order, even when the factor changes sign. Bounding the middle band and the untouched tail exactly as in Theorem 10.5 proves the upper bound in (11.14). The tail beyond $2N$ proves the lower bound. □

At $a = -1$, the formulas give $\rho_a = 2$, $\sigma_a = 1/2$, $\beta_a = 1$, and $s_a = 3/2$. Also $g_a(k) = B_k(2) = k$ and $c = -1$, so $r_m^{(-1)}(k) = 1 - k/N$ on $0 \le k \le N$, the familiar Fejér coefficients. At $a = 1$, every formula reduces to the Thue–Morse theorem. The limit $a \to 0$ is not uniform in the constants: the case $a = 0$ is exactly constant in m and is correctly excluded from the exponent parametrization.

Chapter 12

Computation, verification, and proof engineering

12.1 Exact computation without constructing a large block

To compute a single stabilized finite Fourier coefficient, choose m with $2^m \geq k$, compute $\eta(k)$ from its two adjacent-value recursion, compute $g(k)$ using (3.6), and apply (4.1). The state for η has matrices

$$\begin{pmatrix} \eta(2k) \\ \eta(2k+1) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -1/2 & -1/2 \end{pmatrix} \begin{pmatrix} \eta(k) \\ \eta(k+1) \end{pmatrix}, \qquad \begin{pmatrix} \eta(2k+1) \\ \eta(2k+2) \end{pmatrix} = \begin{pmatrix} -1/2 & -1/2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \eta(k) \\ \eta(k+1) \end{pmatrix}. \tag{12.1}$$

Starting from $(\eta(0), \eta(1)) = (1, -1/3)$ and reading the bits from most to least significant takes $O(\log k)$ arithmetic operations. Leading zero bits leave this initial state unchanged. These are arithmetic-operation counts; bit complexity also accounts for integer sizes. The signed Stern values have $O(\log k)$ bits by $|g(k)| \leq k$, and the rational recursion for η grows its denominator by at most one factor of two per bit.

When all coefficients through N are required, the descending recurrences can be tabulated in $O(N)$ arithmetic operations and memory. Directly summing all correlations takes $O(N^2)$ sign multiplications; an FFT is an alternative for approximation, but the recurrence plus exact boundary term provides exact rational coefficients without rounding.

For the analytic correction, the finite formula (7.8) uses only $1 + \nu_2(n)$ binary partition values. A table of $p_2(0), \ldots, p_2(L)$ can be computed by a coin-change dynamic program, using two colors of each size $1, 2, 4, \ldots$, in $O(L \log L)$ integer additions. Formula (7.7) then computes all κ_n in $O(L)$ rational operations. Positivity prevents large cancellation when evaluating $\mathcal{K}(t)$ for real $0 < t < 1$.

12.2 Avoiding loss of the corrected error

At large N, directly subtracting two nearly equal high-precision expectations can lose most significant digits. Equation (8.4) is a better evaluation formula. Even within that formula, write

$$\mathcal{K}(t) - \mathcal{K}(t^2) = \sum_{j \geq 1} \kappa_j t^j (1 - t^j), \qquad 0 < t < 1, \tag{12.2}$$

instead of subtracting two numbers close to one. Every summand is positive. For very small t, one can factor out $3t$ before accumulating the remaining series. For complex z or shifted observables this particular positivity is lost, but the exact disk factorization still applies.

12.3 The supplied verification program

The companion `verify.py` uses only Python's standard library. Identity checks use integers and `fractions.Fraction`; none uses floating-point equality. The recorded run performs 19,037 checks, including:

Test family	Range or independent representation
Aperiodic and forward corrections	Every $0 \leq k \leq 2^m$ for $0 \leq m \leq 10$.
Finite Fourier products	Independent Laurent-polynomial multiplication of all factors.
Positive correction and first missed mode	Exact low-band agreement and the coefficient at $N+1$.
Signed Stern identity	Direct sign convolution through $k = 1024$, bounds, and valuations.
Energy transfer and fourth moments	Exact rational sums through level 12.
Analytic factorization	Every coefficient through degree 256 at levels 0 through 7.
Partition coefficient formula	Exact coefficients, finite sums, and positivity bounds through 256.
Amplitude family	Nine nonzero rational parameters, every low mode through level 7, and both energy matrices.

The parameters are $-1, -3/4, -2/3, -1/2, -1/7, 1/7, 1/4, 1/2, 1$. These tests are not replacements for proofs of infinite statements. They check independent finite representations and are particularly effective at catching endpoint, sign, normalization, and parameter mistakes.

`make_figures.py` separately uses `mpmath`, `numpy`, and `matplotlib`. The Poisson table uses 160 decimal digits and 2,201 terms for its independent limiting-series evaluation. The data and the exact test report are included as JSON. The verification program itself has no dependency on those optional plotting packages.

12.4 Formalization targets in the supplied repository

No Lean source is supplied or claimed checked here. The following dependency order isolates relatively small formal targets compatible with the atlas's existing finite autocorrelation and polynomial-product development.

The first layer is entirely algebraic: define the convolution sequence g, prove its two parity recurrences, identify it with $B_k(-2)$ if a Stern-polynomial interface is desired, and prove the two exact boundary formulas. The latter need only natural-number induction, finite sums, binary reflection, and rational arithmetic. The central low-band cancellation is then a short consequence.

The second layer packages Fourier coefficients of finite trigonometric products. Positivity of the corrected density is a direct bound on cosine, not a limiting argument. Weak existence of μ can be taken from the existing development or proved from finite coefficient convergence and compactness.

The third layer separates analysis from combinatorics: prove the two energy matrix identities as finite-sum theorems, solve the scalar recurrences, derive shell estimates, and then apply an abstract weighted-shell lemma. The sharp Sobolev thresholds and all three error regimes become applications of that one lemma to an exact norm decomposition. The analytic factorization is a separate Mahler-equation branch; its only infinite-product requirements are local uniform convergence and nonvanishing in the open disk.

This architecture deliberately avoids equating "proved on paper" with "already formalized in the current repository." It also keeps classical spectral-type results out of the dependencies of the new algebraic identities.

Chapter 13

Further questions suggested by the exact mechanism

13.1 Non-dyadic windows and binary boundary states

For a length L that is not a power of two, the finite word decomposes into signed dyadic blocks. The single correction $g(k)$ should then be replaced by several cross-boundary states whose coefficients depend on the binary expansion of L. A concrete next problem is to derive a finite-dimensional exact state system for these defects and determine which subsequences of L admit positive moment-exact combinations. The present theorems prove the one-state collapse on dyadic lengths; they do not assert such a collapse for all lengths.

13.2 Positive corrections for other substitution spectra

The general principle is an affine finite Fourier recursion with a contractive boundary eigenvalue. A negative eigenvalue allows cancellation by positive weights on two consecutive levels, as in the Thue–Morse case. With several boundary eigenvalues, the cancellation conditions become a finite moment problem for the level weights. Determining when these weights, or the resulting density factor, remain nonnegative is a concrete algebraic question. The interval $[-2/3, 1]$ in the amplitude family shows why positivity of the final factor can be a less restrictive condition than positivity of every weight.

13.3 Distribution functions and nonsmooth local probes

Theorems 10.4 and 10.5 describe an entire Sobolev scale, but they do not automatically settle optimal uniform error for cumulative distribution functions. Indicator functions are substantially less smooth than the observables for which the first correction gives a full weak expansion. A next step is to combine the exact band cancellation with sharp local mass bounds for the Thue–Morse measure. This requires additional local information, not another manipulation of the same fixed-mode identity.

13.4 More layers versus more exact modes

Two consecutive products uniquely cancel the one low-band boundary mode. Three or more levels can cancel selected later analytic sectors in (7.10), but the variables at consecutive levels are $z^N, z^{2N}, z^{4N}, \ldots$, not ordinary powers of a common mesh parameter. A useful precise problem is to maximize the exact Fourier band subject to nonnegative level weights, or more generally a nonnegative factored density. The first missed coefficient (7.11) supplies the first nontrivial constraint. No general optimal multi-level scheme is claimed here.

Chapter 14

Exact boundary defects for correlations of every order

The Stern boundary term of Theorem 4.1 concerns the pair correlation. The same mechanism, one dyadic block longer than every shift, controls correlations of every order: a single finite boundary functional, independent of the block level, gives the exact dyadic defect. Even correlations have an alternating defect, whereas odd correlations become an explicit telescoping difference after dyadic blocking, and the same functional gives formulas for arbitrary numbers of complete blocks without a recursive correlation hierarchy. The means and their renormalization belong to the established correlation theory [12]; the contribution here is the explicit finite functional, its independence of the level, and the exact boundary and telescoping formulas. The following elementary identity replaces an infinite correlation recursion by a finite binary-digit expression.

14.1 An exact adjacent-correlation sum

The following elementary identity will replace an infinite correlation recursion by a finite binary-digit expression.

Definition 14.1. For an integer $Q = \sum_{j\geq 0} b_j 2^j \geq 0$, define the alternating digit sum

$$\sigma(Q) = \sum_{j\geq 0} (-1)^j b_j. \tag{14.1}$$

In particular, $\sigma(0) = 0$ and $\sigma(2^k) = (-1)^k$.

Lemma 14.2 (Adjacent correlation at every endpoint)**.** *For every $Q \geq 0$,*

$$C(Q) := \sum_{q=0}^{Q-1} \varepsilon_q \varepsilon_{q+1} = -\frac{Q}{3} - \frac{2}{3}\sigma(Q). \tag{14.2}$$

Consequently $C(Q)/Q \to -1/3$ as $Q \to \infty$.

Proof. Write $c(q) = \varepsilon_q \varepsilon_{q+1}$. The binary recursion gives $c(2q) = -1$ and $c(2q+1) = -c(q)$. Therefore

$$C(2Q) = -Q - C(Q), \qquad C(2Q+1) = -Q - C(Q) - 1.$$

On the other hand, $\sigma(2Q) = -\sigma(Q)$ and $\sigma(2Q+1) = 1 - \sigma(Q)$. Substitution shows that the proposed right-hand side satisfies the same recurrences and agrees at zero; induction proves the formula. The number of nonzero summands in $\sigma(Q)$ is at most the binary length of Q, which is $O(\log(Q+1))$. Division by Q proves the limit. $\square$

14.2 The finite boundary functional

Fix a tuple

$$\boldsymbol{h} = (h_1, \ldots, h_p), \qquad p \geq 1, \qquad 0 = \min_i h_i, \qquad H = \max_i h_i.$$

For $N = 2^m > H$, define the noncyclic and cyclic sums

$$S_m(\boldsymbol{h}) = \sum_{n=0}^{N-1} \prod_{i=1}^{p} \varepsilon_{n+h_i}, \tag{14.3}$$

$$R_m(\boldsymbol{h}) = \sum_{n=0}^{N-1} \prod_{i=1}^{p} \varepsilon_{(n+h_i) \bmod N}. \tag{14.4}$$

The noncyclic sum uses the infinite sequence even when a shifted index exits the first block. The distinction between these two conventions is essential.

For each $1 \leq t \leq H$, set

$$b_t = \#\{i : h_i \geq t\}, \tag{14.5}$$

$$G_t(\boldsymbol{h}) = \prod_{h_i < t} \varepsilon_{t-h_i-1} \prod_{h_i \geq t} \varepsilon_{h_i - t}. \tag{14.6}$$

Define the *boundary functional*

$$D(\boldsymbol{h}) = \sum_{\substack{1 \leq t \leq H \\ b_t \text{ odd}}} G_t(\boldsymbol{h}). \tag{14.7}$$

It is an integer and satisfies $|D(\boldsymbol{h})| \leq H$. All its sign indices lie between zero and $H - 1$, so it is independent of the dyadic level m.

Theorem 14.3 (One boundary functional, all dyadic levels)**.** *Suppose $m \geq 1$ and $N = 2^m > H$.*

If p is even, the correlation mean

$$\eta(\boldsymbol{h}) = \lim_{X \to \infty} \frac{1}{X} \sum_{n=0}^{X-1} \prod_i \varepsilon_{n+h_i} \tag{14.8}$$

exists, and

$$S_m(\boldsymbol{h}) = N\eta(\boldsymbol{h}) - \frac{2}{3}(-1)^m D(\boldsymbol{h}), \tag{14.9}$$

$$R_m(\boldsymbol{h}) = N\eta(\boldsymbol{h}) + \frac{4}{3}(-1)^m D(\boldsymbol{h}). \tag{14.10}$$

If p is odd, the same mean exists and equals zero, while

$$S_m(\boldsymbol{h}) = -2D(\boldsymbol{h}), \qquad R_m(\boldsymbol{h}) = 0. \tag{14.11}$$

Proof. Write an index as $n = qN + r$, with $0 \leq r < N$, and let

$$\chi_i(r) = \mathbf{1}_{\{r+h_i \geq N\}}, \quad b(r) = \sum_i \chi_i(r), \quad g(r) = \prod_i \varepsilon_{(r+h_i) \bmod N}.$$

Because $h_i < N$, each shifted index crosses at most one block boundary. The block identity (1.3) gives the exact factorization

$$\prod_i \varepsilon_{qN+r+h_i} = g(r)\varepsilon_q^{p-b(r)}\varepsilon_{q+1}^{b(r)}. \tag{14.12}$$

Separate the local sums according to the parity of the crossing number:

$$U_m = \sum_{b(r)\text{ even}} g(r), \qquad V_m = \sum_{b(r)\text{ odd}} g(r).$$

Thus $R_m = U_m + V_m$.

We first compute V_m. An odd crossing number can occur only at $r = N - t$, $1 \le t \le H$. A noncrossing index then has the form $N - (t - h_i)$; reflection in (1.3) turns its sign into $(-1)^m \varepsilon_{t-h_i-1}$. A crossing index has the local value $h_i - t$. There are $p - b_t$ noncrossing indices. On the terms selected for V_m, b_t is odd, so

$$V_m = \begin{cases} (-1)^m D(\boldsymbol{h}), & p \text{ even}, \\ D(\boldsymbol{h}), & p \text{ odd}. \end{cases} \tag{14.13}$$

Suppose now that p is even. Summing (14.12) over one block produces

$$\sum_{r=0}^{N-1} \prod_i \varepsilon_{qN+r+h_i} = U_m + V_m \varepsilon_q \varepsilon_{q+1}. \tag{14.14}$$

Over Q complete blocks, division by QN and use of (14.2) give $\eta = (U_m - V_m/3)/N$. An arbitrary endpoint leaves fewer than N terms, whose contribution to the average tends to zero. This proves existence of the full limit. At $q = 0$, $\varepsilon_0\varepsilon_1 = -1$, hence $S_m = U_m - V_m$. Combining this with $R_m = U_m + V_m$, the formula for η, and (14.13) proves (14.9)–(14.10).

If p is odd, the cyclic sum is zero: shifting every summation index by $N/2$ reverses all p signs by (1.4). Consequently $U_m = -V_m = -D(\boldsymbol{h})$. Formula (14.12) now gives

$$\sum_{r=0}^{N-1} \prod_i \varepsilon_{qN+r+h_i} = D(\boldsymbol{h})(\varepsilon_{q+1} - \varepsilon_q). \tag{14.15}$$

This telescopes over q, so the sum over any number of complete blocks is bounded by $2|D|$. A partial final block has fewer than N terms. The correlation mean is therefore zero. Taking $q = 0$ proves (14.11). □

Remark 14.4 (What is and is not new in this statement). The means and their renormalization belong to the established correlation theory [12]. The refinement emphasized here is the explicit finite functional (14.7), its independence of m, and the exact boundary and telescoping formulas. The proof also shows why the number of correlation factors does not introduce additional dyadic defect modes: once a block is longer than every shift, the only coarse signs available are ε_q and ε_{q+1}.

14.3 All block multiples and aligned intervals

Write

$$S_{m,Q}(\boldsymbol{h}) = \sum_{n=0}^{QN-1} \prod_i \varepsilon_{n+h_i}, \qquad Q \ge 0.$$

Corollary 14.5 (Exact endpoint formulas). *Under the assumptions of Theorem 14.3,*

$$S_{m,Q}(\boldsymbol{h}) = \begin{cases} QN\eta(\boldsymbol{h}) - \dfrac{2}{3}(-1)^m D(\boldsymbol{h})\sigma(Q), & p \text{ even}, \\ D(\boldsymbol{h})(\varepsilon_Q - 1), & p \text{ odd}. \end{cases} \tag{14.16}$$

In particular, the sum over the aligned interval $[AN, BN)$ is obtained by subtracting the same expression at $Q = A$ from that at $Q = B$.

Proof. For even p, sum (14.14) and use $U_m = N\eta + V_m/3$, (14.2), and (14.13). For odd p, telescope (14.15). Interval subtraction is exact. □

Thus finite endpoints are controlled either by the alternating digit sum of Q or by its Thue–Morse sign. For a fixed tuple, the odd-order partial sums are bounded uniformly in the endpoint. Even-order centered partial sums have an elementary logarithmic bound.

Corollary 14.6 (Discrepancy bounds). *For fixed* $\boldsymbol{h}$*, let* $X = QN + R$*,* $0 \le R < N$*. If* p *is even, then*

$$\left|\sum_{n<X}\prod_i \varepsilon_{n+h_i} - X\eta(\boldsymbol{h})\right| \le \frac{2}{3}|D(\boldsymbol{h})|\,|\sigma(Q)| + 2N. \tag{14.17}$$

If p *is odd, then*

$$\left|\sum_{n<X}\prod_i \varepsilon_{n+h_i}\right| \le 2|D(\boldsymbol{h})| + N. \tag{14.18}$$

Proof. Apply (14.16). A partial block contains at most N signs of absolute value one, and $|\eta| \le 1$, so centering that partial block costs at most $2N$. In the odd case no centering is necessary. □

The even bound is $O_{\boldsymbol{h}}(\log(X+2))$, while the odd bound is $O_{\boldsymbol{h}}(1)$. Neither statement is asserted to have an optimal constant. For shifts growing with X, the explicit dependence on H and N must be retained; a fixed-shift estimate must not be used as a uniform growing-shift estimate.

14.4 Two-level extraction and rationality

Corollary 14.7 (Exact extrapolation, not an asymptotic approximation). *If* p *is even and* $N = 2^m > H$*,*

$$\eta(\boldsymbol{h}) = \frac{S_m(\boldsymbol{h}) + S_{m+1}(\boldsymbol{h})}{3N} = \frac{R_m(\boldsymbol{h}) + R_{m+1}(\boldsymbol{h})}{3N}. \tag{14.19}$$

Moreover,

$$\eta(\boldsymbol{h}) \in \frac{1}{3N}\mathbb{Z}, \qquad \left|\frac{S_m}{N} - \eta\right| \le \frac{2H}{3N}, \qquad \left|\frac{R_m}{N} - \eta\right| \le \frac{4H}{3N}. \tag{14.20}$$

Proof. The defects in consecutive dyadic levels have opposite signs and equal magnitudes. Adding the two exact formulas cancels them, while the main terms sum to $3N\eta$. The rationality and inequalities follow from integrality of S_m, D and $|D| \le H$. □

The frequency module with denominators consisting of a factor three and powers of two is already part of the classical structure [12]. Here it is recovered together with an explicit finite extraction formula.

14.5 Pairs, a convolution coefficient, and examples

For $\boldsymbol{h} = (0, h)$, the odd crossing number is one throughout the boundary, so

$$D(0,h) = \sum_{j=0}^{h-1} \varepsilon_j \varepsilon_{h-1-j} = [z^{h-1}]E(z)^2 \qquad (h \ge 1). \tag{14.21}$$

This is a useful bridge between a finite correlation defect and the ordinary generating series.

Proposition 14.8 (Binary recursion for the pair defect). *With* $D_h = D(0,h)$*, set* $D_0 = 0$*. Then* $D_1 = 1$*, and*

$$D_{2h} = -2D_h, \qquad D_{2h+1} = D_h + D_{h+1}. \tag{14.22}$$

Proof. Write $a_n = [z^n]E(z)^2$, with $a_{-1} = 0$. Squaring $E(z) = (1-z)E(z^2)$ gives $a_{2n} = a_n + a_{n-1}$ and $a_{2n+1} = -2a_n$. Since $D_h = a_{h-1}$, the claimed formulas follow, including the boundary values. □

Remark 14.9 (The pair defect is the signed Stern sequence). The recursion (14.22) is the recursion (3.3) of Theorem 3.2, and (14.21) is (3.1): $D(0,h) = g(h) = B_h(-2)$. For $p = 2$, $\boldsymbol{h} = (0,h)$ and $h \le N$, the noncyclic formula (14.9) is the forward-correlation formula (4.5), $A_m(h) = N\eta(h) - \frac{2}{3}(-1)^m g(h)$, and the two-level extraction (14.19) is the positive correction of Theorem 5.1 read at one mode. The boundary functional therefore extends the Stern boundary term of the diffraction measure to correlations of every order, at the price of replacing the closed Stern recursion by the finite sum (14.7).

Table 14.1: Exact even-order examples. Here $N = 2^m > H$, and cyclic and noncyclic sums use the conventions in (14.3)–(14.4).

Shifts $\boldsymbol{h}$	m	D	η	S_m	R_m
$(0,1)$	1	1	$-1/3$	0	-2
$(0,2)$	2	-2	$-1/3$	0	-4
$(0,3)$	2	-1	$1/3$	2	0
$(0,1,2,3)$	2	2	$1/3$	0	4
$(0,1,3,7)$	3	-3	0	-2	4
$(0,2,5,9)$	4	-2	$1/6$	4	0
$(0,1,2,4,7,11)$	4	-5	$-1/3$	-2	-12

Example 14.10 (A zero mean with a persistent dyadic boundary signal)**.** For $\boldsymbol{h} = (0,1,3,7)$, Table 14.1 gives $\eta = 0$, $D = -3$. For every $m \ge 3$,

$$S_m = 2(-1)^m, \qquad R_m = -4(-1)^m.$$

The unnormalized finite signal does not tend to zero even though its normalized mean does. Cyclic and noncyclic experiments detect different boundary signals.

Example 14.11 (An odd-order telescoping observable)**.** For $\boldsymbol{h} = (0,1,2)$, $D = -1$. Every dyadic block of length $N \ge 4$ satisfies

$$\sum_{r=0}^{N-1} \varepsilon_{qN+r}\varepsilon_{qN+r+1}\varepsilon_{qN+r+2} = \varepsilon_q - \varepsilon_{q+1}.$$

Thus the sum up to QN is exactly $1 - \varepsilon_Q$, taking only the values zero and two. This is stronger information than vanishing of the limiting mean.

14.6 A nonrecursive algorithm

To evaluate an even correlation mean, choose $m = \max(1, \lceil \log_2(H+1) \rceil)$. Compute one block sum and the boundary functional, then use

$$\eta(\boldsymbol{h}) = \frac{3S_m(\boldsymbol{h}) + 2(-1)^m D(\boldsymbol{h})}{3 \cdot 2^m}.$$

For fixed order p, a direct implementation uses $O(pH)$ sign evaluations, up to the harmless factor coming from $2^m < 2(H+1)$. The integer digit-parity operation makes each sign inexpensive. This is a transparent alternative to recursively reducing a large family of correlation tuples. It is not a claim that this direct algorithm beats every specialized recursion.

Part II

The dyadic completion, Mellin renormalization, and the gamma tower

Chapter 15

Scope and principal conclusions

This part connects three pieces of the atlas's framework: the lacunary Laplace product P, the compactly supported uniform-convolution law underlying the Fabius function, and the Dirichlet continuation at large negative arguments. Throughout, with $L = \log 2$,

$$D(s,a) = \sum_{n\geq 0} \frac{\varepsilon_n}{(n+a)^s}, \qquad M(s,a) = \int_0^\infty t^{s-1} e^{-at} P(t)\,\mathrm{d}t, \tag{15.1}$$

where the shift a is a positive real number unless explicitly stated otherwise, a prime on D means differentiation with respect to s, and the Mellin transform uses the measure $t^{s-1}\,\mathrm{d}t$. The series converges locally uniformly in $\operatorname{Re} s > 0$ by summation by parts, since the partial sums of ε_n have absolute value at most one, and $D(s,a) = M(s,a)/\Gamma(s)$ gives its entire continuation (Equation (18.2)).

Entireness of the Thue–Morse Dirichlet continuation is classical: see Allouche–Cohen [6], the general automatic-series theory [8], and the subsequent treatment [29]. The uniform series and its Fabius connection are also established background [10, 23, 1]. Lognormal moment indeterminacy, size-bias scaling, and geometric-lattice representations are classical [3]; log-periodic effects in Mellin analysis and the lognormal moment problem have established precedents [21, 24]. None of those subjects is claimed as newly discovered here. The research contributions of this part are the combined completion–Mellin–zero mechanism, the coefficientwise factorization of the scaled Dirichlet expansion with a global remainder constant, the signed remainder theorem at the half-shift, the classification of completely monotone solutions of the dyadic equation $f(a) = f(a/2) - f((a+1)/2)$ by scale-invariant measures, the construction of distinct normalized gamma towers sharing every formal high-order correction, and the boundary-slope characterization of the canonical tower. The literature search did not establish priority, and one relevant 2001 preprint by Alkauskas [4] could not be retrieved; "proved here" never means "proved here for the first time in the literature."

Two limiting parameters occur and must not be conflated: $\rho \to \infty$ describes high negative order of the Dirichlet function, whereas $a \downarrow 0$ describes the boundary of the shift variable. The first governs Chapters 19 to 24 and Chapter 20; the second is what selects the canonical tower in Chapter 30.

The zero theorem has a specific historical consequence. Section 3.2 of Allouche's survey [5] asks whether the nontrivial zeros of $D(s,1)$ consist only of $2\pi \mathrm{i} k/L$. Theorem 21.1 supplies zeros with negative real part, and hence a negative answer to that exact question, as a consequence of the proof.

15.1 The main theorem in one place

Define

$$P(t) = \prod_{j\geq 0}(1 - e^{-2^j t}), \qquad \Phi(t) = \prod_{j\geq 1} \frac{1 - e^{-t/2^j}}{t/2^j}, \qquad K(t) = P(t)\Phi(t). \tag{15.2}$$

For $t > 0$ all three functions are positive. Put

$$C(s) = \int_0^\infty t^{s-1} K(t)\,\mathrm{d}t, \qquad C_0 = C(0), \qquad Q(s) = \frac{C(s)}{C_0\, 2^{s(s+1)/2}}. \tag{15.3}$$

The powers of 2 always mean e^{Lz}, with no branch choice.

Theorem 15.1 (Completion and horizontal zero strings)**.** *The integral $C(s)$ is entire; Q is entire, 1-periodic, positive on the real axis, nonconstant, and of order at most two. In particular, C has a nonreal zero.*

If s_0 is any zero of C of multiplicity d, then for each fixed $a > 0$ and all sufficiently large integers m, $D(\cdot, a)$ has a cluster of d zeros near $s_0 - m$. For every $A > 0$ these zeros lie in

$$|s - (s_0 - m)| < 2^{-Am} \tag{15.4}$$

for all sufficiently large m depending on A, a, s_0. The clusters for different sufficiently large m are disjoint. For $a = 1/2$, the stronger bound

$$|s - (s_0 - m)| \le \exp\left(-\frac{L}{2d} m^2 + O_{s_0}(m)\right) \tag{15.5}$$

holds. Every $a > 0$ therefore has infinitely many nonreal zeros in the left half-plane, including the unshifted-denominator case $a = 1$.

The ingredients and proof of the first bound occupy Chapters 16 to 19 and 21. The quantitative refinement is proved in Chapter 24. The use of "cluster" allows multiple zeros of C; simplicity is not assumed. No numerical zero enclosure is needed for the theorem.

Chapter 16

From binary cancellation to a completed product

The Laplace product P of (1.11) satisfies $P(2t) = P(t)/(1-e^{-t})$ and is flat at the origin by Theorem 1.4; the Fabius transform Φ of (1.18) satisfies $\Phi(2t) = \psi(t)\Phi(t)$ with $\psi(t) = (1-e^{-t})/t$ by Theorem 1.5. The nonmonomial factor in the first dilation law is exactly the inverse of the factor in the second. Multiplying the two products therefore produces a function whose dyadic scaling is a pure monomial.

Theorem 16.1 (Exact dyadic completion)**.** *For every $t > 0$,*

$$\boxed{K(2t) = \frac{K(t)}{t}.} \tag{16.1}$$

For every integer k, positive or negative,

$$K(2^k t) = 2^{-k(k-1)/2} t^{-k} K(t). \tag{16.2}$$

There is a positive, real-analytic, L-periodic function H such that

$$\boxed{K(t) = t^{1/2} \exp\left(-\frac{(\log t)^2}{2L}\right) H(\log t).} \tag{16.3}$$

In particular, constants $0 < h_- \le h_+ < \infty$ exist with

$$h_- t^{1/2} e^{-(\log t)^2/(2L)} \le K(t) \le h_+ t^{1/2} e^{-(\log t)^2/(2L)}. \tag{16.4}$$

Proof. Multiply Equation (1.12) and Equation (1.19) to obtain Equation (16.1). Induction upward and downward proves Equation (16.2). Set

$$H(v) = e^{v^2/(2L) - v/2} K(e^v).$$

Since $K(e^{v+L}) = e^{-v} K(e^v)$, direct substitution shows $H(v+L) = H(v)$. Positivity and continuity on $[0, L]$ give the two positive bounds. Holomorphy of $K(e^v)$ in $|\mathrm{Im}\, v| < \pi/2$ gives real analyticity. The displayed representation and estimates are now identities and consequences of compactness, respectively. $\square$

16.1 The periodic factor is genuinely present

It is tempting to omit H, because its oscillation is numerically small. Doing so would destroy the zero theorem and would replace the actual moment measure by its lognormal twin.

Proposition 16.2 (Nonconstancy). *The function H in Equation (16.3) is not constant.*

Proof. Suppose that $H \equiv c > 0$. Both $K(z)$ and

$$c\, z^{1/2} \exp\left(-\frac{(\operatorname{Log} z)^2}{2L}\right)$$

are holomorphic for $\operatorname{Re} z > 0$, where Log is the principal logarithm. Equality on the positive axis implies equality throughout that half-plane. Now take $z = x + 2\pi\mathrm{i}$ with $x > 0$. Since 2^j is an integer,

$$P(x + 2\pi\mathrm{i}) = P(x).$$

By Equation (1.13), this tends to zero as $x \downarrow 0$. The entire function Φ remains bounded near $2\pi\mathrm{i}$, so $K(x + 2\pi\mathrm{i}) \to 0$. The proposed elementary expression, however, has a finite nonzero limit at $2\pi\mathrm{i}$. This is a contradiction. $\square$

This proof uses only a boundary value and an identity theorem. In particular, it does not depend on numerical detection of a tiny Fourier coefficient.

Chapter 17

The entire Mellin transform and its periodic factor

17.1 Convergence and exact shifts

Theorem 17.1 (Entire Mellin law)**.** *The function C in Equation (15.3) is entire and satisfies*

$$C(s+1) = 2^{s+1}C(s), \tag{17.1}$$

$$C(s+j) = 2^{js+j(j+1)/2}C(s) \qquad (j \in \mathbb{Z}). \tag{17.2}$$

For real σ, $C(\sigma) > 0$. Moreover,

$$|C(\sigma + \mathrm{i}\tau)| \le h_+\sqrt{2\pi L}\, e^{L(\sigma+1/2)^2/2}. \tag{17.3}$$

Proof. With $t = e^v$, the integrand becomes

$$C(s) = \int_{-\infty}^{\infty} e^{-v^2/(2L)+(s+1/2)v} H(v)\,\mathrm{d}v. \tag{17.4}$$

On every compact set of s, this integrand and every derivative in s are bounded by an integrable Gaussian times a polynomial in v. Differentiation under the integral proves entireness. Completing the square proves Equation (17.3), and positivity is immediate. Changing variables $t = 2u$ in $C(s+1)$ gives

$$C(s+1) = 2^{s+1}\int_0^{\infty} u^s K(2u)\,\mathrm{d}u = 2^{s+1}C(s).$$

Iteration in either direction gives Equation (17.2). □

It follows at once that $Q(s+1) = Q(s)$ and $Q(\sigma) > 0$ for real σ. Dividing Equation (17.3) by the absolute value of $C_0 e^{L(s^2+s)/2}$ yields a useful stronger bound:

$$|Q(\sigma + \mathrm{i}\tau)| \le B_Q e^{L\tau^2/2}, \qquad B_Q = \frac{h_+\sqrt{2\pi L}\, e^{L/8}}{C_0}. \tag{17.5}$$

The constant does not depend on σ.

17.2 Fourier–Gaussian representation

Let

$$h_k = \frac{1}{L}\int_0^L H(v)e^{-2\pi \mathrm{i}kv/L}\,\mathrm{d}v, \qquad H(v) = \sum_{k\in\mathbb{Z}} h_k e^{2\pi \mathrm{i}kv/L}. \tag{17.6}$$

Analyticity in a strip implies exponential decay of h_k; in particular, the series converges absolutely and uniformly on the real line.

Proposition 17.2 (Gaussian smoothing in Mellin space). *For every $s \in \mathbb{C}$,*

$$C(s) = \sqrt{2\pi L}\, e^{L(s+1/2)^2/2} \sum_{k\in\mathbb{Z}} h_k e^{-2\pi^2 k^2/L} e^{2\pi \mathrm{i} k(s+1/2)}. \tag{17.7}$$

The series on the right converges normally on the entire s-plane. The periodic entire function Q is nonconstant.

Proof. Insert Equation (17.6) in Equation (17.4). Absolute summability and the Gaussian bound permit interchange. For each k, the elementary Gaussian integral gives

$$\int_{\mathbb{R}} e^{-v^2/(2L)+(s+1/2+2\pi \mathrm{i}k/L)v}\, \mathrm{d}v = \sqrt{2\pi L}\, e^{L(s+1/2+2\pi \mathrm{i}k/L)^2/2}.$$

Expanding the square proves the formula. The quadratic decay in k proves normal convergence for all complex s.

By Theorem 16.2, some h_k with $k \neq 0$ is nonzero. Multiplication by $e^{-2\pi^2 k^2/L}$ never annihilates a coefficient. Uniqueness of Fourier coefficients on the real period therefore shows that the sum in Equation (17.7), and hence Q, is nonconstant. □

The distinction between H and Q is important. The first harmonic of H is small, but the corresponding harmonic of Q is further multiplied by $e^{-2\pi^2/L}$. On the real axis Q is consequently extraordinarily close to a constant. At complex arguments with a moderate imaginary part, the factor $e^{2\pi \mathrm{i}ks}$ can undo this suppression. That is where the nonreal zeros become visible.

17.3 Why a nonreal zero must exist

We record the elementary finite-order fact needed for the main argument rather than assume a zero-location conjecture.

Lemma 17.3 (A zero-free periodic function of finite order). *Let F be an entire, 1-periodic function satisfying $|F(z)| \le e^{A+B|z|^2}$ for some constants. If F is positive on the real axis and has no zeros, then F is constant.*

Proof. Since $\mathbb{C}$ is simply connected and F has no zeros, there is an entire function g with $F = e^g$. The growth hypothesis bounds $\operatorname{Re} g(z)$ above by $A + B\,|z|^2$. The Borel–Carathéodory inequality on concentric disks of radii R and $R/2$ therefore gives

$$\max_{|z|\le R/2} |g(z) - g(0)| = O(R^2).$$

For completeness, the form used here follows by applying the maximum principle and Schwarz's lemma to $w/(2M - w)$, where $w = g - g(0)$ and $M = \max_{|z|\le R} \operatorname{Re} w$; it gives $\max_{|z|\le R/2} |w| \le 2M$ when $M > 0$. Cauchy's estimates now show that every coefficient of degree greater than two in the Taylor series of g vanishes. Thus $g(z) = az^2 + bz + c$.

Periodicity implies that $g(z+1) - g(z)$ takes values in the discrete set $2\pi\mathrm{i}\mathbb{Z}$. It is continuous, so it is constant. Therefore $a = 0$ and $b = 2\pi\mathrm{i}k$ for an integer k. Since $e^{c+2\pi\mathrm{i}kx}$ is positive for every real x, $k = 0$. Thus F is constant. □

Corollary 17.4 (Nonreal zeros of the completion). *The functions Q and C have nonreal zeros. Their zero sets agree, are invariant under complex conjugation and translation by integers, and contain no real point.*

Proof. The Gaussian factor relating Q and C never vanishes. Apply Theorem 17.3 to Equation (17.5) and use nonconstancy. Positivity excludes real zeros. Reality of K gives $C(\bar{s}) = \overline{C(s)}$, and Equation (17.2) gives the translation invariance. □

17.4 The periodic amplitude at real order

For a real order ρ it is convenient to write

$$\Theta(\rho) = 2^{-\rho(\rho-1)/2} C(-\rho) = C_0\, Q(-\rho), \tag{17.8}$$

which by (17.2) is entire, one-periodic, and positive for real ρ; at integers, $\Theta(m) = C_0$. Substituting $s = -\rho$ in (17.7) and combining the two Gaussian factors gives

$$\Theta(\rho) = \sqrt{2\pi L}\, 2^{1/8} \sum_{k\in\mathbb{Z}} (-1)^k h_k e^{-2\pi^2 k^2/L} e^{-2\pi \mathrm{i} k\rho}. \tag{17.9}$$

Mellin transformation thus suppresses the kth oscillatory mode of the profile by a Gaussian factor in k; the numerical experiments of Section 33.4 find relative variations of Θ on the real axis around 10^{-18}. That smallness is a numerical observation. The nonconstancy of Q, and hence of Θ, is proved in Theorem 17.2 and is what produces the complex zeros.

Chapter 18

The Dirichlet family: continuation and real zeros

18.1 Entire continuation with a positive Mellin numerator

The partial sums of ε_n have absolute value at most one:

$$\sum_{n<2N} \varepsilon_n = 0, \qquad \sum_{n<2N+1} \varepsilon_n = \varepsilon_N.$$

Summation by parts therefore gives locally uniform convergence of Equation (15.1) in $\operatorname{Re} s > 0$. Define

$$M(s,a) = \int_0^\infty t^{s-1} e^{-at} P(t)\, \mathrm{d}t. \tag{18.1}$$

The flatness bound Equation (1.13) handles every compact set of s at the origin, and e^{-at} handles infinity. Thus M is entire. In $\operatorname{Re} s > 1$, absolute convergence permits termwise integration, giving

$$\boxed{D(s,a) = \frac{M(s,a)}{\Gamma(s)}.} \tag{18.2}$$

Because $1/\Gamma$ is entire, this is the entire continuation. This recovers a classical result in the normalization needed here.

Proposition 18.1 (All real zeros, their signs, and their slopes)**.** *For every $a > 0$, the real zeros of $D(s,a)$ are exactly $0, -1, -2, \ldots$, and every one is simple. Writing*

$$Z_m(a) = \frac{(-1)^m}{m!} D'(-m,a), \qquad m \geq 0, \tag{18.3}$$

one has

$$\boxed{Z_m(a) = M(-m,a) > 0.} \tag{18.4}$$

For every $k \geq 0$,

$$(-1)^k \frac{\partial^k}{\partial a^k} Z_m(a) = M(k-m,a) > 0. \tag{18.5}$$

Proof. For every real s, the integrand defining $M(s,a)$ is strictly positive. The only real zeros of $1/\Gamma(s)$ are the simple zeros at $s = -m$. The residue formula $\operatorname{Res}_{s=-m} \Gamma(s) = (-1)^m/m!$ gives

$$\frac{1}{\Gamma(s)} = (-1)^m m!(s+m) + O((s+m)^2).$$

Multiply by $M(s,a)$ to obtain Equation (18.4) and simplicity. Differentiation in a is justified by the same endpoint bounds and introduces $(-t)^k$, proving Equation (18.5). ☐

These observations also show that the sign of $D(s,a)$ between successive real zeros is exactly the sign of $1/\Gamma(s)$. The sequence of slopes has a separate moment positivity property.

Corollary 18.2 (Strict moment inequalities for zero derivatives)**.** *For each fixed $a>0$, every finite Hankel matrix $[Z_{i+j}(a)]_{i,j=0}^{r}$ is positive definite. In particular,*

$$Z_m(a)Z_{m+2}(a) > Z_{m+1}(a)^2 \qquad (m \geq 0). \tag{18.6}$$

Also $Z_m'(a) = -Z_{m-1}(a)$ for $m \geq 1$.

Proof. Substitute $u = 1/t$ in Equation (18.4):

$$Z_m(a) = \int_0^\infty u^m e^{-a/u} P(1/u) \frac{\mathrm{d}u}{u}.$$

The measure is positive and has support all of $(0,\infty)$. The integral of the square of a nonzero polynomial is therefore strictly positive. This proves the matrix statement. Strict Cauchy–Schwarz applied to $u^{m/2}$ and $u^{m/2+1}$ proves the inequality. The derivative identity follows from Equation (18.5). □

18.2 Entire growth

Proposition 18.3 (Quadratic growth type)**.** *For each fixed $a>0$, both $M(\cdot,a)$ and $D(\cdot,a)$ have entire order two and type $L/2$, in the convention*

$$\limsup_{r\to\infty} \frac{\log \max_{|s|=r} |F(s)|}{r^2} = \frac{L}{2}.$$

Proof. For $0<t\leq 1$, $\Phi(t) \geq e^{-t} \geq e^{-1}$ since $X \leq 1$. Thus Equation (16.4) bounds the small-t part of M by a constant times a Gaussian integral in $\log t$, at most $\exp(Lr^2/2 + O(r))$ when $|s| \leq r$. For $t \geq 1$, $P(t) \leq 1$ and

$$\int_1^\infty t^{\operatorname{Re} s-1} e^{-at}\,\mathrm{d}t \leq \int_0^\infty t^r e^{-at}\,\mathrm{d}t = a^{-r-1}\Gamma(r+1).$$

Its logarithm is $O_a(r\log r)$. These estimates give the required upper type for M. The standard reciprocal-Gamma product gives $\log\max_{|s|\leq r} |1/\Gamma(s)| = O(r\log r)$, so the same upper type holds for D.

For the lower bound, take $s=-r$. In the logarithmic integral for M, restrict v to a fixed-length interval around $v = L(1/2-r)$. On this interval $e^{-ae^v}/\Phi(e^v)$ tends uniformly to one, and $H(v) \geq h_-$. Completing the square yields

$$M(-r,a) \geq c_a \exp\left(\frac{L}{2}(r-1/2)^2\right)$$

for all sufficiently large r. This proves the lower type for M. For D, use $r = n+1/2$. The reflection formula gives $|1/\Gamma(-r)| = \Gamma(1+r)/\pi$, which does not reduce the quadratic lower bound. The asserted order and type follow. □

18.3 Real orders, complete monotonicity, and the dyadic equation

The slopes $Z_m(a) = M(-m,a)$ of Theorem 18.1 are integer samples of a family defined at every real order. For real ρ put

$$Z_\rho(a) = M(-\rho,a) = \int_0^\infty t^{-\rho-1} e^{-at} P(t)\,\mathrm{d}t, \tag{18.7}$$

so that, for an integer $m \geq 0$, $D'(-m, a) = (-1)^m m!\, Z_m(a)$.

It is useful to keep ρ real rather than immediately restrict it to an integer. The positive quantities Z_ρ in (18.7) exist for every real ρ, although the asymptotic statements below will use $\rho \geq 0$. For every integer $\ell \geq 0$,

$$(-1)^\ell \partial_a^\ell Z_\rho(a) = \int_0^\infty t^{\ell-\rho-1} e^{-at} P(t)\, \mathrm{d}t > 0. \tag{18.8}$$

Thus every Z_ρ is completely monotone. In particular,

$$Z'_m(a) = -Z_{m-1}(a) \quad (m \geq 1). \tag{18.9}$$

Using (1.12) and substituting $t = 2u$ gives

$$Z_\rho(a) = 2^\rho \left\{ Z_\rho(a/2) - Z_\rho((a+1)/2) \right\}. \tag{18.10}$$

All of these identities follow from absolutely convergent integrals; regularized sums are unnecessary here.

Chapter 19

Exact rescaling and all-orders coefficient formulas

19.1 A bounded reciprocal transform

Define the analytic germ

$$B_a(z) = \frac{e^{-az}}{\Phi(z)} = \sum_{j\geq 0} \beta_j(a) z^j. \tag{19.1}$$

The nearest zeros of Φ are at $z = \pm 4\pi\mathrm{i}$, so the radius of this Taylor series is exactly 4π. To verify that no extra zero is hidden inside that disk, note that each factor in the normally convergent product for Φ is nonzero there and that the sum of its deviations from one converges normally. At $4\pi\mathrm{i}$ the first factor has a zero, while the numerator in Equation (19.1) is nonzero.

Lemma 19.1 (Global Taylor remainder on the positive axis)**.** *For each $a > 0$ and integer $N \geq 0$, there is a finite constant $B_{a,N}$ such that*

$$\left| B_a(t) - \sum_{j=0}^{N-1} \beta_j(a) t^j \right| \leq B_{a,N} t^N \qquad (t \geq 0). \tag{19.2}$$

The constants can be chosen uniformly when a belongs to a compact subinterval of $(0,\infty)$.

Proof. First, $\mathbb{P}(X \leq \delta) > 0$ for every $\delta > 0$. Choose J so that $\sum_{j>J} 2^{-j} \leq \delta/2$, and restrict the finitely many U_j, $j \leq J$, to a sufficiently small interval. This event has positive probability and forces $X \leq \delta$. Hence, for $0 < \delta < a$,

$$\Phi(t) \geq \mathbb{P}(X \leq \delta) e^{-\delta t}, \qquad 0 < B_a(t) \leq \frac{e^{-(a-\delta)t}}{\mathbb{P}(X \leq \delta)}.$$

Thus B_a is bounded and tends exponentially to zero on the positive axis. Near zero, analyticity bounds the remainder divided by t^N. On a compact interval away from zero the quotient is continuous; at infinity boundedness of B_a and the degree of the subtracted polynomial bound the quotient. Taking the supremum proves the claim. For a compact shift interval, fix δ smaller than its left endpoint and use compactness for the local Taylor bounds. □

This lemma does not assert convergence of the Taylor series for all positive t. It states a separate global inequality for each finite truncation. That distinction is essential when the integral samples arbitrarily large t.

19.2 The rescaling identity

Theorem 19.2 (Exact renormalization). *For $m \geq 0$ an integer, put $h = 2^{-m}$ and*

$$F_{m,a}(s) = 2^{-m(m-1)/2+ms} M(s-m, a). \tag{19.3}$$

Then, for every $s \in \mathbb{C}$,

$$\boxed{F_{m,a}(s) = \int_0^\infty u^{s-1} K(u) B_a(hu)\,\mathrm{d}u.} \tag{19.4}$$

For every compact $\mathcal{K} \subset \mathbb{C}$ and fixed integer $N \geq 1$,

$$\boxed{F_{m,a}(s) = C(s) \sum_{j=0}^{N-1} \beta_j(a)\, 2^{js+j(j+1)/2} h^j + O_{\mathcal{K},a,N}(h^N).} \tag{19.5}$$

The remainder is locally uniform in s and in positive a.

Proof. Write $e^{-at}P(t) = K(t)B_a(t)$ in Equation (18.1), and set $t = hu$. The negative-index case of Equation (16.2) says

$$K(hu) = 2^{-m(m+1)/2} u^m K(u).$$

Consequently

$$M(s-m, a) = 2^{m(m-1)/2-ms} \int_0^\infty u^{s-1} K(u) B_a(hu)\,\mathrm{d}u,$$

which is the exact identity.

Insert the finite Taylor expansion from Theorem 19.1. The jth integral is $\beta_j(a) h^j C(s+j)$; use Equation (17.2). If $s = \sigma + i\tau$, the absolute value of the remainder integral is at most

$$B_{a,N} h^N C(\sigma + N). \tag{19.6}$$

Since C is continuous on the real line, this is uniform for s in a compact set. The uniform shift statement follows from the last part of Theorem 19.1. □

The factorization in Equation (19.5) is stronger than ordinary asymptotic convergence. Every coefficient vanishes at *every* zero of C. This is precisely why the zero displacement is smaller than any fixed geometric scale.

19.3 Nonrecursive finite formulas for the coefficients

The elementary Bernoulli expansion

$$\log \frac{1-e^{-z}}{z} = -\frac{z}{2} + \sum_{r \geq 1} \frac{B_{2r}}{2r(2r)!} z^{2r}$$

yields, after summing over dyadic scales,

$$\log B_a(z) = \left(\frac{1}{2} - a\right) z - \sum_{r \geq 1} \frac{B_{2r}}{2r(2r)!(2^{2r}-1)} z^{2r}. \tag{19.7}$$

This converges for $|z| < 4\pi$. Let

$$\lambda_1 = \frac{1}{2} - a, \qquad \lambda_{2r} = -\frac{B_{2r}}{2r(2r)!(2^{2r}-1)}, \qquad \lambda_{2r+1} = 0 \quad (r \geq 1). \tag{19.8}$$

Then the ordinary coefficient, with no auxiliary recurrence, is

$$\boxed{\beta_j(a) = \sum_{\substack{n_1,\dots,n_j \ge 0 \\ n_1+2n_2+\cdots+jn_j=j}} \prod_{r=1}^{j} \frac{\lambda_r^{n_r}}{n_r!}.} \tag{19.9}$$

It follows by multiplying the exponential series $\prod_r \exp(\lambda_r z^r)$. Equivalently, $j!\beta_j$ is a complete exponential Bell polynomial evaluated at $r!\lambda_r$. The derivative relation $\partial_a \beta_j(a) = -\beta_{j-1}(a)$ for $j \ge 1$ is immediate from Equation (19.1).

With $b = 1/2 - a$, the first coefficients are

$$\begin{aligned} &\beta_0 = 1, && \beta_1 = b, && \beta_2 = \frac{b^2}{2} - \frac{1}{72}, \\ &\beta_3 = \frac{b^3}{6} - \frac{b}{72}, && \beta_4 = \frac{b^4}{24} - \frac{b^2}{144} + \frac{31}{259200}. \end{aligned} \tag{19.10}$$

19.4 All-orders slopes at the real zeros

Taking $s = 0$ in Equation (19.5) gives the promised derivative formula.

Corollary 19.3 (Zero-derivative expansion)**.** *For every fixed $a > 0$ and $N \ge 1$,*

$$\boxed{\frac{(-1)^m D'(-m,a)}{m!} = C_0\, 2^{m(m-1)/2} \left\{ \sum_{j=0}^{N-1} 2^{j(j+1)/2} \beta_j(a) 2^{-mj} + O_{a,N}(2^{-mN}) \right\}.} \tag{19.11}$$

In particular,

$$\begin{aligned} \frac{Z_m(a)}{C_0\, 2^{m(m-1)/2}} &= 1 + 2b\, 2^{-m} + \left(4b^2 - \frac{1}{9}\right) 2^{-2m} \\ &\quad + \left(\frac{32}{3} b^3 - \frac{8}{9} b\right) 2^{-3m} + O_a(2^{-4m}). \end{aligned} \tag{19.12}$$

The half-shift is the unique value of a for which all the odd coefficients vanish. It is not merely a convenient numerical choice; it corresponds exactly to centering the compact random variable X.

19.5 Two exact expansions from the same periodic factor

The identity $P = K/\Phi$ converts the profile representation (16.3) into two convergent expansions of the original product, one at each endpoint.

The identity $P = K/\Phi$, together with (1.26), gives the convergent small-t correction formula

$$\begin{aligned} \log P(t) = &- \frac{(\log t)^2}{2L} + \frac{1}{2} \log t + \log H(\log t) \\ &+ \frac{t}{2} - \sum_{j \ge 1} b_j t^{2j}, \qquad 0 < t < 4\pi. \end{aligned} \tag{19.13}$$

At the other end, expanding $-\log P(t)$ in exponentials gives

$$\begin{aligned} \log \Phi(t) = &- \frac{(\log t)^2}{2L} + \frac{1}{2} \log t + \log H(\log t) \\ &+ \sum_{n \ge 1} \frac{2^{v_2(n)+1} - 1}{n} e^{-nt}, \qquad t > 0, \end{aligned} \tag{19.14}$$

where $v_2(n)$ is the exponent of two dividing n. Indeed, in $\sum_{j\geq 0}\sum_{r\geq 1} e^{-r2^j t}/r$, the coefficient of e^{-nt} is $n^{-1}\sum_{j=0}^{v_2(n)} 2^j$. All terms are nonnegative, so the rearrangement is justified by Tonelli's theorem. Thus the same periodic amplitude links a Bernoulli correction at one endpoint to a convergent exponential correction at the other.

19.5.1 The scaled coefficients

In the expansions below the coefficients occur multiplied by the Mellin shifts, in the combination

$$\alpha_j(a) = 2^{j(j+1)/2}\beta_j(a), \tag{19.15}$$

for which Section B.1 gives a direct finite formula. The ordinary recurrence $n\beta_n = \sum_{\ell=1}^{n} \ell\lambda_\ell\beta_{n-\ell}$, $\beta_0 = 1$, with the λ_ℓ of (19.8), is often more economical for computation; it is a convenience, not a prerequisite to the closed expression.

With $b = 1/2 - a$ as in (19.10), the first coefficients are

$$\begin{aligned}
\alpha_0 &= 1, & \alpha_1 &= 2b,\\
\alpha_2 &= 4b^2 - \frac{1}{9}, & \alpha_3 &= \frac{32}{3}b^3 - \frac{8}{9}b,\\
\alpha_4 &= \frac{8}{2025}(10800b^4 - 1800b^2 + 31), &&\\
\alpha_5 &= \frac{256b}{2025}(2160b^4 - 600b^2 + 31), &&\\
\alpha_6 &= \frac{2048}{2679075}(3810240b^6 - 1587600b^4 + 164052b^2 - 2347). &&
\end{aligned} \tag{19.16}$$

Differentiation and reflection give useful independent checks:

$$\alpha_k'(a) = -2^k\alpha_{k-1}(a), \qquad \alpha_k(1-a) = (-1)^k\alpha_k(a). \tag{19.17}$$

For the second identity, use $B_{1-a}(z) = B_a(-z)$.

Chapter 20

A uniform all-orders expansion with an explicit remainder

Theorem 19.2 gives the all-orders factorization with a remainder $O_{\mathcal{K},a,N}(h^N)$ whose constant is not specified. This chapter sharpens it, at real orders $\rho \geq 0$ and for the normalized slopes $Z_\rho(a)/(2^{\rho(\rho-1)/2}\Theta(\rho))$, to an inequality with a completely explicit constant: the supremum of B_a on a tube around the positive axis. The proof replaces the global Taylor lemma Theorem 19.1 by a Cauchy estimate on that tube, which needs a lower bound for $|\Phi|$ there; that bound comes from the tilted mean of X. The same estimate then yields a dyadic q-Gevrey bound on the coefficients and an optimal-truncation error of Gaussian size in the order.

20.1 Statement of the uniform theorem

For a compact interval $I \subset (0, \infty)$ and $0 < r < 4\pi$, set

$$\mathcal{T}_r = \{z \in \mathbb{C} : \mathrm{dist}(z, [0, \infty)) \leq r\}, \qquad M_{I,r} = \sup_{a \in I,\ z \in \mathcal{T}_r} |B_a(z)|. \tag{20.1}$$

The finiteness of this constant is part of the proof below. It is a mathematically specified bound; no numerical enclosure for $M_{I,r}$ is claimed in the computational tables.

Theorem 20.1 (Uniform all-orders expansion). *For every compact $I \subset (0, \infty)$, every $0 < r < 4\pi$, every integer $N \geq 1$, every $a \in I$, and every real $\rho \geq 0$,*

$$\frac{Z_\rho(a)}{2^{\rho(\rho-1)/2}\Theta(\rho)} = \sum_{k=0}^{N-1} \alpha_k(a) 2^{-k\rho} + \mathcal{R}_N(\rho, a), \tag{20.2}$$

where

$$|\mathcal{R}_N(\rho, a)| \leq M_{I,r}\, r^{-N} 2^{N(N+1)/2 - N\rho}. \tag{20.3}$$

In particular, for each fixed N the error is uniformly $O_{I,N}(2^{-N\rho})$ as $\rho \to \infty$.

The estimate is useful both with fixed N and with N increasing with ρ. The latter use is the reason for keeping the dependence on N explicit.

20.2 Step one: exact finite-level renormalization

Lemma 20.2. *Write $\rho = m + \theta$, where $m = \lfloor \rho \rfloor$ and $0 \leq \theta < 1$, and set $h = 2^{-m}$. Then*

$$Z_{m+\theta}(a) = 2^{m(m-1)/2 + m\theta} \int_0^\infty t^{-\theta-1} K(t) B_a(ht)\, \mathrm{d}t. \tag{20.4}$$

Proof. Iteration of the product equation gives

$$P(2^{-m}t) = P(t) \prod_{j=1}^{m} (1 - e^{-t/2^j}).$$

Separate each factor into $(t/2^j)\psi(t/2^j)$ and use the product for Φ:

$$P(2^{-m}t) = 2^{-m(m+1)/2} t^m Q(t) \frac{\Phi(t)}{\Phi(2^{-m}t)}. \tag{20.5}$$

Substituting the old variable $2^{-m}t$ in the definition of $Z_{m+\theta}$ now gives (20.4), with all powers of two displayed. □

The point of this formula is that the rapidly changing order m has been removed from the Mellin weight. It occurs only in the explicit prefactor and in the small argument of B_a.

20.3 Step two: a Taylor bound valid on the entire positive axis

Lemma 20.3 (Boundedness on a pole-free tube)**.** *For $I \subset (0, \infty)$ compact and $r < 4\pi$, the constant $M_{I,r}$ in (20.1) is finite. Consequently, for every real $x \geq 0$,*

$$\left| B_a(x) - \sum_{k=0}^{N-1} \beta_k(a) x^k \right| \leq M_{I,r} r^{-N} x^N \qquad (a \in I). \tag{20.6}$$

Proof. The tube contains no zero of Φ. Thus B_a is jointly continuous and uniformly bounded when $a \in I$ and z stays in a compact portion of the tube. It remains to control its unbounded right-hand end.

For real $x > 0$, let

$$\mu_x = \frac{\mathbb{E}[X e^{-xX}]}{\mathbb{E}[e^{-xX}]} = -\frac{\Phi'(x)}{\Phi(x)}.$$

Logarithmic differentiation of the independent factors gives

$$\mu_x = \sum_{j \geq 1} \left(\frac{1}{x} - \frac{2^{-j}}{e^{x 2^{-j}} - 1} \right). \tag{20.7}$$

Each summand is the mean of a uniform variable on $[0, 2^{-j}]$ under a positive exponential tilt; it lies between zero and $2^{-j}/2$, and is also at most $1/x$. The upper bound by the untilted mean follows because the derivative of a tilted mean is the negative tilted variance. Splitting at $J = \lfloor \log_2 x \rfloor$ for $x \geq 1$ therefore gives

$$0 \leq \mu_x \leq \frac{\log_2 x + 2}{x}. \tag{20.8}$$

For $|y| \leq r$,

$$|\Phi(x+iy) - \Phi(x)| \leq \mathbb{E}[e^{-xX} |e^{-iyX} - 1|] \leq |y| \mu_x H(x).$$

The rightmost multiplier is at most $1/2$ for sufficiently large x, uniformly in $|y| \leq r$. Thus $|\Phi(x+iy)| \geq \Phi(x)/2$ there.

Since $P(x) \leq 1$, we have $\Phi(x) = K(x)/P(x) \geq K(x)$. The lower bound in Theorem 16.1, with $a_0 = \min I > 0$, yields

$$|B_a(x+iy)| \leq \frac{2}{h_-} \exp\left[-a_0 x + \frac{(\log x)^2}{2L} - \frac{1}{2} \log x \right] \longrightarrow 0.$$

The remaining part of the tube with negative real part is compact. This proves $M_{I,r} < \infty$.

Every closed disc of radius r centered on $[0, \infty)$ lies in the tube. Cauchy's estimate gives $|B_a^{(N)}(x)|/N! \leq M_{I,r} r^{-N}$ for all $x \geq 0$. Taylor's integral remainder along the real interval from 0 to x gives (20.6). Notice that this argument does not require $x < 4\pi$; the radius restriction is on the complex discs used to control derivatives, not on the real argument of the final estimate. □

20.4 Step three: integration of the global remainder

Proof of Theorem 20.1. Insert the finite Taylor polynomial from (20.6) into (20.4). The kth term integrates to $\beta_k(a)h^k K(k-\theta)$. The shift identity gives

$$\frac{C(k-\theta)}{C(-\theta)} = 2^{-k\theta+k(k+1)/2}. \tag{20.9}$$

Also,

$$2^{m(m-1)/2+m\theta}C(-\theta) = 2^{\rho(\rho-1)/2}\Theta(\rho). \tag{20.10}$$

Dividing the finite sum by this leading factor yields $\alpha_k(a)2^{-k\rho}$ term by term.

Positivity of the weight $t^{-\theta-1}K(t)$ permits direct integration of the absolute remainder bound. The resulting relative error is at most

$$M_{I,r}r^{-N}h^N\frac{C(N-\theta)}{C(-\theta)} = M_{I,r}r^{-N}2^{N(N+1)/2-N\rho}.$$

This proves the estimate without interchanging an infinite Taylor series and an improper integral. □

20.5 Divergence, optimal truncation, and centering

The Taylor series for B_a has radius 4π, so

$$\limsup_{k\to\infty} |\beta_k(a)|^{1/k} = \frac{1}{4\pi}. \tag{20.11}$$

Along an infinite subsequence the additional factor $2^{k(k+1)/2}$ forces $|\alpha_k(a)|^{1/k} \to \infty$. Hence the formal power series $\sum \alpha_k(a)h^k$ has radius zero for every $a > 0$. Its use in (20.2) is asymptotic, not convergent. For any $r < 4\pi$, Cauchy's estimate also gives $|\alpha_k(a)| \leq M_{I,r}r^{-k}2^{k(k+1)/2}$. We call this specific bound *dyadic q-Gevrey growth*; the displayed formula fixes the convention, rather than relying on differing definitions of q-Gevrey order in the literature.

Let

$$\tau = \rho + \log_2 r - \frac{1}{2}. \tag{20.12}$$

The base-two logarithm of the order-dependent factor in (20.3) is $\frac{1}{2}(N-\tau)^2 - \frac{1}{2}\tau^2$. If $\tau \geq 1$, choose a nearest integer $N \geq 1$. Then

$$|\mathcal{R}_N(\rho,a)| \leq 2^{1/8}M_{I,r}\,2^{-\frac{1}{2}(\rho+\log_2 r-1/2)^2}. \tag{20.13}$$

This is an achievable upper bound after a specified truncation, not a claim that the true error has exactly this size or that the estimate is asymptotically sharp.

When $a = 1/2$, the function $B_{1/2}$ is even. Therefore all odd coefficients vanish and the small scale in (22.12) is 4^{-m} instead of 2^{-m}. Fixed-order errors may accordingly be improved by retaining the next vanishing odd coefficient before applying the theorem. For example, the error after the displayed 2^{-6m} term is $O(2^{-8m})$.

20.6 Growth of the entire function and its gamma tower

Define the gamma hierarchy by

$$\Gamma_m^{\mathrm{TM}}(a) = \exp\bigl(\partial_s D(-m,a)\bigr) = \exp\bigl((-1)^m m! Z_m(a)\bigr). \tag{20.14}$$

Thus the expansion describes the *signed logarithms* of these functions. Exponentiating a truncated logarithmic expansion without tracking the very large factor $m!2^{m(m-1)/2}$ would not automatically produce a small relative error for Γ_m^{TM} itself. This distinction matters at high order.

Chapter 21

Horizontal strings of nonreal Dirichlet zeros

21.1 Transferring a zero of the limiting transform

Theorem 21.1 (Supergeometric localization)**.** *Let s_0 be a zero of C of multiplicity d. Fix $a > 0$. For every $A > 0$, and every sufficiently large integer m, $D(\cdot, a)$ has exactly d zeros, counted with multiplicity, in*

$$|s - (s_0 - m)| < 2^{-Am}. \tag{21.1}$$

These zeros are nonreal. In particular, there are infinitely many nonreal zeros with negative real part.

Proof. Choose a fixed disk $\mathcal{K}$ about s_0 containing no other zero of C and not meeting the real axis. In a sufficiently small disk, write

$$C(s) = (s - s_0)^d G(s), \qquad |G(s)| \geq g_0 > 0.$$

Let $h = 2^{-m}$ and choose an integer $N > Ad$. Define

$$S_N(s, h) = \sum_{j=0}^{N-1} \beta_j(a) 2^{js + j(j+1)/2} h^j.$$

For this fixed N, $S_N(s, h) \to 1$ uniformly on $\mathcal{K}$. It is consequently nonzero there and has modulus at least $1/2$ for large m. By Theorem 19.2,

$$F_{m,a}(s) = C(s) S_N(s, h) + R_{m,N}(s), \qquad \sup_{s \in \mathcal{K}} |R_{m,N}(s)| \leq c_N h^N.$$

On the circle $|s - s_0| = h^A$,

$$|C(s) S_N(s, h)| \geq \frac{g_0}{2} h^{Ad}.$$

Because $N > Ad$, this is strictly larger than $c_N h^N$ for large m. Rouché's theorem gives exactly d zeros of $F_{m,a}$ inside the circle. The nonzero prefactor in Equation (19.3) shows that the corresponding points shifted by $-m$ are zeros of $M(\cdot, a)$. They remain nonreal, so $1/\Gamma$ is nonzero there and they are also zeros of $D(\cdot, a)$, with the same multiplicities.

Finally, Theorem 17.4 supplies at least one such s_0. The centers $s_0 - m$ are separated by distance one, while the radii tend to zero. The resulting disks are disjoint and lie in the left half-plane for all sufficiently large m. □

The threshold in this theorem depends on A. The assertion is not that one fixed numerical error bound works simultaneously for all A and all m. Rather, every fixed geometric scale is eventually too coarse to resolve the displacement.

A standard $N = 1$ application of Rouché would only give convergence to a translated zero. The all-orders factorization proves much more: there is no nonzero formal power series in h describing the movement of the zero. Its displacement is a genuinely beyond-all-orders effect.

21.2 The historical zero question

For $a = 1$, the series is exactly

$$D(s, 1) = \sum_{n \geq 0} \frac{(-1)^{s_2(n)}}{(n+1)^s},$$

the series in Section 3.2 of [5]. Theorem 21.1 gives infinitely many zeros with negative real part and nonzero imaginary part. They are neither nonpositive integers nor points $2\pi \mathrm{i} k/L$. Thus the proposed exhaustion of the nontrivial zero set by that imaginary lattice is false.

The proof does not rely on the known imaginary-axis zeros, on their multiplicities, or on any unproved property of $\zeta(s)$. Its main inputs are a positive product, an entire periodic function of finite order, and Rouché's theorem. The particular zero displayed numerically later is an illustration; existence is already forced by Theorem 17.3.

For comparisons between the centered and the original family, one also has the exact elementary identity

$$\boxed{D(s, 1/2) = (1 + 2^s) D(s, 1).} \tag{21.2}$$

Indeed, separating even and odd indices in $D(s, 1)$ gives $D(s, 1) = 2^{-s} D(s, 1/2) - 2^{-s} D(s, 1)$ in the region of absolute convergence, and entireness extends the identity to all s. The multiplier has zeros only on the imaginary axis. Hence every nonreal zero off that axis in the half-shift family is also a zero of $D(s, 1)$, and conversely. This provides an additional normalization check on the numerical examples.

21.3 What the theorem does not classify

The theorem does not enumerate all zeros of Q in a fundamental strip, prove that they are simple, or exclude other families of zeros of D. It also does not assert that every zero in a left half-plane lies in one of the strings when the imaginary part is allowed to grow with m. The asymptotic theorem is locally uniform in the translated variable s, so it controls bounded-height windows. Those distinctions are useful when formulating further questions rather than overstating the result.

Chapter 22

Centering, complete monotonicity, and signed remainders

22.1 The centered reciprocal transform

At $a = 1/2$, symmetry of X gives

$$\Psi(z) := B_{1/2}(z) = \prod_{j\geq 1} \frac{z/2^{j+1}}{\sinh(z/2^{j+1})}. \tag{22.1}$$

This is an even meromorphic function. We next represent its restriction to the positive axis by a positive Laplace transform. That positivity will certify the sign and size of every finite remainder.

Lemma 22.1 (An exponential-sum representation)**.** *Let $E_{j,n}$, $j, n \geq 1$, be independent exponential random variables of mean one, and put*

$$W = \sum_{j,n\geq 1} \frac{E_{j,n}}{\pi^2 2^{2j+2} n^2}. \tag{22.2}$$

Then W is finite and positive almost surely, $\mathbb{E}W = 1/72$, and

$$\boxed{\Psi(\sqrt{x}) = \mathbb{E}e^{-xW} \qquad (x \geq 0).} \tag{22.3}$$

All positive moments of W are finite. If

$$c_r = \frac{\mathbb{E}W^r}{r!}, \tag{22.4}$$

then $c_r > 0$, and $\Psi(z) = \sum_{r\geq 0}(-1)^r c_r z^{2r}$ for $|z| < 4\pi$.

Proof. The sum of the expectations in Equation (22.2) is

$$\frac{\zeta(2)}{\pi^2} \sum_{j\geq 1} 2^{-2j-2} = \frac{1}{6} \cdot \frac{1}{12} = \frac{1}{72}.$$

A nonnegative random sum with finite expectation is finite almost surely. Euler's product $\sinh z/z = \prod_{n\geq 1}(1 + z^2/(\pi^2 n^2))$ now gives

$$\Psi(\sqrt{x}) = \prod_{j,n\geq 1} \left(1 + \frac{x}{\pi^2 2^{2j+2} n^2}\right)^{-1}.$$

Each factor is the Laplace transform of the corresponding term in Equation (22.2). First multiply finitely many factors, then pass to the almost sure limit by dominated convergence.

The least denominator in Equation (22.2) is $16\pi^2$. For $0 < y < 16\pi^2$, the product $\prod_{j,n}(1 - y/(\pi^2 2^{2j+2} n^2))^{-1}$ converges, because the sum of the inverse denominators converges and only finitely many can be close to y. Thus $\mathbb{E}e^{yW} < \infty$, so every moment is finite. Expanding this moment generating function gives the claimed Taylor coefficients and their positivity. ☐

The coefficients also have a finite positive expression. Define

$$\gamma_r = \frac{|B_{2r}|}{2r(2r)!(2^{2r}-1)}. \tag{22.5}$$

Then

$$\boxed{c_r = \sum_{\substack{n_1,\dots,n_r \geq 0 \\ n_1+2n_2+\cdots+rn_r=r}} \prod_{k=1}^{r} \frac{\gamma_k^{n_k}}{n_k!}.} \tag{22.6}$$

Indeed, substituting $z = \mathrm{i}\sqrt{y}$ in Equation (19.7) gives $\log \mathbb{E}e^{yW} = \sum_{r\geq 1} \gamma_r y^r$. Thus neither the definition of the coefficients nor their positivity requires solving a recurrence.

22.2 One positive function contains every centered slope

Introduce the probability measure

$$\mu(\,\mathrm{d}t) = \frac{K(t)}{C_0}\frac{\mathrm{d}t}{t}, \qquad t > 0, \tag{22.7}$$

and let T have this distribution, independently of W. Define

$$\mathcal{H}(x) = \mathbb{E}e^{-xT^2W} \qquad (x \geq 0), \qquad A_r = 2^{2r^2+r}c_r. \tag{22.8}$$

The exact integer moments of T follow immediately from Equation (17.2):

$$\mathbb{E}T^n = \frac{C(n)}{C_0} = 2^{n(n+1)/2} \qquad (n \in \mathbb{Z}). \tag{22.9}$$

These moments will be developed further in Chapter 25.

Theorem 22.2 (Exact centered interpolation and global remainder)**.** *The function $\mathcal{H}$ is completely monotone on $[0,\infty)$, with finite right derivatives of every order at zero, and*

$$\boxed{\frac{Z_m(1/2)}{C_0\, 2^{m(m-1)/2}} = \mathcal{H}(4^{-m}) \qquad (m \geq 0).} \tag{22.10}$$

For every integer $N \geq 0$ and every $x \geq 0$,

$$\boxed{0 \leq (-1)^N \left[\mathcal{H}(x) - \sum_{r=0}^{N-1} (-1)^r A_r x^r\right] \leq A_N x^N.} \tag{22.11}$$

For $x > 0$, the lower inequality is strict; the upper inequality is strict when $N \geq 1$. In particular, the normalized centered slopes are enclosed between successive even and odd truncations of the same expansion.

Proof. The Laplace representation makes complete monotonicity immediate:

$$(-1)^r \mathcal{H}^{(r)}(x) = \mathbb{E}[(T^2W)^r e^{-xT^2W}] > 0.$$

At zero the expectation is finite by Equation (22.9) and Theorem 22.1; dominated convergence justifies the right derivatives. Furthermore,

$$\frac{\mathbb{E}(T^2W)^r}{r!} = \mathbb{E}T^{2r}\,c_r = 2^{2r^2+r}c_r = A_r.$$

Taking $s = 0$, $a = 1/2$ in Equation (19.4), dividing by C_0, and using Equation (22.3) gives Equation (22.10).

For $y \geq 0$ and $N \geq 1$, the integral Taylor remainder is

$$e^{-y} - \sum_{r=0}^{N-1} \frac{(-y)^r}{r!} = \frac{(-1)^N}{(N-1)!} \int_0^y e^{-v}(y-v)^{N-1}\,\mathrm{d}v.$$

Its signed value is positive for $y > 0$ and at most $y^N/N!$, strictly less when $y > 0$. Substitute $y = xT^2W$ and take expectations. For $N = 0$, the statement is simply $0 < \mathcal{H}(x) \leq 1$. □

The first terms are

$$\begin{aligned}\mathcal{H}(x) \sim 1 - \frac{x}{9} + \frac{248}{2025}x^2 - \frac{4806656}{2679075}x^3 \\ + \frac{3993751257088}{10247461875}x^4 - \cdots .\end{aligned} \tag{22.12}$$

For example, for every $m \geq 0$,

$$1 - \frac{4^{-m}}{9} < \frac{Z_m(1/2)}{C_0\,2^{m(m-1)/2}} < 1 - \frac{4^{-m}}{9} + \frac{248}{2025}16^{-m}. \tag{22.13}$$

The inequalities are useful even though the full infinite series will be shown to diverge at every nonzero x.

22.3 Additional inequalities without asymptotic notation

Since $Y = T^2W$ is positive and $\mathbb{E}Y = A_1 = 1/9$, Jensen's inequality for the convex function $y \mapsto e^{-xy}$ gives

$$e^{-x/9} \leq \mathcal{H}(x) \leq 1 \qquad (x \geq 0). \tag{22.14}$$

For $x > 0$ the first inequality is strict because Y is not constant. The function $\mathcal{H}$ is also strictly log-convex:

$$\mathcal{H}(x)\mathcal{H}''(x) - \mathcal{H}'(x)^2 > 0. \tag{22.15}$$

To see this, apply strict Cauchy–Schwarz to 1 and Y under the positive measure $e^{-xY}\,\mathrm{d}\mathbb{P}$. Consequently the quantity $-\mathcal{H}'(x)/\mathcal{H}(x)$ decreases strictly with x and lies between zero and $1/9$. These are genuine global constraints on the centered slope interpolant, not only statements as $m \to \infty$.

Chapter 23

Coefficient asymptotics and optimized error bounds

23.1 The first pole controls the coefficients

Set

$$R = 16\pi^2, \qquad B = 2\prod_{k\geq 1}\frac{\pi/2^k}{\sin(\pi/2^k)}. \tag{23.1}$$

The product for B converges, because its logarithmic terms are $O(4^{-k})$.

Theorem 23.1 (Sharp leading coefficient growth)**.** *The positive coefficients c_r satisfy*

$$c_r R^r \nearrow B, \qquad c_r \sim BR^{-r}. \tag{23.2}$$

Consequently,

$$A_r \sim B\,\frac{2^{2r^2+r}}{(16\pi^2)^r}, \qquad 0 < A_r \leq B\,\frac{2^{2r^2+r}}{(16\pi^2)^r}. \tag{23.3}$$

The series $\sum_{r\geq 0}(-1)^r A_r x^r$ has radius of convergence zero.

Proof. Write the moment generating function from Theorem 22.1 as

$$\sum_{r\geq 0} c_r w^r = \prod_{j,n\geq 1}\Big(1-\frac{w}{\pi^2 2^{2j+2}n^2}\Big)^{-1} = \frac{G(w)}{1-w/R},$$

where the factor with $j = n = 1$ has been removed to define G. Every Taylor coefficient g_r of G is nonnegative. All its remaining pole parameters are at least $4R$, so G is analytic on $|w| < 4R$, and

$$c_r R^r = \sum_{k=0}^{r} g_k R^k \nearrow G(R).$$

The increase is strict because G has positive coefficients of positive degree. To evaluate $G(R)$, put $z = \mathrm{i}\sqrt{w}$ in Equation (22.1). The first factor satisfies

$$\lim_{w\to R}(1-w/R)\,\frac{\sqrt{w}/4}{\sin(\sqrt{w}/4)} = 2.$$

The remaining factors tend to those in Equation (23.1), so $G(R) = B$. This proves both the asymptotic and the upper bound. Finally, $A_r^{1/r} \sim 2^{2r+1}/R \to \infty$, proving zero radius. $\square$

Thus Equation (22.12) is an all-orders asymptotic expansion, not a convergent power series. Increasing the number of terms helps only up to a truncation order that depends on x.

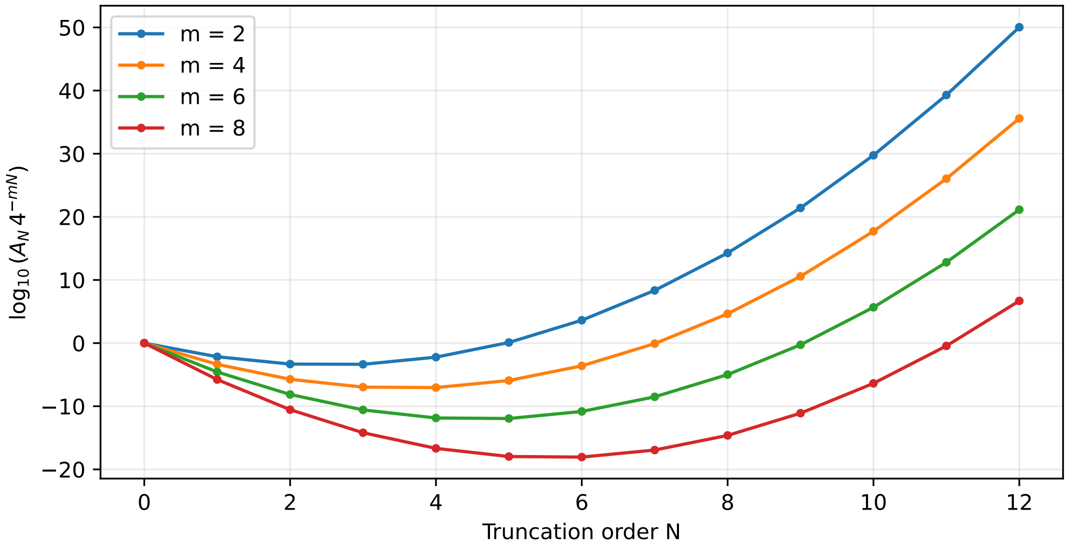

Figure 23.1: The proved first-omitted-term bounds $A_N 4^{-mN}$. Each curve first decreases and then increases. These are values of explicit bounds, not measured numerical errors.

23.2 An explicit optimal-truncation estimate

Let $\ell = \log(1/x)$, and suppose $x > 0$ is sufficiently small that the nearest integer to the following expression is nonnegative:

$$N_*(x) = \frac{\ell + \log R - L}{4L}. \tag{23.4}$$

Choose any nearest nonnegative integer N to $N_*(x)$.

Corollary 23.2 (Certified logarithmic-Gaussian error)**.** *For this truncation,*

$$\boxed{\left| \mathcal{H}(x) - \sum_{r=0}^{N-1} (-1)^r A_r x^r \right| \le B \exp\left\{ \frac{L}{2} - \frac{[\log(1/x) + \log(8\pi^2)]^2}{8L} \right\}.} \tag{23.5}$$

At the dyadic samples $x = 4^{-m}$, this is a relative error bound for Equation (22.10) of size $\exp[-Lm^2/2 + O(m)]$.

Proof. By Equations (22.11) and (23.3), the error is at most

$$B \exp\{2LN^2 + (L - \log R - \ell)N\}.$$

Completing the square gives the minimum at Equation (23.4), with value $-(\ell + \log R - L)^2/(8L)$ in the exponent. Rounding to a nearest integer increases the quadratic by at most $2L(1/2)^2 = L/2$. Since $\log R - L = \log(8\pi^2)$, the result follows. □

This is a rigorous upper bound, including its explicit constant. It is not an assertion that the actual error has that leading asymptotic size. The distinction matters, especially at complex zeros where extra cancellation occurs.

Chapter 24

Gaussian localization of the centered zero strings

The alternating inequality also controls complex translated arguments, although the complex remainder no longer has a sign. This is the quantitative refinement asserted in Theorem 15.1.

Proposition 24.1 (Complex centered remainder)**.** *Let $h = 2^{-m}$, $s = \sigma + \mathrm{i}\tau$, and $N \geq 0$. Then*

$$F_{m,1/2}(s) = C(s) \sum_{r=0}^{N-1} (-1)^r c_r 2^{2rs+2r^2+r} h^{2r} + R_{m,N}(s), \tag{24.1}$$

$$|R_{m,N}(s)| \leq c_N h^{2N} C(\sigma + 2N) = c_N h^{2N} 2^{2N\sigma+2N^2+N} C(\sigma). \tag{24.2}$$

For s in any fixed compact set, an integer $N = N(m) = m/2 + O(1)$ can be chosen so that

$$\sup_s \left|R_{m,N(m)}(s)\right| \leq \exp[-Lm^2/2 + O(m)]. \tag{24.3}$$

Proof. Use the Taylor remainder of $\Psi(t) = \mathbb{E}e^{-t^2 W}$ for real $t \geq 0$: its absolute value after N even terms is at most $c_N t^{2N}$. Insert it into Equation (19.4); taking absolute values replaces u^{s-1} by $u^{\sigma-1}$. This proves the first inequality. The Mellin shift identity proves the equality.

On a compact set let $\sigma \leq b$, and bound $C(\sigma)$ by a constant. Using $c_N \leq BR^{-N}$, the logarithm of the bound is at most

$$2LN^2 + \{L(1 + 2b - 2m) - \log R\}N + O(1).$$

Its minimizing integer is $m/2 + O(1)$; substitution gives Equation (24.3). □

Theorem 24.2 (Centered Gaussian zero clusters)**.** *If s_0 is a zero of C of multiplicity d, then the d zeros of $D(\cdot, 1/2)$ in the cluster approaching $s_0 - m$ satisfy*

$$|s - (s_0 - m)| \leq \exp[-Lm^2/(2d) + O_{s_0}(m)]. \tag{24.4}$$

The same conclusion holds for the corresponding off-axis zeros of $D(\cdot, 1)$.

Proof. Choose a small fixed disk $\mathcal{K}$ about s_0 as in Theorem 21.1. Use the truncation $N(m)$ of Theorem 24.1. We must check that the finite multiplier

$$S_m(s) = \sum_{r=0}^{N(m)-1} (-1)^r c_r 2^{2rs+2r^2+r} h^{2r}$$

remains nonzero near s_0 even though its number of terms grows. For $1 \leq r < N(m)$, the bound on the logarithm of the rth term is

$$2Lr^2 + \{L(1 + 2b - 2m) - \log R\}r + \log B.$$

This convex quadratic decreases until a point $m/2+O(1)$. On the integer interval from 1 to $N(m)-1$ its maximum is at an endpoint. The first endpoint is $-2Lm+O(1)$, while the last is $-Lm^2/2+O(m)$. Thus $\sup_{s\in\mathcal{K}} |S_m(s)-1| = O(m2^{-2m}) \to 0$. In particular $|S_m| \geq 1/2$ for large m.

The local lower bound $|C(s)| \geq g_0 |s-s_0|^d$ and Equation (24.3) now allow Rouché on a circle of radius $\exp[-Lm^2/(2d)+B_1 m]$, for a sufficiently large fixed B_1. The comparison term strictly dominates the remainder. This gives the asserted d zeros. Identity Equation (21.2) transfers the statement to $a=1$, since these disks are eventually disjoint from the imaginary axis. □

The proof is deliberately insensitive to the multiplicity of the limiting zero. For a simple zero the coefficient $1/d$ is absent; for a multiple zero it is the natural loss produced by a local factor $(s-s_0)^d$.

Chapter 25

A non-lognormal measure with exact lognormal moments

25.1 Moment equality and genuine distinctness

Let V be lognormal with

$$\log V \sim N(L/2, L). \tag{25.1}$$

Its moments are $\mathbb{E}V^s = \exp[Ls(s+1)/2]$ for real s, and indeed for complex s. By Equation (22.9),

$$\mathbb{E}T^n = \mathbb{E}V^n \qquad (n \in \mathbb{Z}). \tag{25.2}$$

In logarithmic coordinates, the two densities are

$$f_{\log T}(v) = \frac{1}{C_0} e^{-v^2/(2L)+v/2} H(v), \tag{25.3}$$

$$f_{\log V}(v) = \frac{1}{\sqrt{2\pi L}} e^{-(v-L/2)^2/(2L)}. \tag{25.4}$$

Their density ratio is

$$\frac{\mathrm{d}\mathbb{P}_T}{\mathrm{d}\mathbb{P}_V}(e^v) = \frac{\sqrt{2\pi L}\, e^{L/8}}{C_0} H(v). \tag{25.5}$$

It is bounded above and below by positive constants, but it is not constant. Thus the measures are distinct, despite matching every positive and negative integer moment.

This is an explicit realization of a classical indeterminate moment problem. The new role of this particular representing measure is that it is built from the Thue–Morse product and supplies the scaling limit for the Dirichlet zero problem. The abstract fact that lognormal moments do not determine a measure is not new [3, Section 9].

25.2 Size bias and the periodic degree of freedom

The size-biased law of a positive random variable T with finite mean is defined by

$$\mathbb{E}f(T^*) = \frac{\mathbb{E}[Tf(T)]}{\mathbb{E}T}$$

for bounded measurable f. In the present normalization, $\mathbb{E}T = 2$. A change of variables and Equation (16.1) give

$$T^* \stackrel{d}{=} 2T. \tag{25.6}$$

Indeed, the density of T^* is $K(t)/(2C_0)$, and the density of $2T$ is $K(t/2)/(C_0 t) = K(t)/(2C_0)$.

More generally, let p be a bounded L-periodic measurable function. The same Mellin substitution used for Equation (17.1) shows that

$$\int_0^\infty t^n p(\log t)\mu(\,\mathrm{d}t) = 2^{n(n+1)/2}\int_0^\infty p(\log t)\mu(\,\mathrm{d}t) \quad (n \in \mathbb{Z}). \tag{25.7}$$

If p has mean zero under μ, then $(1 + \theta p(\log t))\mu(\,\mathrm{d}t)$ is a probability measure with the same integer moments whenever $|\theta|\;\|p\|_\infty \le 1$. Choosing a nonconstant p produces infinitely many such measures.

To prove Equation (25.7) for all integer n, define $C_p(s) = \int t^{s-1}K(t)p(\log t)\,\mathrm{d}t$. Boundedness of p permits every integral, and its periodicity gives $C_p(s+1) = 2^{s+1}C_p(s)$ by the same substitution. Iteration at integer arguments proves the formula, including negative indices.

25.3 An exact mixture over geometric lattices

Write every $t > 0$ uniquely as $t = v2^k$ with $1 \le v < 2$ and $k \in \mathbb{Z}$. Define

$$\vartheta(v) = \sum_{k\in\mathbb{Z}} v^{-k}2^{-k(k-1)/2}. \tag{25.8}$$

The series converges uniformly for $1 \le v \le 2$. Then

$$\mu = \int_1^2 \nu_v \,\frac{K(v)\vartheta(v)}{C_0}\,\frac{\mathrm{d}v}{v}, \tag{25.9}$$

where the discrete probability measure ν_v is supported on $\{v2^k : k \in \mathbb{Z}\}$ and has weights

$$\nu_v(\{v2^k\}) = \frac{v^{-k}2^{-k(k-1)/2}}{\vartheta(v)}. \tag{25.10}$$

To verify the identity, split the integral defining μ into the intervals $[2^k, 2^{k+1})$, substitute $t = 2^k v$, and apply Equation (16.2). This also proves that the mixing density integrates to one.

Every single ν_v has the same integer moments:

$$\int t^n \nu_v(\,\mathrm{d}t) = 2^{n(n+1)/2} \qquad (n \in \mathbb{Z}). \tag{25.11}$$

In fact, shift the summation index by n in the numerator obtained from Equation (25.10); completing the exponent gives the factor $v^{-n}2^{n(n+1)/2}$, which cancels the external v^n. These geometric-lattice moment representatives are classical Chihara–Leipnik-type laws; see [3]. Formula Equation (25.9) identifies the particular phase mixture selected by the Thue–Morse completion.

25.4 Hankel determinants of the slopes and their growth

Theorem 18.2 proved that the Hankel matrices of the slopes $Z_m(a)$ are positive definite. The uniform expansion of Theorem 20.1 makes their growth explicit.

25.4.1 Strict positivity and quantitative Hankel growth

The substitution $x = 1/t$ in Z_m gives

$$Z_m(a) = \int_0^\infty x^m w_a(x)\,\mathrm{d}x, \qquad w_a(x) = x^{-1}e^{-a/x}P(1/x) > 0. \tag{25.12}$$

Every integer moment exists. Thus the following positivity concerns the *derivative data* of D, not an unrelated Hankel matrix formed straight from the signs ε_n.

Proposition 25.1 (Strict Hankel positivity). *For integers $n, \ell \geq 0$,*

$$\det[Z_{i+j+\ell}(a)]_{i,j=0}^{n} > 0. \tag{25.13}$$

In particular,

$$Z_m(a)Z_{m+2}(a) > Z_{m+1}(a)^2, \qquad \frac{Z_m(a)Z_{m+2}(a)}{Z_{m+1}(a)^2} \longrightarrow 2. \tag{25.14}$$

More generally, for fixed n as $m \to \infty$,

$$\begin{aligned}\det[Z_{m+i+j}(a)]_{i,j=0}^{n} \sim C_0^{n+1} 2^{\frac{n+1}{2}m(m-1)+mn(n+1)+\frac{n(n+1)(n-1)}{3}} \\ \times \prod_{0\leq i<j\leq n} (2^j - 2^i).\end{aligned} \tag{25.15}$$

The relative error in (25.15) is $O(2^{-m})$ for fixed a, n, and is $O(2^{-2m})$ at $a = 1/2$.

Proof. The matrix in (25.13) is the Gram matrix of $1, x, \ldots, x^n$ for the strictly positive measure $x^\ell w_a(x)\,\mathrm{d}x$. A nonzero polynomial cannot vanish almost everywhere for this measure, so the matrix is positive definite.

For the asymptotic determinant, apply Theorem 20.1 to its finitely many entries. Factor $C_0 2^{m(m-1)/2}$ from each row, $2^{mi+i(i-1)/2}$ from row i, and $2^{mj+j(j-1)/2}$ from column j. The remaining matrix is $[2^{ij} + O(2^{-m})]$. Its limiting determinant is the nonzero Vandermonde product $\prod_{i<j}(2^j - 2^i)$. Summing the row and column exponents gives (25.15); the centered improvement follows from $\alpha_1(1/2) = 0$. The limiting ratio in (25.14) follows directly from the leading terms of the three entries. □

Chapter 26

Exact moment algebra: determinants and orthogonal polynomials

The integer moments in Equation (22.9) make a small amount of exact algebra available essentially for free. It provides useful checks on both the normalization and the relation to the Stieltjes–Wigert moment problem. The algebraic identities themselves are standard consequences of geometric moments, not independent priority claims.

Proposition 26.1 (Hankel determinant)**.** *For every integer $d \geq 1$,*

$$\boxed{\det \left[2^{(i+j)(i+j+1)/2}\right]_{i,j=0}^{d-1} = 2^{d^2(d-1)/2} \prod_{k=1}^{d-1} (2^k - 1)^{d-k}.} \tag{26.1}$$

Proof. Factor $2^{i(i+1)/2}$ from row i and $2^{j(j+1)/2}$ from column j. The remaining matrix is $[(2^i)^j]$, whose determinant is the Vandermonde product $\prod_{0 \leq i < j < d}(2^j - 2^i)$. In this product factor 2^i from each difference. The powers of 2 from the rows and columns have total exponent $d(d-1)(d+1)/3$; the extra exponent is $d(d-1)(d-2)/6$. Their sum is $d^2(d-1)/2$. A difference $j - i = k$ occurs $d - k$ times, giving the remaining product. □

Define the Gaussian binomial coefficient at $q = 2$ by

$$\begin{bmatrix} n \\ k \end{bmatrix}_2 = \prod_{j=0}^{k-1} \frac{2^{n-j} - 1}{2^{j+1} - 1} \qquad (0 \leq k \leq n). \tag{26.2}$$

Empty products equal one.

Proposition 26.2 (Monic orthogonal polynomials)**.** *The monic degree-n polynomial orthogonal to $1, t, \ldots, t^{n-1}$ under μ is*

$$\boxed{p_n(t) = \sum_{k=0}^{n} (-1)^{n-k} 2^{n(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_2 t^k.} \tag{26.3}$$

Its squared norm is

$$\int p_n(t)^2 \mu(\,\mathrm{d}t) = 2^{n(3n+1)/2} \prod_{k=1}^{n} (2^k - 1). \tag{26.4}$$

The same polynomials and norms apply to every other representing measure discussed in Chapter 25.

Proof. For a polynomial $p(t) = \sum_{k=0}^{n} a_k t^k$,

$$\int t^j p(t) \mu(\,\mathrm{d}t) = 2^{j(j+1)/2} \sum_{k=0}^{n} a_k 2^{k(k+1)/2} (2^j)^k.$$

Thus orthogonality for $j = 0, \ldots, n-1$ is equivalent to the polynomial identity

$$\sum_{k=0}^{n} a_k 2^{k(k+1)/2} z^k = 2^{n(n+1)/2} \prod_{j=0}^{n-1} (z - 2^j).$$

Expanding the product by the finite q-binomial identity gives Equation (26.3). This proves both orthogonality and monicity. Strict positivity of the Hankel matrices guarantees uniqueness. The norm is the ratio of the size-$(n+1)$ and size-n Hankel determinants; substitution of Equation (26.1) gives Equation (26.4). Alternatively, the same ratio follows by triangular Gram–Schmidt elimination. Every integral used here is a polynomial moment, so it is unchanged for any representing measure with those moments. □

For example,

$$p_0(t) = 1, \qquad p_1(t) = t - 2, \qquad p_2(t) = t^2 - 12t + 16.$$

The norm of p_1 is $8 - 4 = 4$, agreeing with Equation (26.4). These low-degree checks are sensitive to the easily confused alternatives $2^{n(n-1)/2}$ and $2^{n(n+1)/2}$ in the moment normalization.

Chapter 27

Identical asymptotic series, different positive functions

The moment interpretation gives a concrete nonuniqueness result for the all-orders expansion itself. Let V be the lognormal variable in Equation (25.1), independent of the same W, and set

$$\widetilde{\mathcal{H}}(x) = \mathbb{E}e^{-xV^2W}. \tag{27.1}$$

Theorem 27.1 (A completely monotone beyond-all-orders ambiguity)**.** *The functions $\mathcal{H}$ and $\widetilde{\mathcal{H}}$ are distinct, completely monotone functions with identical derivatives of every order at zero. Both satisfy the same alternating bound Equation (22.11). Consequently, for every $N \geq 0$,*

$$\left|\mathcal{H}(x) - \widetilde{\mathcal{H}}(x)\right| \leq 2A_N x^N \qquad (x \geq 0). \tag{27.2}$$

Optimizing N as in Theorem 23.2 gives a logarithmic-Gaussian upper bound for their difference at small x.

Proof. Equality of the integer moments of T and V gives

$$\frac{\mathbb{E}(T^2W)^r}{r!} = \frac{\mathbb{E}(V^2W)^r}{r!} = A_r$$

for every integer $r \geq 0$. The proof of Theorem 22.2 applies unchanged to both Laplace transforms, so the derivatives and remainder bounds agree. Subtracting the same finite polynomial gives Equation (27.2).

Suppose nevertheless that the two functions were identical. Uniqueness of Laplace transforms for probability measures on $[0, \infty)$ would imply $T^2W \overset{d}{=} V^2W$. For every real $r > 0$, all moments involved are finite and $\mathbb{E}W^r > 0$, so independence permits cancellation:

$$\mathbb{E}T^{2r}\,\mathbb{E}W^r = \mathbb{E}V^{2r}\,\mathbb{E}W^r \quad \Longrightarrow \quad Q(2r) = 1.$$

The identity theorem would then force $Q \equiv 1$, contradicting Theorem 17.2. Thus the functions are distinct. □

The factor 2 in Equation (27.2) is convenient rather than optimal: since both functions lie in the same interval between successive alternating truncations, one can in fact replace it by 1. Keeping the weaker bound makes clear that even the triangle inequality already proves flatness. Either version is enough to show that every coefficient of the asymptotic expansion can be known exactly while the function still is not uniquely specified.

This phenomenon is not an arbitrary addition of a small oscillatory term. Both realizations are positive Laplace transforms and satisfy all complete-monotonicity inequalities. Thus positivity, smoothness, and a full asymptotic series still leave a nontrivial ambiguity. The Thue–Morse product supplies the extra information selecting $\mathcal{H}$ rather than its lognormal replacement.

Chapter 28

Classification of completely monotone dyadic solutions

28.1 The measure theorem

We use the Bernstein representation theorem: a completely monotone function on $(0, \infty)$ is the Laplace transform of a unique positive Radon measure on $[0, \infty)$, with finite Laplace transform at every positive argument. One reference stating both representation and uniqueness is Aguech–Jedidi [2, Theorem 1].

Theorem 28.1 (All positive dyadic solutions). *A function $f : (0, \infty) \to \mathbb{R}$ is completely monotone and satisfies*

$$f(a) = f(a/2) - f((a + 1)/2) \tag{28.1}$$

if and only if there is a positive Radon measure ν on $(0, \infty)$ with $\nu(2B) = \nu(B)$ such that

$$f(a) = \int_0^\infty e^{-at} P(t)\, \nu(\,\mathrm{d}t). \tag{28.2}$$

The measure is unique. It is determined by an arbitrary finite positive measure ν_0 on the half-open fundamental interval $[1, 2)$, and the correspondence is explicitly

$$f(a) = \int_{[1,2)} \sum_{j\in\mathbb{Z}} e^{-a2^j v} P(2^j v)\, \nu_0(\,\mathrm{d}v). \tag{28.3}$$

The zero measure corresponds to the zero solution.

Proof. Suppose first that f is completely monotone. Write $f(a) = \int_{[0,\infty)} e^{-at}\, \mu(\,\mathrm{d}t)$ by Bernstein's theorem. For a measure μ, denote by $\mu(2\,\mathrm{d}u)$ the pushforward under $t \mapsto t/2$, so it assigns mass $\mu(2B)$ to a Borel set B. Equation (28.1) and uniqueness of Laplace transforms give

$$\mu(\,\mathrm{d}u) = (1 - e^{-u})\mu(2\,\mathrm{d}u). \tag{28.4}$$

This also proves $\mu(\{0\}) = 0$. Since $P(u) > 0$ for $u > 0$, define $\nu(\,\mathrm{d}u) = P(u)^{-1}\mu(\,\mathrm{d}u)$. It is locally finite on $(0, \infty)$. Using $P(u) = (1 - e^{-u})P(2u)$ in (28.4) yields $\nu(\,\mathrm{d}u) = \nu(2\,\mathrm{d}u)$, exactly the stated invariance. Uniqueness follows from uniqueness of μ.

Conversely, let ν be positive and invariant. Its mass on $[1, 2)$ is finite by local finiteness. Partitioning $(0, \infty)$ into the disjoint half-open intervals $[2^j, 2^{j+1})$ gives (28.3). At $j \to -\infty$, the bound (1.13) with arbitrarily large N makes the shell sum converge, even after multiplication by any fixed power of t. At $j \to +\infty$, the factor e^{-at} dominates every power. The convergence is locally uniform for $a > 0$ and persists under all a-derivatives. Thus (28.2) is finite and completely monotone. The scaling equation of P and invariance of ν verify (28.1) directly. Finally, a finite measure on $[1, 2)$ extends uniquely by scaling to such a measure on $(0, \infty)$. □

This classification includes discrete, absolutely continuous, and singular measures. It does not secretly assume that a density with respect to $\mathrm{d}t/t$ exists.

28.2 Normalization fixes mass, not phase

Corollary 28.2 (The normalized solution simplex). *For the solution in Theorem 28.1,*

$$f(1/2) = \nu_0([1,2)). \tag{28.5}$$

Consequently, the normalized solutions $f(1/2) = L$ are exactly the mixtures of

$$f_\theta(a) = L \sum_{j\in\mathbb{Z}} e^{-a2^{j+\theta}} P(2^{j+\theta}), \qquad 0 \le \theta < 1. \tag{28.6}$$

Distinct phases give distinct functions. Their uniform phase average is canonical:

$$Z_0(a) = \int_0^1 f_\theta(a)\,\mathrm{d}\theta. \tag{28.7}$$

Proof. The identity

$$e^{-t/2}P(t) = P(t) - P(t/2) \tag{28.8}$$

gives, for every fixed $v > 0$,

$$\sum_{j\in\mathbb{Z}} e^{-2^{j-1}v} P(2^j v) = \lim_{J\to\infty} \{P(2^J v) - P(2^{-J-1}v)\} = 1.$$

Insert this into (28.3) and use positivity to interchange sum and integral. This proves (28.5). The extremal measures with mass L are point masses $L\delta_{2^\theta}$; the representation is affine and injective, so their functions are the extremal normalized solutions. Distinct phases have disjoint multiplicative lattices, and uniqueness of Laplace transforms distinguishes the functions. For the last identity, substitute $t = 2^{j+\theta}$ on each shell, so $\mathrm{d}t/t = L\,\mathrm{d}\theta$. ☐

In particular,

$$Z_0(1/2) = L, \qquad Z_0(1) = L/2, \qquad \Gamma_0^{\mathrm{TM}}(1/2) = 2. \tag{28.9}$$

Every normalized solution also has $f(1) = L/2$, because the equation at $a = 1$ reads $f(1) = f(1/2) - f(1)$. Thus evaluating only at either of these two points conceals the nonuniqueness. The computations below use $a = 1/3$ instead.

28.3 Nonuniqueness persists for entire differential towers

Given an invariant positive measure ν, define, for $\rho \in \mathbb{R}$,

$$Z_\rho^\nu(a) = \int_0^\infty t^{-\rho} e^{-at} P(t)\,\nu(\mathrm{d}t). \tag{28.10}$$

The endpoint argument in the classification proof shows that these integrals are finite and completely monotone for every fixed ρ. For integer levels,

$$(Z_m^\nu)' = -Z_{m-1}^\nu \quad (m \ge 1), \qquad Z_m^\nu(a) = 2^m\{Z_m^\nu(a/2) - Z_m^\nu((a+1)/2)\}. \tag{28.11}$$

Thus the phase freedom is compatible with every level of the differential ladder and every dyadic equation. Moreover, $Z_m^\nu(a) \to 0$ as $a \to \infty$, by dominated convergence with any fixed positive lower bound on a. Vanishing at infinity does not eliminate this freedom.

Chapter 29

Universal coefficients and beyond-all-orders ambiguity

29.1 The same coefficient calculus works for every phase measure

For a nonzero invariant positive measure ν, set

$$C_\nu(s) = \int_0^\infty t^s K(t)\, \nu(\,\mathrm{d}t), \qquad \Theta_\nu(\rho) = 2^{-\rho(\rho-1)/2} C_\nu(-\rho). \tag{29.1}$$

The Gaussian envelope on each shell proves that C_ν is entire and positive on the real axis. Invariance and (16.1) imply

$$C_\nu(s+1) = 2^{s+1} C_\nu(s). \tag{29.2}$$

Consequently Θ_ν is one-periodic and positive on the real axis.

Theorem 29.1 (Universal high-order coefficients)**.** *For every nonzero invariant positive measure ν, the expansion*

$$\frac{Z_\rho^\nu(a)}{2^{\rho(\rho-1)/2}\Theta_\nu(\rho)} = \sum_{k=0}^{N-1} \alpha_k(a) 2^{-k\rho} + \mathcal{R}_N^\nu(\rho, a) \tag{29.3}$$

has exactly the same coefficient polynomials as in Theorem 20.1, and

$$|\mathcal{R}_N^\nu(\rho, a)| \le M_{I,r} r^{-N} 2^{N(N+1)/2 - N\rho}. \tag{29.4}$$

In particular, the relative remainder bound is independent of ν.

Proof. Write $\rho = m + \theta$ as before. Scaling by 2^{-m} in the invariant measure and using (20.5) gives

$$Z_{m+\theta}^\nu(a) = 2^{m(m-1)/2 + m\theta} \int_0^\infty t^{-\theta} K(t) B_a(2^{-m} t)\, \nu(\,\mathrm{d}t).$$

Apply the global Taylor estimate and replace the K-recurrence in the proof of Theorem 20.1 by (29.2). Positivity permits the same absolute remainder estimate. The moment ratios are identical, so no coefficient changes. □

29.2 Normalized towers can agree to every formal order

Theorem 29.2 (Beyond-all-orders nonuniqueness)**.** *There are infinitely many invariant positive measures $\nu \ne \mathrm{d}t/t$ for which*

$$Z_0^\nu(1/2) = L, \qquad C_\nu(0) = C_0. \tag{29.5}$$

For each of them and every fixed $a > 0$, the two sequences $Z_m^\nu(a)$ and $Z_m(a)$ have the same complete formal expansion in powers of 2^{-m}, with the same leading constant. More quantitatively, they may be constructed so that, for every compact $I \subset (0, \infty)$, $0 < r < 4\pi$, and integer $N \geq 1$,

$$\frac{|Z_m^\nu(a) - Z_m(a)|}{C_0\, 2^{m(m-1)/2}} \leq |\eta| M_{I,r} r^{-N} 2^{N(N+1)/2 - Nm}, \qquad a \in I, \tag{29.6}$$

where $0 < |\eta| < 1$ is a freely chosen perturbation parameter. The corresponding functions $a \mapsto Z_0^\nu(a)$ are nevertheless distinct.

Proof. Choose a nonzero real trigonometric polynomial p of period one satisfying

$$\int_0^1 p(y)\,\mathrm{d}y = 0, \qquad \int_0^\infty K(t) p(\log_2 t) \frac{\mathrm{d}t}{t} = 0, \qquad \|p\|_\infty \leq 1. \tag{29.7}$$

Such choices exist in an infinite-dimensional family. To see this concretely, take any finite-dimensional span of nonconstant sines and cosines. Every element already has zero ordinary mean; the weighted condition imposes at most one linear equation. With two or more independent modes there is a nonzero solution, which can be rescaled. Increasing the number of modes gives arbitrarily large families.

Define

$$\nu(\mathrm{d}t) = \{1 + \eta p(\log_2 t)\} \frac{\mathrm{d}t}{t}, \qquad 0 < |\eta| < 1. \tag{29.8}$$

This is positive and dyadically invariant. The first zero mean gives $\nu([1, 2)) = L$; the second gives $C_\nu(0) = C_0$. The periodic-perturbation identity (25.7) gives equality of all integer Mellin moments. Theorem 29.1, at integer orders, therefore gives the same formal expansion and leading constant.

For the stronger bound, use the exact renormalization of the difference:

$$Z_m^\nu(a) - Z_m(a) = \eta\, 2^{m(m-1)/2} \int_0^\infty K(t) p(\log_2 t) B_a(2^{-m} t) \frac{\mathrm{d}t}{t}.$$

Every term of the Taylor polynomial integrates to zero by (25.7). The global Taylor remainder, $|p| \leq 1$, and $C(N) = 2^{N(N+1)/2} C_0$ give (29.6). Finally, the Laplace measures $P(t)\nu(\mathrm{d}t)$ are different because p is nonzero and $P > 0$. Laplace uniqueness implies that the functions Z_0^ν are different. □

Choosing N as in (20.12)–(20.13), with $\rho = m$, makes the relative difference in (29.6) at most

$$|\eta| 2^{1/8} M_{I,r}\, 2^{-\frac{1}{2}(m + \log_2 r - 1/2)^2}. \tag{29.9}$$

This gives a concrete mathematical meaning to “invisible to all formal high-order corrections.” It does not say that two distinct towers have the same exact values. Nor does it rule out characterization by a different asymptotic variable: the next section uses $a \downarrow 0$ precisely to recover the missing information.

Chapter 30

A boundary characterization of the canonical tower

30.1 A single slope limit selects the invariant measure

Theorem 30.1 (Boundary uniqueness). *Let f be completely monotone on $(0,\infty)$ and satisfy (28.1). Then*

$$\lim_{a\downarrow 0}(-af'(a)) = 1 \tag{30.1}$$

holds if and only if

$$f(a) = Z_0(a) = \int_0^\infty e^{-at}P(t)\frac{\mathrm{d}t}{t}. \tag{30.2}$$

No separate point normalization is needed. More generally, replacing the limit one by a nonnegative constant c selects $f = cC_0$.

Proof. Let ν be the invariant representing measure from Theorem 28.1. Fix $s > 0$ and set $a_N = s2^{-N}$. Scaling by 2^N in the measure gives

$$-a_N f'(a_N) = \int_0^\infty ste^{-st}P(2^N t)\,\nu(\,\mathrm{d}t). \tag{30.3}$$

For each fixed $t > 0$, $P(2^N t) \to 1$. The dominating kernel ste^{-st} is integrable against ν: on shells tending to zero it is $O_s(2^j)$, and on shells tending to infinity it has exponential decay. Each shell has the same finite mass. Dominated convergence therefore yields

$$\lim_{N\to\infty}(-a_N f'(a_N)) = s\int_0^\infty te^{-st}\,\nu(\,\mathrm{d}t). \tag{30.4}$$

If (30.1) holds, the right side equals one for every $s > 0$. Hence the measure $t\nu(\,\mathrm{d}t)$ has Laplace transform $1/s$. This is the Laplace transform of Lebesgue measure $\,\mathrm{d}t$ on $(0,\infty)$, and uniqueness gives $t\nu(\,\mathrm{d}t) = \,\mathrm{d}t$. Thus $\nu(\,\mathrm{d}t) = \,\mathrm{d}t/t$, proving necessity.

For the canonical measure, substitution $u = at$ gives

$$-aC_0'(a) = a\int_0^\infty e^{-at}P(t)\,\mathrm{d}t = \int_0^\infty e^{-u}P(u/a)\,\mathrm{d}u \longrightarrow 1$$

by $0 < P \le 1$ and dominated convergence. The proof with limit c identifies $t\nu(\,\mathrm{d}t) = c\,\mathrm{d}t$ in exactly the same way, including $c = 0$. □

Equation (30.4) explains the obstruction in the other solutions: in general, the dyadic boundary limits retain the phase information of ν. A full limit as $a \downarrow 0$, rather than a limit along one selected dyadic sequence, removes that information. The hypothesis genuinely means a limit through *all* positive a.

30.2 A finite-part condition is sufficient

Corollary 30.2 (Finite-part characterization)**.** *Within the completely monotone solutions of* (28.1)*, the existence of a finite limit*

$$\lim_{a\downarrow 0}\{f(a)+\log a\}\in\mathbb{R} \tag{30.5}$$

is equivalent to $f=Z_0$*. For the canonical function, the limit is*

$$\kappa=-\gamma+\int_0^1\frac{P(t)}{t}\,\mathrm{d}t+\int_1^\infty\frac{P(t)-1}{t}\,\mathrm{d}t, \tag{30.6}$$

where γ *is Euler's constant.*

Proof. Complete monotonicity implies convexity. For $0<h<1$, the two secant bounds around a give

$$\frac{f(a)-f((1+h)a)}{h}\le -af'(a)\le\frac{f((1-h)a)-f(a)}{h}. \tag{30.7}$$

Under (30.5), the left and right expressions tend, as $a\downarrow 0$, to $\log(1+h)/h$ and $-\log(1-h)/h$ respectively. Letting $h\downarrow 0$ proves (30.1), and Theorem 30.1 applies.

For the converse, split the canonical integral as

$$Z_0(a)=\int_0^1 e^{-at}\frac{P(t)}{t}\,\mathrm{d}t+\int_1^\infty e^{-at}\frac{P(t)-1}{t}\,\mathrm{d}t+\int_1^\infty e^{-at}\frac{\mathrm{d}t}{t}.$$

The first two integrals converge to their values at $a=0$ by Lemma 1.4. The last is the exponential integral $E_1(a)=-\log a-\gamma+o(1)$, which gives (30.6). □

The finite-part hypothesis is stronger than the slope hypothesis as a condition on arbitrary functions, but both select exactly the same member of the dyadic completely monotone family.

30.3 Uniqueness of the full differential hierarchy

Corollary 30.3 (Canonical gamma-tower characterization)**.** *Suppose* $f_0,f_1,\dots$ *are completely monotone functions on* $(0,\infty)$ *satisfying*

$$\begin{aligned} f_m(a)&=2^m\{f_m(a/2)-f_m((a+1)/2)\} && (m\ge 0),\\ f_m'(a)&=-f_{m-1}(a) && (m\ge 1),\end{aligned} \tag{30.8}$$

and $-af_0'(a)\to 1$ *as* $a\downarrow 0$*. Then* $f_m=Z_m$ *for every* $m\ge 0$*. Equivalently, the functions* $\exp((-1)^m m!f_m)$ *are exactly the hierarchy in* (20.14)*.*

Proof. Theorem 30.1 gives $f_0=Z_0$. Inductively, if $f_{m-1}=Z_{m-1}$, the differential ladder says that f_m-Z_m is a constant d_m. Subtracting the two level-m dyadic equations gives $d_m=2^m(d_m-d_m)=0$. This completes the induction. □

This supplies a normalization and a proof of uniqueness in a precise positive class. It does not establish every possible Weierstrass-product or complex-analytic assertion one might include in a broader Barnes-type program. The two issues should be kept separate.

Chapter 31

Extension to sparse digit sequences in other integer bases

The completion is not restricted to base two. This section records the extension with enough detail to identify exactly where integrality of the base is used.

For a real number $b > 1$, define

$$P_b(t) = \prod_{j \geq 0}(1 - e^{-b^j t}), \qquad X_b = \sum_{j \geq 1} b^{-j} U_j, \qquad \Phi_b(t) = \mathbb{E}e^{-tX_b}, \qquad K_b = P_b \Phi_b. \tag{31.1}$$

Now X_b is supported on $[0, 1/(b-1)]$ and has mean $a_b = 1/[2(b-1)]$. The same cancellation gives

$$K_b(bt) = \frac{K_b(t)}{t}, \qquad C_b(s+1) = b^{s+1} C_b(s), \tag{31.2}$$

where $C_b(s) = \int_0^\infty t^{s-1} K_b(t)\,\mathrm{d}t$. Thus the normalized measure has moments $b^{n(n+1)/2}$ for every integer n, and the exact logarithmic profile is periodic with period $\log b$.

When $b \geq 2$ is an integer, define $\mathcal{S}_b$ to be the nonnegative integers whose base-b digits are all zero or one. For $n \in \mathcal{S}_b$, let $w_b(n)$ be the number of nonzero digits. The product expansion gives

$$P_b(t) = \sum_{n \in \mathcal{S}_b} (-1)^{w_b(n)} e^{-nt}, \tag{31.3}$$

and hence the sparse signed Dirichlet series

$$D_b(s, a) = \sum_{n \in \mathcal{S}_b} \frac{(-1)^{w_b(n)}}{(n+a)^s}. \tag{31.4}$$

Absolute convergence holds at least in $\operatorname{Re} s > \log_b 2$: the integers in $\mathcal{S}_b$ below b^k are exactly 2^k in number, and comparison by base-b blocks proves convergence there. The Mellin integral continues the series to an entire function just as before.

Theorem 31.1 (Integer-base zero strings). *For every integer $b \geq 2$ and every $a > 0$, $D_b(\cdot, a)$ has infinitely many nonreal zeros in the left half-plane. More precisely, if s_0 is a zero of C_b of multiplicity d, the corresponding clusters approach $s_0 - m$ with error smaller than b^{-Am} for every fixed $A > 0$. At the centered shift $a = a_b$, their error is bounded by*

$$\exp\left[-\frac{\log b}{2d} m^2 + O(m)\right]. \tag{31.5}$$

Proof. All the completion and Mellin estimates replace L by $\log b$. The nonconstancy proof is still valid: because b^j is an integer, $P_b(x + 2\pi \mathrm{i}) = P_b(x) \to 0$, whereas a constant-profile elementary Gaussian expression would have a nonzero limit there. The normalized periodic Mellin function is therefore nonconstant, entire, positive on the real axis, and of order at most two. Lemma 17.3 supplies a nonreal zero.

Put $B_{b,a}(t) = e^{-at}/\Phi_b(t)$. Since X_b has positive probability in every neighborhood of zero, the proof of the global Taylor remainder in Theorem 19.1 applies. With $h = b^{-m}$, exact rescaling gives

$$b^{-m(m-1)/2+ms}\Gamma(s-m)D_b(s-m,a) = \int_0^\infty u^{s-1}K_b(u)B_{b,a}(hu)\,\mathrm{d}u.$$

At poles of the Gamma factor, the left side is interpreted through its entire Mellin numerator. The jth coefficient is again a multiple of $C_b(s+j)$, and Equation (31.2) factors out $C_b(s)$. The proof of Theorem 21.1 now applies verbatim.

For the centered bound,

$$B_{b,a_b}(z) = \prod_{j\geq 1} \frac{z/(2b^j)}{\sinh(z/(2b^j))}$$

is the Laplace transform in z^2 of

$$W_b = \sum_{j,n\geq 1} \frac{E_{j,n}}{4\pi^2 b^{2j} n^2}.$$

Its first moment-generating pole is at $R_b = 4\pi^2 b^2$. Removing the first pole as in Theorem 23.1 bounds its coefficients by a constant times R_b^{-r}. The counterpart of Equation (24.2) then has exponent $2(\log b)N^2 - 2m(\log b)N + O(N)$. Optimizing at $N = m/2 + O(1)$ and repeating the Rouché argument proves Equation (31.5). $\square$

For noninteger real $b > 1$, the positive product, Mellin identity, and moment construction still hold. The boundary-periodicity step used to prove nonconstancy does not automatically survive, and Equation (31.3) is no longer an ordinary integer-indexed Dirichlet series. We do not silently extend the integer-base zero claim to that setting.

Chapter 32

An explicit Fourier formula for the logarithmic profile

The proofs above need neither zeta values nor an evaluated constant. For reproducible computation, however, it is useful to make the profile explicit. Mellin analysis of harmonic sums is a standard method [21]; the calculation is included here to fix every sign and normalization.

Let γ denote Euler's constant and γ_1 the first Stieltjes constant in

$$\zeta(1+z) = \frac{1}{z} + \gamma - \gamma_1 z + O(z^2). \tag{32.1}$$

Define

$$\chi_k = \frac{2\pi\mathrm{i}k}{L}, \qquad c_* = -\frac{L}{12} - \frac{\pi^2/12 - \gamma^2/2 - \gamma_1}{L}. \tag{32.2}$$

Proposition 32.1 (Exact profile formula)**.** *For real v,*

$$\boxed{\log H(v) = c_* - \frac{1}{L}\sum_{k\in\mathbb{Z}\setminus\{0\}} \Gamma(\chi_k)\zeta(1+\chi_k)e^{-\chi_k v}.} \tag{32.3}$$

The Fourier series and every fixed derivative converge absolutely.

Proof. For $\operatorname{Re} z > 0$, expanding $\log(1-e^{-t})$ into exponentials and integrating termwise gives

$$\int_0^\infty t^{z-1}\log(1-e^{-t})\,\mathrm{d}t = -\Gamma(z)\zeta(z+1).$$

Summing the dyadic dilates, also absolutely for $\operatorname{Re} z > 0$, gives

$$\log P(t) = -\frac{1}{2\pi\mathrm{i}}\int_{c-\mathrm{i}\infty}^{c+\mathrm{i}\infty} \frac{\Gamma(z)\zeta(z+1)}{1-2^{-z}} t^{-z}\,\mathrm{d}z, \qquad c > 0. \tag{32.4}$$

This Mellin inversion is justified by exponential decay of the Gamma factor on vertical lines.

Move the line to $\operatorname{Re} z = -\delta$, with $0 < \delta < 1$. The crossed poles are $z = 0$ and $z = \chi_k$, $k \neq 0$. One may use horizontal sides at heights $2\pi(N+1/2)/L$, which stay away from the zeros of the denominator. Stirling's estimate for Γ, polynomial bounds for ζ on this fixed vertical strip, and the bound on the denominator show that the horizontal integrals tend to zero. On the new vertical line the integral is $O(t^\delta)$ as $t \downarrow 0$. The residue sum converges absolutely because $|\Gamma(\chi_k)|$ decays exponentially in $|k|$.

At a nonzero χ_k, the derivative of $1-2^{-z}$ is L, so its residue contributes $-\Gamma(\chi_k)\zeta(1+\chi_k)t^{-\chi_k}/L$. At zero, use

$$\Gamma(z)\zeta(1+z) = \frac{1}{z^2} + \left(\frac{\pi^2}{12} - \frac{\gamma^2}{2} - \gamma_1\right) + O(z),$$

$$\frac{1}{1-e^{-Lz}} = \frac{1}{Lz} + \frac{1}{2} + \frac{Lz}{12} + O(z^3).$$

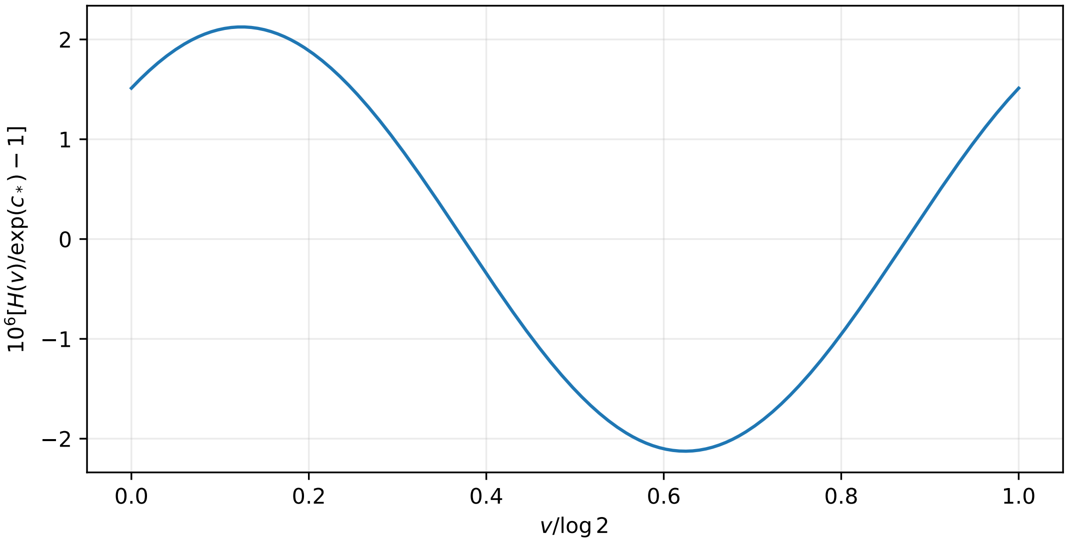

Figure 32.1: The nonconstant profile H over one logarithmic period. The vertical scale is in parts per million relative to e^{c_*}. This small oscillation is not a negligible detail for the complex zero problem.

Multiplication by t^{-z} gives the contribution

$$-\frac{(\log t)^2}{2L} + \frac{\log t}{2} + c_*.$$

Consequently,

$$\log P(t) = -\frac{(\log t)^2}{2L} + \frac{\log t}{2} + c_* - \frac{1}{L}\sum_{k\neq 0}\Gamma(\chi_k)\zeta(1+\chi_k)t^{-\chi_k} + O(t^\delta).$$

Since $\log\Phi(t) = O(t)$, comparison with the definition of H gives the claimed Fourier expression plus $O(t^\delta)$ for small t. Finally replace t by $2^{-m}t$. Both the profile and the Fourier sum are invariant under this substitution, while the error tends to zero. This turns the asymptotic identity into the exact formula Equation (32.3). Exponential Gamma decay absorbs any fixed derivative multiplier χ_k^r. □

The Fourier series for $\log H$ and the Fourier series for H should not be confused. To use Equation (17.7), first exponentiate Equation (32.3), then calculate the coefficients h_k of the resulting positive function. Substituting the coefficients of $\log H$ directly into Equation (17.7) gives an incorrect normalization and shifts the computed zeros.

Chapter 33

Numerical experiments and reproducibility

33.1 What was computed

The accompanying script `verification/source2/code/verify.py` performs exact rational coefficient and determinant checks, evaluates K by its products, evaluates the profile by Equation (32.3), and compares direct Mellin quadrature with the Fourier–Gaussian representation. It also solves for selected zeros in the normalized variable, using Equation (19.4) rather than a very large unnormalized Gamma factor. The files under `verification/source2/data/` contain the numerical outputs. The completion constant and the displayed limiting zero agree in separate 60- and 80-digit runs. The script `verification/source2/code/check_original_mellin.py` additionally tests the first listed Dirichlet zero directly in the original integral Equation (18.1), for both $a = 1/2$ and $a = 1$, without using the completion to construct its integrand.

These are high-precision floating-point experiments. Their reported residuals are not interval-arithmetic error certificates. The existence theorems and the signed remainder bounds are proved analytically above and do not depend on interpreting a small floating-point residual as a proof.

The normalizing constant is approximately

$$C_0 = 0.750721980407870959209707914050634694 7658\ldots. \tag{33.1}$$

One numerically observed simple zero of C, chosen in a real fundamental strip, is

$$s_0 \approx 0.1240623273169621334141320413937 + 6.7214865334392714744741792585671\,\mathrm{i}. \tag{33.2}$$

The word "one" is intentional: the computation does not establish that this is the lowest-height nonreal zero, nor that all zeros are simple. The normalized derivative $Q'(s_0)$ is numerically extremely close to $2\pi\mathrm{i}$, consistent with a leading two-term Fourier balance, but that observation is not used in the proof.

m	$\operatorname{Re} s_m$	$\operatorname{Im} s_m$	$\lvert s_m - (s_0 - m)\rvert$
4	-3.8759680374867420	6.7214963407429050	$3.19093 \cdot 10^{-5}$
6	-5.8759376731062212	6.7214865336192914	$4.59882 \cdot 10^{-10}$
8	-7.8759376726830382	6.7214865334392717	$4.19307 \cdot 10^{-16}$

Table 33.1: Numerical zeros of $D(s, 1/2)$ and, by Equation (21.2), of $D(s, 1)$. The distances refer to the higher-precision value of s_0 stored in the data file.

The centered slope ratios provide an independent positive-axis check:

m	$Z_m(1/2)/(C_0\, 2^{m(m-1)/2})$
0	0.9233074275823854568421676411
2	0.9934028797641805193051831301
4	0.9995677618356360314893011938
6	0.9999728805385416834067380219
8	0.9999983046075012694941809752

Table 33.2: Direct Mellin quadrature of the exact samples $\mathcal{H}(4^{-m})$. The script checks the signed inequalities for truncation orders one through seven.

33.2 Stable product evaluation

For positive t, compute $\log P(t)$ as a sum of $\log(-\operatorname{expm1}(-2^j t))$ rather than by subtracting nearly equal floating-point numbers. Its omitted large-scale terms decrease doubly exponentially.

For $\Phi(t)$, first apply enough halvings that the remaining argument v lies in $(0,1]$. Multiply the finitely many removed factors exactly at working precision. The infinite remaining tail is then computed from

$$\log \Phi(v) = -\frac{v}{2} + \sum_{r\geq 1} \frac{B_{2r} v^{2r}}{2r(2r)!(2^{2r}-1)}. \tag{33.3}$$

This avoids multiplying hundreds of factors close to one. Both the product route and the Fourier route are retained in the script so that they can cross-check one another.

For the zeta values in Equation (32.3), the points $1 + 2\pi i k/L$ are exact dyadic resonances. An implementation using $\zeta(s) = \eta(s)/(1-2^{1-s})$ naively can lose precision there, since both numerator and denominator vanish. The supplied code explicitly requests Euler–Maclaurin evaluation of ζ. In the tested `mpmath` environment this avoided a visible loss of accuracy in the profile. This is an implementation detail, not a mathematical singularity of ζ at these points.

33.3 Why the Mellin quadrature is manageable

In logarithmic coordinates, the integrand for C is a Gaussian multiplied by a bounded periodic function and e^{sv}. For a fixed complex s, the center of its absolute Gaussian majorant is $L(\operatorname{Re} s + 1/2)$. The code chooses its finite integration interval using that center and the requested precision. At the centered shift, $0 < \Psi(2^{-m}e^v) \leq 1$, so the same Gaussian tail estimate also controls $F_{m,1/2}$.

The finite interval and quadrature discretization are not interval-certified in this implementation. Repeating at increased precision and comparing two analytic representations gives a strong consistency check, while the proofs supply the logical foundation. The exact rational calculations, by contrast, are exact: the coefficient formulas and Hankel determinants are evaluated with `SymPy` rationals and integers.

33.4 Independent evaluation of the slopes and of the phases

A second program, retained as `verification/source3/verify_results.py`, evaluates the slopes $Z_\rho(a)$ from their original integrals, the amplitude Θ on one multiplicative period, and the explicit extremal solutions of Theorem 28.2, independently of the first program. Below, $\mathcal{H}_m$ abbreviates the exact sample $\mathcal{H}(4^{-m})$ of Theorem 22.2.

33.4.1 Independent integral evaluation

The archive contains `verify_results.py` and two JSON reports. The coefficient identities were checked exactly with rational symbolic arithmetic through $k = 12$. The numerical calculations were performed with 48 Gauss–Legendre nodes at 60 decimal digits and repeated with 80 nodes at 80 digits. Both the precision and the quadrature size were increased; agreement is not based solely on changing the displayed number of digits.

For C_0 and Θ, the program evaluates a single multiplicative period. Combining (16.3) with $t = 2^{y+j}$ gives

$$\Theta(\rho) = L\,2^{1/8}\int_0^1 H(Ly)\sum_{j\in\mathbb{Z}} 2^{-(j+y+\rho-1/2)^2/2}\,\mathrm{d}y. \tag{33.4}$$

Periodicity permits reduction of ρ modulo one. The rapidly convergent Gaussian shell sum makes this inexpensive.

The Z_ρ checks are evaluated separately from their *original* P-integrals, not by inserting the claimed coefficient expansion or the B_a-renormalization. On logarithmic shells the computation uses

$$\begin{aligned}\frac{Z_\rho(a)}{2^{\rho(\rho-1)/2}} = L\int_0^1 \sum_{j\in\mathbb{Z}} \exp\big(&-\rho L(y+j) - a2^{y+j} \\ &+ \log P(2^{y+j}) - \tfrac{L}{2}\rho(\rho-1)\big)\,\mathrm{d}y.\end{aligned} \tag{33.5}$$

Small factors are evaluated with `expm1` to avoid cancellation. The shell values of P use its own product recurrence. This comparison can detect an incorrect prefactor or a wrong coefficient in the theorem.

The two runs agree far beyond the digits printed below. For example, the reported C_0 values agree in the first 49 decimal places; their stored difference is less than $4\cdot 10^{-51}$. The calculations are *not interval-certified*: quadrature stability and exact symbolic checks support the implementation, while the proofs establish the theorems. The numerical work does not establish a sharp constant in the remainder bound or historical novelty.

33.4.2 Centered derivative data

Let

$$\mathcal{H}_m = \frac{Z_m(1/2)}{C_0\,2^{m(m-1)/2}},\quad P_2 = 1-\frac{2^{-2m}}{9},\quad P_4 = P_2 + \frac{248}{2025}2^{-4m},\quad P_6 = P_4 - \frac{4806656}{2679075}2^{-6m}.$$

Table 33.3 compares successive approximations. For fixed truncation order the improvement agrees with the even powers predicted by symmetry.

m	$\lvert\mathcal{H}_m-1\rvert$	$\lvert\mathcal{H}_m-P_2\rvert$	$\lvert\mathcal{H}_m-P_4\rvert$	$\lvert\mathcal{H}_m-P_6\rvert$
4	$4.322382\,10^{-4}$	$1.789613\,10^{-6}$	$7.911730\,10^{-8}$	$2.782223\,10^{-8}$
8	$1.695392\,10^{-6}$	$2.850821\,10^{-11}$	$6.353778\,10^{-15}$	$2.031475\,10^{-17}$
12	$6.622738\,10^{-9}$	$4.350974\,10^{-16}$	$3.799206\,10^{-22}$	$4.918108\,10^{-27}$
16	$2.587007\,10^{-11}$	$6.639065\,10^{-21}$	$2.264533\,10^{-29}$	$1.145315\,10^{-36}$

Table 33.3: Absolute errors in the normalized centered expansion. The notation $x\,10^{-j}$ means $x\times 10^{-j}$.

At $m=4$, the benefit of further terms is already much less spectacular than at $m=16$. This is consistent with the fact that the expansion is divergent: adding indefinitely many terms at a fixed order is not justified. The full data files also include nonsymmetric shifts and noninteger orders, for example $a=1$ and $\rho = 8.25, 8.5, 12.75$.

33.4.3 The tiny Mellin phase and the visible gamma phase

The observed relative modulation of Θ is shown in Table 33.4. Its small size is consistent with the exact Fourier smoothing factor in (17.9).

θ	$\Theta(\theta)/\Theta(0) - 1$
$1/4$	$1.289029564918397699 \times 10^{-18}$
$1/2$	$1.296624062281340296 \times 10^{-18}$
$3/4$	$7.594497362942597195 \times 10^{-21}$

Table 33.4: Numerical phase variation. Nonconstancy is numerical evidence here, not an additional theorem assumed elsewhere.

θ	$f_\theta(1/3)$	$-af'_\theta(a)$ at $a = 2^{-24}$
0	0.956669839913615728	1.00000628193280958
$1/4$	0.956669741884495862	0.99999235648832259
$1/2$	0.956665362383835818	0.99999357544426848
$3/4$	0.956665460410626991	1.00000750091845969

Table 33.5: Distinct positive normalized solutions. The canonical value is $Z_0(1/3) = 0.956667601148143600\ldots$. The slope column is a finite-a evaluation, not a claimed exact limiting value.

For comparison, Table 33.5 gives the explicit extremal solutions of (28.6). Their differences are much easier to see, although all have $f_\theta(1/2) = L$ and $f_\theta(1) = L/2$.

The program also checks the dyadic equation away from these special normalization points, using $a = 1.3$ and $\theta = 0.27$. The residual is below 10^{-70} in the 80-digit run. Agreement at $a = 1$ alone would be an inadequate test because that value is forced by the normalization.

Chapter 34

A formalization route and further questions

34.1 A proof-oriented dependency chain

A formal development can be organized into four layers, without requiring the numerical Fourier formula for the principal theorem. First prove positivity and normal convergence of P and Φ, then their two scaling equations. This yields $K(2t) = K(t)/t$ and positive upper and lower bounds for its logarithmic profile.

Next establish the Gaussian domination needed for the entire Mellin transform and its derivatives. The Mellin shift, periodicity of Q, and the boundary contradiction proving nonconstancy can then be formalized independently of coefficient calculations. The zero-free finite-order lemma and a version of Rouché with multiplicities complete the complex-analytic layer.

The asymptotic layer consists of the positive-axis global Taylor bound for B_a and the exact change of variables in Equation (19.4). Once these are available, the all-orders factorization is a finite-sum integration lemma, and the supergeometric zero-cluster theorem is a uniform remainder estimate followed by Rouché.

Finally, the centered layer adds Euler's $\sinh$ product, the positive exponential-sum law W, and an integral Taylor remainder. It supplies the signed bounds, the coefficient-pole estimate, and the Gaussian refinement. The finite Bell-polynomial formula and the Hankel determinant can be developed in a separate exact algebra module.

The existing atlas names modules such as `ThueMorseBoundaryFlatness.lean` and `ThueMorseEntireContinuation.lean` for relevant background. No new Lean file is supplied here, and no theorem in this extension is claimed to have been machine-checked. The dependency chain is a technical implementation guide, not a formalization status claim.

34.2 Further questions suggested by the proved results

The following questions concern refinements not needed for any preceding theorem.

Question 34.1 (Zeros in a fundamental strip). Can the nonreal zeros of Q in $0 \le \operatorname{Re} s < 1$ be counted and localized effectively as their imaginary parts grow? Are they all simple?

The Fourier–Gaussian representation is a natural starting point. At increasing height, different Fourier harmonics compete. A rigorous two-term dominance argument on suitable contours could produce explicit zero enclosures, but the present computation is not a proof of such a global pattern.

Question 34.2 (Leading exponentially small displacement). For a simple zero s_0 of C, determine an asymptotic equivalent, not just an upper bound, for $s_m - (s_0 - m)$ at the centered shift.

All algebraic powers of 2^{-m} vanish from the displacement. The nearest poles of Ψ and the logarithmic-Gaussian saddle suggest a more detailed exponentially small analysis. Such an analysis must retain the phase information in H; replacing the completion by its lognormal twin would erase the zeros of the limiting periodic factor.

Question 34.3 (Growing-height zero windows). How far may $|\operatorname{Im} s|$ grow with m while the translated asymptotic expansion remains useful for zero localization?

The remainder bound in Equation (19.6) depends on $\operatorname{Re} s$ but not directly on $\operatorname{Im} s$. The difficulty is instead a lower bound for $C(s)$ on the contours used by Rouché. Thus this question couples growth of the periodic entire factor to the geometry of its zeros.

Question 34.4 (Noninteger dilation). For every real $b > 1$, is the logarithmic profile of K_b nonconstant? What replaces the integer-frequency boundary argument when b is not an integer?

An affirmative answer would extend the Mellin zero mechanism to a continuous family of generalized exponential sums. The exact completion and moment identities already hold; the missing step in the proof given here is the nonconstancy argument in that greater generality.

Closing perspective

The most useful organizing fact is a cancellation between an expanding product and a contracting one. The Thue–Morse product alone carries a complicated dyadic scale dependence; the uniform-convolution transform removes its nonmonomial factor. The resulting exact law $K(2t) = K(t)/t$ simultaneously explains lognormal integer moments, the negative-argument scale of the Dirichlet continuation, the universal factor in every asymptotic coefficient, and the occurrence of nonreal zero strings. At the half-shift, symmetry supplies an additional positive Laplace representation and converts the expansion into certified inequalities. The log-periodic information is small on the real axis, but it is precisely the information that survives beyond all orders and controls the complex zeros.

34.3 Further directions for the uniform remainder and the classification

The classification theorem gives a lesson complementary to the zero strings. The dyadic functional equation controls how a measure moves between scales but does not prescribe the distribution of mass within a scale; one-point normalization determines only that scale's total mass, and even a full formal expansion at integer orders can fail to recover the missing phase information. The boundary slope recovers the whole Laplace transform of the invariant measure and is therefore strong enough to characterize it. The Fourier formula of Theorem 32.1 supplies the closed expression for the amplitude that the uniqueness analysis leaves implicit; the leading exponentially small discrepancy in Theorem 29.2, rather than only its Gaussian-in-order upper bound, remains to be determined, and the poles of B_a at imaginary multiples of 4π together with the chosen phase measure should enter such a refinement.

A separate quantitative task is to replace the tube supremum $M_{I,r}$ by simple computable constants optimized for a given shift interval. The proof of Lemma 20.3 already separates a compact region from a controlled tail, so interval arithmetic could turn it into a numerical certificate. The present tables should not be mistaken for that certificate.

The positive-measure classification and the proof of the uniform remainder also have clear formalization targets: product convergence, pushforward identities, the Mellin recurrence, Cauchy estimates on a tube, and Laplace uniqueness. None of these targets requires asserting that the results already appear in the repository's Lean code. Finally, analogous product pairs for other dilation factors may produce related invariant-measure classifications; a radix change must be made at the level of the product equation rather than by replacing every two in the final formulas without checking the underlying sequence.

Part III

Rational resonances and coalescing roots

Chapter 35

Scope and the main results

The finite Thue–Morse polynomial $P_m(w) = \prod_{j<m}(1-w^{2^j})$ contains more information than its values at roots of unity. This part investigates its complete local profile in a window of width 2^{-m} around each rational frequency. The atlas already treats the binary product, Prouhet cancellation, ordinary power moments, root-of-unity filters, rarefied sums, and the finite-product bridge to the Fabius–Rvachev system; those are starting points here. The closest external comparisons are Sobolewski's root-of-unity recurrence theory for automatic-sequence polynomials [28] and Duke–Nguyen's asymptotic theory of infinite cyclotomic products [19]; neither recurrence at a rational frequency nor the use of cyclotomic products is itself new. The intended extensions are the explicit moving-window factorization, its full derivative calculus, and the coalescing-root limits. Baake and Coons study higher correlations [12], Gohlke, Kesseböhmer and Schindler generalized spectral measures [22], and Cloitre iterated transforms [15]; these concern neighboring problems.

35.1 A guide to the contributions

There are four main groups of results.

Exact local profiles. For a nontrivial odd-order root ζ, define

$$\mathcal{Q}_{m,\zeta}(z) = \frac{P_m(\zeta e^{z/2^m})}{P_m(\zeta)}.$$

Theorem 37.1 gives entire functions $\mathcal{H}_{\zeta,s}$, periodic in the depth phase s, with the exact identity

$$\mathcal{Q}_{m,\zeta}(z)\mathcal{H}_{\zeta,0}(z/2^m) = \mathcal{H}_{\zeta,s}(z).$$

Theorem 37.3 turns this into a convergent, all-orders expansion with an explicit remainder bound. Theorem 39.1 differentiates it into a closed formula for every twisted power moment.

Coalescing-root universality. At a primitive dyadic root of order 2^r, a polynomial of depth $m = r + M$ has a zero of multiplicity M. Removing that zero gives a normalized function $\mathcal{N}_{r,M}$. For a fixed root, its limit is $\mathcal{U}$, the Fabius transform. Theorem 41.2 determines the different limits when r grows with M. In particular,

$$r4^{-M} \longrightarrow \lambda \quad \Longrightarrow \quad e^{-r2^{-M}z/(2\pi i)}\mathcal{N}_{r,M}(z) \longrightarrow \mathcal{U}(z)e^{\lambda z^2/(8\pi^2)}.$$

This Gaussian–Fabius limit is the most distinctive new derivation in the part.

Cyclotomic aggregation. Theorem 42.1 multiplies the rational-frequency profiles into an integer dilation quotient of $\mathcal{U}$. Theorem 43.1 classifies the associated probability laws: odd dilation factors give singular continuous laws; even factors give densities of an explicitly determined, finite differentiability class.

Geometry and exact arithmetic. Theorem 38.1 determines the entire zero divisor of each profile. Theorem 39.2 gives eventual recurrences for weighted residue sums, with completely explicit characteristic roots. These results supply independent checks on the analytic constructions.

Chapter 36

The finite product and the Fabius transform

36.1 The positive comparison function

This part works with the Fabius transform in the moment-generating normalization

$$\mathcal{U}(z) = \mathbb{E}e^{zX} = \Phi(-z), \tag{36.1}$$

where $X = \sum_{k\geq 1} 2^{-k}V_k$ is the uniform sum (1.16) (the letters V_k are used for the uniform variables here) and Φ is the transform of (1.18). Define the removable entire function $g(z) = (e^z-1)/z = \psi(-z)$, $g(0) = 1$, and its finite and infinite dyadic products

$$\mathcal{U}_M(z) = \prod_{k=1}^{M} g(z/2^k), \qquad \mathcal{U}(z) = \prod_{k=1}^{\infty} g(z/2^k). \tag{36.2}$$

By Theorem 1.5, $\mathcal{U}$ is entire, its zeros are the points $4\pi i n$, $n \neq 0$, with multiplicity $1 + v_2(n)$, and it obeys the denominator-free identities

$$\mathcal{U}_M(z)\mathcal{U}(z/2^M) = \mathcal{U}(z), \qquad \mathcal{U}(2z) = g(z)\mathcal{U}(z). \tag{36.3}$$

We use the bounded Fabius normalization $F(x) = \mathbb{P}(X \leq x)$ of (1.23), which agrees on $[0, 1]$ with the classical Fabius construction [20]; X has a smooth compactly supported density. The centered product is

$$\mathcal{U}(z) = e^{z/2} \prod_{k\geq 1} \frac{\sinh(z/2^{k+1})}{z/2^{k+1}}, \tag{36.4}$$

by (1.30) with $Y = 2X - 1$. Thus the familiar sinc product is already present, but the rational profiles below will not individually be transforms of positive measures.

36.2 An explicit coefficient convention

Let b_n denote Bernoulli numbers with $b_1 = -1/2$, and define

$$\beta_1 = \frac{1}{2}, \qquad \beta_n = \frac{b_n}{n} \quad (n \geq 2). \tag{36.5}$$

Then

$$\log g(z) = \sum_{n\geq 1} \beta_n \frac{z^n}{n!}, \qquad \log \mathcal{U}(z) = \sum_{n\geq 1} \frac{\beta_n}{2^n - 1} \frac{z^n}{n!}. \tag{36.6}$$

The first equality follows by differentiating the logarithm and using the generating function of the Bernoulli numbers; summing the dyadic geometric series gives the second. These logarithms are the branches that vanish at zero. The second series has radius 4π, the distance to the first zero of $\mathcal{U}$.

Chapter 37

Odd-order rational resonance profiles

37.1 Backward phases

Fix an odd integer $q \geq 3$, an integer a with $1 \leq a < q$, and

$$\zeta = \mathrm{e}^{2\pi \mathrm{i} a/q}, \qquad L = \mathrm{ord}_q(2). \tag{37.1}$$

The root need not be primitive. Negative exponents of 2 in exponents of ζ always mean inverses modulo q. Equivalently, reduce the exponent of 2 modulo L. Define

$$a_{s,k} = \zeta^{2^{s-k}}, \qquad \mathcal{H}_{\zeta,s}(z) = \prod_{k \geq 1} \frac{1 - a_{s,k}\mathrm{e}^{z/2^k}}{1 - a_{s,k}}, \qquad s \in \mathbb{Z}/L\mathbb{Z}. \tag{37.2}$$

Every denominator is nonzero. The index k follows the binary orbit backwards because the last, largest-scale factor of P_m becomes the first factor after rescaling.

Theorem 37.1 (Exact profile factorization). *The products $\mathcal{H}_{\zeta,s}$ are entire and satisfy $\mathcal{H}_{\zeta,s}(0) = 1$. If $m \geq 0$ and $s \equiv m \pmod{L}$, then*

$$\boxed{\frac{P_m(\zeta \mathrm{e}^{z/2^m})}{P_m(\zeta)} \mathcal{H}_{\zeta,0}(z/2^m) = \mathcal{H}_{\zeta,s}(z).} \tag{37.3}$$

In particular, along $m \equiv s \pmod{L}$,

$$\mathcal{Q}_{m,\zeta}(z) \longrightarrow \mathcal{H}_{\zeta,s}(z) \tag{37.4}$$

uniformly on every compact subset of $\mathbb{C}$, together with all derivatives.

Proof. Put $d = \min_{0 \leq j < L} |1 - \zeta^{2^j}| > 0$. For $|z| \leq R$, the k-th factor differs from one by at most

$$\frac{R\mathrm{e}^{R/2^k}}{d2^k} \leq \frac{R\mathrm{e}^{R/2}}{d2^k}. \tag{37.5}$$

The sum converges. The standard product criterion, which follows directly from the Cauchy criterion and $\prod(1 + u_k) \leq \exp(\sum u_k)$ for $u_k \geq 0$, gives compact-uniform convergence and entireness.

Reindex the finite product by $k = m - j$. Its normalized factors are exactly the first m factors of $\mathcal{H}_{\zeta,s}$. For the omitted factors write $k = m + \ell$. Then

$$a_{s,m+\ell} = \zeta^{2^{-\ell}}, \qquad z/2^{m+\ell} = (z/2^m)/2^\ell.$$

Their product is $\mathcal{H}_{\zeta,0}(z/2^m)$, proving (37.3). This tail tends to one uniformly on compact sets and is nonzero there for all sufficiently large m. The limit follows, and Cauchy's integral formula on slightly larger compact sets gives convergence of every derivative. ☐

The denominator-free form (37.3) is important. A quotient formula without that qualification would appear to have poles at zeros of the tail even though the finite polynomial profile is entire.

37.2 Closed cumulants

For $n \geq 1$ and $w \neq 1$, define the rational function

$$\operatorname{Li}_{1-n}(w) = \left(w\frac{\mathrm{d}}{\mathrm{d}w}\right)^{n-1} \frac{w}{1-w}. \tag{37.6}$$

This is the rational continuation of the usual nonpositive-order polylogarithm; no infinite series at a root of unity is intended. For example,

$$\operatorname{Li}_0(w) = \frac{w}{1-w}, \quad \operatorname{Li}_{-1}(w) = \frac{w}{(1-w)^2}, \quad \operatorname{Li}_{-2}(w) = \frac{w(1+w)}{(1-w)^3}.$$

Theorem 37.2 (Finite formulas for all cumulants). *Write, near zero,*

$$\log \mathcal{H}_{\zeta,s}(z) = \sum_{n\geq 1} \kappa_{\zeta,s,n} \frac{z^n}{n!}. \tag{37.7}$$

Then

$$\boxed{\kappa_{\zeta,s,n} = -\frac{\sum_{k=1}^{L} 2^{-kn} \operatorname{Li}_{1-n}(\zeta^{2^{s-k}})}{1 - 2^{-Ln}}.} \tag{37.8}$$

In particular, all cumulants and Taylor coefficients belong to $\mathbb{Q}(\zeta)$.

Proof. Repeated differentiation gives

$$\frac{\mathrm{d}^n}{\mathrm{d}z^n} \log(1 - w\mathrm{e}^z)\bigg|_{z=0} = -\operatorname{Li}_{1-n}(w).$$

The compact convergence estimates used above justify summing derivatives in a zero-free neighborhood of the origin. Thus the cumulant is $-\sum_{k\geq 1} 2^{-kn} \operatorname{Li}_{1-n}(a_{s,k})$. The phase is periodic with period L; summing each residue class of k yields (37.8). Every summand is a rational function of ζ. □

Define two coefficient families explicitly by

$$h_{\zeta,s,j} = \frac{1}{j!}\mathcal{B}_j(\kappa_{\zeta,s,1}, \ldots, \kappa_{\zeta,s,j}), \tag{37.9}$$

$$c_{\zeta,j} = \frac{1}{j!}\mathcal{B}_j(-\kappa_{\zeta,0,1}, \ldots, -\kappa_{\zeta,0,j}). \tag{37.10}$$

Then $\mathcal{H}_{\zeta,s}(z) = \sum h_{\zeta,s,j} z^j$ and $1/\mathcal{H}_{\zeta,0}(z) = \sum c_{\zeta,j} z^j$ near zero. In particular,

$$c_0 = 1, \qquad c_1 = -\kappa_1, \tag{37.11}$$

$$c_2 = \tfrac{1}{2}(\kappa_1^2 - \kappa_2), \qquad c_3 = \tfrac{1}{6}(-\kappa_1^3 + 3\kappa_1\kappa_2 - \kappa_3), \tag{37.12}$$

where the suppressed cumulants have phase zero.

37.3 All orders with a certified analytic bound

Theorem 37.3 (Convergent depth expansion). *For* $s \equiv m \pmod{L}$ *and every fixed* $N \geq 0$,

$$\mathcal{Q}_{m,\zeta}(z) = \mathcal{H}_{\zeta,s}(z) \sum_{j=0}^{N} c_{\zeta,j} z^j 2^{-mj} + O_{R,N,\zeta}(2^{-(N+1)m}), \qquad |z| \leq R. \tag{37.13}$$

More explicitly, put

$$d = \min_j |1 - \zeta^{2^j}|, \quad \rho = d/4, \quad M_R = \exp(Re^{R/2}/d), \quad J_\rho = \exp(2\rho e^{\rho/2}/d).$$

Whenever $R2^{-m} < \rho$, the absolute remainder in (37.13) is at most

$$M_R J_\rho \frac{(R2^{-m}/\rho)^{N+1}}{1 - R2^{-m}/\rho}. \tag{37.14}$$

The infinite expansion converges whenever $|z|2^{-m}$ is smaller than the distance from zero to the nearest zero of $\mathcal{H}_{\zeta,0}$.

Proof. Equation (37.5) bounds $|\mathcal{H}_{\zeta,s}(z)|$ by M_R. On $|w| \leq \rho$, write each factor of $\mathcal{H}_{\zeta,0}$ as $1 + v_k$. The estimates give

$$\sum |v_k| \leq \rho e^{\rho/2}/d, \qquad |v_k| \leq \rho e^{\rho/2}/(2d) < 1/2,$$

since $d \leq 2$. The inequality $(1-u)^{-1} \leq e^{2u}$ for $0 \leq u \leq 1/2$ bounds the reciprocal product by J_ρ. Cauchy's coefficient estimate therefore gives $|c_j| \leq J_\rho \rho^{-j}$. Summing the geometric tail and multiplying by M_R proves (37.14). The exact factorization proves convergence throughout the zero-free Taylor disc of the reciprocal. □

This is stronger than a merely formal asymptotic series. The error correction is the Taylor expansion of one fixed analytic reciprocal, evaluated at the shrinking argument $z2^{-m}$.

Chapter 38

Zeros, symmetry, and refinement of the profiles

38.1 The complete zero divisor

A normal product is especially informative when all its factors have explicit zeros. Write $\zeta = \mathrm{e}^{2\pi\mathrm{i}a/q}$, with q odd and $1 \le a < q$, as before. We use $v_2(t) = v_2(|t|)$ for nonzero integers, and take the phase s modulo L.

Theorem 38.1 (Zeros and their exact multiplicities)**.** *The zeros of $\mathcal{H}_{\zeta,s}$ are precisely*

$$z = \frac{2\pi\mathrm{i}t}{q}, \qquad t \in 2\mathbb{Z} \setminus \{0\}, \qquad t \equiv -a2^s \pmod q. \tag{38.1}$$

The multiplicity of this zero is $v_2(t)$. Equivalently, the zeros are

$$z = \frac{4\pi\mathrm{i}\ell}{q}, \qquad \ell \equiv -a2^{s-1} \pmod q, \quad \textit{with multiplicity } 1 + v_2(\ell). \tag{38.2}$$

Here negative powers of two in congruences mean inverses modulo q.

Proof. Choose an integer v_k representing $a2^{s-k} \pmod q$. The k-th factor vanishes exactly when

$$z = \frac{2\pi\mathrm{i}}{q}\, 2^k(qj - v_k), \qquad j \in \mathbb{Z}.$$

Consequently, at a point $z = 2\pi\mathrm{i}t/q$, this factor vanishes if and only if $2^k \mid t$ and $t \equiv -a2^s \pmod q$. Each such zero of the factor is simple. For a fixed nonzero t, precisely $k = 1, \dots, v_2(t)$ contribute. The congruence excludes $t = 0$, since $a \not\equiv 0 \pmod q$.

It remains to rule out zeros introduced by the infinite tail. On every compact set the sum of the distances of the factors from 1 converges. A sufficiently late tail consists of nonzero factors close to 1, so its logarithm converges normally and its product never vanishes. Thus the zeros and multiplicities are exactly those already counted. ☐

Corollary 38.2 (The exact radius of a reciprocal expansion)**.** *Let $b \in \{1, \dots, q-1\}$ represent $-a2^{s-1}$ $\pmod q$. The Taylor series of $1/\mathcal{H}_{\zeta,s}(z)$ at zero has radius*

$$\frac{4\pi}{q} \min\{b, q-b\}. \tag{38.3}$$

Proof. The closest integers to zero in the progression $b + q\mathbb{Z}$ are b and $b - q$. The reciprocal has a pole at each zero of $\mathcal{H}_{\zeta,s}$, so the nearest one determines the radius. ☐

Corollary 38.3 (Non-holonomicity)**.** *No $\mathcal{H}_{\zeta,s}$ satisfies a nontrivial homogeneous linear differential equation with polynomial coefficients.*

Proof. For arbitrarily large k, choose an odd integer u in the residue class $-a2^{s-k} \pmod q$; odd representatives exist because q is odd. Then $t = 2^k u$ gives a zero of multiplicity k. In particular the multiplicities are unbounded, at infinitely many distinct points.

Suppose a nonzero entire function satisfies a homogeneous differential equation of order r, with nonzero leading polynomial. At a point where that polynomial does not vanish, a zero of multiplicity at least r sets all r initial data to zero. Local uniqueness for the corresponding first-order system forces the function to vanish identically. Only finitely many exceptional points are zeros of the leading polynomial, contradicting the unbounded multiplicities just established. An order-zero equation is impossible for a nonzero entire function as well. □

This conclusion is only non-holonomicity, often called non-D-finiteness. It does not establish differential transcendence with respect to nonlinear differential equations.

38.2 A real centered Fourier profile

Proposition 38.4 (Reflection and strict concavity)**.** *For every phase,*

$$\mathcal{H}_{\zeta,s}(z) = \mathrm{e}^{z}\overline{\mathcal{H}_{\zeta,s}(-\overline{z})}. \tag{38.4}$$

Hence $\operatorname{Re}\kappa_{\zeta,s,1} = 1/2$*, every even cumulant is real, and every odd cumulant of order at least three is purely imaginary.*

The function

$$G_{\zeta,s}(t) = \mathrm{e}^{-\mathrm{i}t/2}\mathcal{H}_{\zeta,s}(\mathrm{i}t) \tag{38.5}$$

is real on the real axis. Between any two consecutive distinct real zeros, $\log|G_{\zeta,s}(t)|$ *is strictly concave, and* $|G_{\zeta,s}|$ *has exactly one critical point, a strict maximum.*

Proof. For $|a| = 1$, $a \neq 1$, direct algebra gives

$$\frac{1 - a\mathrm{e}^{u}}{1-a} = \mathrm{e}^{u}\,\overline{\frac{1 - a\mathrm{e}^{-\overline{u}}}{1-a}}.$$

Multiplying over the profile factors and using $\sum_{k\geq 1} 2^{-k} = 1$ gives (38.4). Comparing Taylor coefficients of the logarithms near zero proves the cumulant assertions.

Write $a_{s,k} = \mathrm{e}^{\mathrm{i}\theta_k}$, where θ_k is chosen periodically. The elementary sine identity gives

$$G_{\zeta,s}(t) = \prod_{k\geq 1} \frac{\sin(\theta_k/2 + t/2^{k+1})}{\sin(\theta_k/2)}. \tag{38.6}$$

All factors are real for real t. On compact subintervals avoiding the zeros, the logarithmic derivatives converge normally. Indeed, in the tail the sine denominators remain uniformly bounded away from zero, while each differentiation supplies a factor 2^{-k-1}. Thus

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}\log|G_{\zeta,s}(t)| = -\sum_{k\geq 1} 4^{-k-1}\csc^2(\theta_k/2 + t/2^{k+1}) < 0. \tag{38.7}$$

If consecutive distinct zeros are $\alpha < \beta$, the logarithmic derivative tends to $+\infty$ as $t \downarrow \alpha$, and to $-\infty$ as $t \uparrow \beta$, with the corresponding zero multiplicities as positive coefficients. By strict monotonicity it crosses zero exactly once. Since G does not vanish inside the interval, this proves the assertion about its absolute value. □

38.3 A cyclic refinement equation

Set

$$A_\zeta = \prod_{j=0}^{L-1} (1 - \zeta^{2^j}), \qquad b = 2^L. \tag{38.8}$$

The value of this product is unchanged by a cyclic shift of the exponents.

Proposition 38.5 (One step and one complete cycle)**.** *The profiles satisfy*

$$\mathcal{H}_{\zeta,s}(2z) = \frac{1 - \zeta^{2^{s-1}} \mathrm{e}^z}{1 - \zeta^{2^{s-1}}} \mathcal{H}_{\zeta,s-1}(z), \tag{38.9}$$

$$\mathcal{H}_{\zeta,s}(bz) = \frac{P_L(\zeta^{2^s} \mathrm{e}^z)}{A_\zeta} \mathcal{H}_{\zeta,s}(z). \tag{38.10}$$

Proof. In the first formula, split off the first factor and shift k to $k - 1$ in the tail. For the second, split off the first L factors and put $j = L - k$. Periodicity gives $\zeta^{2^{s-L+j}} = \zeta^{2^{s+j}}$, identifying the numerator with the finite Thue–Morse product. The remaining product is the original phase. □

The coefficients of this refinement equation are generally complex. The existence of such an equation does not by itself make an individual profile the transform of a positive measure.

Chapter 39

Exact twisted moments and arithmetic progressions

39.1 All moments from a finite Bell polynomial

For $h \geq 0$, define

$$T_{m,h}(\zeta) = \sum_{n<2^m} \varepsilon_n \zeta^n n^h. \tag{39.1}$$

The coefficients $h_{\zeta,s,j}$ and $c_{\zeta,j}$ were defined in (37.9) and (37.10).

Theorem 39.1 (Closed twisted-moment formulas)**.** *For every $m, h \geq 0$, with $s \equiv m \pmod L$,*

$$\frac{T_{m,h}(\zeta)}{P_m(\zeta)} = h! \sum_{j=0}^{h} h_{\zeta,s,j} c_{\zeta,h-j} 2^{mj} \tag{39.2}$$

$$= \mathcal{B}_h\big(2^m \kappa_{\zeta,s,1} - \kappa_{\zeta,0,1}, \ldots, 2^{mh} \kappa_{\zeta,s,h} - \kappa_{\zeta,0,h}\big). \tag{39.3}$$

All sums defining the cumulants and Bell polynomials are finite. Thus these formulas require no auxiliary recurrence in m or h.

Proof. The exponential generating function of the left side is

$$\sum_{h\geq 0} \frac{T_{m,h}(\zeta)}{P_m(\zeta)} \frac{t^h}{h!} = \frac{P_m(\zeta e^t)}{P_m(\zeta)} = \frac{\mathcal{H}_{\zeta,s}(2^m t)}{\mathcal{H}_{\zeta,0}(t)}$$

in a neighborhood of zero, by Theorem 37.1. Multiplying the two Taylor series gives (39.2). Taking a logarithm first, then applying the defining generating function of the complete Bell polynomials, gives (39.3). □

In particular,

$$\frac{T_{m,1}}{P_m} = 2^m \kappa_{s,1} - \kappa_{0,1}, \tag{39.4}$$

$$\frac{T_{m,2}}{P_m} = (2^m \kappa_{s,1} - \kappa_{0,1})^2 + 2^{2m} \kappa_{s,2} - \kappa_{0,2}. \tag{39.5}$$

The two parts of the result have different uses. Formula (39.3) is compact and suitable for symbolic evaluation. Formula (39.2) exhibits the exact finite collection of depth scales 2^{mj}.

39.2 The first nontrivial example: order three

Let $\omega = \mathrm{e}^{2\pi\mathrm{i}/3}$. Then $L = 2$, $A_\omega = 3$, and

$$P_{2t}(\omega) = 3^t, \qquad P_{2t+1}(\omega) = 3^t(1-\omega). \tag{39.6}$$

Substitution into the finite cumulant formula gives

$$\kappa_{\omega,0,1} = \frac{1}{2} + \frac{\mathrm{i}\sqrt{3}}{18}, \qquad \kappa_{\omega,1,1} = \frac{1}{2} - \frac{\mathrm{i}\sqrt{3}}{18}, \tag{39.7}$$

$$\kappa_{\omega,s,2} = \frac{1}{9}, \qquad \kappa_{\omega,s,4} = -\frac{1}{45}, \tag{39.8}$$

$$\kappa_{\omega,0,3} = -\frac{\mathrm{i}\sqrt{3}}{81}, \qquad \kappa_{\omega,1,3} = \frac{\mathrm{i}\sqrt{3}}{81}. \tag{39.9}$$

For example, $\mathrm{Li}_0(\omega) = -1/2 + \mathrm{i}\sqrt{3}/6$, and the order-one weighted sum is

$$-\frac{\frac{1}{2}\,\mathrm{Li}_0(\omega^2) + \frac{1}{4}\,\mathrm{Li}_0(\omega)}{1-1/4} = \frac{1}{2} + \frac{\mathrm{i}\sqrt{3}}{18}.$$

The first centered moment consequently has the exact form

$$\frac{T_{m,1}(\omega)}{P_m(\omega)} - \frac{2^m-1}{2} = \frac{\mathrm{i}\sqrt{3}}{18}\big((-1)^m 2^m - 1\big). \tag{39.10}$$

The imaginary term is not an error: the weights $\varepsilon_n \omega^n$ are complex.

39.3 Weighted sums in an arithmetic progression

For an odd $q \geq 3$, a residue $b \in \mathbb{Z}$, and $h \geq 0$, put

$$S^{(q)}_{m,b,h} = \sum_{\substack{0 \leq n < 2^m \\ n \equiv b \pmod q}} \varepsilon_n n^h. \tag{39.11}$$

These are integers. Let $\zeta_q = \mathrm{e}^{2\pi\mathrm{i}/q}$, $L = \mathrm{ord}_q\, 2$, and

$$A_a = \prod_{j=0}^{L-1} (1 - \zeta_q^{a2^j}), \qquad 1 \leq a < q.$$

Use the same L for all roots, even when one has smaller order.

Theorem 39.2 (An explicit eventual recurrence in the block depth). *Fix q, b, h, and fix $s \in \{0, \dots, L-1\}$. Whenever $Lt + s > h$,*

$$S^{(q)}_{Lt+s,b,h} = \sum_{a=1}^{q-1} \sum_{j=0}^{h} C_{a,s,b,h,j}\,(A_a 2^{Lj})^t, \tag{39.12}$$

where, writing the subscripts a for the root ζ_q^a,

$$C_{a,s,b,h,j} = \frac{\zeta_q^{-ab}}{q}\, P_s(\zeta_q^a)\, h!\, c_{a,h-j} h_{a,s,j} 2^{sj}. \tag{39.13}$$

In particular, the ordinary generating series in t is rational, with denominator dividing

$$\prod_{a=1}^{q-1} \prod_{j=0}^{h} (1 - A_a 2^{Lj} T) \in \mathbb{Z}[T]. \tag{39.14}$$

Equivalently, the sequence satisfies an eventual integer-coefficient linear recurrence whose characteristic polynomial can be taken to be $\prod_{a,j}(X - A_a 2^{Lj})$.

Proof. The elementary root-of-unity filter is

$$\frac{1}{q}\sum_{a=0}^{q-1}\zeta_q^{a(n-b)} = \begin{cases} 1, & n \equiv b \pmod q, \\ 0, & \text{otherwise.} \end{cases}$$

After multiplication by $\varepsilon_n n^h$ and summation, it yields

$$S^{(q)}_{m,b,h} = \frac{1}{q}\sum_{a=0}^{q-1}\zeta_q^{-ab}T_{m,h}(\zeta_q^a).$$

For $m > h$, the term $a = 0$ vanishes by Prouhet cancellation. The product periodicity gives $P_{Lt+s}(\zeta_q^a) = A_a^t P_s(\zeta_q^a)$. Insert (39.2) in the remaining terms and collect powers of 2^{Lt}. This proves (39.12) and (39.13).

Each geometric sequence in (39.12) has a rational generating series with the indicated linear denominator. Discarding or adding finitely many initial terms changes only a polynomial, so the same denominator works for the full generating series. Finally, each A_a is an algebraic integer, and every automorphism $\zeta_q \mapsto \zeta_q^u$, $\gcd(u, q) = 1$, permutes the factors of (39.14). Its coefficients are therefore rational algebraic integers, hence integers. □

The denominator is an explicit bound, not necessarily the minimal denominator. Cycle products may coincide, coefficients may vanish, and conjugate terms may cancel. General recurrence mechanisms are already present in automatic-sequence theory [28]; the point here is the direct formula for every weighted coefficient and every candidate characteristic root.

39.4 Closed residue formulas modulo three

Let $N = 4^t$, so the binary depth is $m = 2t$. Substituting (39.9) in the theorem and adding conjugates gives

$$S^{(3)}_{2t,0,0} = 2\cdot 3^{t-1}, \qquad t \geq 1, \tag{39.15}$$

$$S^{(3)}_{2t,0,1} = 3^{t-1}(N-1), \qquad t \geq 1, \tag{39.16}$$

$$S^{(3)}_{2t,0,2} = \frac{3^{t-1}}{27}(19N^2 - 26N + 7), \qquad t \geq 2, \tag{39.17}$$

$$S^{(3)}_{2t,0,3} = \frac{3^{t-1}}{9}(N-1)(5N^2 - 4N - 1), \qquad t \geq 2. \tag{39.18}$$

For clarity, the degree-two computation before multiplication by the root filter is

$$\frac{T_{2t,2}(\omega)}{3^t} = (N-1)^2\left(\frac{1}{2} + \frac{\mathrm{i}\sqrt{3}}{18}\right)^2 + \frac{N^2-1}{9}.$$

Its real part is $(19N^2 - 26N + 7)/54$, and the filter contributes twice this value divided by three. This proves (39.17). The cubic identity follows by using $\mathcal{B}_3(x_1, x_2, x_3) = x_1^3 + 3x_1x_2 + x_3$; expansion gives (39.18).

t	N	S_0	S_1	S_2	S_3
1	4	2	3	9	27
2	16	6	45	495	6075
3	64	18	567	25389	1274049
4	256	54	6885	1238535	249891075
5	1024	162	82863	59688981	48233475081

Here $S_h = S^{(3)}_{2t,0,h}$. The first row was calculated directly; the last two displayed formulas do not apply to it. At $m = h$, the unweighted-frequency term in the root filter survives. Omitting that term would incorrectly predict $S_2 = 23/3$ instead of 9 when $t = 1$.

For example, if $R_t = S^{(3)}_{2t,0,2}$, the three roots in (39.17) are $3, 12, 48$. Hence

$$R_{t+3} = 63R_{t+2} - 756R_{t+1} + 1728R_t, \qquad t \geq 2. \tag{39.19}$$

This concrete recurrence is one useful consequence of the more general derivative calculus.

Chapter 40

All rational frequencies: transients and dyadic zeros

40.1 A finite transient followed by an odd cycle

Let ζ have exact order $2^r q$, where $q > 1$ is odd. Put

$$\xi = \zeta^{2^r}, \qquad m = r + M, \qquad G_{r,\zeta}(w) = \frac{P_r(\zeta e^w)}{P_r(\zeta)}.$$

The denominator is nonzero. For $r = 0$, this definition means $G_{0,\zeta} = 1$.

Proposition 40.1 (Separation of transient and periodic parts)**.** *For $s \equiv M \pmod{\operatorname{ord}_q 2}$,*

$$\mathcal{Q}_{r+M,\zeta}(z)\mathcal{H}_{\xi,0}(z/2^M) = G_{r,\zeta}(z/2^{r+M})\mathcal{H}_{\xi,s}(z). \tag{40.1}$$

Consequently, for fixed ζ and fixed phase, the normalized local profile tends to $\mathcal{H}_{\xi,s}$. Every finite-depth correction is obtained from the single analytic germ

$$C_{r,\zeta}(w) = \frac{G_{r,\zeta}(w)}{\mathcal{H}_{\xi,0}(2^r w)}. \tag{40.2}$$

Proof. Split the defining product after r factors:

$$P_{r+M}(v) = P_r(v)P_M(v^{2^r}).$$

Substitute $v = \zeta e^{z/2^{r+M}}$ and divide by the corresponding value at $z = 0$. The remaining normalized M-factor product is $\mathcal{Q}_{M,\xi}(z)$. Apply Theorem 37.1. Near zero, the multiplier is exactly $C_{r,\zeta}(z/2^{r+M})$. Since $C_{r,\zeta}(0) = 1$, convergence and all Taylor corrections follow. Equation (40.1) itself is entire and does not require division by a possibly vanishing profile. □

Thus rational points of non-dyadic order have no new limiting objects beyond the odd cycles. Their 2-primary part produces a finite, explicitly computable transient.

40.2 Removing the growing zero at a dyadic root

Suppose now that ζ is a primitive 2^r-th root, with $r \geq 1$; we also allow $r = 0, \zeta = 1$. If $m = r + M$, precisely the factors with indices $j = r, \ldots, r + M - 1$ vanish at ζ. Their number is M.

Definition 40.2 (Zero-removed dyadic profile)**.** Define

$$\mathcal{N}_{r,M,\zeta}(z) = \frac{(-1)^M 2^{M(M+1)/2}}{P_r(\zeta)z^M} P_{r+M}\big(\zeta e^{z/2^{r+M}}\big). \tag{40.3}$$

The value at $z = 0$ is defined by continuity.

Theorem 40.3 (Exact dyadic factorization). *The function in* (40.3) *is entire, equals* 1 *at zero, and satisfies*

$$\mathcal{N}_{r,M,\zeta}(z) = G_{r,\zeta}(z/2^{r+M})\mathcal{U}_M(z). \tag{40.4}$$

In denominator-free form,

$$\mathcal{N}_{r,M,\zeta}(z)\mathcal{U}(z/2^M) = \mathcal{U}(z)G_{r,\zeta}(z/2^{r+M}). \tag{40.5}$$

For every fixed dyadic root ζ*, the functions* $\mathcal{N}_{r,M,\zeta}$ *tend to* $\mathcal{U}$ *locally uniformly, with all derivatives, as* $M \to \infty$.

Proof. The factors following the first r are

$$\prod_{j=r}^{r+M-1} \left(1 - \mathrm{e}^{2^j z/2^{r+M}}\right) = \prod_{k=1}^{M}(1 - \mathrm{e}^{z/2^k}) = (-1)^M z^M 2^{-M(M+1)/2}\mathcal{U}_M(z).$$

The leading factors form $P_r(\zeta)G_{r,\zeta}(z/2^{r+M})$, proving (40.4). It also proves removal of the zero and the normalization at the origin. Multiplying by $\mathcal{U}(z/2^M)$ gives (40.5). For fixed r, ζ, the finite head tends uniformly to 1 on compact sets, while $\mathcal{U}_M \to \mathcal{U}$. Derivative convergence follows from Cauchy's integral formula on slightly larger compact discs. □

For $r = 0$, this is the usual Thue–Morse–Fabius finite-product bridge. The added information is its exact form at every dyadic root, and, in the next section, its loss of uniformity when the root moves toward 1.

40.3 Every surviving dyadic moment

The zero removal also gives a closed formula for the first nonzero moment and all moments beyond it. Define

$$C_M = (-1)^M 2^{M(2r+M-1)/2} P_r(\zeta) \tag{40.6}$$

and, for $n \geq 1$,

$$K_n = -\sum_{j=0}^{r-1} 2^{jn}\,\mathrm{Li}_{1-n}(\zeta^{2^j}) + \beta_n \frac{2^{(r+M)n} - 2^{rn}}{2^n - 1}. \tag{40.7}$$

The first sum is empty when $r = 0$. None of its arguments equals 1.

Theorem 40.4 (Dyadic moments after cancellation). *For* $0 \leq d < M$, $T_{r+M,d}(\zeta) = 0$*, and for* $h \geq 0$,

$$T_{r+M,M+h}(\zeta) = C_M \frac{(M+h)!}{h!}\mathcal{B}_h(K_1, \ldots, K_h). \tag{40.8}$$

In particular, the first surviving moment is $C_M M!$.

Proof. Use the moment generating variable t, without the local-window scaling. The factorization is

$$P_{r+M}(\zeta\mathrm{e}^t) = C_M t^M \frac{P_r(\zeta\mathrm{e}^t)}{P_r(\zeta)} \prod_{j=r}^{r+M-1} g(2^j t).$$

For the finite head, the n-th logarithmic derivative at zero is $-\sum_{j<r} 2^{jn}\,\mathrm{Li}_{1-n}(\zeta^{2^j})$. For the remaining factors, $\log g(t) = \sum_{n\geq 1} \beta_n t^n/n!$; summing the geometric progression $\sum_{j=r}^{r+M-1} 2^{jn}$ gives the second term in (40.7). The normalized analytic factor therefore has exponential generating coefficients $\mathcal{B}_h(K)/h!$. Multiplication by $C_M t^M$ and differentiation proves the theorem. □

40.4 A uniform, but subcritical, moving-root estimate

A useful preliminary observation works uniformly over all primitive dyadic roots.

Proposition 40.5 (A sufficient condition for the fixed-root limit to persist)**.** *Let $r = r_M \geq 1$ and choose any primitive 2^r-th root ζ_M. If $r2^{-M} \to 0$, then $\mathcal{N}_{r,M,\zeta_M} \to \mathcal{U}$ locally uniformly.*

Proof. For $0 \leq j < r$, primitivity implies

$$|1 - \zeta_M^{2^j}| \geq 2\sin(\pi/2^{r-j}) \geq 4/2^{r-j}.$$

Here $\sin x \geq 2x/\pi$ on $[0, \pi/2]$. The deviation from 1 of the corresponding normalized head factor, at $|z| \leq R$, is bounded by

$$\frac{|\mathrm{e}^{2^j z/2^{r+M}} - 1|}{|1 - \zeta_M^{2^j}|} \leq R2^{-M-2}\mathrm{e}^{R2^{-M-1}}.$$

Summing over the r factors and using $|\prod(1 + u_j) - 1| \leq \exp(\sum |u_j|) - 1$ gives

$$\left|G_{r,\zeta_M}(z/2^{r+M}) - 1\right| \leq \exp\left(rR2^{-M-2}\mathrm{e}^{R2^{-M-1}}\right) - 1. \tag{40.9}$$

The bound tends to zero under the stated hypothesis. Combine it with (40.4). □

The condition is sufficient, not necessary for every moving sequence of roots. The following theorem identifies the critical scales for the roots closest to 1.

Chapter 41

Coalescing roots: a Gaussian–Fabius hierarchy

41.1 The exact two-parameter formula

For the rest of this section fix

$$c = 2\pi \mathrm{i}, \qquad \zeta_r = \mathrm{e}^{c/2^r}, \qquad \epsilon = 2^{-M}, \qquad \mathcal{N}_{r,M} = \mathcal{N}_{r,M,\zeta_r}, \qquad r \geq 1. \tag{41.1}$$

The letter ϵ here is a small positive real parameter, not the Thue–Morse sign ε_n.

Lemma 41.1 (An exact factorization at the closest dyadic root)**.** *For all $r \geq 1$, $M \geq 0$,*

$$\mathcal{N}_{r,M}(z) = \mathcal{U}_M(z) \left(1 + \frac{\epsilon z}{c}\right)^r \frac{\mathcal{U}_r(c + \epsilon z)}{\mathcal{U}_r(c)}. \tag{41.2}$$

In particular, on any fixed compact disc, for all sufficiently large M, this is also

$$\mathcal{N}_{r,M}(z) = \frac{\mathcal{U}(z)}{\mathcal{U}(\epsilon z)} \left(1 + \frac{\epsilon z}{c}\right)^r \frac{\mathcal{U}_r(c + \epsilon z)}{\mathcal{U}_r(c)}. \tag{41.3}$$

The factors other than the power satisfy, uniformly in $r \geq 1$,

$$\frac{1}{\mathcal{U}(\epsilon z)} = 1 + O_R(\epsilon), \qquad \frac{\mathcal{U}_r(c + \epsilon z)}{\mathcal{U}_r(c)} = 1 + O_R(\epsilon) \quad (|z| \leq R). \tag{41.4}$$

Proof. In the head $G_{r,\zeta_r}(z/2^{r+M})$, replace j by $r - k$. Each normalized factor becomes

$$\frac{1 - \mathrm{e}^{(c+\epsilon z)/2^k}}{1 - \mathrm{e}^{c/2^k}} = \left(1 + \frac{\epsilon z}{c}\right) \frac{g((c + \epsilon z)/2^k)}{g(c/2^k)}.$$

Multiplication gives (41.2) by (40.4). No denominator $g(c/2^k)$ vanishes: its argument is $2\pi\mathrm{i}/2^k$, not a nonzero integer multiple of $2\pi\mathrm{i}$.

For the uniform estimate, choose a closed disc about c of radius less than 2π. All the functions $\mathcal{U}_r$, as well as their locally uniform limit $\mathcal{U}$, are nonzero there, since their possible zeros begin at $\pm 4\pi\mathrm{i}$. The normally convergent logarithmic derivative products are uniformly bounded on a slightly smaller disc. Equivalently, one may split finitely many factors from a uniformly nonvanishing tail. Integration of the logarithmic derivative along the segment from c to $c + \epsilon z$ proves the second estimate in (41.4), uniformly in r. The first follows from analyticity and $\mathcal{U}(0) = 1$. Finally use the exact tail identity for $\mathcal{U}_M$. □

Formula (41.2) isolates the only factor that can accumulate a macroscopic correction: $(1 + \epsilon z/c)^r$. In particular, the double-scaling limit is governed by a logarithm, not by an uncontrolled approximation of the original polynomial's exponentially many coefficients.

41.2 The complete hierarchy

Theorem 41.2 (Renormalized coalescing-root limits). *Fix an integer $p \geq 0$. Let $M \to \infty$, let $r = r_M \geq 1$ be integers, and suppose*

$$r\epsilon^{p+1} \longrightarrow \lambda \in [0, \infty), \qquad \epsilon = 2^{-M}. \tag{41.5}$$

Define the entire renormalized functions

$$\widetilde{\mathcal{N}}^{(p)}_{r,M}(z) = \exp\left(-r \sum_{j=1}^{p} \frac{(-1)^{j-1}}{j} \left(\frac{\epsilon z}{c}\right)^j \right) \mathcal{N}_{r,M}(z), \tag{41.6}$$

with an empty sum when $p = 0$. Then, locally uniformly on $\mathbb{C}$ and with every fixed derivative,

$$\widetilde{\mathcal{N}}^{(p)}_{r,M}(z) \longrightarrow \mathcal{U}(z) \exp\left(\frac{(-1)^p \lambda}{p+1} \left(\frac{z}{c}\right)^{p+1} \right). \tag{41.7}$$

For fixed R, p, B, if $r\epsilon^{p+1} \leq B$ and $0 \leq \lambda \leq B$, the difference from the right side of (41.7), on $|z| \leq R$, is

$$O_{R,p,B}\left(\epsilon + |r\epsilon^{p+1} - \lambda|\right). \tag{41.8}$$

Proof. For sufficiently small ϵ, the Taylor logarithm of $1 + \epsilon z/c$ is defined throughout $|z| \leq R$. Put $u = \epsilon z/c$. Taylor's formula, uniformly on that disc, gives

$$\log(1+u) - \sum_{j=1}^{p} \frac{(-1)^{j-1} u^j}{j} = \frac{(-1)^p u^{p+1}}{p+1} + O_{R,p}(\epsilon^{p+2}).$$

After multiplication by r, the remainder is $O_{R,p,B}(\epsilon)$, and the leading term differs from $(-1)^p \lambda (z/c)^{p+1}/(p+1)$ by $O_{R,p}(|r\epsilon^{p+1} - \lambda|)$. The exponents are uniformly bounded on the compact disc, so exponentiation preserves this additive error estimate. Multiply by the two uniformly controlled ratios in (41.4) and by $\mathcal{U}(z)$. This proves (41.8), including at zeros of $\mathcal{U}$: only multiplication by the bounded function $\mathcal{U}$, not division by it, was used. Convergence follows immediately. Cauchy's formula on a larger disc gives convergence of all derivatives. ☐

41.3 The Gaussian member and its interpretation

The first two levels deserve separate statements.

Corollary 41.3 (Drift and Gaussian correction). *If $r2^{-M} \to \lambda$, then without further renormalization,*

$$\mathcal{N}_{r,M}(z) \longrightarrow \mathcal{U}(z) \mathrm{e}^{\lambda z/(2\pi \mathrm{i})}. \tag{41.9}$$

If instead $r4^{-M} \to \lambda$, then

$$\mathrm{e}^{-r2^{-M} z/(2\pi \mathrm{i})} \mathcal{N}_{r,M}(z) \longrightarrow \mathcal{U}(z) \mathrm{e}^{\lambda z^2/(8\pi^2)}. \tag{41.10}$$

For $\lambda \geq 0$, the limit in (41.10) is the moment generating function of

$$X + Z_\lambda, \qquad Z_\lambda \sim N\left(0, \frac{\lambda}{4\pi^2}\right), \qquad Z_\lambda \text{ independent of } X. \tag{41.11}$$

Proof. Take $p = 0$ or $p = 1$ in Theorem 41.2. Since $c^2 = -4\pi^2$, the quadratic exponent is $\lambda z^2/(8\pi^2)$. A centered normal variable of variance σ^2 has moment generating function $\mathrm{e}^{\sigma^2 z^2/2}$, so independence and (36.1) prove the last assertion. ☐

The distinction between an analytic limit and a probabilistic limit is essential. The finite functions $\mathcal{N}_{r,M}$ have complex coefficients in general. The linear exponential renormalization corresponds to a complex displacement, not to centering a sequence of real random variables. The theorem therefore does not invoke a classical central limit theorem or claim weak convergence of positive finite-level laws. Rather, positivity emerges in this particular limiting transform.

Its limiting mean is $1/2$ and its variance is

$$\frac{1}{36} + \frac{\lambda}{4\pi^2}. \tag{41.12}$$

When $\lambda > 0$, the limiting distribution is the convolution of the compactly supported Fabius density with a Gaussian density, hence has support all of $\mathbb{R}$.

For $p = 2$, removal of the linear and quadratic terms gives a cubic exponential factor. Higher levels similarly retain the first logarithmic coefficient not removed. No positivity is asserted for these higher limiting transforms.

41.4 Why the scales matter

The exact factorization supplies three qualitatively different regimes. If $r = o(2^M)$, the head becomes invisible and the ordinary Fabius limit survives. At $r \asymp 2^M$, a nontrivial linear exponential remains. At $r \asymp 4^M$, removal of the linear term leaves a Gaussian correction. More generally, after subtracting p logarithmic terms, the critical depth is $r \asymp 2^{(p+1)M}$.

These are genuine limits in two independently growing parameters: the root order is 2^r, while the number of vanishing factors beyond that root is M. A fixed-root theorem alone cannot detect them. The hierarchy also makes computation feasible: one evaluates a short convergent product and a renormalized logarithm instead of expanding a polynomial of degree $2^{r+M} - 1$.

Chapter 42

Cyclotomic aggregation and Fabius-transform quotients

42.1 Multiplying all nontrivial rational profiles

Let $q \ge 3$ be odd, $\zeta_q = \mathrm{e}^{2\pi \mathrm{i}/q}$, and use a common phase period $L = \mathrm{ord}_q\, 2$ for the profiles associated with ζ_q^a, $1 \le a < q$.

Theorem 42.1 (Cyclotomic aggregation identity)**.** *For every phase s,*

$$\prod_{a=1}^{q-1} \mathcal{H}_{\zeta_q^a, s}(z) = \mathcal{R}_q(z), \qquad \mathcal{R}_q(z) := \frac{\mathcal{U}(qz)}{\mathcal{U}(z)}. \tag{42.1}$$

The apparent quotient extends to an entire function, and the product is independent of s. It has the positive-factor representation

$$\mathcal{R}_q(z) = \prod_{k \ge 1} \left(\frac{1}{q} \sum_{j=0}^{q-1} \mathrm{e}^{jz/2^k} \right). \tag{42.2}$$

At finite depth there is also the exact identity

$$\prod_{a=1}^{q-1} \mathcal{Q}_{m,\zeta_q^a}(z) = \frac{\mathcal{U}_m(qz)}{\mathcal{U}_m(z)}, \tag{42.3}$$

again with removable singularities on the right.

Proof. For a variable x, factorization of $x^q - 1$ yields

$$\prod_{a=1}^{q-1} (1 - \zeta_q^a x) = 1 + x + \cdots + x^{q-1}, \qquad \prod_{a=1}^{q-1} (1 - \zeta_q^a) = q.$$

Since multiplication by any power of two permutes the nonzero residues modulo the odd integer q, for every fixed profile factor k we obtain

$$\prod_{a=1}^{q-1} \frac{1 - \zeta_q^{a 2^{s-k}} \mathrm{e}^u}{1 - \zeta_q^{a 2^{s-k}}} = \frac{1}{q} \sum_{j=0}^{q-1} \mathrm{e}^{ju} = \frac{g(qu)}{g(u)}.$$

The middle expression is entire and defines the quotient at its removable singularities. Multiply over $u = z/2^k$. All products converge normally, so rearrangement of the finitely many profile families is valid. This proves (42.1) and (42.2); stopping at $k = m$ proves (42.3). $\square$

The word "aggregation" avoids a small but important algebraic ambiguity. When q is prime, the left side is the full cyclotomic Galois norm of a single profile. When q is composite, the nonzero residues include nonunits, so the product includes roots of several exact orders and is not one field norm.

Corollary 42.2 (Primitive cyclotomic norm)**.** *For odd $q > 1$,*

$$\prod_{\substack{1\le a<q \\ \gcd(a,q)=1}} \mathcal{H}_{\zeta_q^a,0}(z) = \prod_{d|q} \mathcal{U}(dz)^{\mu(q/d)}, \tag{42.4}$$

where μ is the Möbius function. This meromorphic-looking expression has an entire continuation. For $q = p^v$, an odd prime power, it reduces to

$$\frac{\mathcal{U}(p^v z)}{\mathcal{U}(p^{v-1} z)} = \mathcal{R}_p(p^{v-1} z). \tag{42.5}$$

Proof. Let F_q denote the primitive product on the left. Sorting the nontrivial roots by exact order gives $\mathcal{R}_q = \prod_{d|q,\, d>1} F_d$, with $F_1 = 1$. Multiplicative Möbius inversion therefore gives $F_q = \prod_{d|q} \mathcal{R}_d^{\mu(q/d)}$. Substitute $\mathcal{R}_d = \mathcal{U}(dz)/\mathcal{U}(z)$. Because $\sum_{d|q} \mu(q/d) = 0$ for $q > 1$, the common denominator powers cancel, proving (42.4). These manipulations first hold near zero, where all factors are nonzero; the primitive product supplies the entire continuation. For a prime power only the two displayed divisors contribute. □

No general positivity claim for the primitive norm at composite non-prime-power orders is needed here.

42.2 Cumulant traces and an independent check

Taking logarithms of (42.1) near zero and using (36.6) gives

Corollary 42.3 (Trace of every profile cumulant)**.** *For all $n \ge 1$,*

$$\sum_{a=1}^{q-1} \kappa_{\zeta_q^a,s,n} = \beta_n \frac{q^n - 1}{2^n - 1}. \tag{42.6}$$

For example, at $q = 3$, the two second cumulants sum to $2/9$, while the fourth cumulants sum to $-2/45$, in agreement with (39.9). This trace is independent of the phase even though individual odd cumulants need not be.

For comparison, the zeros of $\mathcal{U}$ are

$$z = 4\pi i n, \qquad n \in \mathbb{Z} \setminus \{0\}, \quad \text{with multiplicity } 1 + v_2(n). \tag{42.7}$$

Indeed, the k-th factor $g(z/2^k)$ vanishes at $z = 2^{k+1}\pi i j$, $j \neq 0$; count the contributing factors exactly as in Theorem 38.1. This also justifies the zero-free neighborhoods used in the coalescing-root argument. The product identity explains arithmetically how the zeros in the denominator $\mathcal{U}(z)$ cancel inside $\mathcal{U}(qz)$.

42.3 A probability law for every integer dilation

The expression in (42.2) makes sense for every integer $q \ge 1$, even though the profile aggregation theorem required odd q. Let $D_1, D_2, \ldots$ be independent, each uniform on $\{0, 1, \ldots, q-1\}$, and define

$$Y_q = \sum_{k\ge1} 2^{-k} D_k, \qquad \nu_q = \mathcal{L}(Y_q). \tag{42.8}$$

The letter $\mathcal{L}$ denotes probability law. The series converges absolutely and lies in $[0, q-1]$.

Proposition 42.4 (Dilation quotients are digit-series transforms)**.** *For every integer $q \geq 1$,*

$$\mathcal{R}_q(z) = \mathbb{E}e^{zY_q} = \frac{\mathcal{U}(qz)}{\mathcal{U}(z)}, \tag{42.9}$$

with an entire interpretation of the quotient. Moreover,

$$\mathbb{E}Y_q = \frac{q-1}{2}, \qquad \mathrm{Var}(Y_q) = \frac{q^2-1}{36}, \qquad Y_q \overset{d}{=} \frac{D+Y_q'}{2}. \tag{42.10}$$

Here D is an independent uniform digit and Y_q' is an independent copy of Y_q. The law is symmetric about $(q-1)/2$.

Proof. Independence identifies the moment generating function with (42.2); bounded support permits passage to the infinite product for all complex z. The elementary identity $q^{-1}\sum_{j=0}^{q-1} e^{ju} = g(qu)/g(u)$ then proves the quotient relation first near zero, and hence everywhere by continuation. The mean follows from $\mathbb{E}D = (q-1)/2$ and $\sum 2^{-k} = 1$. The variance follows from $\mathrm{Var}(D) = (q^2-1)/12$ and $\sum 4^{-k} = 1/3$. Shifting the digit sequence proves the self-similarity equation. Replacing every digit D_k by $q-1-D_k$ proves symmetry. □

These positive laws should not be confused with the classical Thue–Morse diffraction measure. They arise here from a product of rational-frequency profiles, not from a limit of their squared absolute values.

Chapter 43

A sharp singularity and differentiability classification

The representation $\mathcal{R}_q = \mathcal{U}(qz)/\mathcal{U}(z)$ might suggest a smooth function for every q, since $\mathcal{U}$ itself is the transform of a smooth density. That suggestion is false. Odd dilation factors produce singular continuous laws. Each additional factor of two contributes a uniform convolution, and the resulting differentiability threshold can be determined exactly.

Theorem 43.1 (Regularity determined by the 2-adic part)**.** *Write $q = 2^r d$, where d is odd and $r \geq 0$. Then:*

(i) If $q = 1$, ν_q is the point mass at zero.

(ii) If $r = 0$ and $d \geq 3$, ν_q is singular continuous.

(iii) If $r \geq 1$ and $d \geq 3$, ν_q has a density belonging to $C^{r-1}(\mathbb{R})$, but no density representative belongs to $C^r(\mathbb{R})$.

(iv) If $d = 1$ and $r = 1$, ν_q is uniform on $[0, 1]$ and has no continuous density representative on $\mathbb{R}$. If $d = 1$ and $r \geq 2$, it has a density in $C^{r-2}(\mathbb{R})$, but not in $C^{r-1}(\mathbb{R})$.

These are global differentiability assertions; they do not specify all local Hölder exponents.

43.1 Nondecaying Fourier coefficients at odd dilation factors

We first prove singularity without making an invalid inference from Fourier nondecay alone. A measure can have both an absolutely continuous part and a singular part, so nondecay by itself would not prove pure singularity.

Let $d \geq 3$ be odd, and write

$$c_d = \mathbb{E}e^{2\pi i Y_d} = \prod_{k \geq 1} \left(\frac{1}{d} \sum_{j=0}^{d-1} e^{2\pi i j/2^k} \right).$$

Every factor is nonzero. Indeed, the geometric sum would vanish precisely if its nontrivial 2^k-th root of unity had d-th power 1, requiring $2^k \mid d$. The deviations of the factors from 1 in the tail are bounded by a constant times 2^{-k}. Thus the product converges to a nonzero number:

$$c_d \neq 0. \tag{43.1}$$

The digit self-similarity also gives

$$\mathbb{E}e^{2\pi i 2^n Y_d} = c_d, \qquad n \geq 0. \tag{43.2}$$

To promote this average identity to pure singularity we compare almost-sure orbit averages.

Lemma 43.2 (Two incompatible full-measure orbit averages)**.** *For ν_d-almost every x,*

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} \mathrm{e}^{2\pi \mathrm{i} 2^n x} = c_d. \tag{43.3}$$

For Lebesgue-almost every $x \in \mathbb{R}$, the same limit equals zero.

Proof. First consider a random digit sequence. The first n terms of $2^n Y_d$ are integers, so

$$\mathrm{e}^{2\pi \mathrm{i} 2^n Y_d} = \exp\left(2\pi \mathrm{i} \sum_{k\geq 1} 2^{-k} D_{n+k}\right).$$

Truncate the sum on the right after K terms. The resulting bounded function depends only on $D_{n+1}, \ldots, D_{n+K}$, and its error is at most $2\pi(d-1)2^{-K}$, uniformly in n and in the digit sequence. For each fixed residue class of n modulo K, these blocks are disjoint, hence independent and identically distributed. The strong law of large numbers applied to their real and imaginary parts proves convergence of each residue-class average to the expectation of the truncated function. Combining the finitely many classes gives convergence of the full average. There are only countably many K, so these conclusions hold simultaneously almost surely. Let $K \to \infty$ and use the uniform truncation bound to obtain (43.3) with expectation c_d.

For the Lebesgue assertion, set $A_N(x) = N^{-1} \sum_{n=0}^{N-1} \mathrm{e}^{2\pi \mathrm{i} 2^n x}$. Orthogonality of the distinct integer-frequency exponentials on $[0, 1]$ gives

$$\int_0^1 |A_N(x)|^2 \, \mathrm{d}x = \frac{1}{N}.$$

For $N = j^2$, Chebyshev's inequality and the convergence of $\sum_j j^{-2}$ imply, by the Borel–Cantelli lemma, that $A_{j^2}(x) \to 0$ almost everywhere. To see convergence rather than only boundedness at one threshold, apply the argument at thresholds $1, 1/2, 1/3, \ldots$. If $j^2 \leq N < (j+1)^2$, boundedness of the summands gives

$$|A_N(x) - A_{j^2}(x)| \leq \frac{2(N - j^2)}{N} \leq \frac{2(2j+1)}{j^2},$$

which tends to zero uniformly. Thus $A_N \to 0$ for Lebesgue-almost every $x \in [0, 1]$. Periodicity extends the conclusion to $\mathbb{R}$. □

Since $c_d \neq 0$, the set where the limit equals c_d has full ν_d-measure and zero Lebesgue measure. This proves that ν_d is purely singular.

43.2 Absence of atoms

Lemma 43.3 (Odd digit laws have no atoms)**.** *For odd $d \geq 3$, $\nu_d(\{x\}) = 0$ for every real x.*

Proof. Represent a uniform digit as a mixture: with probability $2/d$, select a fair digit from $\{0, 1\}$; with the remaining probability, select uniformly from $\{2, \ldots, d-1\}$. Choose all mixture selectors and all component digits independently. Each possible digit has probability $1/d$, so this is the required law.

There are almost surely infinitely many selectors of the first kind. Condition on all selectors and on all digits of the second kind. The remaining randomness is a series of independent fair bits at an infinite set of binary positions. For any fixed value of the series, there are at most two possible full binary expansions, after setting the unselected bit positions to zero. Each particular infinite fair-bit sequence has conditional probability zero, since fixing its first J active bits has probability 2^{-J}. The conditional probability of any specified value is therefore zero, and averaging proves the assertion. □

Combining this lemma with the preceding singularity argument proves part (ii) of Theorem 43.1.

43.3 Uniform convolutions and the upper regularity bound

Iterating $\mathcal{U}(2z) = g(z)\mathcal{U}(z)$ gives the exact factorization

$$\mathcal{R}_{2^r d}(z) = \mathcal{R}_d(z) \prod_{j=0}^{r-1} g(d2^j z). \tag{43.4}$$

Since $g(\ell z)$ is the moment generating function of a uniform law on $[0, \ell]$, uniqueness of characteristic functions gives

$$\nu_{2^r d} = \nu_d * \lambda_{\ell_0} * \cdots * \lambda_{\ell_{r-1}}, \qquad \ell_j = d2^j, \tag{43.5}$$

where λ_ℓ is the uniform probability measure on $[0, \ell]$. For $d = 1$, take $\nu_1 = \delta_0$.

For a subset $S \subseteq \{0, \dots, r-1\}$, write $\ell_S = \sum_{j \in S} \ell_j$, and let $\tau_a \nu$ denote translation of a measure to the right by a. Put $L_r = \prod_{j=0}^{r-1} \ell_j$. In the sense of distributions,

$$\mathrm{D}^r f_q = \frac{1}{L_r} \sum_{S \subseteq \{0, \dots, r-1\}} (-1)^{|S|} \tau_{\ell_S} \nu_d, \tag{43.6}$$

where f_q is a density of ν_q, which exists when $r \geq 1$. To prove the identity, differentiate each uniform density once: $\mathrm{D}(\mathbf{1}_{[0,\ell]}/\ell) = (\delta_0 - \delta_\ell)/\ell$, and convolve the resulting signed measures.

If $d \geq 3$ is odd, the right side of (43.6) is singular, being supported on a finite union of translates of a Lebesgue-null carrier of ν_d. It is also nonzero. Otherwise $\mathrm{D}^r f_q = 0$, so f_q, as a distribution, would be a polynomial of degree at most $r - 1$. This elementary distributional fact follows inductively from the fact that a distribution with zero first derivative is constant. But a polynomial distribution with compact support is zero, whereas $\int f_q = 1$. This contradiction proves nonvanishing.

Were there a C^r density representative, its r-th distributional derivative would be the ordinary continuous function $f_q^{(r)}$, hence an absolutely continuous measure on compact sets. It cannot equal a nonzero singular measure. Therefore the density is not C^r.

43.4 The matching lower bound

Let $F_d(x) = \nu_d((-\infty, x])$. The absence of atoms makes F_d continuous when $d \geq 3$ is odd. Integrating (43.6) from $-\infty$ gives

$$f_q^{(r-1)}(x) = \frac{1}{L_r} \sum_{S \subseteq \{0, \dots, r-1\}} (-1)^{|S|} F_d(x - \ell_S). \tag{43.7}$$

The integration constant is zero because the density and the expression vanish sufficiently far to the left. The right side is continuous. More explicitly, a representative of the density is

$$f_q(x) = \frac{1}{(r-1)! L_r} \sum_S (-1)^{|S|} \int (x - \ell_S - y)_+^{r-1} \, \nu_d(\mathrm{d}y), \tag{43.8}$$

where for $r = 1$ the zero-th truncated power is the indicator of the nonnegative half-line. Repeated convolution of uniform densities, or repeated integration of (43.6), proves this formula. Differentiating at most $r - 2$ times is justified by bounded support and dominated convergence; the final derivative is the continuous expression (43.7). Thus $f_q \in C^{r-1}(\mathbb{R})$. Together with the upper bound, this proves part (iii).

When $d = 1$, replace F_d in (43.7) by a step function. The subset sums of $1, 2, \dots, 2^{r-1}$ are all distinct, so the resulting piecewise constant function has nonzero jumps. For $r \geq 2$, its integral is continuous, proving that the density is C^{r-2} but not C^{r-1}. For $r = 1$, the density is exactly $\mathbf{1}_{[0,1]}$, up to changes at endpoints, and cannot have a continuous representative on the whole real line. This proves part (iv); part (i) is immediate from the digit definition. The proof of Theorem 43.1 is complete.

43.5 Examples and limitations

The first cases illustrate why the odd part matters:

Dilation q	Type of the law with transform $\mathcal{U}(qz)/\mathcal{U}(z)$
2	Uniform density; no continuous representative on $\mathbb{R}$
3	Singular continuous
4	Continuous density, not C^1
6	Continuous density, not C^1
8	C^1 density, not C^2
12	C^1 density, not C^2
24	C^2 density, not C^3

The power-of-two rows are finite uniform-convolution splines, a classical construction. The other even rows replace the last jump-bearing derivative by a continuous singular distribution function. Their next derivative remains singular, which is why the differentiability threshold is sharp. This argument does not calculate Hausdorff dimensions or establish a multifractal spectrum for the odd laws.

Chapter 44

Exact checks, numerical experiments, and reproducibility

44.1 What was checked exactly

The accompanying program `verify.py` works in the cyclotomic quotient field $\mathbb{Q}[x]/\Phi_q(x)$, using rational coefficients, rather than substituting floating-point approximations to roots of unity. It independently evaluates the exponential coefficients from their defining differential identity and compares the predicted moments with direct Thue–Morse sums.

For the odd orders $q = 3, 5, 7, 9$, it checks the moment formula for every $m = 0, \dots, 8$ and $h = 0, \dots, 6$: this gives $4 \cdot 9 \cdot 7 = 252$ exact equalities. For primitive roots of orders 2^r, $r = 1, \dots, 4$, and $M = 0, \dots, 5$, it checks all vanishing moments below order M and the four moments of orders $M, M+1, M+2, M+3$: this gives another 156 exact equalities. All 408 passed. The modulo-three table and its four closed formulas were also checked directly on the displayed range.

These tests detect algebraic errors in signs, powers of two, phase conventions, and factorial normalization. They do not constitute formal verification of the quantified theorems. In particular, a computer algebra check in a finite set of cyclotomic fields is not a Lean proof.

44.2 Convergence after cubic depth correction

The numerical part uses 65 decimal digits of working precision and 270 factors in the normally convergent products. For $q = 3$, $\zeta = \omega$, $z = 1 + \mathrm{i}/3$, and even m, define

$$E_m^{(0)} = |\mathcal{Q}_{m,\omega}(z) - \mathcal{H}_{\omega,0}(z)|, \quad E_m^{(3)} = \left| \mathcal{Q}_{m,\omega}(z) - \mathcal{H}_{\omega,0}(z) \sum_{j=0}^{3} c_{\omega,j} z^j 2^{-mj} \right|.$$

The observed errors are:

m	$E_m^{(0)}$	$E_m^{(3)}$
4	$5.60097403435 \times 10^{-2}$	$9.96991319447 \times 10^{-8}$
8	$3.52559385875 \times 10^{-3}$	$1.51299266718 \times 10^{-12}$
12	$2.20446499346 \times 10^{-4}$	$2.30783803001 \times 10^{-17}$
16	$1.37782844844 \times 10^{-5}$	$3.52140445937 \times 10^{-22}$

Increasing m by four reduces the uncorrected error asymptotically by a factor of 16, while the cubic correction reduces it by a factor close to $16^4 = 65536$. This is the behavior proved in Theorem 37.3, not a fitted convergence law.

For the tested odd orders $3, 5, 7, 9, 15$, the maximum residual in the exact profile-tail identities was approximately 2.62×10^{-65}. The maximum residual in the cyclotomic aggregation identity was approximately 1.61×10^{-60}. These are floating-point residuals, not interval-arithmetic certificates. The analytic bounds in the proofs, rather than numerical residuals, justify the limiting identities.

44.3 The Gaussian double-scaling experiment

For $r = 4^M$, $\lambda = 1$, and $z = 1 + \mathrm{i}/3$, evaluate the left side of (41.10) using the factorization (41.2). The absolute difference from $\mathcal{U}(z)\mathrm{e}^{z^2/(8\pi^2)}$ is:

M	r	Absolute error
2	16	$9.09055797472 \times 10^{-2}$
3	64	$4.55831847811 \times 10^{-2}$
4	256	$2.28234397433 \times 10^{-2}$
5	1024	1.1419588×10^{-2}
6	4096	$5.71174922749 \times 10^{-3}$
7	16384	$2.85636193939 \times 10^{-3}$
8	65536	$1.42830261560 \times 10^{-3}$

The error is consistent with the proved $O(2^{-M})$ estimate. The data files also include the linear and cubic members $p = 0, 2$ of the hierarchy. The largest cubic experiment has $r = 2^{24}$; the product-tail method avoids any attempt to expand P_{r+M}.

Chapter 45

Further questions suggested by the results

45.1 Beyond periodic backward orbits

The periodicity of the odd rational orbit supplies a uniform lower bound for $|1 - a_{s,k}|$. More generally, for any sequence a_k on the unit circle, with $a_k \neq 1$, the condition

$$\sum_{k \geq 1} \frac{2^{-k}}{|1 - a_k|} < \infty \tag{45.1}$$

implies normal convergence of $\prod_{k \geq 1}(1 - a_k e^{z/2^k})/(1 - a_k)$. Indeed, on $|z| \leq R$, the sum of the deviations of its factors from 1 is bounded by $Re^{R/2}$ times (45.1). Thus a sufficient small-divisor criterion follows immediately from the proof of Theorem 37.1.

A more difficult classification problem concerns coherent nonperiodic backward orbits $a_{k+1}^2 = a_k$, especially when their approaches to 1 violate this summability condition. Which renormalizations produce entire limits? The coalescing-root hierarchy gives one explicit model of a failure of uniform separation, but it does not settle the general classification.

45.2 Positivity of primitive cyclotomic norms

The all-nontrivial-root product is positive definite on the imaginary axis because each factor in (42.2) is a characteristic function. The primitive-root product (42.4) has this interpretation for prime powers, but the same factorwise argument does not apply to every composite order: a general cyclotomic polynomial need not have nonnegative coefficients.

It is natural to ask for which odd integers q the primitive product is itself a probability transform, and, when it is not, what signed or complex object represents it. No complete criterion, or untested prime-power-only conjecture, is asserted here.

45.3 Minimal recurrences and exact arithmetic complexity

Theorem 39.2 gives a universal denominator. Determining its minimal factor after cancellations requires information about equalities between cycle products and about vanishing Bell-polynomial coefficients. The explicit coefficient formula turns this into a finite algebraic problem for each q, h, s, b, but a uniform classification in q remains to be developed.

There is also an algorithmic question: how much of the moment calculus can be computed in a proper subfield of $\mathbb{Q}(\zeta_q)$, using conjugation and the trace identity before expanding individual terms? The order-three formulas show the possible simplification, but no optimal bit-complexity bound is claimed.

45.4 A route toward formalization

The results separate naturally into layers. The finite polynomial product, root-of-unity filters, Bell-polynomial coefficient extraction, and cyclotomic exact arithmetic form the first layer. Normal convergence, zero divisors, and the coalescing-root theorem form an analytic second layer. The probability laws and sharp regularity theorem add independent digit sequences, convolution, and distributional derivatives.

This separation suggests a practical formalization order: first certify the finite identities and coefficient formulas, then the compact-uniform product bounds, and finally the probabilistic interpretation. The present archive does not contain a completed Lean formalization of any of these new results.

Part IV

Spline corrections, dyadic reconstruction, lattice limits, and energy moments

Chapter 46

Research setting and the main contributions

The apparent irregularity of the signed sequence coexists with exact cancellation on every complete binary block. This part investigates what quantitative information survives when those finite cancellations are integrated repeatedly and then passed to a smooth limit. The atlas already contains the finite-block factorization, the Fabius–Rvachev bridge, spectral correction coefficients, and a weak spline expansion; its Part II leaves a strong spline-correction statement away from knots as a conjecture. The first main theorem below proves a global fixed-order statement, including the knots, and identifies the exact leading norm constant. The other principal developments concern finite stopping on dyadic arguments, the local behavior of repeated-prefix coefficients, and a moment-theoretic interpretation of Fourier energy. The underlying compactly supported function is not new; its probability interpretation, derivative identities, and rational values at dyadic points belong to the established theory, particularly the work of Arias de Reyna [9, 10]. The derivative-copy geometry, rather than only the formal product expansion, is what is exploited here. The Fourier-energy identification is presented as an additional connection, with the lognormal background credited to the classical moment literature [24]; the same phenomenon, in the Mellin setting, occurs in Part II with a different variance.

46.1 A compact statement of the results

Let $V_1, V_2, \ldots$ be independent, uniformly distributed random variables on $[-1, 1]$. Let u_m and u be the densities of

$$Y_m = \sum_{j=1}^{m} 2^{-j} V_j, \qquad Y = \sum_{j=1}^{\infty} 2^{-j} V_j, \qquad h = 2^{-m}. \tag{46.1}$$

Thus u is the Rvachev up-function in the normalization $\operatorname{supp} u = [-1, 1]$ and $u(0) = 1$. Write

$$a_j = \frac{\mathbb{E} Y^{2j}}{(2j)!}, \qquad K_r = 2^{r(r+1)/2}. \tag{46.2}$$

The four main conclusions are as follows.

Global correction with a sharp constant. For fixed $k, N \geq 0$, and $m \geq k + 2N + 4$,

$$\left\| \mathrm{D}^k \left(u - \sum_{j=0}^{N} a_j 4^{-jm} u_m^{(2j)} \right) \right\|_\infty \leq a_{N+1} K_{k+2N+2} 4^{-(N+1)m}. \tag{46.3}$$

After multiplication by $4^{(N+1)m}$, the norm on the left converges to the constant on the right. In particular, $4^m \|u - u_m\|_\infty \to 4/9$. The proof is in Chapter 50.

Exact dyadic reconstruction. If $x = a/2^d \in (-1, 1)$, then, once $m \geq d + 2$,

$$u_m(x) = \sum_{j=0}^{\lfloor d/2 \rfloor} b_j 4^{-jm} u^{(2j)}(x), \qquad \sum_{j \geq 0} b_j z^{2j} = \frac{1}{\mathbb{E}e^{zY}}. \tag{46.4}$$

This is an equality, not an asymptotic series. Consequently $\lfloor d/2 \rfloor + 1$ consecutive spline levels determine $u(x)$ exactly. The reconstruction weights have total absolute value less than two. See Chapter 51.

A local-limit expansion with finite stopping. Let $c_m(k)$ be the coefficients of

$$A_m(z) = \frac{\prod_{j=0}^{m-1}(1 - z^{2^j})}{(1 - z)^m} = \prod_{j=0}^{m-1}(1 + z + \cdots + z^{2^j - 1}). \tag{46.5}$$

After the centering and scaling in Chapter 52, their lattice density ρ_m satisfies uniformly on its entire lattice

$$\rho_m(x) = u(x) - \left(\frac{m}{6} + \frac{1}{18}\right) 4^{-m} u''(x) + O(m^2 16^{-m}). \tag{46.6}$$

There are explicit polynomial coefficients at every order. At fixed dyadic points, on the odd-level lattice subsequence, the expansion terminates exactly. The sharp uncorrected uniform error constant is $4/3$ when the error is divided by $m4^{-m}$.

Moment twins of Fourier energy. With angular-frequency Fourier transform $\widehat{u}(t) = \int e^{\mathrm{i}tx} u(x)\,\mathrm{d}x$, the probability measure

$$\mathrm{d}\nu(t) = \frac{|\widehat{u}(t)|^2\,\mathrm{d}t}{\int_{\mathbb{R}} |\widehat{u}(s)|^2\,\mathrm{d}s} \tag{46.7}$$

has odd moments zero and even moments $2^{r(r+1)}$. It has the same polynomial moments as Se^L, where S is an independent uniform sign and

$$L \sim N\left(\frac{\log 2}{2}, \frac{\log 2}{2}\right). \tag{46.8}$$

An explicit one-parameter family of purely atomic measures has these moments as well. The identification of this spectral measure, rather than the classical indeterminacy of lognormal moments itself, is the point of Chapter 55.

Chapter 47

From binary cancellation to compactly supported splines

The finite product (1.6) and the Prouhet cancellation (1.10) are the algebraic source of all compact-support cancellations used below. The variables V_j, Y_m, Y, $h = 2^{-m}$, the densities u_m, u, and the transform $\widehat{u}$ are those of Chapter 1, and $a_j = \mathbb{E}Y^{2j}/(2j)!$, $K_r = 2^{r(r+1)/2}$ as in (46.2).

47.1 The continuous convolution model

The jth term $2^{-j}V_j$ has density

$$f_j(x) = 2^{j-1}\mathbf{1}_{[-2^{-j},2^{-j}]}(x). \tag{47.1}$$

Therefore $u_m = f_1 * \cdots * f_m$. Its support is $[-1+h, 1-h]$, and for $m \geq 2$ it is a C^{m-2} piecewise polynomial of degree at most $m-1$.

The infinite random series in (46.1) converges absolutely for every outcome. The distributional tail is an exact scaled copy of the entire series:

$$Y \overset{\mathrm{d}}{=} Y_m + hY', \qquad Y' \text{ independent of } Y_m, \quad \mathcal{L}(Y') = \mathcal{L}(Y). \tag{47.2}$$

Its characteristic function is

$$\widehat{u}(t) = \prod_{j=1}^{\infty} \operatorname{sinc}(2^{-j}t), \qquad \operatorname{sinc} t = \begin{cases} \sin t/t, & t \neq 0, \\ 1, & t = 0. \end{cases} \tag{47.3}$$

Our Fourier convention is

$$\widehat{u}(t) = \int_{\mathbb{R}} e^{\mathrm{i}tx} u(x)\,\mathrm{d}x, \qquad u(x) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-\mathrm{i}tx}\widehat{u}(t)\,\mathrm{d}t. \tag{47.4}$$

No factor of 2π is hidden inside the definition of sinc.

Lemma 47.1 (Smoothness and a useful Fourier bound)**.** *For every integer $L \geq 1$,*

$$|\widehat{u}(t)| \leq \min\{1, K_L|t|^{-L}\}, \qquad K_L = 2^{L(L+1)/2}. \tag{47.5}$$

The law of Y has an even density $u \in C_c^{\infty}(\mathbb{R})$, supported on $[-1,1]$, with $0 \leq u \leq 1$ and $u(0) = 1$.

Proof. For the first L factors, use $|\operatorname{sinc}(2^{-j}t)| \leq 2^j/|t|$; all remaining factors have modulus at most one. This proves (47.5). Consequently $|t|^r Q(t)$ is integrable for every r, by choosing $L > r+1$. Fourier inversion yields a smooth density. Compact support and evenness follow from the random series. Finally write $Y = V_1/2 + R$, where R is independent and supported on $[-1/2, 1/2]$. Convolution with the unit-height density of $V_1/2$ proves $u \leq 1$ and $u(0) = 1$. The same reasoning shows $\|u_m\|_{\infty} = 1$. $\square$

47.2 A finite signed formula

For $y \in \mathbb{R}$ write $y_+ = \max\{y, 0\}$.

Proposition 47.2 (Thue–Morse positive-part formula)**.** *For $m \geq 2$,*

$$u_m(x) = \frac{2^{m(m-1)/2}}{(m-1)!} \sum_{n=0}^{2^m-1} \varepsilon_n (x + 1 - h - 2hn)_+^{m-1}. \tag{47.6}$$

In particular the knots are among $-1 + h + 2hn$. For $0 \leq r \leq m-2$, differentiation gives

$$u_m^{(r)}(x) = \frac{2^{m(m-1)/2}}{(m-1-r)!} \sum_{n=0}^{2^m-1} \varepsilon_n (x + 1 - h - 2hn)_+^{m-1-r}. \tag{47.7}$$

The analogous formula for $r = m-1$ holds on open cells.

Proof. The distributional derivative of f_j is $2^{j-1}(\delta_{-2^{-j}} - \delta_{2^{-j}})$. Hence

$$u_m^{(m)} = 2^{m(m-1)/2} \sum_{n=0}^{2^m-1} \varepsilon_n \delta_{-1+h+2hn}. \tag{47.8}$$

To see the locations and signs, select an upper endpoint in any subset of the m factors. The lengths 2^{1-j}, when put in reverse order, are $2h, 4h, \dots, 2^m h$. Their subset sums give $2hn$ and their signs are ε_n. Integrating m times from $-\infty$ proves (47.6). The polynomial terms to the right of the support cancel by (1.10). The derivative formula follows. □

The signed formula is excellent for exact rational computation, but its large cancelling summands make naive floating-point evaluation unreliable. A stability statement for extrapolation is not a stability statement for this inner signed sum.

Chapter 48

The derivative-copy geometry

The key analytic fact is that, after differentiation, the rescaled copies have disjoint interiors. Thus differentiating a fixed number of times does *not* introduce a factor growing with m.

Proposition 48.1 (Exact derivative copies and norm identities)**.** *For integers $r \geq 0$ and $m \geq r+2$,*

$$u_m^{(r)}(x) = K_r \sum_{n=0}^{2^r-1} \varepsilon_n u_{m-r}(2^r x + 2^r - 1 - 2n), \tag{48.1}$$

$$u^{(r)}(x) = K_r \sum_{n=0}^{2^r-1} \varepsilon_n u(2^r x + 2^r - 1 - 2n). \tag{48.2}$$

For every $1 \leq p \leq \infty$,

$$\left\| u_m^{(r)} \right\|_p = K_r \left\| u_{m-r} \right\|_p, \qquad \left\| u^{(r)} \right\|_p = K_r \left\| u \right\|_p, \tag{48.3}$$

$$\left\| u_m^{(r)} \right\|_\infty = K_r, \qquad \left\| u^{(r)} \right\|_\infty = K_r. \tag{48.4}$$

Proof. Differentiating the first box and rescaling the tail gives

$$u_m'(x) = 2\{u_{m-1}(2x+1) - u_{m-1}(2x-1)\}. \tag{48.5}$$

The same argument applies to u. On iterating, the accumulated factor is $2^{1+2+\cdots+r} = K_r$. At every step, endpoint choices contribute binary digit signs, proving the two copy identities.

The nth infinite copy is supported on

$$\left[-1 + \frac{2n}{2^r},\ -1 + \frac{2n+2}{2^r}\right]. \tag{48.6}$$

These intervals meet only at their endpoints. Finite copies have still smaller supports. Thus there is no cancellation in the integral of the pth power of the absolute value. A change of variable contributes 2^{-r} for each of the 2^r copies, giving (48.3). The supremum statement uses $\|u_j\|_\infty = \|u\|_\infty = 1$. $\square$

Remark 48.2 (What is classical, and what is used here)*.* Derivative self-similarity is an established part of the Fabius theory [9, 10]. We emphasize the exact norm consequence and its uniformity in the finite level. That simple observation is the mechanism behind the global remainder theorem and the spectral-moment calculation below.

Corollary 48.3 (Vanishing high derivatives at a dyadic point)**.** *If $x = a/2^d \in (-1,1)$ and $r > d$, then $u^{(r)}(x) = 0$.*

Proof. In (48.2), $2^r x$ is an even integer and $2^r - 1 - 2n$ is odd. Every argument is therefore an odd integer. The density u is zero at and outside ± 1. $\square$

This does not say that u agrees with its Taylor polynomial on a neighborhood of x. Vanishing higher derivatives at one point and local polynomiality are very different assertions for a C^∞ function.

Chapter 49

Explicit coefficient calculus

The moment generating function

$$M(z) = \mathbb{E}e^{zY} = \prod_{j=1}^{\infty} \frac{\sinh(2^{-j}z)}{2^{-j}z} = \sum_{j=0}^{\infty} a_j z^{2j} \tag{49.1}$$

is entire. Its reciprocal is analytic near the origin. We shall need both its coefficients and those of two closely related functions.

Define

$$\lambda_r = \frac{\zeta(2r)}{r\pi^{2r}}, \qquad r \geq 1. \tag{49.2}$$

The sine product [18, Sec. 4.22], or its logarithmic derivative followed by integration, gives

$$-\log \operatorname{sinc} z = \sum_{r\geq 1} \lambda_r z^{2r}, \tag{49.3}$$

$$\log M(z) = \sum_{r\geq 1} \frac{(-1)^{r+1}\lambda_r}{4^r - 1} z^{2r}. \tag{49.4}$$

The first coefficients are $\lambda_1 = 1/6$, $\lambda_2 = 1/180$, and $\lambda_3 = 1/2835$. Euler's even-zeta formula makes every λ_r rational.

Proposition 49.1 (The four coefficient families)**.** *Let*

$$\frac{1}{M(z)} = \sum_{j\geq 0} b_j z^{2j}, \tag{49.5}$$

$$\left(\frac{z}{\sinh z}\right)^m = \sum_{j\geq 0} \gamma_j(m) z^{2j}, \tag{49.6}$$

$$\exp\left(\sum_{r\geq 1} \lambda_r \left(m + \frac{1}{4^r - 1}\right) z^r\right) = \sum_{j\geq 0} H_j(m) z^j. \tag{49.7}$$

Then

$$a_j = \mathcal{E}_j\left(\frac{(-1)^{r+1}\lambda_r}{4^r - 1}\right), \tag{49.8}$$

$$b_j = (-1)^j \mathcal{E}_j\left(\frac{\lambda_r}{4^r - 1}\right), \tag{49.9}$$

$$\gamma_j(m) = (-1)^j \mathcal{E}_j(m\lambda_r), \tag{49.10}$$

$$H_j(m) = \mathcal{E}_j\left(\lambda_r\left(m + \frac{1}{4^r - 1}\right)\right). \tag{49.11}$$

For each j, $H_j(m)$ is a rational polynomial of degree j with positive coefficients and leading term $m^j/(6^j j!)$. Moreover,

$$\sum_{\ell=0}^{j} \gamma_\ell(m) b_{j-\ell} = (-1)^j H_j(m). \tag{49.12}$$

Proof. Expand the exponentials in (49.4)–(49.7). For the alternating families, the sign of each partition monomial is $(-1)^{\sum rk_r} = (-1)^j$. The only degree-$j$ term in m comes from $k_1 = j$. Multiplication of the two reciprocal generating functions gives (49.12). □

For reference, the first a_j and b_j are

j	a_j	b_j
0	1	1
1	$1/18$	$-1/18$
2	$19/16200$	$31/16200$
3	$583/42865200$	$-2347/42865200$
4	$132809/1311675120000$	$1904369/1311675120000$

The first lattice correction polynomials are

$$\begin{aligned}
H_0(m) &= 1, \\
H_1(m) &= \frac{m}{6} + \frac{1}{18}, \\
H_2(m) &= \frac{(m+1/3)^2}{72} + \frac{m+1/15}{180}, \\
H_3(m) &= \frac{(m+1/3)^3}{1296} + \frac{(m+1/3)(m+1/15)}{1080} + \frac{m+1/63}{2835}.
\end{aligned} \tag{49.13}$$

49.1 An optional rational recurrence for computation

Writing $\mu_{2j} = \mathbb{E}Y^{2j}$, one may alternatively use

$$(2j+1)(4^j - 1)\mu_{2j} = \sum_{\ell=0}^{j-1} \binom{2j+1}{2\ell} \mu_{2\ell}, \qquad \mu_0 = 1. \tag{49.14}$$

For completeness, $M(2z) = M(z)\sinh z/z$. Multiplying by z, comparing the coefficient of z^{2j+1}, and moving the last term to the left proves (49.14). This recurrence is convenient for exact arithmetic, but (49.8) is a nonrecursive alternative.

Chapter 50

Global spline corrections, sharpness, and increasing order

Define the corrected spline

$$C_{m,N}(x) = \sum_{j=0}^{N} a_j h^{2j} u_m^{(2j)}(x). \tag{50.1}$$

Theorem 50.1 (Global all-orders correction)**.** *Let $k, N \geq 0$ be integers, put $r = k + 2N + 2$, and suppose $m \geq r + 2$. Then*

$$\boxed{\left\| \mathrm{D}^k(u - C_{m,N}) \right\|_\infty \leq a_{N+1} K_r h^{2N+2}.} \tag{50.2}$$

The same right-hand side bounds $\|u - C_{m,N}\|_{C^k} := \max_{0\leq \ell \leq k} \left\| \mathrm{D}^\ell(u - C_{m,N}) \right\|_\infty$. The estimate is global on $\mathbb{R}$, including all spline knots and the endpoints of the supports.

Proof. The exact tail identity (47.2) gives

$$u^{(k)}(x) = \mathbb{E}\, u_m^{(k)}(x - hY). \tag{50.3}$$

Set $g = u_m^{(k)}$. By the hypothesis on m, g has $2N + 2$ continuous derivatives. Taylor's formula with the integral remainder implies, for real y,

$$\left| g(x - hy) - \sum_{\ell=0}^{2N+1} \frac{(-hy)^\ell}{\ell!} g^{(\ell)}(x) \right| \leq \frac{h^{2N+2}|y|^{2N+2}}{(2N+2)!} \left\| g^{(2N+2)} \right\|_\infty . \tag{50.4}$$

Average over Y. All odd powers disappear because Y is symmetric. The remaining finite sum is precisely $\mathrm{D}^k C_{m,N}$, and the remainder is bounded by $a_{N+1} h^{2N+2} \left\| u_m^{(r)} \right\|_\infty$. Now apply (48.4). For lower derivatives use $K_{\ell+2N+2} \leq K_r$. □

Corollary 50.2 (Global Hölder control)**.** *Let $0 < \alpha \leq 1$, $r = k + 2N + 2$, and $m \geq r + 3$. With $[f]_\alpha = \sup_{x\neq y} |f(x) - f(y)|/|x - y|^\alpha$,*

$$[\mathrm{D}^k(u - C_{m,N})]_\alpha \leq 2^{1+\alpha r} a_{N+1} K_r h^{2N+2}. \tag{50.5}$$

Thus a global fixed-order Hölder version is available as well.

Proof. For a bounded differentiable function, the two estimates $|f(x) - f(y)| \leq 2 \|f\|_\infty$ and $|f(x) - f(y)| \leq \|f'\|_\infty |x - y|$ imply $[f]_\alpha \leq (2 \|f\|_\infty)^{1-\alpha} \|f'\|_\infty^\alpha$. Apply Theorem 50.1 at derivative orders k and $k + 1$, and use $K_{r+1} = 2^{r+1} K_r$. □

Why the knots are harmless here. The differentiation order is fixed before the level m tends to infinity. The spline degree and its classical smoothness grow with m, while the norm of each admissible fixed derivative is exactly constant in m. Thus there is no knot singularity in the derivatives used in the Taylor remainder. A different argument would be required for arbitrary derivative orders beyond the stated regularity threshold.

Theorem 50.3 (Uniform leading remainder and sharp norm constant)**.** *For fixed* $k, N \geq 0$,

$$h^{-2N-2}\mathrm{D}^k(u - C_{m,N}) \longrightarrow a_{N+1}u^{(k+2N+2)} \quad \textit{uniformly on } \mathbb{R}. \tag{50.6}$$

Consequently

$$\lim_{m\to\infty} 4^{(N+1)m} \left\| \mathrm{D}^k(u - C_{m,N}) \right\|_\infty = a_{N+1}K_{k+2N+2}. \tag{50.7}$$

Proof. Apply Theorem 50.1 with order $N+1$ to obtain

$$\mathrm{D}^k(u - C_{m,N}) = a_{N+1}h^{2N+2}u_m^{(k+2N+2)} + O_{k,N}(h^{2N+4}) \tag{50.8}$$

in the supremum norm. The same theorem at correction order zero, with derivative order $k+2N+2$, proves convergence of the displayed spline derivative to $u^{(k+2N+2)}$. Uniform convergence implies convergence of the supremum norms, and Theorem 48.1 evaluates the limiting norm. □

For the first two un-differentiated cases,

$$\lim_{m\to\infty} 4^m \left\| u - u_m \right\|_\infty = \frac{4}{9}, \tag{50.9}$$

$$\lim_{m\to\infty} 16^m \left\| u - u_m - \frac{4^{-m}}{18}u_m'' \right\|_\infty = \frac{2432}{2025}. \tag{50.10}$$

These constants are not estimates fitted to numerical data. They are forced by the exact derivative norms.

50.1 A certified increasing-order construction

The next result allows the correction order to grow with the spline level. It is an absolute error statement, not a relative endpoint asymptotic.

Corollary 50.4 (Supergeometric absolute accuracy)**.** *Fix* $k \geq 0$*, let* $m \geq k+4$*, and choose* r *to be the largest integer with* $r \leq m-2$ *and* $r \equiv k \pmod 2$*. Set* $N = (r-k-2)/2$*. Then*

$$\left\| \mathrm{D}^k(u - C_{m,N}) \right\|_\infty \leq \frac{2^{-m^2/2+(k+1/2)m+3}}{(m-k-3)!}. \tag{50.11}$$

Proof. Since $|Y| \leq 1$, $a_{N+1} \leq 1/(r-k)!$. The bound in Theorem 50.1 becomes

$$\frac{2^{r(r+1)/2-(r-k)m}}{(r-k)!}. \tag{50.12}$$

Write $r = m - \delta$, where $\delta \in \{2, 3\}$. The exponent equals $-m^2/2 + (k+1/2)m + \delta(\delta-1)/2$, and $(r-k)! \geq (m-k-3)!$. This proves the claim. □

The large positive and negative terms in an actual evaluation still need controlled arithmetic. The error certificate establishes the quality of the exact corrected expression; it does not provide a complete floating-point or bit-complexity analysis.

Chapter 51

Exact stabilization on dyadic arguments

Fix $x = a/2^d \in (-1, 1)$. It is harmless to choose a nonminimal d; a smaller denominator exponent simply gives a sharper degree bound.

Lemma 51.1 (A low-degree cell around a dyadic point)**.** *For $m \geq d + 2$, the restriction of u_m to $[x - h, x + h]$ agrees with one polynomial of degree at most d.*

Proof. The knots have coordinates $-1 + h + 2hn$, which are odd multiples of h when $m \geq 1$. Since $2^m x$ is even, x is the center of an open cell whose endpoints are $x - h$ and $x + h$.

Apply the derivative-copy formula at order $d + 1$ on this open cell. It remains valid there in the borderline case as a piecewise derivative. Put $\ell = m - d - 1 \geq 1$. Each argument of u_ℓ has the form

$$\text{an odd integer} + 2^{d+1}(y - x), \qquad |y - x| < h. \tag{51.1}$$

Its distance from zero is larger than $1 - 2^{-\ell}$, unless it is farther away still. But u_ℓ is supported on $[-1 + 2^{-\ell}, 1 - 2^{-\ell}]$. Thus $u_m^{(d+1)} = 0$ on the cell. The spline is a polynomial there, so its degree is at most d. Continuity gives the endpoint equality as well. □

Theorem 51.2 (Exact finite tail calculus at dyadic points)**.** *For $x = a/2^d \in (-1, 1)$, $m \geq d + 2$, and $0 \leq r \leq d$,*

$$u^{(r)}(x) = \sum_{j=0}^{\lfloor (d-r)/2 \rfloor} a_j h^{2j} u_m^{(r+2j)}(x), \tag{51.2}$$

$$u_m^{(r)}(x) = \sum_{j=0}^{\lfloor (d-r)/2 \rfloor} b_j h^{2j} u^{(r+2j)}(x). \tag{51.3}$$

Both formulas are exact.

Proof. The tail formula (50.3) averages the derivative of u_m at points $x - hY \in [x - h, x + h]$. On this cell the derivative is a polynomial of degree at most $d - r$. Its Taylor formula has no remainder. Symmetry gives (51.2).

View the vector of derivatives of orders $r, r + 2, \ldots$ at x as a finite vector. The first formula is a triangular transformation with diagonal entries one. The power-series identity $M(z)/M(z) = 1$ shows that its inverse has coefficients b_j, which proves (51.3). This is inversion of a finite triangular system, not an assertion that an infinite differential operator converges on an arbitrary smooth function. □

Corollary 51.3 (Polynomial stabilization)**.** *For fixed $x = a/2^d$, there is a polynomial R_x of degree at most $p = \lfloor d/2 \rfloor$ such that*

$$u_m(x) = R_x(4^{-m}), \quad m \geq d + 2, \qquad R_x(0) = u(x). \tag{51.4}$$

Its coefficient of z^j is $b_j u^{(2j)}(x)$.

Corollary 51.4 (The degree is exact in lowest terms). *Suppose $d \geq 1$ and a is odd, so $a/2^d$ is in lowest terms. The polynomial R_x has degree exactly $p = \lfloor d/2 \rfloor$. More explicitly, its highest derivative factor is*

$$u^{(2p)}(a/2^d) = \begin{cases} K_d \varepsilon_{(a+2^d-1)/2}, & d \text{ even}, \\ \frac{1}{2} K_{d-1} \varepsilon_{\lfloor (a+2^d)/4 \rfloor}, & d \text{ odd}. \end{cases} \tag{51.5}$$

Proof. For even d, precisely one argument in the order-d copy formula is zero, and all the others are outside the support. For odd d, at order $d-1$ precisely one argument is $1/2$ or $-1/2$. The corresponding values are $u(0) = 1$ and $u(\pm 1/2) = 1/2$. The latter equality follows directly by convolving the first box with its symmetric tail. The indices in (51.5) identify these copies. Both values are nonzero. Finally $b_p \neq 0$ by (49.9), so the highest coefficient of R_x is nonzero. □

The crucial word is "polynomial." An ordinary all-orders asymptotic expansion, by itself, would not justify finite exact reconstruction. Here the geometry of the dyadic cell supplies the missing exactness.

51.1 Finite Richardson reconstruction

For $0 < q < 1$ and $p \geq 0$, let

$$(q;q)_j = \prod_{\ell=1}^{j} (1 - q^\ell), \qquad (q;q)_0 = 1, \tag{51.6}$$

and define

$$w_{p,j}(q) = \prod_{\substack{0 \leq \ell \leq p \\ \ell \neq j}} \frac{-q^\ell}{q^j - q^\ell} = \frac{(-1)^{p-j} q^{(p-j)(p-j+1)/2}}{(q;q)_j (q;q)_{p-j}}. \tag{51.7}$$

These are simply the Lagrange interpolation weights for evaluation at zero from the nodes $1, q, \ldots, q^p$.

Theorem 51.5 (Exact reconstruction and uniform stability). *Let $x = a/2^d \in (-1,1)$, $p = \lfloor d/2 \rfloor$, and $m \geq d+2$. Then*

$$\boxed{u(x) = \sum_{j=0}^{p} w_{p,j}(1/4)\, u_{m+j}(x).} \tag{51.8}$$

At every order,

$$\sum_{j=0}^{p} |w_{p,j}(1/4)| = \prod_{\ell=1}^{p} \frac{1 + 4^{-\ell}}{1 - 4^{-\ell}} < 2. \tag{51.9}$$

Thus absolute errors bounded by δ in each input spline value produce an absolute output error smaller than 2δ.

Proof. Apply Lagrange interpolation to the polynomial $z \mapsto R_x(4^{-m} z)$ from Theorem 51.3. This proves the equality.

Taking absolute values in (51.7), and using the finite q-binomial product identity, gives

$$\begin{aligned} \sum_{j=0}^{p} |w_{p,j}(q)| &= \frac{1}{(q;q)_p} \sum_{k=0}^{p} q^{k(k+1)/2} \frac{(q;q)_p}{(q;q)_k (q;q)_{p-k}} \\ &= \frac{\prod_{\ell=1}^{p} (1 + q^\ell)}{(q;q)_p}. \end{aligned} \tag{51.10}$$

The finite product identity follows by expanding $\prod_{\ell=1}^{p} (1 + tq^\ell)$ and comparing the coefficient of t^k; alternatively its coefficients satisfy the same two-term recurrence on adjoining the last factor.

For the strict bound, $p = 0, 1$ are immediate. At $q = 1/4$, the first two factors multiply to $17/9$. For the remaining factors use

$$\sum_{\ell=3}^{\infty} \log \frac{1+q^\ell}{1-q^\ell} \le \frac{2q^3}{(1-q)(1-q^6)} = \frac{512}{12285}. \tag{51.11}$$

Since $e^t \le (1-t)^{-1}$ for $0 \le t < 1$, the entire product is at most

$$\frac{17}{9}\frac{1}{1-512/12285} = \frac{208845}{105957} < 2. \tag{51.12}$$

Finally apply the triangle inequality to the input perturbations. □

The first three weight rows are

$$(1), \qquad \left(-\frac{1}{3}, \frac{4}{3}\right), \qquad \left(\frac{1}{45}, -\frac{4}{9}, \frac{64}{45}\right). \tag{51.13}$$

These familiar extrapolation weights are not new in isolation. The point is the exact stopping theorem that makes them recover physical-space dyadic values in finitely many levels.

51.2 Examples and a closed finite formula

Example 51.6. The following identities hold at every indicated sufficiently large level:

$$\begin{aligned} u_m(1/2) &= \frac{1}{2}, && m \ge 3, \\ u_m(1/4) &= \frac{67}{72} + \frac{4}{9}4^{-m}, && m \ge 4, \\ u_m(1/8) &= \frac{287}{288} + \frac{2}{9}4^{-m}, && m \ge 5, \\ u_m(1/16) &= \frac{2073457}{2073600} + \frac{5}{162}4^{-m} - \frac{3968}{2025}16^{-m}, && m \ge 6. \end{aligned} \tag{51.14}$$

For example, $u''(1/16) = -5/9$ and $u^{(4)}(1/16) = -1024$; substitution into (51.3) gives the last line. The stated thresholds are uniform sufficient thresholds, not claims of minimality for each example.

Combining (51.8) with (47.6) gives a closed finite expression for a rational dyadic value. Put $m = d + 2$ and $p = \lfloor d/2 \rfloor$:

$$\begin{aligned} u(x) = \sum_{j=0}^{p} & \frac{w_{p,j}(1/4)\, 2^{(m+j)(m+j-1)/2}}{(m+j-1)!} \\ & \times \sum_{n=0}^{2^{m+j}-1} \varepsilon_n \left(x + 1 - 2^{-(m+j)} - 2^{1-m-j} n\right)_+^{m+j-1}. \end{aligned} \tag{51.15}$$

Every quantity on the right is rational. This gives another proof of classical dyadic rationality, and more importantly exhibits a finite extrapolation mechanism with a uniform condition bound. The formula is not advertised as the fastest existing algorithm.

Chapter 52

Repeated-prefix coefficients: an exact differential identity

52.1 The positive coefficient array and its natural normalization

Let $c_m(k)$ be the coefficient of z^k in $A_m(z)$ from (46.5), extended by zero outside its support. Put

$$D_m = 2^m - m - 1, \qquad B_m = A_m(1) = 2^{m(m-1)/2}. \tag{52.1}$$

The support is $0 \le k \le D_m$. The coefficients are positive integers on this interval and are symmetric about $D_m/2$.

Repeated inclusive summation multiplies a generating function by $(1-z)^{-1}$. Therefore these are precisely the m-fold cumulative coefficients of the finite signed block, and

$$c_m(k) = \sum_{n=0}^{\min(k,2^m-1)} \varepsilon_n \binom{k-n+m-1}{m-1}, \qquad k \ge 0. \tag{52.2}$$

For $k < 2^m$ this agrees with the m-fold inclusive prefix of the infinite Thue–Morse sequence. Beyond that range, the finite block and infinite sequence should not be conflated.

Let $D_0^*, \dots, D_{m-1}^*$ be independent random variables with D_j^* uniform on $\{0, \dots, 2^j - 1\}$. Then

$$\mathbb{P}\left(\sum_{j=0}^{m-1} D_j^* = k\right) = \frac{c_m(k)}{B_m}. \tag{52.3}$$

Set

$$\Delta = 2h = 2^{1-m}, \qquad x_{m,k} = \Delta(k - D_m/2) = -1 + (m+1)h + 2hk, \tag{52.4}$$

and define the scaled lattice mass

$$\rho_m(x_{m,k}) = \frac{c_m(k)}{\Delta B_m}. \tag{52.5}$$

The division by Δ is essential: ρ_m is a density-scale quantity, not the probability of one lattice point.

52.2 A centered factorial polynomial identity

The next lemma is a useful form of the generalized Bernoulli polynomial calculus; the standard generating function is recorded in [18]. Its short proof is included to fix all normalizations.

Lemma 52.1 (Centered factorial conversion). *For $m \geq 2$ define*

$$p_{m-1}(w) = \prod_{r=1}^{m-1} (w + r - m/2). \tag{52.6}$$

As an identity of polynomials,

$$p_{m-1}(w) = \left(\frac{\mathrm{D}_w/2}{\sinh(\mathrm{D}_w/2)} \right)^m w^{m-1}. \tag{52.7}$$

The power series on the right acts finitely: every derivative of order greater than $m-1$ annihilates the polynomial.

Proof. By coefficient extraction and the formal change of variable $y = e^z - 1$,

$$\begin{aligned}
[z^{m-1}] \left(\frac{z}{2\sinh(z/2)} \right)^m e^{wz} &= \mathrm{Res}_{z=0} \frac{e^{(w+m/2)z}}{(e^z-1)^m} \,\mathrm{d}z \\
&= \mathrm{Res}_{y=0} \frac{(1+y)^{w+m/2-1}}{y^m} \,\mathrm{d}y \\
&= \binom{w+m/2-1}{m-1} = \frac{p_{m-1}(w)}{(m-1)!}.
\end{aligned} \tag{52.8}$$

The first coefficient equals $1/(m-1)!$ times the differential operator in (52.7) applied to w^{m-1}. The change of variable is valid formally because $e^z - 1 = z + O(z^2)$. ∎

Theorem 52.2 (Exact lattice-to-spline differential conversion). *For $m \geq 2$ and every $k \in \mathbb{Z}$,*

$$\boxed{\rho_m(x_{m,k}) = \sum_{j=0}^{\lfloor (m-1)/2 \rfloor} \gamma_j(m) h^{2j} u_m^{(2j)}(x_{m,k}).} \tag{52.9}$$

When m is odd the highest derivative is a cellwise derivative, but the lattice nodes are then cell centers. When m is even the highest derivative used is of order $m-2$ and is continuous. Thus every displayed evaluation is unambiguous.

Proof. At $x = x_{m,k}$ put

$$t = \frac{x+1-h}{2h} = k + \frac{m}{2}, \qquad J_m = 2^{-(m-1)(m-2)/2}. \tag{52.10}$$

The signed spline formula becomes

$$u_m(x) = \frac{J_m}{(m-1)!} \sum_{\substack{0 \leq n < 2^m \\ n < t}} \varepsilon_n (t-n)^{m-1}. \tag{52.11}$$

On the other hand, (52.2) and $1/(\Delta B_m) = J_m$ give

$$\rho_m(x) = \frac{J_m}{(m-1)!} \sum_{\substack{0 \leq n < 2^m \\ n \leq k}} \varepsilon_n p_{m-1}(t-n). \tag{52.12}$$

One can extend this last sum from $n \leq k$ to $n < t$: every new integer has $n = k + i$ with $1 \leq i < m/2$, and then $p_{m-1}(m/2 - i) = 0$. The same observation works if the original sum is empty or if the finite block truncates the range.

Now apply Theorem 52.1 term by term to (52.11). Since $\mathrm{D}_t = 2h\mathrm{D}_x$, its operator becomes $(h\mathrm{D}_x/\sinh(h\mathrm{D}_x))^m$, whose coefficients are $\gamma_j(m)h^{2j}$. This proves the formula. At an even-level knot, the derivatives involved have positive remaining polynomial degree; the limiting formula agrees from both sides. ∎

The first terms are

$$\rho_m(x) = u_m(x) - \frac{m}{6} h^2 u_m''(x) + \left(\frac{m^2}{72} + \frac{m}{180} \right) h^4 u_m^{(4)}(x) + \cdots , \tag{52.13}$$

where the complete expression is the finite sum in (52.9). This identity is stronger than a formal differential approximation.

Chapter 53

The all-orders uniform local-limit theorem

The exact operator in Theorem 52.2 is useful algebraically. To obtain a uniform fixed-order remainder as m grows, it is cleaner to work on the lattice Fourier interval.

53.1 Exact removal of the lattice

Proposition 53.1 (Continuous noise representation)**.** *Let*

$$X_m = \Delta\left(\sum_{j=0}^{m-1} D_j^* - D_m/2\right). \tag{53.1}$$

There is a random variable W_m, independent of X_m, such that

$$Y \overset{\mathrm{d}}{=} X_m + hW_m, \qquad W_m = V_1 + \cdots + V_m + Y', \tag{53.2}$$

where all summands on the right are independent, the V_j are uniform on $[-1,1]$, and Y' is a copy of Y. In particular

$$\operatorname{Var}(W_m) = \frac{m}{3} + \frac{1}{9}. \tag{53.3}$$

The characteristic function of X_m satisfies

$$\psi_m(t) = \frac{\widehat{u}(t)}{\widehat{u}(ht)\,\operatorname{sinc}(ht)^m} \qquad \left(|t| \le \frac{\pi}{2h}\right). \tag{53.4}$$

Proof. Adjoin independent U_j uniform on $[0,1]$ to D_j^*. Then $D_j^* + U_j$ is uniform on $[0, 2^j]$, and

$$\Delta\left(D_j^* + U_j - 2^{j-1}\right) = \Delta\left(D_j^* - \frac{2^j - 1}{2}\right) + h(2U_j - 1). \tag{53.5}$$

Summing over j produces X_m plus m independent uniform noises of half-width h. The continuous sum has the law of Y_m after reversing the order of its widths. Adjoin the tail hY'. The variance follows from $\mu_2 = 1/9$. Taking characteristic functions proves the quotient wherever its denominator is nonzero, in particular on the stated interval. $\square$

This calculation explains the leading factor m. Continuous spline truncation omits one self-similar tail of scale h, whereas converting a lattice sum to its continuous counterpart requires m additional small uniforms. Their combined variance is of order mh^2.

53.2 Uniform expansion and an explicit remainder constant

For use in the proof, write

$$\mathcal{H}_m(z) = \sum_{j\geq 0} H_j(m) z^j = \exp\left(\sum_{r\geq 1} \lambda_r \left(m + \frac{1}{4^r - 1} \right) z^r \right). \tag{53.6}$$

By (53.4), $\psi_m(t) = \widehat{u}(t)\mathcal{H}_m(h^2t^2)$ on the fundamental interval.

Theorem 53.2 (Uniform all-orders local-limit expansion)**.** *For each fixed integer $N \geq 0$, there exists an explicit constant C_N such that for every $m \geq 4$,*

$$\boxed{\sup_{k\in\mathbb{Z}} \left| \rho_m(x_{m,k}) - \sum_{j=0}^{N} (-1)^j H_j(m) h^{2j} u^{(2j)}(x_{m,k}) \right| \leq C_N (m+1)^{N+1} h^{2N+2}.} \tag{53.7}$$

One permissible, deliberately nonoptimized choice is obtained by putting $L = 4N+8$ and taking

$$C_N = \frac{2}{\pi} \left\{ \frac{1}{2N+3} + \frac{K_L}{2N+5} + \frac{K_L}{4N+7} + K_L \sum_{j=0}^{N} \frac{1}{4N+7-2j} \right\}. \tag{53.8}$$

Proof. Lattice Fourier inversion and our normalization give

$$\rho_m(x_{m,k}) = \frac{1}{2\pi} \int_{-\pi/(2h)}^{\pi/(2h)} e^{-\mathrm{i}tx_{m,k}} \psi_m(t)\,\mathrm{d}t. \tag{53.9}$$

The finite proposed approximation is the full-line inverse Fourier transform of $\widehat{u}(t)\sum_{j=0}^{N} H_j(m)h^{2j}t^{2j}$, because multiplication by t^{2j} corresponds to $(-1)^j \mathrm{D}^{2j}$. We estimate the difference of these transforms in L^1; the resulting bound is independent of k.

Low frequencies. Put $T = h^{-1/2}$ and $R = (m+1)^{-1}$. Since $\lambda_r \leq 2/(r\pi^{2r})$, positivity gives

$$\log \mathcal{H}_m(R) \leq \frac{2}{\pi^2 - R} \leq \frac{2}{\pi^2 - 1} < \log 2. \tag{53.10}$$

Hence

$$\mathcal{H}_m(R) < 2, \qquad H_j(m) \leq 2(m+1)^j. \tag{53.11}$$

For $|t| \leq T$, put $z = h^2t^2 \leq h$. Since $m \geq 4$, $z/R \leq (m+1)h \leq 5/16 < 1$. Positivity of all coefficients implies

$$\left| \mathcal{H}_m(z) - \sum_{j=0}^{N} H_j(m) z^j \right| \leq 2\big((m+1)z\big)^{N+1}. \tag{53.12}$$

Therefore the low-frequency error before division by 2π is at most

$$2(m+1)^{N+1} h^{2N+2} \int_{\mathbb{R}} |t|^{2N+2} |\widehat{u}(t)|\,\mathrm{d}t. \tag{53.13}$$

Using Theorem 47.1 with $L = 4N+8$ bounds the integral by

$$2\left(\frac{1}{2N+3} + \frac{K_L}{2N+5} \right). \tag{53.14}$$

High frequencies of the exact lattice transform. For $|s| \leq \pi/2$, $\operatorname{sinc} s \geq 2/\pi$. Also, multiplying the bounds $\operatorname{sinc}(2^{-j}s) \geq 1 - 2^{-2j}s^2/6$ gives

$$\widehat{u}(s) \geq 1 - \frac{s^2}{18} \geq 1 - \frac{\pi^2}{72} > \frac{1}{2}. \tag{53.15}$$

Thus on the fundamental interval,

$$|\psi_m(t)| \le 2(\pi/2)^m |\widehat{u}(t)|. \tag{53.16}$$

Write $\beta = \log_2(\pi/2) < 1$. Integration over $|t| > T$ yields

$$\int_{T<|t|\le\pi/(2h)} |\psi_m(t)|\,\mathrm{d}t \le \frac{4K_L}{L-1} h^{(L-1)/2-\beta} \le \frac{4K_L}{4N+7} h^{2N+2}. \tag{53.17}$$

High frequencies of the finite approximation. By (53.11) and Theorem 47.1,

$$\begin{aligned}
&\int_{|t|>T} |\widehat{u}(t)| \sum_{j=0}^{N} H_j(m) h^{2j} |t|^{2j}\,\mathrm{d}t \\
&\le 4K_L \sum_{j=0}^{N} \frac{(m+1)^j h^{(L-1)/2+j}}{L-2j-1} \\
&\le 4K_L (m+1)^{N+1} h^{2N+2} \sum_{j=0}^{N} \frac{1}{4N+7-2j}.
\end{aligned} \tag{53.18}$$

Add (53.13)–(53.18), divide by 2π, and use $(m+1)^{N+1} \ge 1$ in the exact-transform tail. The result is precisely (53.7) with (53.8). □

Remark 53.3 (Meaning of the local limit). The supremum in (53.7) includes nodes outside the finite support, where $c_m(k) = 0$. This is a uniform comparison of scaled point masses with a density evaluated at the same nodes. It is not convergence in total variation of a discrete measure to an absolutely continuous one.

Corollary 53.4 (Sharp first local-limit error)**.** *Uniformly for $k \in \mathbb{Z}$,*

$$\frac{\rho_m(x_{m,k}) - u(x_{m,k})}{m4^{-m}} + \frac{1}{6} u''(x_{m,k}) \longrightarrow 0. \tag{53.19}$$

In particular,

$$\lim_{m\to\infty} \frac{\sup_{k\in\mathbb{Z}} |\rho_m(x_{m,k}) - u(x_{m,k})|}{m4^{-m}} = \frac{4}{3}. \tag{53.20}$$

Proof. Use Theorem 53.2 with $N = 1$ and $H_1(m)/m \to 1/6$. Its remainder divided by $m4^{-m}$ tends to zero. The node spacing tends to zero, so the sampled supremum of the continuous compactly supported function $|u''|$ tends to $\|u''\|_\infty = 8$ by Theorem 48.1. □

Chapter 54

Exact dyadic stopping for the integer coefficient array

The local-limit expansion has an unexpectedly exact specialization. It is obtained from the finite operator, not by declaring an asymptotic series to terminate.

Theorem 54.1 (Exact dyadic lattice expansion). *Fix $x = a/2^d \in (-1, 1)$. If m is odd and $m \geq d + 2$, then*

$$k_m = 2^{m-1}(1 + x) - \frac{m+1}{2} \in \mathbb{Z}, \qquad x_{m,k_m} = x, \tag{54.1}$$

and

$$\boxed{\rho_m(x) = \sum_{j=0}^{\lfloor d/2 \rfloor} (-1)^j H_j(m) 4^{-jm} u^{(2j)}(x).} \tag{54.2}$$

The right-hand side is an exact finite sum of polynomials in m times powers of 4^{-m}.

Proof. The stated parity makes k_m an integer. On the cell about x, u_m has degree at most d by Theorem 51.1. Hence the finite differential conversion in Theorem 52.2 has no nonzero terms with $2j > d$ at this point. Substitute (51.3) into its remaining terms. The coefficient of $h^{2j} u^{(2j)}(x)$ is the convolution $\sum_{\ell=0}^{j} \gamma_\ell(m) b_{j-\ell}$, which is $(-1)^j H_j(m)$ by (49.12). □

Example 54.2 (A closed repeated-prefix identity). For every odd $m \geq 5$,

$$\rho_m(1/4) = \frac{67}{72} + \left(\frac{4m}{3} + \frac{4}{9}\right) 4^{-m}. \tag{54.3}$$

Equivalently, an explicitly indexed coefficient of the positive Thue–Morse quotient is

$$\boxed{c_m\left(5 \cdot 2^{m-3} - \frac{m+1}{2}\right) = 2^{m(m-1)/2-m+1} \left\{\frac{67}{72} + \left(\frac{4m}{3} + \frac{4}{9}\right) 4^{-m}\right\}.} \tag{54.4}$$

This also evaluates the corresponding m-fold inclusive prefix of the infinite sequence, since the index is below 2^m.

Proof. At $x = 1/4$, $u = 67/72$, $u'' = -8$, and all derivatives of order above two vanish. Apply Theorem 54.1 and then undo the density normalization. □

At $x = 1/16$ the exact identity has three terms:

$$\rho_m(1/16) = \frac{2073457}{2073600} + \frac{5}{9} H_1(m) 4^{-m} - 1024 H_2(m) 16^{-m}, \qquad m \geq 7 \text{ odd}. \tag{54.5}$$

The increasing polynomial factors $H_j(m)$ are therefore not merely artifacts of a Fourier proof; they occur in exact rational coefficient identities.

A useful distinction. For the spline values at a fixed dyadic point, the coefficients of 4^{-jm} are constants depending on that point. For the discrete array values, the coefficients are polynomials in m. The extra polynomials come from the m independent lattice-removal noises. This distinction prevents the incorrect use of the simple spline Richardson weights as an exact annihilator of all lattice errors.

Chapter 55

Fourier-energy moments and lognormal twins

We now use the L^2 version of the derivative-copy identity. This has a different flavor from either approximation theorem.

Definition 55.1 (The normalized energy measure)**.** Let

$$I_0 = \int_{\mathbb{R}} |\widehat{u}(t)|^2 \, \mathrm{d}t = 2\pi \, \|u\|_2^2 , \qquad \mathrm{d}\nu(t) = I_0^{-1} |\widehat{u}(t)|^2 \, \mathrm{d}t. \tag{55.1}$$

This is an absolutely continuous probability measure on the angular frequency line. It is not the singular diffraction measure of the unscaled Thue–Morse substitution system.

Theorem 55.2 (Exact energy moments)**.** *Every moment of ν exists, and for $r \geq 0$,*

$$\int_{\mathbb{R}} t^{2r} \, \mathrm{d}\nu(t) = 2^{r(r+1)}, \qquad \int_{\mathbb{R}} t^{2r+1} \, \mathrm{d}\nu(t) = 0. \tag{55.2}$$

Proof. The decay estimate (47.5) gives existence of every moment. Evenness gives the odd moments. Plancherel's identity and (48.3) at $p = 2$ give

$$\frac{\int t^{2r} |\widehat{u}(t)|^2 \, \mathrm{d}t}{\int |\widehat{u}(t)|^2 \, \mathrm{d}t} = \frac{\left\| u^{(r)} \right\|_2^2}{\|u\|_2^2} = K_r^2 = 2^{r(r+1)}. \tag{55.3}$$

$\square$

55.1 An explicit absolutely continuous twin

Theorem 55.3 (Lognormal moment twin)**.** *Let S take values $-1, 1$ with equal probabilities and let $L \sim N((\log 2)/2, (\log 2)/2)$ be independent of S. Then the law of $T = Se^L$ has the same moments as ν, but it is a different probability measure. Its density is*

$$g(t) = \frac{1}{2|t|\sqrt{\pi \log 2}} \exp\left(-\frac{(\log|t| - (\log 2)/2)^2}{\log 2} \right), \quad t \neq 0, \qquad g(0) = 0. \tag{55.4}$$

Consequently the Hamburger moment problem for (55.2) *is indeterminate.*

Proof. For a normal random variable of mean μ and variance σ^2, completion of the square in its density gives $\mathbb{E}e^{sL} = e^{s\mu + s^2\sigma^2/2}$. At $s = 2r$ and the stated parameters, this is $2^{r(r+1)}$. The independent sign kills all odd moments. Change of variable $t = e^L$, followed by symmetrization, gives (55.4).

The density g extends continuously to zero with value zero, whereas the density of ν has value $1/I_0 > 0$ there. They therefore differ on a neighborhood of zero, not merely at a single point. Thus two distinct measures have the same complete moment sequence. $\square$

Lognormal moment indeterminacy is classical [24]. The contribution here is the exact identification of a compactly supported smooth function's Fourier-energy measure with that moment class. No inference from the failure of a sufficient uniqueness criterion is needed: two distinct measures have been constructed explicitly.

55.2 A continuum of atomic twins

For $\alpha \in [0, 1)$ set

$$\Theta(\alpha) = \sum_{k\in\mathbb{Z}} 2^{-(k+\alpha)^2}. \tag{55.5}$$

It is finite and strictly positive.

Theorem 55.4 (Theta-supported moment twins)**.** *Every measure*

$$\nu_\alpha = \frac{1}{2\Theta(\alpha)} \sum_{k\in\mathbb{Z}} 2^{-(k+\alpha)^2} \left(\delta_{2^{k+\alpha+1/2}} + \delta_{-2^{k+\alpha+1/2}}\right) \tag{55.6}$$

is a probability measure with the moment sequence (55.2).

Proof. The total mass is one and the odd moments vanish. For the even moments, complete the square in the exponent:

$$\begin{aligned}\int t^{2r}\,\mathrm{d}\nu_\alpha(t) &= \frac{1}{\Theta(\alpha)} \sum_{k\in\mathbb{Z}} 2^{-(k+\alpha)^2+2r(k+\alpha+1/2)} \\ &= \frac{2^{r^2+r}}{\Theta(\alpha)} \sum_{k\in\mathbb{Z}} 2^{-(k+\alpha-r)^2} = 2^{r(r+1)}. \end{aligned}\tag{55.7}$$

The last equality shifts the integer summation index by r. All sums converge absolutely. ☐

The theta construction is a standard type of discrete lognormal moment mechanism; the displayed proof is included rather than claiming that mechanism itself as a new discovery. Its combination with the energy identity gives the exact polynomial quadrature

$$\frac{1}{I_0}\int_{\mathbb{R}} p(t)|\widehat{u}(t)|^2\,\mathrm{d}t = \int_{\mathbb{R}} p(t)g(t)\,\mathrm{d}t = \int_{\mathbb{R}} p(t)\,\mathrm{d}\nu_\alpha(t) \quad \text{for every polynomial } p. \tag{55.8}$$

This equality cannot be extended to all bounded continuous test functions, since the measures are distinct.

55.3 Identical jets, and what restores uniqueness

Define the normalized autocorrelation

$$R_u(x) = \frac{\int_{\mathbb{R}} u(y)u(y+x)\,\mathrm{d}y}{\|u\|_2^2}. \tag{55.9}$$

It is a C^∞ function supported on $[-2, 2]$ and is the characteristic function of ν. The energy moments give

$$R_u^{(2r)}(0) = (-1)^r 2^{r(r+1)}, \qquad R_u^{(2r+1)}(0) = 0. \tag{55.10}$$

Its formal Taylor series at zero has radius of convergence zero, since $2^{r(r+1)}/(2r)!$ has unbounded $2r$th root. The characteristic functions of the lognormal and atomic twins have exactly the same derivatives at zero. Thus their differences from R_u are smooth and flat at zero, but not identically zero.

Moments alone lose information, but the original scaling relation retains it.

Proposition 55.5 (Refinement restores uniqueness). *Suppose a probability density f is continuous at zero and satisfies pointwise*

$$f(2t) = \operatorname{sinc}^2(t) f(t), \qquad t \in \mathbb{R}. \tag{55.11}$$

Then $f(t) = I_0^{-1} \widehat{u}(t)^2$.

Proof. Iterating toward zero gives

$$f(t) = f(2^{-n} t) \prod_{j=1}^{n} \operatorname{sinc}^2(2^{-j} t). \tag{55.12}$$

Continuity at zero and convergence of the product imply $f(t) = f(0)\widehat{u}(t)^2$. Integration fixes $f(0) = 1/I_0$. □

Chapter 56

A separated geometric extension

The moment phenomenon is not an accidental equality special to one choice of units. It follows from separation of derivative copies. Here is a one-parameter extension that makes this explicit.

For $0 < \theta \le 1/2$, let v_θ be the density of

$$Y_\theta = (1-\theta)\sum_{j=0}^{\infty} \theta^j V_j. \tag{56.1}$$

It is supported on $[-1, 1]$ and is smooth by the same Fourier argument as before. At $\theta = 1/2$ it equals u.

Theorem 56.1 (Separated-copy norms and their moment class)**.** *For $r \ge 0$ and $1 \le p \le \infty$,*

$$\left\| v_\theta^{(r)} \right\|_p = (2\theta)^{r/p}[2(1-\theta)]^{-r}\theta^{-r(r+1)/2} \left\| v_\theta \right\|_p , \tag{56.2}$$

where $(2\theta)^{r/p} = 1$ at $p = \infty$. If ν_θ is its normalized Fourier-energy measure, then

$$\int t^{2r}\, \mathrm{d}\nu_\theta(t) = [2(1-\theta)^2]^{-r}\theta^{-r^2}, \qquad \int t^{2r+1}\, \mathrm{d}\nu_\theta(t) = 0. \tag{56.3}$$

These are the moments of a symmetric lognormal variable with logarithmic mean and variance

$$\mu_\theta = -\frac{1}{2}\log\bigl(2(1-\theta)^2\bigr), \qquad \sigma_\theta^2 = -\frac{1}{2}\log\theta. \tag{56.4}$$

Proof. Differentiating the first uniform factor yields

$$v_\theta'(x) = \frac{1}{2(1-\theta)\theta}\left\{ v_\theta\left(\frac{x+1-\theta}{\theta}\right) - v_\theta\left(\frac{x-1+\theta}{\theta}\right)\right\}. \tag{56.5}$$

The two copies have disjoint interiors because $\theta \le 1/2$. At depth r there are 2^r disjoint copies, each of horizontal scale θ^r, with signs given by binary endpoint parity. Their common vertical multiplier is

$$[2(1-\theta)]^{-r}\theta^{-1-2-\cdots-r}. \tag{56.6}$$

Integrating the pth power gives (56.2). At $p = 2$, squaring the norm ratio simplifies to $[2(1-\theta)^2]^{-r}\theta^{-r^2}$. Plancherel proves the moment formula. The normal exponential-moment calculation gives (56.4). □

For a completely explicit atomic version, put $a = -\log\theta > 0$, $\mu = \mu_\theta$, and

$$\Theta_a(\alpha) = \sum_{k\in\mathbb{Z}} e^{-a(k+\alpha)^2}. \tag{56.7}$$

Then

$$\frac{1}{2\Theta_a(\alpha)} \sum_{k\in\mathbb{Z}} e^{-a(k+\alpha)^2} \left(\delta_{e^{\mu+a(k+\alpha)}} + \delta_{-e^{\mu+a(k+\alpha)}}\right) \tag{56.8}$$

has the same moments, by shifting k after completing the square. This generalization concerns separated copies. For $\theta > 1/2$ the copies overlap, the simple norm calculation no longer applies, and (56.3) is not asserted.

Chapter 57

Exact computations and reproducibility

The accompanying `verify.py` uses only the Python standard library. Algebraic tests use arbitrary-size integers and `fractions.Fraction`, not floating-point agreement. The saved `verification_results.json` records the actual run used in preparing this part.

57.1 What was checked

The script performed 3,796 categorized exact checks: 35 Prouhet moment identities, 135 derivative-copy identities, 2,307 dyadic polynomial identities, 127 exact Richardson reconstructions, 510 prefix-convolution identities, 15 extrapolation-norm identities, 501 lattice differential conversions, and 166 dyadic lattice identities. The tested ranges and loop bounds are visible in the script. Some checks compare formulas obtained from the same underlying theory; they are useful for catching sign, indexing, and normalization errors, not substitutes for the general proofs.

The implementation constructs the positive coefficient array by sliding-window convolution with all-ones vectors. At level j the vector length is $O(2^j)$, so all levels through m use $O(2^m)$ integer operations. This is an arithmetic-operation count, not a bit-complexity bound: the integers themselves grow.

57.2 A nondyadic corrected approximation

For $x = 1/3$, the reference value was computed from $C_{18,6}(x)$. The exact global certificate from Theorem 50.1 is less than 3.469×10^{-59}, and the reference starts

$$u(1/3) = 0.8198348851985181\ldots. \tag{57.1}$$

The rational reference and its rational error bound are included in the JSON output. The entries in Table 57.1 are rounded absolute differences from that reference; its certified uncertainty is negligible at the digits displayed.

m	$\lvert u - u_m\rvert$	$\lvert u - C_{m,1}\rvert$	$\lvert u - C_{m,2}\rvert$
6	$8.90547 \cdot 10^{-5}$	$5.96017 \cdot 10^{-8}$	–
8	$5.56024 \cdot 10^{-6}$	$2.29567 \cdot 10^{-10}$	$8.44219 \cdot 10^{-14}$
10	$3.47493 \cdot 10^{-7}$	$8.95576 \cdot 10^{-13}$	$2.03133 \cdot 10^{-17}$
12	$2.17182 \cdot 10^{-8}$	$3.49806 \cdot 10^{-15}$	$4.95223 \cdot 10^{-21}$

Table 57.1: Corrected spline errors at $x = 1/3$. The omitted entry has not been used because the uniform classical-regularity threshold for $N = 2$ is $m \geq 8$.

For example, at $m = 12$, $N = 2$, the actual displayed error is approximately 4.95223×10^{-21}, while the rigorous global bound is approximately 6.03996×10^{-21}. The same certificate works at every real x, not only at this sample point.

57.3 An exact discrete-array experiment

At $x = 1/4$ and odd levels the normalized error is exactly $4/3 + 4/(9m)$ by (54.3). The script computed the coefficient arrays independently by positive convolution and obtained Table 57.2. In every row the residual after subtracting the two terms of (54.3) is exactly zero.

m	$\rho_m(1/4)$	$\dfrac{\rho_m(1/4) - 67/72}{m4^{-m}}$
5	$15/16$	$64/45$
7	$1907/2048$	$88/63$
9	$15247/16384$	$112/81$
11	$487881/524288$	$136/99$
13	$1951517/2097152$	$160/117$
15	$124897055/134217728$	$184/135$

Table 57.2: Exact repeated-prefix coefficients after density normalization. The last column tends to $4/3$.

The script also makes finite numerical checks of the theta moment formulas at several values of α. Those checks are explicitly numerical; the completion-of-the-square proof, not truncation of a theta sum, establishes the exact identity for every moment.

Chapter 58

Consequences, limitations, and concrete next problems

The strongest conclusion for the motivating approximation question is that fixed-order strong spline corrections need no deleted knot neighborhoods. The sufficient condition is a regularity threshold on the spline level. This directly addresses the strong-spline conjecture in the supplied atlas [1]; it does not establish that document's separate relative-error endpoint saddle evaluator.

The exact dyadic results exhibit a second phenomenon. A function can be globally nonanalytic and yet admit finite exact extraction from an approximation sequence at every dyadic point. The source of this exactness is a low-degree *finite spline cell*, not analyticity of the limit. The highest nonzero derivative even has an explicit Thue–Morse sign in (51.5).

On the discrete side, the exact centered-factorial identity separates two sources of correction: $\gamma_j(m)$ accounts for lattice removal, and b_j accounts for the infinite continuous tail. Their convolution produces $H_j(m)$. This is an effective way to organize further coefficient identities without repeatedly rediscovering their signs and centering terms.

Finally, the spectral calculation shows that an entire family of energy moments can conceal major differences between probability measures. A continuous density positive at zero, a lognormal-type density flat at zero, and a purely atomic measure can share every polynomial moment. The refinement equation restores information that the moments alone discard.

Research problem 58.1 (Relative endpoint accuracy). Develop bounds for $C_{m,N}(x) - u(x)$ relative to $u(x)$ as x approaches an endpoint while m and N vary. The present global absolute bounds remain true there, but division by the very small value of $u(x)$ can make them ineffective. A proof will need local information beyond a global derivative norm.

Research problem 58.2 (Efficient exact extraction). Determine the best bit complexity of dyadic reconstruction from positive coefficient arrays, local spline cells, or the signed formula (51.15). The extrapolation condition bound is uniform, but the cost of producing the inputs and the cost of large integer cancellation must be analyzed separately.

Research problem 58.3 (Overlapping geometric copies). For $1/2 < \theta < 1$, compute or characterize the cross terms in $\left\|v_\theta^{(r)}\right\|_2^2$. The separated calculation in Theorem 56.1 identifies exactly where the argument changes: the derivative-copy intervals now overlap. It is natural to ask which parameters, if any, retain an explicit moment class.

58.1 A formalization route without a formalization claim

The main dependency chain is short. Finite convolution and binary endpoint enumeration yield the derivative-copy identity. Disjoint supports give its norm formula. A Taylor remainder bound and the probability-tail identity then yield the global theorem. The dyadic cell lemma and finite triangular inversion are algebraic once that bridge is available. Separately, the centered-factorial identity can be formalized in a polynomial ring, and the lattice local-limit theorem requires Fourier inversion with explicit integrable majorants. Plancherel plus disjoint supports proves the energy moments.

These are proposed proof components, not assertions that new Lean theorems have been compiled or added to the repository. The manuscript and the exact-arithmetic script are the deliverables of this investigation.

Part V

Nonlinear Prouhet geometry and flat automatic q-products

Chapter 59

Scope and the contribution map

The signed Thue–Morse sequence connects binary carries, cancellation of moments, and products across geometric scales. The source of this part developed three refinements of these connections. The first, a finite boundary functional giving the exact dyadic correlation defect at every order, is Chapter 14 of Part I. The second transports the signs to nonlinear Boolean-cube coordinates, and the third studies the Thue–Morse-weighted geometric product $\Phi_q(z) = \prod_{j\geq 0}(1 - zq^j)^{\varepsilon_j}$. The nonlinear moment section uses classical Boolean inclusion–exclusion and the analytic section a classical saddle-point method; new research lies in the exact objects these methods expose. Generalized Prouhet constructions are discussed by Bolker, Offner, Richman and Zara [14]; products weighted by Thue–Morse signs already form an established subject [11, 26], the distinction here being the geometric argument q^j and its singular limit; the harmonic-sum viewpoint of Flajolet, Gourdon and Dumas [21] provides analytic context, and Cloitre's transform [15] a different direction.

59.1 The three directions

Boundary defects. For an arbitrary finite tuple of shifts, a signed sum of at most as many terms as the largest shift determines all finite dyadic correlation corrections. For even order, the only remaining scale dependence is $(-1)^m$. For odd order, the sum over the q-th sufficiently large dyadic block is a constant multiple of $\varepsilon_{q+1} - \varepsilon_q$.

Nonlinear Prouhet geometry. The signs stay fixed, but the point with binary digits b is moved to a multilinear polynomial $x(b)$. The first surviving moment is governed by covers of the bit positions. Pair interactions produce perfect matchings and hafnians; weak pair interactions produce a full matching polynomial.

Flat automatic products. A Thue–Morse-weighted q-Pochhammer-type product tends to one faster than every power of $-\log q$. Its small deviation is nevertheless computable: a growing family of terms, centered at an explicit Lambert-W saddle, is modulated by an exact log-periodic function.

These three directions share a useful principle: the dyadic structure can often be isolated into a finite combinatorial quantity or a bounded periodic function. What remains can then be analyzed without repeatedly expanding the entire Thue–Morse word.

Chapter 60

Nonlinear Prouhet cancellation as a covering problem

60.1 Moving the points while keeping the signs

The usual Prouhet sum assigns the sign $(-1)^{b_0+\cdots+b_{m-1}}$ to the point $\sum_i 2^i b_i$. We retain the signs but allow nonlinear coordinates. Let $V = \{0, \dots, m-1\}$, let $\mathcal{H}$ be a finite family of nonempty subsets of V, and put

$$x(b) = c_0 + \sum_{S \in \mathcal{H}} c_S b_S, \qquad b_S = \prod_{i \in S} b_i, \qquad b \in \{0,1\}^m. \tag{60.1}$$

Terms with zero coefficient can be removed. Every function on the Boolean cube has a unique multilinear representation, but sparsity of $\mathcal{H}$ is what makes the following description informative.

Define the signed exponential generating function

$$Z_x(t) = \sum_{b \in \{0,1\}^m} (-1)^{|b|} \mathrm{e}^{tx(b)} = \sum_{r \geq 0} \frac{t^r}{r!} \sum_{b \in \{0,1\}^m} (-1)^{|b|} x(b)^r. \tag{60.2}$$

A subfamily $\mathcal{C} \subseteq \mathcal{H}$ is a *cover* if $\bigcup_{S \in \mathcal{C}} S = V$. Let $\tau(\mathcal{H})$ be the minimum number of members in a cover, and set $\tau = \infty$ if there is no cover.

Theorem 60.1 (Exact covering-family identity)**.** *For every complex t and all complex coefficients in* (60.1),

$$Z_x(t) = (-1)^m \mathrm{e}^{tc_0} \sum_{\substack{\mathcal{C} \subseteq \mathcal{H} \\ \bigcup_{S \in \mathcal{C}} S = V}} \prod_{S \in \mathcal{C}} (\mathrm{e}^{tc_S} - 1). \tag{60.3}$$

Proof. On the Boolean cube, b_S takes only the values zero and one, hence

$$\mathrm{e}^{tc_S b_S} = 1 + (\mathrm{e}^{tc_S} - 1) b_S.$$

Multiply these identities over $S \in \mathcal{H}$. Choosing the nonconstant term for the subfamily $\mathcal{C}$ contributes $\prod_{S \in \mathcal{C}} (\mathrm{e}^{tc_S} - 1) b_U$, where $U = \bigcup_{S \in \mathcal{C}} S$; repeated bit factors collapse because $b_i^2 = b_i$. The signed cube sum of b_U factors over coordinates. If any coordinate is missing from U, its factor is $1 - 1 = 0$. If $U = V$, only the all-one vertex contributes, and its sign is $(-1)^m$. Multiplying by e^{tc_0} proves (60.3). □

Corollary 60.2 (Cover number and the first surviving moment)**.** *If $\tau < \infty$, then*

$$\sum_b (-1)^{|b|} x(b)^r = 0 \quad (0 \leq r < \tau), \tag{60.4}$$

and

$$\sum_b (-1)^{|b|} x(b)^\tau = (-1)^m \tau! \sum_{\substack{\mathcal{C}\ \text{cover} \\ |\mathcal{C}|=\tau}} \prod_{S\in\mathcal{C}} c_S. \tag{60.5}$$

If all $c_S > 0$, this is nonzero, so the vanishing order of Z_x at zero is exactly τ. If there is no cover, then $Z_x \equiv 0$.

Proof. Every factor $\mathrm{e}^{tc_S} - 1$ begins with tc_S, so a cover with k members contributes first at order t^k. At the minimum order only its linear terms contribute. The constant-coordinate factor starts with one and does not affect that coefficient. For positive weights all minimum-cover contributions have the same sign. If there is no cover, the sum in (60.3) is empty. □

Remark 60.3 (A cancellation caveat). For negative or complex weights, the minimum-cover sum may vanish. Then τ is a lower bound on the vanishing order, not necessarily its exact value. This distinction matters when designing enhanced cancellation by tuning interaction coefficients.

60.2 All moments in a finite closed form

Extracting coefficients from (60.3) gives a formula that involves no auxiliary moment recurrence:

$$\sum_b (-1)^{|b|} x(b)^r = (-1)^m r! \sum_{\mathcal{C}\ \text{cover}} \sum_{\substack{k_0\ge 0,\ k_S\ge 1\ (S\in\mathcal{C}) \\ k_0+\sum_{S\in\mathcal{C}} k_S = r}} \frac{c_0^{k_0}}{k_0!} \prod_{S\in\mathcal{C}} \frac{c_S^{k_S}}{k_S!}. \tag{60.6}$$

The inner sum is finite for each r; covers with more than r members can be discarded. The proof is simply the absolutely convergent exponential series expansion of the finite expression (60.3).

Corollary 60.4 (A degree bound, with a sharper sparse refinement)**.** *If every edge $S \in \mathcal{H}$ has at most d vertices, then*

$$\sum_b (-1)^{|b|} x(b)^r = 0 \quad \textit{whenever} \quad r < \left\lceil \frac{m}{d} \right\rceil. \tag{60.7}$$

The exact cover number can be larger than this degree bound.

Proof. A union of r edges of size at most d contains at most rd vertices. It cannot cover V when $rd < m$. Apply Corollary 60.2. □

For the ordinary linear coordinate $x(b) = c_0 + \sum_i a_i b_i$, the only cover uses all singleton edges. Formula (60.3) becomes

$$Z_x(t) = (-1)^m \mathrm{e}^{tc_0} \prod_{i=0}^{m-1} (\mathrm{e}^{ta_i} - 1), \tag{60.8}$$

recovering classical Prouhet cancellation and the leading moment $(-1)^m m! \prod_i a_i$. This recovery is a consistency check, not a new version of the classical theorem.

60.3 Quadratic geometry and hafnians

Consider

$$x(b) = c_0 + \sum_{i\in V} a_i b_i + \sum_{i<j} J_{ij} b_i b_j, \tag{60.9}$$

where the symmetric matrix J has zero diagonal. For an even set of vertices U, define its hafnian by

$$\mathrm{Haf}(J_U) = \sum_{M\ \text{perfect matching of}\ U} \prod_{\{i,j\}\in M} J_{ij}, \qquad \mathrm{Haf}(J_\varnothing) = 1. \tag{60.10}$$

A perfect matching partitions the vertices into disjoint pairs. Zero entries of J simply remove the corresponding edges.

Theorem 60.5 (Quadratic leading moments). *For $m = 2r$, all moments of degree less than r vanish, and*

$$\sum_b (-1)^{|b|} x(b)^r = r! \operatorname{Haf}(J). \tag{60.11}$$

For $m = 2r + 1$, all moments of degree less than $r + 1$ vanish, and

$$\begin{aligned}\sum_b (-1)^{|b|} x(b)^{r+1} = -(r+1)! \Bigg[&\sum_i a_i \operatorname{Haf}(J_{V\setminus\{i\}}) \\ &+ \sum_{i<j<k} (J_{ij}J_{ik} + J_{ij}J_{jk} + J_{ik}J_{jk}) \operatorname{Haf}(J_{V\setminus\{i,j,k\}}) \Bigg].\end{aligned} \tag{60.12}$$

The second sum is empty when $m = 1$.

Proof. For $m = 2r$, a cover using r sets of size at most two must use only pairs, with no repeated vertices. These covers are precisely perfect matchings. The factor $(-1)^m$ is positive. This proves (60.11).

For $m = 2r + 1$, a cover with $r + 1$ members has only two possible shapes. It can use one singleton and r disjoint pairs, or it can use $r + 1$ pairs with exactly one repeated vertex. In the latter case, two pairs form a path on three vertices, while all remaining pairs form a perfect matching on the complement. A three-element vertex set has three possible two-edge paths, with weights displayed in (60.12). There can be no other shape: the maximum total incidence count is $2r + 2$, so covering $2r + 1$ vertices permits only one incidence in excess of a disjoint cover. The sign $(-1)^{2r+1}$ is negative. The degree bound and Corollary 60.2 complete the proof. □

For even m, the first potentially nonzero moment is completely independent of the linear coefficients and the translation c_0. The quadratic interaction graph alone controls it. This makes a hafnian an observable of a nonlinearly deformed Thue–Morse partition.

60.4 An explicit nearest-neighbor family

Set

$$x_\lambda(b) = \sum_{i=0}^{m-1} 2^i b_i + \lambda \sum_{i=0}^{m-2} b_i b_{i+1}, \qquad \lambda > 0. \tag{60.13}$$

The interaction graph is a path, and every allowed coefficient is positive.

Corollary 60.6 (Exact path moments). *If $m = 2r$, the first nonzero moment has degree r, and*

$$\sum_b (-1)^{|b|} x_\lambda(b)^r = r!\lambda^r. \tag{60.14}$$

If $m = 2r + 1$, the first nonzero moment has degree $r + 1$, and

$$\sum_b (-1)^{|b|} x_\lambda(b)^{r+1} = -(r+1)! \left[\lambda^r \frac{4^{r+1} - 1}{3} + r\lambda^{r+1} \right]. \tag{60.15}$$

Proof. An even path has a unique perfect matching, proving (60.14). For an odd path, removal of a singleton leaves two even paths precisely when the removed vertex has even index $2j$, $0 \le j \le r$. The corresponding linear coefficients sum to $\sum_{j=0}^r 2^{2j} = (4^{r+1} - 1)/3$, and the matching weight is λ^r.

A two-edge path that leaves an even path on both sides must start at an even index $0, 2, \ldots, 2r - 2$. There are r choices. Including the matching on the complement, each such cover has weight λ^{r+1}. Theorem 60.5 gives the formula. Positivity ensures that these leading moments do not vanish. □

At $\lambda = 0$, the coordinate is linear and has cancellation through degree $m - 1$. For every $\lambda > 0$, the first surviving degree falls to $\lceil m/2 \rceil$. Thus the *order* of cancellation is not stable under arbitrarily small quadratic perturbations, even though the newly surviving moments can have small coefficients.

For example, when $m = 4$,

$$\sum_{n=0}^{15} \varepsilon_n x_\lambda(n)^2 = 2\lambda^2,$$

where $x_\lambda(n)$ means (60.13) evaluated at the four binary digits of n. This is an exact polynomial identity, not merely a small-λ approximation.

Chapter 61

Weak interactions and a matching-polynomial crossover

The previous phenomenon has a useful scaling resolution. Let the quadratic coupling shrink at the same rate as the generating-function variable. Define

$$x_{\theta t}(b) = c_0 + \sum_i a_i b_i + \theta t \sum_{i<j} J_{ij} b_i b_j. \tag{61.1}$$

A singleton now contributes one power of t to the exponent, while an edge contributes two. This balances its ability to cover two bit positions.

Definition 61.1 (Weighted matching polynomial)**.** Let G be the graph of nonzero entries of J. Set

$$\mathcal{M}_G(\theta; \boldsymbol{a}, J) = \sum_{M \text{ matching in } G} \theta^{|M|} \prod_{\{i,j\} \in M} J_{ij} \prod_{i \text{ unmatched by } M} a_i. \tag{61.2}$$

Here a matching need not cover all vertices; unmatched vertices carry their singleton weights.

Theorem 61.2 (The weak-coupling crossover)**.** *For fixed coefficients and m, as $t \to 0$,*

$$\sum_{b \in \{0,1\}^m} (-1)^{|b|} \exp\bigl(t x_{\theta t}(b)\bigr) = (-1)^m t^m \mathcal{M}_G(\theta; \boldsymbol{a}, J) + O(t^{m+1}), \tag{61.3}$$

uniformly for θ in a compact subset of $\mathbb{C}$.

Proof. Apply the covering identity to singleton factors $\mathrm{e}^{ta_i} - 1 = ta_i + O(t^2)$ and pair factors $\mathrm{e}^{\theta t^2 J_{ij}} - 1 = \theta t^2 J_{ij} + O(t^4)$. If a cover has s singleton members and k pair members, its first possible order is $s + 2k$. A cover of all m vertices has $s + 2k \geq m$, with equality exactly when all its members are disjoint. Such covers consist of a matching and singleton choices for its unmatched vertices. Their leading coefficients give (61.2). All other terms have order at least $m+1$. There are finitely many covers, and their Taylor estimates are uniform for bounded θ, giving the stated remainder. □

This polynomial simultaneously records the purely linear Prouhet coefficient $\prod_i a_i$ at $\theta = 0$ and, for even m, the hafnian coefficient of $\theta^{m/2}$. It provides a precise transition between the two cancellation regimes.

61.1 A path recurrence and a complete-graph formula

For a path on vertices $0, \dots, k-1$, let M_k denote (61.2). Then

$$M_0 = 1, \quad M_1 = a_0, \quad M_k = a_{k-1} M_{k-1} + \theta J_{k-2,k-1} M_{k-2} \quad (k \geq 2). \tag{61.4}$$

Indeed, the last vertex is either unmatched or paired with its only available neighbor. These alternatives are disjoint and exhaustive.

If the graph is complete and all $a_i = a$, $J_{ij} = J$, then

$$M_m = \sum_{k=0}^{\lfloor m/2 \rfloor} \frac{m!}{2^k k!(m-2k)!} (\theta J)^k a^{m-2k}. \tag{61.5}$$

Choose $2k$ matched vertices and partition them into k unordered pairs; the number of choices is the coefficient displayed. Equivalently,

$$\sum_{m \geq 0} M_m \frac{u^m}{m!} = \exp\left(au + \frac{\theta J u^2}{2} \right). \tag{61.6}$$

No square-root convention or special-function normalization is needed for this finite formula.

61.2 Cancellation design and its limits

The positivity assumption in Corollary 60.2 is useful for proving exact orders, but it rules out cancellations among minimum covers. Allowing signed interactions changes the problem: one can seek a nonzero matrix J with $\mathrm{Haf}(J) = 0$, thereby suppressing the first quadratic moment. For four vertices the condition is simply

$$J_{01}J_{23} + J_{02}J_{13} + J_{03}J_{12} = 0.$$

This is a concrete algebraic design equation. Satisfying it suppresses the moment of degree two, but does not by itself suppress the degree-three moment; that requires the next coefficient of (60.3). The covering identity therefore provides a hierarchy of *finite polynomial constraints*, rather than an unjustified promise that one canceled hafnian restores full Prouhet regularity.

Chapter 62

A Thue–Morse-weighted geometric product

62.1 Definition and exact logarithmic series

For $0 < q < 1$ and $|z| < 1$, define

$$\Phi_q(z) = \prod_{j=0}^{\infty} (1 - zq^j)^{\varepsilon_j}. \tag{62.1}$$

This resembles the ordinary q-Pochhammer product, but each factor is put in the numerator or denominator according to a Thue–Morse sign. It is not a fractional-power product: the exponents are the integers ± 1. On the unit disk, we use the logarithm specified by the Taylor series at $z = 0$.

Theorem 62.1 (Convergence, a dyadic equation, and a positive series)**.** *The product* (62.1) *converges locally uniformly in* $|z| < 1$*, defines a nonvanishing holomorphic function there, and satisfies*

$$\log \Phi_q(z) = -\sum_{r \geq 1} \frac{z^r}{r} E(q^r), \tag{62.2}$$

$$\Phi_q(z) = \frac{\Phi_{q^2}(z)}{\Phi_{q^2}(qz)}. \tag{62.3}$$

For $0 < z < 1$*, one has* $0 < \Phi_q(z) < 1$*.*

Proof. For $|z| \leq \rho < 1$,

$$\sum_{j \geq 0} |\log(1 - zq^j)| \leq \frac{\rho}{1 - \rho} \sum_{j \geq 0} q^j < \infty.$$

The logarithmic sum therefore converges uniformly, and its exponential is a nonzero holomorphic function. Expanding each logarithm is justified by

$$\sum_{j \geq 0} \sum_{r \geq 1} \frac{|z|^r q^{jr}}{r} = \sum_{r \geq 1} \frac{|z|^r}{r(1 - q^r)} < \infty.$$

Interchanging sums and using (1.8) gives (62.2). Separating even and odd values of j in the original product gives (62.3). Finally, every factor of $E(q^r) = \prod_{k \geq 0} (1 - q^{r2^k})$ lies in $(0, 1)$, and its convergent product is positive. Thus the series on the right of (62.2) is strictly negative for real $0 < z < 1$, proving the inequalities. □

The sign-weighted product literature includes extensive families in which the factor is a rational function of the index [11, 26]. Formula (62.1) instead samples a geometric progression. Our purpose is not to reclassify all such products, but to resolve its singular limit at $q = 1$.

62.2 Two useful truncation bounds

The defining logarithmic product, stopped before $j = J$, has an error bounded by

$$\left|\sum_{j\geq J} \varepsilon_j \log(1 - zq^j)\right| \leq \frac{|z|q^J}{(1-|z|)(1-q)}. \tag{62.4}$$

This follows from the same absolute logarithm bound as in the proof above. For the series (62.2), real $q \in (0,1)$ gives

$$\left|\sum_{r>R} \frac{z^r}{r} E(q^r)\right| \leq \frac{|z|^{R+1}}{(R+1)(1-|z|)}. \tag{62.5}$$

The latter estimate uses $0 < E(q^r) < 1$ and a geometric series. Near $q = 1$, (62.2) is usually the more stable representation: its terms are positive after a change of sign when z is real and positive, while the original logarithmic sum exhibits severe cancellation.

62.3 Flatness at the endpoint

Put

$$t = -\log q > 0, \qquad P(x) = E(\mathrm{e}^{-x}) = \prod_{k\geq 0}(1 - \mathrm{e}^{-2^k x}), \tag{62.6}$$

and write

$$F(t,z) = -\log \Phi_{\mathrm{e}^{-t}}(z) = \sum_{r\geq 1} \frac{z^r}{r} P(rt). \tag{62.7}$$

For real $0 < z < 1$, this is a positive sum.

Theorem 62.2 (Uniform superalgebraic flatness)**.** *For every integer $M \geq 1$, every $0 < \rho < 1$, and all $t > 0$,*

$$\sup_{|z|\leq\rho} |F(t,z)| \leq 2^{M(M-1)/2} t^M \sum_{r\geq 1} r^{M-1}\rho^r. \tag{62.8}$$

In particular, on compact subsets of $|z| < 1$, $\Phi_{\mathrm{e}^{-t}}(z) - 1 = O(t^M)$ for every M. For fixed real $0 < z < 1$, extension by $\Phi_1(z) = 1$ cannot be real analytic in t at zero.

Proof. Keep only the first M factors of $P(x)$, bound every remaining factor by one, and use $1 - \mathrm{e}^{-u} \leq u$:

$$0 < P(x) \leq \prod_{k=0}^{M-1}(2^k x) = 2^{M(M-1)/2} x^M. \tag{62.9}$$

Substitute this in (62.7) and take absolute values. The resulting series in r converges because $\rho < 1$. Since $F \to 0$, exponentiation preserves the every-order estimate for $\Phi - 1 = \mathrm{e}^{-F} - 1$.

If the real extension were analytic, the every-order estimate would force all its positive-degree Taylor coefficients to vanish. It would then equal one in a neighborhood of zero. But Theorem 62.1 gives $\Phi_{\mathrm{e}^{-t}}(z) < 1$ for every $t > 0$. This contradiction proves nonanalyticity. ☐

The statement is stronger than any fixed power law, but it does not yet tell us how small F actually is. Optimizing (62.8) is possible, but a precise answer requires the logarithmic oscillation hidden inside A, together with a moving saddle over the index r.

Chapter 63

An exact periodic amplitude and its Fabius connection

63.1 The complementary Laplace product and the scaling invariant

The flatness bound of Theorem 62.2 does not yet say how small F actually is. A precise answer requires the logarithmic oscillation hidden inside $P(x) = E(e^{-x})$, together with a moving saddle over the index r. The oscillation is isolated by the completion of Part II. In this part's notation, with $a = \log 2$, the two dilation laws (1.12) and (1.19) read

$$P(x/2) = \left(1 - e^{-x/2}\right)P(x), \qquad \Phi(x/2) = \frac{\Phi(x)}{\psi(x/2)}, \tag{63.1}$$

so that the completed product $K(x) = P(x)\Phi(x)$ of Theorem 16.1 satisfies the simple equation

$$K(x/2) = \frac{x}{2}K(x). \tag{63.2}$$

The purpose of Φ is not merely to approximate a product tail: multiplying by it turns the dyadic functional equation into an exact scaling law with a linear factor, and the remaining dependence on logarithmic scale is entirely periodic.

Theorem 63.1 (Exact positive periodic factorization)**.** *Define, for real v,*

$$\Psi(v) = \exp\Big(\frac{a}{2}(v^2 + v)\Big)\, P(2^{-v})\Phi(2^{-v}) = \exp\Big(\frac{a}{2}(v^2 + v)\Big)\, K(2^{-v}). \tag{63.3}$$

Then Ψ is positive, real-analytic, and one-periodic; it and all its derivatives are bounded, and its minimum is positive. For every $x > 0$, writing $u = \log(1/x)$, one has the exact identity

$$P(x) = \exp\left(-\frac{u^2}{2a} - \frac{u}{2}\right)\frac{\Psi(u/a)}{\Phi(x)}. \tag{63.4}$$

In terms of the logarithmic profile H of (16.3),

$$\Psi(v) = H(-av). \tag{63.5}$$

Proof. By (16.3), $K(t) = t^{1/2}\exp(-(\log t)^2/(2a))H(\log t)$ with H positive, real-analytic and a-periodic. Substituting $t = 2^{-v}$, so that $\log t = -av$ and $t^{1/2} = e^{-av/2}$, gives

$$K(2^{-v}) = e^{-av/2}e^{-av^2/2}H(-av),$$

and multiplication by $\exp(a(v^2 + v)/2)$ leaves exactly $H(-av)$. This proves (63.5); periodicity with period one, positivity, analyticity, and the bounds on $[0, 1]$ follow. Solving (63.3) for $P(x)$ with $x = 2^{-v}$, $v = u/a$, gives (63.4). □

Combining (63.4) with the convergent Bernoulli expansion (1.26) of $\log\Phi$ gives an exact periodic factor times a convergent analytic correction near the endpoint.

63.2 Smooth flatness, not just a value estimate

Corollary 63.2 (The full zero jet)**.** *For every $|z| < 1$, the function $F(t, z)$, extended by $F(0, z) = 0$, is C^∞ for real $t \geq 0$ near zero. All its t-derivatives at zero vanish. The assertions and every-order bounds for each derivative are uniform for $|z| \leq \rho < 1$. Consequently $\Phi_{\mathrm{e}^{-t}}(z)$, extended by one at zero, has the constant jet one there.*

Proof. Differentiate (63.4) a fixed number k of times. Since Ψ has bounded derivatives and $1/\Phi$ is analytic near zero, every resulting term is bounded, for small positive x, by

$$C_k x^{-k}(1 + |\log x|)^{d_k} \exp\left(-\frac{\log^2(1/x)}{2a} - \frac{\log(1/x)}{2}\right)$$

for some integer d_k. This is $O(x^M)$ for every M, because a negative quadratic in $\log(1/x)$ dominates every linear term and logarithm. On $x \geq \delta > 0$, product differentiation in (62.6) shows that $P^{(k)}(x)$ is bounded; the sums of derivative bounds are dominated by series of the form $\sum_j 2^{jk}\mathrm{e}^{-\delta 2^j}$. Enlarging the constant therefore gives, for every integer $M \geq 1$,

$$|P^{(k)}(x)| \leq C_{k,M} x^M \qquad (x > 0).$$

Termwise differentiation of (62.7) is valid away from zero, and

$$|\partial_t^k F(t, z)| \leq C_{k,M} t^M \sum_{r \geq 1} \rho^r r^{M+k-1}.$$

All these derivatives tend to zero. Extending them by zero and applying the fundamental theorem of calculus successively proves smoothness at the endpoint and the claimed derivatives. Exponentiation gives the final assertion. $\square$

Chapter 64

An all-order Lambert-W saddle expansion

64.1 Statement of the asymptotic law

We now fix $z \in (0,1)$ and put

$$\beta = -\log z > 0, \qquad \mathsf{L} = \log(1/t), \qquad a = \log 2. \tag{64.1}$$

The principal Lambert function W_0 is the positive solution of $we^w = y$ when $y > 0$; this convention agrees with DLMF §4.13 [17]. Define

$$w = W_0\left(\frac{a\beta}{\sqrt{2}\,t}\right), \qquad r_* = \frac{w}{a\beta}, \qquad \theta = \frac{w}{a} + \frac{1}{2}. \tag{64.2}$$

All these parameters are real, and $w, r_* \to \infty$ as $t \to 0^+$. Set

$$\Pi(t,z) = \frac{te^{a/8}}{\beta}\sqrt{\frac{2\pi w}{a}}\exp\left(-\frac{w^2+2w}{2a}\right). \tag{64.3}$$

For real θ, define the smooth amplitude on $y > 0$

$$\mathcal{A}_\theta(y) = \exp\left(-\frac{(\log y)^2}{2a}\right)\Psi\left(\theta - \frac{\log y}{a}\right). \tag{64.4}$$

Its derivative $\mathcal{A}_\theta^{(\ell)}(1)$ always means differentiation with respect to y; primes on Ψ mean differentiation with respect to its phase.

Theorem 64.1 (All-order saddle coefficients). *For every fixed integer $K \geq 1$, as $t \to 0^+$,*

$$F(t,z) = \Pi(t,z)\left[\sum_{n=0}^{K-1}\left(\frac{a}{w}\right)^n c_n(\theta) + O(w^{-K})\right]. \tag{64.5}$$

The estimate is uniform for z in a compact subset of $(0,1)$. Each c_n is a smooth one-periodic function. A finite formula for every coefficient is

$$c_n(\theta) = \sum_{\substack{\ell,k_3,\dots,k_{2n+2}\geq 0 \\ \ell+\sum_{j=3}^{2n+2}(j-2)k_j=2n}} \frac{\mathcal{A}_\theta^{(\ell)}(1)}{\ell!}\prod_{j=3}^{2n+2}\frac{1}{k_j!}\left(\frac{(-1)^{j+1}}{j}\right)^{k_j} \cdot \left(\ell + \sum_{j=3}^{2n+2} jk_j - 1\right)!!. \tag{64.6}$$

Here $(-1)!! = 1$*, and empty products have value one. In particular,*

$$c_0(\theta) = \Psi(\theta), \tag{64.7}$$

and

$$F(t, z) = \Pi(t, z)\left[\Psi(\theta) + \frac{1}{w}\left\{\left(\frac{a}{12} - \frac{1}{2}\right)\Psi(\theta) - \frac{1}{2}\Psi'(\theta) + \frac{1}{2a}\Psi''(\theta)\right\} + O(w^{-2})\right]. \tag{64.8}$$

This is an all-order Poincaré expansion in $1/w$, with a bounded periodic coefficient field. It is not asserted to be a complete resurgent transseries: contributions smaller than every power of $1/w$ are absorbed into the remainder analysis, rather than individually enumerated.

64.2 Locating the saddle exactly

The proof begins by discarding only terms that are beyond every algebraic order in $1/w$. For $r \le t^{-1/2}$, (1.26) gives $\Phi(rt)^{-1} = 1 + O(\sqrt{t})$, uniformly. For $r > t^{-1/2}$, $P(rt) \le 1$ and the geometric tail in (62.5) is bounded by $O(\exp(-\beta/\sqrt{t}))$, uniformly when β is in a compact positive interval. The eventual saddle main term has logarithm of order $-\mathsf{L}^2$, so this tail is negligible to every relative power of $1/w$. A lower bound of that size can already be obtained from the summand nearest r_*, using the positive minimum of Ψ.

Using (63.4), the main part of the summand is

$$\exp(f(r))\Psi\left(\frac{\mathsf{L} - \log r}{a}\right), \qquad f(r) = -\beta r - \frac{(\mathsf{L} - \log r)^2}{2a} - \frac{\mathsf{L} - \log r}{2} - \log r. \tag{64.9}$$

Its exponential phase has derivative

$$f'(r) = -\beta + \frac{\mathsf{L} - \log r}{ar} - \frac{1}{2r}.$$

Thus $f'(r) = 0$ is equivalent to

$$a\beta r + \log r = \mathsf{L} - \frac{a}{2}. \tag{64.10}$$

The left side is strictly increasing on $(0, \infty)$, so there is exactly one solution. With $w = a\beta r$, exponentiation turns (64.10) into $we^w = a\beta/(\sqrt{2}t)$, giving (64.2). Notice that we locate the saddle of the large exponential factor; the periodic amplitude is retained in the coefficient calculus and need not be absorbed into a shifted saddle.

At the saddle,

$$\mathsf{L} - \log r_* = w + \frac{a}{2}, \qquad f(r_*) = -\frac{w^2 + 2w}{2a} + \frac{a}{8} - \mathsf{L}. \tag{64.11}$$

An especially useful identity holds for every $y > 0$:

$$f(r_* y) - f(r_*) = \frac{w}{a}(\log y - y + 1) - \frac{(\log y)^2}{2a}. \tag{64.12}$$

Unlike a local Taylor expansion, this is an exact global identity. The function $H(y) = \log y - y + 1$ is nonpositive, has its unique maximum zero at $y = 1$, and satisfies $H''(1) = -1$. It therefore gives both localization and the Gaussian scale.

64.3 Why the lattice does not change the algebraic coefficients

For clarity, the sum-to-integral step is justified explicitly. Let χ be a smooth function of y supported in $(1/3, 3)$, equal to one on $[1/2, 2]$, and define

$$g(r) = \chi(r/r_*)\mathrm{e}^{f(r)}\Psi\left(\frac{\mathsf{L} - \log r}{a}\right).$$

Outside the central interval, (64.12) bounds both the sum and integral tails by $\mathrm{e}^{f(r_*)}$ times a polynomial in w times e^{-cw}, for some positive c. For large y, the term $-wy/a$ provides a geometric or exponential tail; near zero, $y^{w/a}$ provides the corresponding integrability. The periodic amplitude is bounded. These estimates are uniform for β in a compact positive interval.

For a smooth compactly supported function on the real line and any integer $p \geq 2$, the Euler summation identity gives

$$\left|\sum_{n\in\mathbb{Z}} g(n) - \int_{\mathbb{R}} g(r)\,\mathrm{d}r\right| \leq C_p \int_{\mathbb{R}} |g^{(p)}(r)|\,\mathrm{d}r. \tag{64.13}$$

One way to see this bound is to integrate by parts p times against the periodic Bernoulli function in the Euler summation formula. Its boundedness produces the constant C_p; compact support removes all endpoint terms.

The peak width in the r variable is proportional to $\sqrt{w}$, since $r_* \asymp w$ and the y-width is $w^{-1/2}$. Differentiation of (64.12) on the central interval, followed by Gaussian domination, yields

$$\int_{\mathbb{R}} |g^{(p)}(r)|\,\mathrm{d}r \leq C_p' \mathrm{e}^{f(r_*)} w^{(1-p)/2}. \tag{64.14}$$

For completeness, each derivative of the exponential is a sum of products of phase derivatives. In the local variable $v = (r - r_*)/\sqrt{w}$, a derivative with respect to r contributes $w^{-1/2}$, while the resulting factors are bounded by polynomials in $|v|$ times e^{-cv^2}. Integrating $\mathrm{d}r = \sqrt{w}\,\mathrm{d}v$ gives (64.14). Away from a fixed central neighborhood, all cutoff derivatives are exponentially smaller and satisfy the same bound after enlarging the constant. Derivatives of Ψ are bounded and introduce factors $1/r = O(w^{-1})$, no larger than required.

The integral itself is comparable to $\mathrm{e}^{f(r_*)}\sqrt{w}$, using positivity of Ψ. Combining (64.13) and (64.14), the relative lattice error is $O(w^{-p/2})$. Since p can be chosen arbitrarily large, the lattice does not affect any fixed algebraic coefficient of the saddle expansion. This is an all-orders statement, not an assertion that the lattice error is exactly zero.

64.4 The Gaussian coefficient calculation

After the preceding reductions and the substitution $r = r_* y$, it remains to expand

$$\mathrm{e}^{f(r_*)} r_* \int_0^\infty \exp\left(k(\log y - y + 1)\right) \mathcal{A}_\theta(y)\,\mathrm{d}y, \qquad k = \frac{w}{a}. \tag{64.15}$$

Put $y = 1 + v/\sqrt{k}$. Then

$$k(\log y - y + 1) = -\frac{v^2}{2} + \sum_{j\geq 3} \frac{(-1)^{j+1} v^j}{j\, k^{(j-2)/2}}. \tag{64.16}$$

Expand the analytic phase correction and the smooth amplitude to the required finite order. A term from amplitude derivative ℓ and phase multiplicities $k_3, k_4, \ldots$ contributes the power

$$k^{-\frac{1}{2}\{\ell + \sum_{j\geq 3}(j-2)k_j\}}$$

and the monomial $v^{\ell+\sum jk_j}$. Terms of odd degree integrate to zero against the centered Gaussian. At power k^{-n}, the degree constraint is precisely the one in (64.6); the degree of the Gaussian monomial is even because it differs from $2n$ by $2\sum k_j$. Its normalized integral is the double factorial in that formula.

Here is a remainder justification at arbitrary fixed order. First restrict $|y-1|\le\delta$ with small fixed δ, where $H(y)\le -c(y-1)^2$. Outside this interval the integral is exponentially smaller. Inside it, restrict further to $|v|\le k^{\epsilon}$, where $\epsilon>0$ is chosen small enough for the desired finite Taylor order. Taylor's theorem with a bounded derivative remainder applies uniformly in the periodic parameter θ. Expanding sufficiently many phase and amplitude terms and integrating polynomial remainders against e^{-cv^2} gives an error $O(k^{-K})$ after retaining powers through $k^{-(K-1)}$. The region $k^{\epsilon}<|v|\le\delta\sqrt{k}$ has a Gaussian tail smaller than every power of k^{-1}. The phase and amplitude bounds are uniform in θ, so the expansion is uniform. This proves the all-order coefficient formula without treating a divergent Taylor expansion as a global identity.

The Gaussian normalization in (64.15) is

$$\mathrm{e}^{f(r_*)}r_*\sqrt{\frac{2\pi a}{w}}=\Pi(t,z),$$

by (64.2) and (64.11). This proves (64.5). Periodicity and smoothness of c_n follow from those of Ψ and the finite formula (64.6).

64.5 First and second universal coefficients

Writing $A_j=\mathcal{A}_\theta^{(j)}(1)$ temporarily, the coefficient formula gives

$$c_1=\frac{A_0}{12}+A_1+\frac{A_2}{2}, \tag{64.17}$$
$$c_2=\frac{A_0}{288}+\frac{A_1}{12}+\frac{25A_2}{24}+\frac{5A_3}{6}+\frac{A_4}{8}. \tag{64.18}$$

These identities can also be checked directly by Gaussian moments of degrees up to twelve. They are included in the symbolic verification program.

The chain rule gives

$$A_0=\Psi,\qquad A_1=-\frac{\Psi'}{a},\qquad A_2=-\frac{\Psi}{a}+\frac{\Psi'}{a}+\frac{\Psi''}{a^2}, \tag{64.19}$$

where the functions on the right are evaluated at θ. Substitution in (64.17) gives (64.8). This completes the proof of Theorem 64.1.

64.6 Why the first term of the logarithmic series is misleading

A tempting approximation to (62.7) is to keep only $zA(t)$. It has the right qualitative flatness, but not the correct magnitude.

Corollary 64.2 (Collective enhancement of the flat tail)**.** *For fixed $z\in(0,1)$, as $\mathsf{L}=\log(1/t)\to\infty$,*

$$\log F(t,z)=-\frac{\mathsf{L}^2}{2a}+\frac{\mathsf{L}\log\mathsf{L}}{a}+O(\mathsf{L}), \tag{64.20}$$
$$\log\frac{F(t,z)}{P(t)}=\frac{\mathsf{L}\log\mathsf{L}}{a}+O(\mathsf{L}). \tag{64.21}$$

The contributing indices are centered at $r_*=w/(a\beta)\sim\mathsf{L}/(a\beta)$, with width of order $\sqrt{\mathsf{L}}$.

Proof. Taking logarithms in $we^w = a\beta/(\sqrt{2}t)$ gives $w + \log w = \mathsf{L} + \log(a\beta/\sqrt{2})$. It follows first that $w \sim \mathsf{L}$, and then, by substitution, that $w = \mathsf{L} - \log \mathsf{L} + O(1)$. The leading term of Theorem 64.1, together with the positive upper and lower bounds for Ψ, yields

$$\log F = -\mathsf{L} - \frac{w^2 + 2w}{2a} + \frac{1}{2}\log w + O(1).$$

Substituting the estimate for w proves (64.20). Formula (63.4) gives $\log P(t) = -\mathsf{L}^2/(2a) - \mathsf{L}/2 + O(1)$, because $\Phi(t) = 1 + O(t)$. Subtracting proves (64.21). The location and width were derived in (64.2) and the lattice analysis. □

Every individual term is flat, but the effective index drifts to infinity. This is precisely why pointwise asymptotics at a fixed index cannot be interchanged with the infinite sum without a uniform argument.

64.7 Parameter regimes deliberately not covered

The expansion is uniform when z stays in a compact subset of $(0, 1)$. It does not assert uniformity as $z \to 0$, where the saddle may cease to be large under joint scaling, or as $z \to 1$, where $\beta \to 0$ and the geometric cutoff changes scale. For complex z, the defining product and flatness theorem remain valid, but positivity and the real saddle proof no longer apply as stated. Those cases require separate contour and cancellation analysis. In particular, the present proof does not license replacing z by a root of unity in (64.5).

Chapter 65

A joint endpoint limit and classical Thue–Morse products

The fixed-z theorem has a particularly instructive companion. Approach the corner $(q, z) = (1, 1)$ along $z = q^\gamma$, with $\gamma > 0$ fixed. Here the product no longer tends to one. Instead, a classical sign-weighted product emerges, and every algebraic discretization correction vanishes.

Theorem 65.1 (A flat boundary layer)**.** *For $\gamma > 0$, define*

$$\mathcal{H}(\gamma) = \int_0^\infty \mathrm{e}^{-\gamma x} \frac{P(x)}{x}\,\mathrm{d}x, \qquad \mathcal{P}(\gamma) = \exp(-\mathcal{H}(\gamma)). \tag{65.1}$$

The integral is finite and positive. For every integer $M \geq 1$,

$$F(t, \mathrm{e}^{-\gamma t}) = \mathcal{H}(\gamma) + O(t^M), \tag{65.2}$$

$$\Phi_{\mathrm{e}^{-t}}(\mathrm{e}^{-\gamma t}) = \mathcal{P}(\gamma) + O(t^M) \tag{65.3}$$

as $t \to 0^+$, uniformly for γ in compact subsets of $(0, \infty)$. Moreover,

$$\mathcal{P}(\gamma) = \prod_{j=0}^{\infty} \left(\frac{\gamma + 2j}{\gamma + 2j + 1} \right)^{\varepsilon_j}. \tag{65.4}$$

The product on the right is understood in the displayed paired order.

Proof. The function $g_\gamma(x) = \mathrm{e}^{-\gamma x} P(x)/x$, for $x > 0$, extends by zero to a smooth function on the whole real line. Smooth flatness of A, proved in Section 63, handles the origin and the factor $1/x$. At positive infinity, the factor $\mathrm{e}^{-\gamma x}$ ensures integrability of every derivative. All derivative L^1 bounds are uniform for γ in a compact positive interval. In particular, g_γ is integrable, and its integral is strictly positive.

Formula (62.7) becomes the Riemann sum

$$F(t, \mathrm{e}^{-\gamma t}) = t \sum_{r \geq 1} g_\gamma(rt). \tag{65.5}$$

The mesh-t version of (64.13) gives, for any integer $p \geq 2$,

$$\left| t \sum_{r \in \mathbb{Z}} g_\gamma(rt) - \int_{\mathbb{R}} g_\gamma(x)\,\mathrm{d}x \right| \leq C_p t^p \|g_\gamma^{(p)}\|_1.$$

One may first apply it to compactly supported cutoffs and then remove the cutoffs using exponential decay. The terms for nonpositive r are zero. Choosing $p \geq M$ proves (65.2), and exponentiation proves

(65.3). Thus no nonzero Euler–Maclaurin endpoint coefficient survives: every derivative at the left endpoint is zero, and those at infinity vanish as well.

To identify the limiting product, use the finite dyadic polynomial

$$P_m(x) = \prod_{k=0}^{m-1} (1 - \mathrm{e}^{-2^k x}) = \sum_{n=0}^{2^m-1} \varepsilon_n \mathrm{e}^{-nx}.$$

For $m \geq 1$, $0 \leq P_m(x) \leq \min(x, 1)$. These products decrease to $P(x)$, and the displayed bound gives an integrable majorant for $\mathrm{e}^{-\gamma x} P_m(x)/x$. Since $\sum_{n<2^m} \varepsilon_n = 0$, Frullani's identity gives

$$\int_0^\infty \mathrm{e}^{-\gamma x} \frac{P_m(x)}{x}\, \mathrm{d}x = -\sum_{n<2^m} \varepsilon_n \log(\gamma + n). \tag{65.6}$$

For justification, replace each exponential by its difference from $\mathrm{e}^{-\gamma x}$, use the zero sum of signs, and integrate $\int_0^\infty (\mathrm{e}^{-ux} - \mathrm{e}^{-vx})\, \mathrm{d}x/x = \log(v/u)$. The latter identity follows by integrating $\mathrm{e}^{-ux} - \mathrm{e}^{-vx} = \int_u^v x\mathrm{e}^{-sx}\, \mathrm{d}s$ and interchanging a finite parameter integral with the positive exponential integral.

Dominated convergence in (65.6), followed by exponentiation, shows that $\mathcal{P}(\gamma)$ is the limit of $\prod_{n<2^m} (\gamma + n)^{\varepsilon_n}$. Pairing indices $2j$ and $2j+1$ converts it into partial products of (65.4). Their logarithmic series converges: the partial sums of ε_j are bounded, since consecutive even–odd signs cancel, while $\log((\gamma + 2j)/(\gamma + 2j + 1))$ tends monotonically to zero. Dirichlet summation therefore proves convergence in the paired order and identifies its limit with the dyadic subsequence. □

Remark 65.2 (The ordering is essential). The unpaired formal series $\sum_{n\geq 0} \varepsilon_n \log(\gamma + n)$ does not converge in its ordinary term-by-term order: its terms do not tend to zero. Formula (65.4) is a convergent *paired* product, and (65.6) uses balanced dyadic blocks. Omitting this distinction would turn a correct limiting identity into a false convergence claim.

65.1 Two exact constants

The boundary-layer integral permits a short proof of two special values.

Proposition 65.3. *For $\gamma > 0$,*

$$\mathcal{H}(\gamma/2) - \mathcal{H}((\gamma + 1)/2) = \mathcal{H}(\gamma). \tag{65.7}$$

Furthermore,

$$\mathcal{H}(1/2) = \log 2, \qquad \mathcal{H}(1) = \frac{1}{2} \log 2, \qquad \mathcal{P}(1/2) = \frac{1}{2}, \quad \mathcal{P}(1) = \frac{1}{\sqrt{2}}. \tag{65.8}$$

Consequently, for every fixed M,

$$\Phi_q(q) = \frac{1}{\sqrt{2}} + O\big((-\log q)^M\big), \qquad \Phi_q(\sqrt{q}) = \frac{1}{2} + O\big((-\log q)^M\big) \quad (q \to 1^-). \tag{65.9}$$

Proof. Subtract the two integrals on the left of (65.7). The numerator contains $\mathrm{e}^{-\gamma x/2}(1 - \mathrm{e}^{-x/2}) P(x) = \mathrm{e}^{-\gamma x/2} P(x/2)$, by (63.1). The substitution $x = 2y$ gives $\mathcal{H}(\gamma)$.

Also, $P(x) = (1 - \mathrm{e}^{-x}) P(2x)$, so

$$P(2x) - P(x) = \mathrm{e}^{-x} P(2x).$$

Integrate its quotient by x. On $[\delta, R]$, the left side integrates to

$$\int_R^{2R} \frac{P(y)}{y}\, \mathrm{d}y - \int_\delta^{2\delta} \frac{P(y)}{y}\, \mathrm{d}y.$$

As $R \to \infty$ and $\delta \to 0$, this tends to $\log 2$, because $P(y) \to 1$ at infinity and $P(y) \to 0$ at zero. The right side, under $y = 2x$, integrates to $\mathcal{H}(1/2)$. Thus $\mathcal{H}(1/2) = \log 2$. Taking $\gamma = 1$ in (65.7) gives $\mathcal{H}(1/2) = 2\mathcal{H}(1)$. The remaining formulas follow by exponentiation and Theorem 65.1. □

The paired product at $\gamma = 1$ is the classical Woods–Robbins Thue–Morse product, not a newly discovered constant [11, 7]. The derived refinement here is its realization as a joint endpoint value of $\Phi_q(z)$, with an error smaller than every power of $-\log q$. It also explains concretely why the fixed-z limit cannot be uniform up to $z = 1$.

65.2 Monotonicity of the boundary profile

Differentiation under the integral gives, for every $k \geq 1$,

$$(-1)^k \mathcal{H}^{(k)}(\gamma) = \int_0^\infty x^{k-1} \mathrm{e}^{-\gamma x} P(x) \,\mathrm{d}x > 0. \tag{65.10}$$

Thus $\mathcal{H}$ is strictly decreasing and convex, and $\mathcal{P}$ is strictly increasing. Dominated convergence gives $\mathcal{H}(\gamma) \to 0$ as $\gamma \to \infty$. Since $P(x) \to 1$, integrating over a large fixed lower endpoint up to $1/\gamma$ shows $\mathcal{H}(\gamma) \to \infty$ as $\gamma \to 0^+$. The profile $\mathcal{P}$ therefore increases from zero to one. This is a continuous family of limiting constants, rather than an isolated diagonal coincidence.

Chapter 66

Reproducible symbolic and numerical checks

66.1 What was checked exactly

The supplied program `verify.py` uses integers and rational arithmetic for the finite combinatorial identities. The deterministic random seed is 20260904 for the primary tests; the additional block tests use 410926. It performs 1,800 assertions on even-order dyadic correlation identities, using 120 tuples of lengths $2, 4, 6, 8$ and several consecutive levels. A further 960 assertions test odd-order sums, cyclic cancellation, and the block-multiple formulas in (14.16).

The covering identity is compared coefficient by coefficient with direct enumeration of the Boolean cube, giving 35 exact moment equalities in the selected small examples. Nine path examples test (60.14)–(60.15), and six symbolic tests compare the weak-coupling coefficient with the independently enumerated matching polynomial. The general coefficient generator reproduces (64.17)–(64.18).

All these assertions passed. They are finite checks of the implementation and formulas, not replacements for the general proofs.

Table 66.1: The first nonzero signed moment of the path coordinate (60.13) at $\lambda = 1$.

Number of bits m	First degree	First nonzero moment
2	1	1
3	2	-12
4	2	2
5	3	-138
6	3	6
7	4	-2112
8	4	24
9	5	-41400
10	5	120

66.2 Saddle verification at very small scales

Numerical computations use 80 decimal digits of working precision. We evaluate $F = -\log \Phi$, rather than subtracting Φ from one: in the last row of Table 66.2, the deviation has size about 10^{-1824}, far below a direct 80-digit subtraction from one.

The product $P(x)$ is evaluated through the stable logarithmic expression $\log(1 - \mathrm{e}^{-y}) = \log(-\operatorname{expm1}(-y))$, doubling y at successive factors. The positive series (62.7) is truncated with the explicit bound (62.5). The periodic amplitude is evaluated on one period using (63.3) and the convergent series (1.26).

Table 66.2: Saddle approximations for $z = 1/2$, $t = \mathrm{e}^{-\mathsf{L}}$. A signed relative error means approximation divided by the positive series value, minus one.

L	w	$\log_{10} F$	Leading error	With first correction
10	6.977684	-23.376724	$5.8521 \cdot 10^{-2}$	$-8.5644 \cdot 10^{-3}$
20	16.139152	-99.611573	$2.6520 \cdot 10^{-2}$	$-1.6113 \cdot 10^{-3}$
40	35.354962	-430.149058	$1.2320 \cdot 10^{-2}$	$-3.4163 \cdot 10^{-4}$
80	74.608151	-1824.180963	$5.8848 \cdot 10^{-3}$	$-7.7430 \cdot 10^{-5}$

The omitted positive-series tails, divided by the computed values, have bounds below $1.5 \cdot 10^{-75}$ in these examples. This bounds only the omitted r-terms, not every floating-point or product-evaluation error. The computation is not an interval-certified enclosure. Independent checks of the defining logarithmic product against the positive series at moderate parameters agree to at least 65 decimal places. The exact periodic factorization likewise agrees at the working precision in the tested cases; a printed residual of zero means numerical agreement, not an additional exact proof.

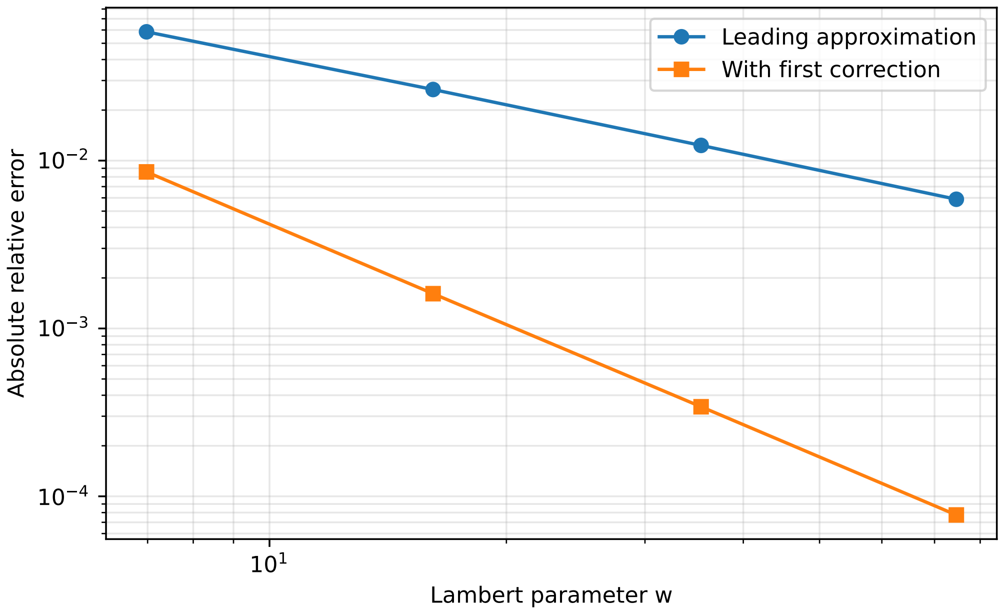

Figure 66.1: Absolute relative errors for the data in Table 66.2. The first correction improves the observed error from the scale $1/w$ to the scale $1/w^2$, as predicted by the theorem. The graph is a numerical check, not a uniform error bound.

66.3 The diagonal boundary layer

For $z = q = \mathrm{e}^{-t}$, direct logarithmic-product evaluation gives Table 66.3. The signs of the errors need not be monotone; the statement is a superalgebraic bound, not monotone convergence of the error.

Table 66.3: The joint endpoint $\Phi_q(q) \to 1/\sqrt{2}$. Values are rounded; the full data are in `boundary_layer.csv`.

t	$\Phi_{e^{-t}}(e^{-t})$	Error from $1/\sqrt{2}$
1	0.758123053523105	$5.1016 \cdot 10^{-2}$
0.5	0.710249881663610	$3.1431 \cdot 10^{-3}$
0.25	0.706267615662706	$-8.3917 \cdot 10^{-4}$
0.125	0.707016842592727	$-8.9939 \cdot 10^{-5}$
0.0625	0.707106345877474	$-4.3531 \cdot 10^{-7}$

Chapter 67

Further deductions, limitations, and research directions

67.1 A unifying finite–infinite correspondence

There are three distinct forms of compression in the proved results. In the correlation problem, all interactions with a dyadic boundary are compressed into the integer $D(\boldsymbol{h})$. In the nonlinear moment problem, all Boolean cancellations are compressed into the covering condition $\bigcup \mathcal{C} = V$. In the product problem, the exact dyadic scaling is compressed into the periodic amplitude Ψ, while the continuous part is handled by one saddle.

The comparison is structural, not an assertion that the three formulas are special cases of a single theorem already proved here. Their common feature is that cancellation is dealt with *before* taking the asymptotic limit. Doing the reverse can lose the boundary term, misidentify the first surviving moment, or retain the wrong index in an infinite sum.

67.2 Classifying vanishing boundary functionals

Question 67.1. Which tuples $\boldsymbol{h}$, modulo translations and deletion of repeated pairs, satisfy $D(\boldsymbol{h}) = 0$? Can their binary structure be characterized without explicitly summing (14.7)?

For even order, $D = 0$ means every sufficiently large dyadic cyclic and noncyclic sum equals its mean times the block length exactly. For odd order, it means every sufficiently large aligned dyadic block sum is zero. This is a finite classification problem with a precise observable consequence. The pair recurrence (14.22) provides a starting model, but the higher-order combinatorics of the threshold parities b_t are richer. No classification is claimed in this part.

67.3 Designed nonlinear cancellation beyond a hafnian

Question 67.2. For a prescribed sparse interaction graph, what is the largest possible vanishing order of $Z_x(t)$ among nontrivial signed quadratic weights, subject to chosen normalization constraints on the linear part?

The positive-weight answer is determined by the minimum cover. Signed weights allow algebraic cancellation among minimum covers, so the problem becomes a finite polynomial system built from (60.6). Useful constraints would exclude degenerate choices that make x independent of a bit and force $Z_x \equiv 0$. Requiring the coordinate map to be injective on the Boolean cube is another natural way to rule out trivial pairwise coincidences. The part supplies the equations but does not assert an optimal solution.

67.4 Uniform matching of the two analytic regimes

The fixed-z saddle corresponds to $\beta = -\log z$ bounded away from zero, while the boundary layer has $\beta = \gamma t$. The intermediate regime $t \ll \beta \ll 1$ should connect the large-γ asymptotics of $\mathcal{H}(\gamma)$ with the moving discrete saddle. A rigorous uniform formula would need to control the relation between saddle width and mesh size throughout that transition. Theorems 64.1 and 65.1 establish the two endpoints but do not prove the full uniform bridge.

Question 67.3. Can one construct a single asymptotic approximation, with an explicit uniform remainder, valid when $\beta/t \to \infty$ while $\beta \to 0$, and matching both (64.5) and (65.2)?

This is a concrete continuation problem rather than a request to speculate about a nonexistent singularity. The exact positive representation (62.7) and the amplitude identity (63.4) remain valid throughout the real parameter region and provide the required starting point.

67.5 Beyond-all-orders corrections and complex parameters

The saddle theorem suppresses lattice effects beyond every algebraic order in $1/w$. The boundary-layer theorem suppresses all Euler summation powers of t. Identifying the leading remaining corrections would require more than extending the coefficient list: one must study the analytic continuation of the amplitude and the relevant complex saddles or Fourier modes. That problem is a natural transseries extension, but it is not solved by the present all-order Poincaré expansion. Similarly, $|z| = 1$ is a boundary of the absolute convergence arguments, and root-of-unity specialization requires a fresh proof.

67.6 Possible formalization targets

The finite correlation and covering theorems have a relatively direct route to formal proof: finite sums, bit decompositions, parity, and Boolean monomial identities suffice. An initial formalization could isolate (14.12), (14.13), and (60.3), then derive the remaining finite identities as corollaries. Such a project should use the actual theorem names and conventions of the current repository after a separate code audit; none are invented here.

The analytic results require additional infrastructure for infinite products, termwise differentiation, dominated convergence, and uniform saddle estimates. They should not be marked formalized merely because their numerical checks pass. The supplied Python code is reproducibility material, not a proof assistant certificate.

Appendix A

Compact formula sheet for Part I

With $N = 2^m$, $g(k) = B_k(-2)$, and $\mathcal{Q}(z) = zE(z)^2$, the central formulas are

$$\begin{aligned}
r_m(k) - \eta(k) &= \frac{(-1)^m}{3N} g(k), && 0 \le k \le N, \\
A_m(k) - N\eta(k) &= -\frac{2}{3}(-1)^m g(k), && 0 \le k \le N, \\
\widetilde{\mu}_m &= (\mu_m + 2\mu_{m+1})/3, && \widetilde{f}_m = f_m(1 - \tfrac{2}{3}\cos(2\pi N x)), \\
\widehat{\widetilde{\mu}_m}(k) &= \eta(k), && |k| \le N, \\
\widehat{\nu}(k) &= g(|k|)/3, && k \ne 0, \\
\mathcal{G}_m - \mathcal{G} &= \frac{(-1)^m}{3N} \mathcal{Q}(z)\mathcal{K}(z^N), && |z| < 1, \\
2\mathcal{K}(z) + \mathcal{K}(z^2) &= 3E(z)^{-2}, && \mathcal{K}(0) = 1, \\
\kappa_n &= \frac{3}{2} \sum_{j=0}^{\nu_2(n)} (-1/2)^j p_2(n/2^j), && n \ge 1.
\end{aligned}$$

The exponents are

$$\sigma = \frac{1}{2} \log_2 \frac{1 + \sqrt{17}}{4}, \qquad s_* = 1 + \sigma.$$

The raw Sobolev rate is $N^{\sigma - s}$ below s_*, $N^{-1}\sqrt{\log N}$ at s_*, and N^{-1} above it. The corrected rate is $N^{\sigma - s}$ throughout $s > \sigma$. Every use of these rates fixes s before taking $N \to \infty$.

Appendix B

Coefficient register and normalization checks for Part II

The first six coefficients in the centered expansion are

r	A_r in $\mathcal{H}(x) \sim \sum_{r\geq 0}(-1)^r A_r x^r$
0	1
1	$1/9$
2	$248/2025$
3	$4806656/2679075$
4	$3993751257088/10247461875$
5	$454890030220640780288/345944065438125$

All are obtained by the finite formula Equation (22.6) followed by multiplication by 2^{2r^2+r}. Their rapid growth is predicted by Theorem 23.1; it is not numerical instability.

The following identities are convenient independent checks:

$$
\begin{aligned}
&\Phi(0) = 1, \quad \Phi'(0) = -\tfrac{1}{2}, \quad [z^2]\log\Phi(z) = \tfrac{1}{72},\\
&K(2t)t = K(t), \quad C(1) = 2C_0, \quad C(-1) = C_0,\\
&\mathbb{E}T^2 = 8, \quad \mathbb{E}W = \tfrac{1}{72}, \quad \mathbb{E}(T^2W) = \tfrac{1}{9},\\
&Z_m(1/2) = C_0\, 2^{m(m-1)/2}\mathcal{H}(4^{-m}),\\
&D(s,1/2) = (1+2^s)D(s,1).
\end{aligned}
$$

There is also an exact normalization check at $m = 0$:

$$
Z_0(1/2) = L, \qquad Z_0(1) = L/2, \qquad C_0\mathcal{H}(1) = L.
$$

Indeed, $P(2u) - P(u) = e^{-u}P(2u)$, so substitution $t = 2u$ gives

$$
M(0,1/2) = \int_0^\infty \frac{P(2u)-P(u)}{u}\,\mathrm{d}u = L.
$$

To justify the last equality without splitting divergent integrals, integrate first over $[\epsilon, R]$. The difference is $\int_R^{2R} P(u)\,\mathrm{d}u/u - \int_\epsilon^{2\epsilon} P(u)\,\mathrm{d}u/u$; use $P(\infty) = 1$ and $P(0) = 0$. Differentiating Equation (21.2) at zero gives the value at $a = 1$. These are classical product-normalization consequences, recorded here as checks rather than new evaluations.

A sign error in the logarithmic centering changes the odd coefficients; a missing factor of 2 in the Mellin scaling changes the moment law; and confusing $\Psi(z)$ with $\Psi(\sqrt{z})$ changes the optimal truncation scale. Each of these errors is detected by at least one of the checks above.

B.1 A direct nonrecursive formula for every asymptotic coefficient

For readers wishing to implement the general-shift expansion without Bell-polynomial conventions, define $b = 1/2 - a$ and

$$d_r = -\frac{B_{2r}}{2r(2r)!(2^{2r}-1)}.$$

Then its coefficient $\alpha_j(a) = 2^{j(j+1)/2}\beta_j(a)$ is

$$\boxed{\alpha_j(a) = 2^{j(j+1)/2} \sum_{\substack{n_0,n_1,\ldots,n_{\lfloor j/2\rfloor}\geq 0 \\ n_0+2n_1+4n_2+\cdots+2\lfloor j/2\rfloor n_{\lfloor j/2\rfloor}=j}} \frac{b^{n_0}}{n_0!}\prod_{r=1}^{\lfloor j/2\rfloor}\frac{d_r^{n_r}}{n_r!}.} \tag{B.1}$$

The sum is finite. At $b = 0$, only even $j = 2r$ survive, and $\alpha_{2r}(1/2) = (-1)^r A_r$. For the complex translated expansion, simply multiply its jth coefficient by 2^{js} before multiplying by $C(s)$. Thus a single finite coefficient formula covers both the real-zero slopes and the complex zero-string expansion.

Appendix C

Finite formulas for every coefficient of Part III

The part uses Bell polynomials, Bernoulli numbers, and rational negative-index polylogarithms as convenient names. This appendix makes explicit that no hidden recurrence is required.

C.1 Eulerian coefficients and rational polylogarithms

For integers $m \geq 1$, $0 \leq j < m$, define

$$A(m,j) = \sum_{\ell=0}^{j+1} (-1)^{\ell} \binom{m+1}{\ell} (j+1-\ell)^{m}. \tag{C.1}$$

Then, for $n \geq 2$,

$$\mathrm{Li}_{1-n}(w) = \frac{\sum_{j=0}^{n-2} A(n-1,j) w^{j+1}}{(1-w)^{n}}, \qquad \mathrm{Li}_{0}(w) = \frac{w}{1-w}. \tag{C.2}$$

To prove this, begin in $|w| < 1$, where repeated application of $w\,\mathrm{d}/\,\mathrm{d}w$ gives $\mathrm{Li}_{-m}(w) = \sum_{k \geq 1} k^{m} w^{k}$. Multiplication by $(1-w)^{m+1}$ produces exactly the coefficients in (C.1). Coefficients beyond degree m vanish by the $(m+1)$-st finite difference of a degree-m polynomial. Both sides are rational functions, so the identity extends to every $w \neq 1$.

C.2 Bernoulli numbers without a recurrence

With the convention $B_1 = -1/2$, one has

$$B_n = \sum_{k=0}^{n} \frac{1}{k+1} \sum_{j=0}^{k} (-1)^{j} \binom{k}{j} j^{n}. \tag{C.3}$$

The convention $0^0 = 1$ applies. Indeed, as a formal power series,

$$\sum_{k \geq 0} \frac{(1-\mathrm{e}^{z})^{k}}{k+1} = \frac{-\log(1-(1-\mathrm{e}^{z}))}{1-\mathrm{e}^{z}} = \frac{z}{\mathrm{e}^{z}-1}.$$

For the coefficient of z^n, only $k \leq n$ contributes. Expanding the finite binomial sum proves (C.3). Use $\beta_1 = 1/2$, $\beta_n = B_n/n$ for $n \geq 2$, as in the main text.

C.3 Bell coefficients as constrained finite sums

For any cumulant list $x_1, \dots, x_h$,

$$\mathcal{B}_h(x_1, \dots, x_h) = h! \sum_{\substack{\nu_1, \dots, \nu_h \geq 0 \\ \nu_1 + 2\nu_2 + \cdots + h\nu_h = h}} \prod_{j=1}^{h} \frac{1}{\nu_j!} \left(\frac{x_j}{j!} \right)^{\nu_j}. \tag{C.4}$$

This follows directly by multiplying the exponential series $\prod_{j \geq 1} \exp(x_j z^j / j!)$ and extracting the coefficient of z^h. Inserting (C.2), (C.3), and the finite cycle sums into (C.4) gives literal nested finite sums for every moment and every correction coefficient appearing in the part.

Appendix D

Normalization and logical checkpoints for Part III

The main identities are sensitive to four conventions. First, P_m has m factors and 2^m coefficients; $P_0 = 1$. Second, the local window is $\zeta \mathrm{e}^{z/2^m}$, not $\zeta \mathrm{e}^{z/2^{m+1}}$. Third, $\mathcal{U}$ is the transform of $X \in [0, 1]$, with mean $1/2$, rather than of its centered or rescaled Rvachev version. Finally, Bell polynomials encode exponential generating coefficients, so the factors $h!$ and $(M + h)!/h!$ cannot be omitted.

There are also three logical distinctions worth retaining. An entire quotient formula is interpreted through a denominator-free identity or a proved removable continuation. Fourier nondecay does not, on its own, prove pure singularity; the orbit-average argument supplies that missing step. And an analytic Gaussian limit does not imply that the finite complex approximants are probability transforms. These distinctions keep the algebraic, analytic, and probabilistic claims aligned.

Appendix E

Normalization dictionary and implementation details for Part IV

E.1 Frequency conventions

The angular-frequency transform used here is $\widehat{u}(t)$ in (47.3). Under the cyclic-frequency convention $\widehat{u}_{\mathrm{c}}(\xi) = \int e^{-2\pi \mathrm{i}\xi x} u(x)\,\mathrm{d}x$, evenness gives

$$\widehat{u}_{\mathrm{c}}(\xi) = \widehat{u}(2\pi\xi) = \prod_{j=0}^{\infty} \operatorname{sinc}(\pi\xi/2^j). \tag{E.1}$$

A finite cyclic product indexed $j = 0, \ldots, n$ therefore corresponds to the spline u_{n+1} in this part. Its omitted-tail scale is $2^{-(n+1)}$, not 2^{-n}. The even moments of its normalized cyclic-frequency energy measure are

$$(2\pi)^{-2r} 2^{r(r+1)}. \tag{E.2}$$

This dictionary is important when comparing formulas with repository documents that use a different Fourier convention or start their product index at zero.

E.2 Integer-only evaluation of a spline derivative

For $x = a/b$ with $b > 0$ and $p = m - 1 - r \geq 1$, put

$$T = a2^m + b(2^m - 1). \tag{E.3}$$

Then (47.7) is

$$u_m^{(r)}(a/b) = \frac{2^{m(m-1)/2}}{p!(b2^m)^p} \sum_{\substack{0 \leq n < 2^m \\ T - 2bn > 0}} \varepsilon_n (T - 2bn)^p. \tag{E.4}$$

The sum is an integer and can be accumulated before one final rational division. For a highest cellwise derivative, $p = 0$, use the same formula away from knots with the positive-part indicator convention. The supplied implementation treats values outside the support as zero before forming the signed sum.

Appendix F

Statement ledger for Part IV

Result	Precisely what is established
Global correction	Theorems 50.1 to 50.3: all fixed orders, global classical and Hölder norms, explicit constants, and sharp leading supremum asymptotics.
Increasing order	Theorem 50.4: a supergeometric absolute error certificate with a specified admissible order; no relative endpoint or numerical-conditioning claim.
Dyadic splines	Theorems 51.2, 51.4 and 51.5: exact polynomial stabilization, exact degree in lowest terms, finite extraction, and an order-independent absolute amplification bound.
Discrete array	Theorems 52.2, 53.2 and 54.1: finite differential conversion, a uniform all-orders local-limit bound, and exact stopping on fixed dyadic odd-level subsequences.
Energy moments	Theorems 55.2 to 55.4: exact moments, explicit distinct continuous and atomic moment twins, and hence moment indeterminacy.
Geometric extension	Theorem 56.1: exact separated-copy norms and the resulting moment class for $0 < \theta \leq 1/2$, not for overlapping copies.
Priority and verification	Proofs and finite exact computations are supplied; universal publication priority, peer review, and Lean formalization are not claimed.

Appendix G

Coefficient generation without hidden recurrences for Part V

For a requested saddle order n, formula (64.6) is a finite weighted partition sum. An implementation can loop over k_j with $0 \le k_j \le \lfloor 2n/(j-2) \rfloor$, discard assignments for which $\sum (j-2)k_j > 2n$, and set $\ell = 2n - \sum (j-2)k_j$. This directly generates the coefficient in the formal symbols $A_0, \ldots, A_{2n}$. The program's recursive enumeration is only a traversal of this finite index set; the mathematical coefficient formula does not depend on previously computed coefficients.

The derivatives A_j can themselves be expressed as finite combinations of $\Psi^{(r)}(\theta)$. Let $s = \log y$ and

$$B_\theta(s) = \mathrm{e}^{-s^2/(2a)} \Psi(\theta - s/a).$$

Since $\mathrm{d}/\mathrm{d}y = \mathrm{e}^{-s}\,\mathrm{d}/\mathrm{d}s$, induction gives

$$\mathcal{A}_\theta^{(\ell)}(1) = \left(\prod_{j=0}^{\ell-1} (D_s - j) \right) B_\theta(s) \Bigg|_{s=0}, \qquad D_s = \frac{\mathrm{d}}{\mathrm{d}s}. \tag{G.1}$$

For $\ell = 0$, the empty operator product is the identity. To prove the formula, write $\mathrm{d}^\ell/\mathrm{d}y^\ell = \mathrm{e}^{-\ell s} \prod_{j=0}^{\ell-1} (D_s - j)$ and differentiate once more; the derivative of $\mathrm{e}^{-\ell s}$ introduces the next factor $D_s - \ell$.

If $s(\ell, j)$ denotes the signed Stirling coefficient defined by $u(u-1)\cdots(u-\ell+1) = \sum_j s(\ell, j) u^j$, then

$$\mathcal{A}_\theta^{(\ell)}(1) = \sum_{j=0}^{\ell} s(\ell, j) \sum_{h=0}^{\lfloor j/2 \rfloor} \frac{j!}{h!(j-2h)!} \left(-\frac{1}{2a} \right)^h \left(-\frac{1}{a} \right)^{j-2h} \Psi^{(j-2h)}(\theta). \tag{G.2}$$

Indeed, expand the Gaussian factor in even powers of s, expand the translated Ψ by its finite derivative coefficient, and extract the j-th derivative at zero. Substituting (G.2) into (64.6) produces a fully explicit finite formula for each saddle coefficient in the derivatives of the periodic function.

Appendix H

Notation and normalization reference for Part V

Symbol	Meaning
ε_n	The sign $(-1)^{s_2(n)}$, with $\varepsilon_0 = 1$.
$E(z)$	The ordinary generating function $\sum_{n\geq 0} \varepsilon_n z^n$.
$\boldsymbol{h}, H, p$	A shift tuple with minimum zero, its maximum, and its length.
$D(\boldsymbol{h})$	The integer boundary functional in (14.7).
S_m, R_m	Noncyclic and cyclic sums over a block of length 2^m.
$\sigma(\Phi)$	The alternating binary digit sum, not the ordinary digit sum.
$Z_x(t)$	The signed exponential sum on a nonlinearly positioned Boolean cube.
$\mathcal{H}, \tau$	The family of nonempty monomial supports and its minimum cover size.
$\mathrm{Haf}(J)$	The sum of products over perfect matchings; no determinant signs.
$\mathcal{M}_G$	The weighted matching polynomial, including unmatched singleton weights.
$\Phi_q(z)$	The geometric product $\prod_{j\geq 0}(1 - zq^j)^{\varepsilon_j}$.
$P(x)$	The positive product $E(\mathrm{e}^{-x})$.
$\Phi(x)$	The Laplace transform of $\sum_{k\geq 1} 2^{-k} U_k$, with U_k uniform on $[0, 1]$.
$\Psi(v)$	The positive one-periodic function in (63.3).
$F(t, z)$	The positive logarithmic deviation $-\log \Phi_{\mathrm{e}^{-t}}(z)$ for real positive z.
a, β, L	Respectively $\log 2$, $-\log z$, and $\log(1/t)$.
w, r_*, θ	The Lambert parameter, saddle location, and periodic phase in (64.2).
$\Pi(t, z), c_n$	The saddle prefactor and its periodic coefficient functions.
$\mathcal{H}, \mathcal{P}$	The joint-endpoint integral and its exponential, distinct from the saddle prefactor $\Pi(t, z)$.

Appendix I

Source provenance, notation dictionary, and reproducibility

I.1 Sources and consolidation

This volume consolidates the six Thue–Morse reports filed in the repository's frontier intake of September 4, 2026, under `drafts/thue-morse/`: `Thue_Morse_Boundary_Corrections`, `Thue_Morse_Dyadic_Completion`, `thue_morse_research_article` (*Thue–Morse Mellin Renormalization and Gamma-Tower Uniqueness*), `Thue_Morse_Rational_Resonances`, `Thue_Mor se_Research` (*New Deductions from Thue–Morse Cancellation*), and `Thue_Morse_New_Directions`. Their LaTeX sources, PDFs, READMEs and audit notes were removed when this volume was filed; the repository history retains them. The verification programs, their recorded outputs, the figures, and the generated table were retained unchanged.

The correspondence between sources and parts is as follows. Part I is the boundary-corrections article, with the correlation chapter of the new-directions article appended, since its pair case is the Stern boundary term. Part II merges the dyadic-completion article (its spine) with the Mellin-renormalization article: the completion, its Mellin transform and periodic factor, the Dirichlet continuation, the real zeros and their slopes, the rescaling identity, the coefficient formulas, and the moment twins were proved in both; the first article's versions were kept because they carried the more complete proofs (the boundary argument for the nonconstancy of the profile, the zero-free periodic lemma, the Gaussian localization), and the second article's contributions were added where they went further: the uniform theorem with an explicit tube constant (Chapter 20), the real-order slopes and their dyadic equation, the Hankel growth (Section 25.4), the classification of completely monotone dyadic solutions, the universal coefficients and the beyond-all-orders ambiguity, the boundary uniqueness and the gamma-tower characterization, and the independent numerical evaluation. Part III is the rational-resonances article, Part IV the new-deductions article, and Part V the remaining chapters of the new-directions article. The preliminaries that all six sources restated (the signed sequence, the finite product, the generating function, Prouhet cancellation, the Laplace product with its flatness, the uniform-sum variable with its transform, its zeros and its Fabius functional equation, the Bernoulli expansions, and the Bell and partition polynomials) are stated once in Chapter 1; the sources' own versions were removed and their references redirected.

Before deduplicating, the following identifications were checked: the two completions $K = P\Phi$ and $W = QH$ are the same function, with the same dilation law, profile and Mellin shift; the two coefficient families $\beta_j(a)$ and $r_k(a)$ agree, as do $\alpha_j = 2^{j(j+1)/2}\beta_j$ and c_k once the sign of $b = 1/2 - a$ is aligned; the centered expansions agree term by term ($1/9$, $248/2025$, $4806656/2679075$); the slopes $Z_m(a) = M(-m, a)$ and $C_m(a)$ are the same integrals; the normalization constants $C_0 = K_0 = 0.75072198040787\ldots$ agree in both reports' runs; the entire-order theorem was proved in both and kept once; and the pair boundary functional $D(0, h)$ of the correlation chapter is the Stern value

$g(h) = B_h(-2)$ of Part I. No discrepancy was found.

I.2 Notation dictionary for the merged part

The merged Part II uses the first source's letters. The second source's letters, which appear in its retained verifier and its data files, are:

This volume	Mellin-renormalization report	Meaning
$P(t)$	$Q(t)$	the Laplace product $\prod_j(1-e^{-2^j t})$
$\Phi(z)$	$H(z)$	the Fabius transform $\mathbb{E}e^{-zX}$
$K(t)$	$\mathcal{W}(t)$	the completion $P\Phi$
$C(s)$, C_0	$K(s)$, K_0	the Mellin transform of K and its value at 0
$H(Lv)$, h_k	$A(v)$, a_k	the periodic profile and its Fourier coefficients
$h_\pm$	$A_\pm$	its bounds
L	λ	$\log 2$
$Z_\rho(a)$	$C_\rho(a)$	the slope $M(-\rho, a)$
$B_a(z)$, $\beta_k(a)$	$R_a(z)$, $r_k(a)$	the reciprocal transform and its coefficients
$\alpha_k(a)$	$c_k(a)$	$2^{k(k+1)/2}\beta_k(a)$
$b = 1/2 - a$	$b = a - 1/2$	the centering parameter (odd coefficients change sign)
$\mathcal{H}(4^{-m})$	Z_m	the normalized centered slope
$\Theta(\rho) = C_0 Q(-\rho)$	$\Theta(\rho)$	the real-order periodic amplitude

In Part I the letters K, H, Q of the boundary-corrections report were changed to $\mathcal{K}$, $\mathcal{G}$, $\mathcal{Q}$ to keep them distinct from Part II's, and its unit disk $\mathbb{D}$ is written out. In Part IV the sinc product $Q(t)$ of the new-deductions report is written $\widehat{u}(t)$, the transform of the up-function. In Part V the letters A, Q, B, $P(t,z)$, L of the new-directions report are P, Φ, K, $\Pi(t,z)$, L, so that its product, Laplace transform and completion are visibly those of Chapter 1 and Part II; its periodic amplitude satisfies $\Psi(v) = H(-av)$. Everything else keeps its source notation, listed in Section 1.6.

I.3 Retained verification programs

The six programs live under `verification/` in the package directory, unchanged, with the outputs of the runs the sources recorded:

- `source1/` (Part I): `verify.py`, standard library only, 19,037 exact rational checks recorded in `verification.json`; `make_figures.py` and `poisson_data.json` regenerate the figures and the Poisson table (`tables/bc_poisson_table.tex`) with `mpmath`, numpy and `matplotlib`.
- `source2/` (Part II, dyadic completion): `code/verify.py` and `code/check_original_mellin.py` with `code/requirements.txt`; outputs in `data/` at 60 and 80 digits, including the completion constant and the displayed zero.
- `source3/` (Part II, Mellin renormalization): `verify_results.py` with `mpmath` and sympy; `results_48.json` and `results_80.json`.
- `source4/` (Part III): `verify.py` working in $\mathbb{Q}[x]/\Phi_q(x)$; `requirements.txt`; `results/` with the verification report and the four error tables.
- `source5/` (Part IV): `verify.py`, standard library only, 3,796 exact checks recorded in `verification_results.json`.
- `source6/` (Parts I and V): `verify.py` with `mpmath` and sympy; `verification_results.json`, `verification_run.txt`, the four CSV tables and `environment.json`.

Each is run with `python` from inside its directory as its part's verification section describes. The programs recompute their certificates; they do not read the saved reports. They are finite checks of the implementation and formulas, not replacements for the proofs.

I.4 Formalization status

No Lean files were added or compiled for this volume. Each part ends with the formalization route its source proposed; the atlas names the modules (`ThueMorseBoundaryFlatness.lean`, `ThueMorseEntireContinuation.lean`, and the finite-product bridges) on which such work would build.

I.5 Files and commands

The package directory contains `Thue_Morse_Frontier_Deductions.tex`, its compiled PDF, `README.md`, `figures/`, `tables/`, and `verification/`. The volume loads `docs/fabius-notation.tex` by relative path; with a TeX installation containing Libertinus (or, as a fallback, Latin Modern) and the listed standard packages, compile with three passes of

```
pdflatex -interaction=nonstopmode -halt-on-error \
  Thue_Morse_Frontier_Deductions.tex
```

Bibliography

[1] Vladimir Reshetnikov, *The Thue–Morse Sequence: Formula Atlas and Fabius–Rvachev Frontier Results*, consolidated August 28, 2026, ProveIt repository. Sources consulted September 4, 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/thue-morse/Thue_Morse_Atlas_and_Frontiers.

[2] R. Aguech and W. Jedidi, "New characterizations of completely monotone functions and Bernstein functions, a converse to Hausdorff's moment characterization theorem," *Arab Journal of Mathematical Sciences* **25** (2019), 57–82.

[3] R. Arratia, L. Goldstein, and F. Kochman, "Size bias for one and all," *Probability Surveys* **16** (2019), 1–61, especially Section 9. https://doi.org/10.1214/13-PS221.

[4] G. Alkauskas, *Dirichlet series associated with the Thue–Morse sequence*, Vilnius University preprint 2001-15 (2001). Bibliographical reference only: the full text was not retrieved by either source that cites it. https://web.vu.lt/mif/g.alkauskas/math/index.html.

[5] J.-P. Allouche, "Thue, combinatorics on words, and conjectures inspired by the Thue–Morse sequence," *Journal de Théorie des Nombres de Bordeaux* **27** (2015), 375–388; arXiv:1401.3727, Section 3.2. https://arxiv.org/abs/1401.3727.

[6] J.-P. Allouche and H. Cohen, "Dirichlet series and curious infinite products," *Bulletin of the London Mathematical Society* **17** (1985), 531–538. https://doi.org/10.1112/blms/17.6.531.

[7] J.-P. Allouche and J. Shallit, "The ubiquitous Prouhet–Thue–Morse sequence," in C. Ding, T. Helleseth, and H. Niederreiter (eds.), *Sequences and Their Applications*, Springer, 1999, pp. 1–16. https://cs.uwaterloo.ca/~shallit/Papers/ubiq15.pdf.

[8] J.-P. Allouche, M. Mendès France, and J. Peyrière, "Automatic Dirichlet series," *Journal of Number Theory* **81** (2000), 359–373. https://doi.org/10.1006/jnth.1999.2487.

[9] J. Arias de Reyna, "An infinitely differentiable function with compact support: definition and properties," *Revista de la Real Academia de Ciencias Exactas, Físicas y Naturales de Madrid* **76** (1982), 21–38; English translation arXiv:1702.05442.

[10] J. Arias de Reyna, "Arithmetic of the Fabius function," *Integers* **18** (2018), Paper A51. https://math.colgate.edu/~integers/s51/s51.pdf.

[11] J.-P. Allouche, S. Riasat, and J. Shallit, "More infinite products: Thue–Morse and the gamma function," *Ramanujan Journal* **49** (2019), 115–128. https://arxiv.org/abs/1709.03398.

[12] M. Baake and M. Coons, "Correlations of the Thue–Morse sequence," *Indagationes Mathematicae* **35** (2024), 914–930. https://arxiv.org/abs/2209.07102.

[13] M. Baake and U. Grimm, "The singular continuous diffraction measure of the Thue–Morse chain," *Journal of Physics A* **41** (2008), 422001. https://arxiv.org/abs/0809.0580.

[14] E. D. Bolker, C. Offner, R. Richman, and C. Zara, "The Prouhet–Tarry–Escott problem and generalized Thue–Morse sequences," *Journal of Combinatorics* **7** (2016), 117–133. https://arxiv.org/abs/1304.6756.

[15] B. Cloitre, *The Thue–Morse transform*, arXiv:2604.06243v2 (May 2026). Cited for related transform and Prouhet-partition context. https://arxiv.org/abs/2604.06243.

[16] M. Coons, J. Mazáč, A. Pincus–Kazmar, and A. Stout, "On the absolute value of the autocorrelations of the Thue–Morse sequence," *Experimental Mathematics*, published online January 31, 2026. https://arxiv.org/abs/2511.06386.

[17] NIST Digital Library of Mathematical Functions, §4.13, *Lambert W-function*. https://dlmf.nist.gov/4.13.

[18] NIST Digital Library of Mathematical Functions, §4.22 and §24.16, infinite trigonometric products and generalized Bernoulli polynomials. https://dlmf.nist.gov/4.22; https://dlmf.nist.gov/24.16.

[19] W. Duke and H. N. Nguyen, "Infinite products of cyclotomic polynomials," *Bulletin of the Australian Mathematical Society* **91** (2015), 400–411.

[20] J. Fabius, "A probabilistic example of a nowhere analytic C^∞-function," *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* **5** (1966), 173–174.

[21] P. Flajolet, X. Gourdon, and P. Dumas, "Mellin transforms and asymptotics: harmonic sums," *Theoretical Computer Science* **144** (1995), 3–58.

[22] P. Gohlke, M. Kesseböhmer, and T. I. Schindler, *Generalized Thue–Morse measures: spectral and fractal analysis*, arXiv:2509.22109 (2025).

[23] J. K. Haugland, *Evaluating the Fabius function*, arXiv:1609.07999, version 2 (2020).

[24] C. C. Heyde, "On a property of the lognormal distribution," *Journal of the Royal Statistical Society, Series B* **25** (1963), 392–393.

[25] S. Klavžar, U. Milutinović, and C. Petr, "Stern polynomials," *Advances in Applied Mathematics* **39** (2007), 86–95.

[26] Y.-Q. Li, *Infinite products related to generalized Thue–Morse sequences*, arXiv:2006.04187 (2020).

[27] C. Mauduit, H. L. Montgomery, and J. Rivat, "Moments of a Thue–Morse generating function," *Journal d'Analyse Mathématique* **135** (2018), 713–724.

[28] B. Sobolewski, "Cyclotomic properties of polynomials associated with automatic sequences," *Advances in Applied Mathematics* **120** (2020), 102063. https://arxiv.org/abs/1909.12052.

[29] L. Tóth, "Linear combinations of Dirichlet series associated with the Thue–Morse sequence," *Integers* **22** (2022), A98. https://arxiv.org/abs/2211.13570.

[30] M. A. Zaks, A. S. Pikovsky, and J. Kurths, "On the correlation dimension of the spectral measure for the Thue–Morse sequence," *Journal of Statistical Physics* **88** (1997), 1387–1392.